

Dokumentation

Documentation

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Gear Unit

H.SH, H.VH, H.HH, H.DH, H.KH, H.FH, H.HM,
H.DM, H.KM, H.FM, H.PH, B.SH, B.VH, B.HH,
B.DH, B.KH, B.FH, B.HM, B.DM, B.KM, B.FM,
T.SH, T.HH, T.KH, T.DH, T.FH
Sizes 1 to 22

Assembly and operating instructions
BA 5010 EN 06/2010

FLENDER gear units

SIEMENS

Gear Unit

H.SH, H.VH, H.HH, H.DH, H.KH, H.FH,
H.HM, H.DM, H.KM, H.FM, H.PH, B.SH,
B.VH, B.HH, B.DH, B.KH, B.FH, B.HM,
B.DM, B.KM, B.FM, T.SH, T.HH, T.KH,
T.DH, T.FH
Sizes 1 to 22

Assembly and operating instructions

Translation of the original assembly and operating instructions

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Notes and symbols in these assembly and operating instructions

Note: The term "Assembly and operating instructions" will in the following also be shortened to "instructions" or "manual".

Legal notes

Warning note concept

This manual comprises notes which must be observed for your personal safety and for preventing material damage. Notes for your personal safety are marked with a warning triangle or an "Ex" symbol (when applying Directive 94/9/EC), those only for preventing material damage with a "STOP" sign.



WARNING! Imminent explosion!

The notes indicated by this symbol are given to prevent **explosion damage**.
Disregarding these notes may result in serious injury or death.



WARNING! Imminent personal injury!

The notes indicated by this symbol are given to prevent **personal injury**.
Disregarding these notes may result in serious injury or death.



WARNING! Imminent damage to the product!

The notes indicated by this symbol are given to prevent **damage to the product**.
Disregarding these notes may result in material damage.



NOTE!

The notes indicated by this symbol must be treated as general **operating information**.
Disregarding these notes may result in undesirable results or conditions.



WARNING! Hot surfaces!

The notes indicated by this symbol are made to prevent **risk of burns due to hot surfaces** and must always be observed.
Disregarding these notes may result in light or serious injury.

Where there is more than one hazard, the warning note for whichever hazard is the most serious is always used. If in a warning note a warning triangle is used to warn of possible personal injury, a warning of material damage may be added to the same warning note.

Qualified personnel

The product or system to which these instructions relate may be handled only by persons qualified for the work concerned and in accordance with the instructions relating to the work concerned, particularly the safety and warning notes contained in those instructions. Qualified personnel must be specially trained and have the experience necessary to recognise risks associated with these products or systems and to avoid possible hazards.

Intended use of Siemens products

Observe also the following:



Siemens products must be used only for the applications provided for in the catalogue and the relevant technical documentation. If products and components of other makes are used, they must be recommended or approved by Siemens. The faultfree, safe operation of the products calls for proper transport, proper storage, erection, assembly, installation, start-up, operation and maintenance. The permissible ambient conditions must be adhered to. Notes in the relevant documentations must be observed.

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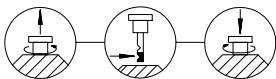
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Exclusion of liability

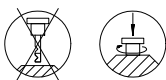
We have checked the content of the instructions for compliance with the hard- and software described. Nevertheless, variances may occur, and so we can offer no warranty for complete agreement. The information given in these instructions is regularly checked, and any necessary corrections are included in subsequent editions.

Symbols

Earth connection point		Air relief point		yellow	
Oil-filling point		yellow	Oil-draining point		white
Oil level		red	Oil level		red
Oil overflow			Connection for vibration-monitoring device		
Lubrication point		red	Apply grease		
Lifting eye			Eye bolt		
Do not unscrew					
Alignment surface, horizontal			Alignment surface, vertical		



These symbols indicate the oil-level checking procedure using the oil dipstick.



These symbols indicate that the oil dipstick must always be firmly screwed in.

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1. Technical data

1.1 General technical data

The most important technical data are shown on the rating plate. These data and the contractual agreements between Siemens and the customer for the gear unit determine the limits of its correct use.

①	
②	
③	④
⑤	⑥
⑦	⑧
⑨	
⑩	⑪
⑫	
⑬	

Fig. 1: Rating plate gear unit

- | | |
|---|---|
| ① Company logo | ⑧ Speed n_2 |
| ② Order number, item, sequence number, year built | ⑨ Oil data
(oil type, oil viscosity, oil quantity) |
| ③ Total weight in kg | ⑩ Instructions number(s) |
| ④ Special information | ⑪ Special information |
| ⑤ Type, size *) | ⑫ Manufacturer and place of manufacture |
| ⑥ Power rating P_2 in kW or torque T_2 in Nm | ⑬ Country of origin |
| ⑦ Speed n_1 | |

*) Example

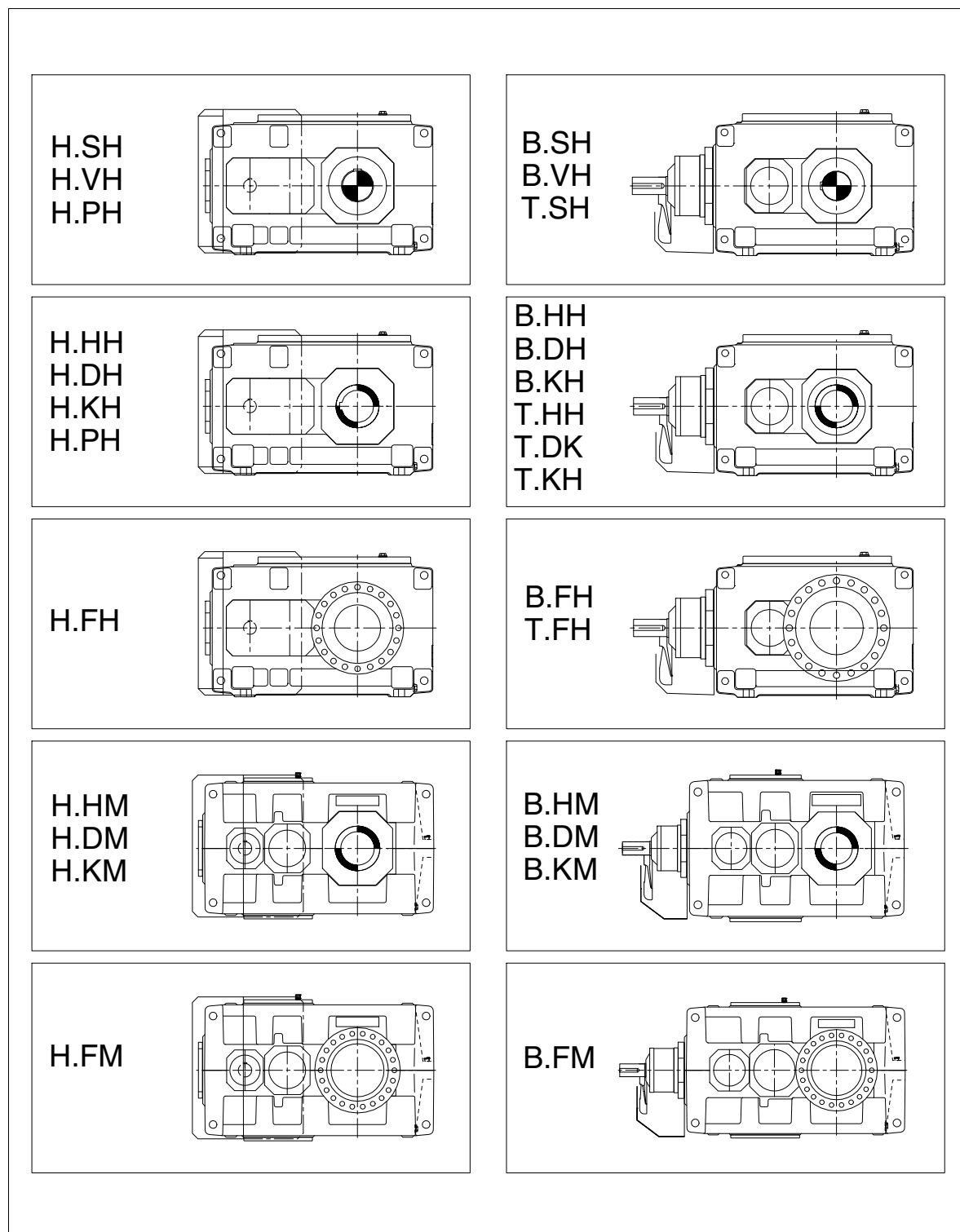
B	3	S	H	13	
					Size 1 ... 22
					Installation H = Horizontal M = Horizontal design without base (from size 13)
					Type of output shaft S = Solid shaft V = Solid shaft, reinforced H = Hollow shaft with parallel keyway D = Hollow shaft for shrink disk K = Hollow shaft with internal spline to DIN 5480 F = Flanged shaft P = Design paper-processing machine
					Number of stages 1, 2, 3 or 4
					Gear-unit type H = Helical gear unit B = Bevel-helical gear unit (2, 3 or 4 stages only) T = Bevel-helical gear unit with split housing (sizes 4 to 12)

Data on weights and measuring-surface sound-pressure levels of the various gear types are given in items 1.2.2 and/or 1.2.3.

For further technical data, refer to the drawings in the gear-unit documentation.

1.2 Configurations and weights

1.2.1 Types



1.2.2 Weights

Table 1: Weights (approximate values)

Type	Approx. weight (kg) for size											
	1	2	3	4	5	6	7	8	9	10	11	12
H1SH	55	-	128	-	302	-	547	-	862	-	1515	-
H2PH	-	-	-	-	340	-	550	-	860	-	1360	-
H2.H	-	-	115	190	300	355	505	590	830	960	1335	1615
H2.M	-	-	-	-	-	-	-	-	-	-	-	-
H3.H	-	-	-	-	320	365	540	625	875	1020	1400	1675
H3.M	-	-	-	-	-	-	-	-	-	-	-	-
H4.H	-	-	-	-	-	-	550	645	875	1010	1460	1725
H4.M	-	-	-	-	-	-	-	-	-	-	-	-
B2.H	50	82	140	235	360	410	615	700	1000	1155	1640	1910
B2.M	-	-	-	-	-	-	-	-	-	-	-	-
B3.H	-	-	130	210	325	380	550	635	890	1020	1455	1730
B3.M	-	-	-	-	-	-	-	-	-	-	-	-
B4.H	-	-	-	-	335	385	555	655	890	1025	1485	1750
B4.M	-	-	-	-	-	-	-	-	-	-	-	-

Type	Approx. weight (kg) for size									
	13	14	15	16	17	18	19	20	21	22
H1SH	2395	-	3200	-	4250	-	5800	-	-	-
H2PH	-	-	-	-	-	-	-	-	-	-
H2.H	2000	2570	3430	3655	4650	5125	6600	7500	8900	9600
H2.M	1880	2430	3240	3465	4420	4870	6300	7200	8400	9200
H3.H	2295	2625	3475	3875	4560	5030	6700	8100	9100	9800
H3.M	2155	2490	3260	3625	4250	4740	6200	7600	8500	9300
H4.H	2390	2730	3635	3965	4680	5185	6800	8200	9200	9900
H4.M	2270	2600	3440	3740	4445	4915	6300	7700	8600	9400
B2.H	2450	2825	3990	4345	5620	6150	-	-	-	-
B2.M	2350	2725	3795	4160	5320	5860	-	-	-	-
B3.H	2380	2750	3730	3955	4990	5495	7000	8100	9200	9900
B3.M	2260	2615	3540	3765	4760	5240	6500	7600	8600	9400
B4.H	2395	2735	3630	3985	4695	5200	6800	8200	9200	9900
B4.M	2280	2605	3435	3765	4460	4930	6300	7700	8600	9400



All weights are for units without oil filling and add-on parts. For the exact weights, refer to the drawings in the gear-unit documentation.

Table 2: Total weights (approximate values) for gear units including auxiliary drive (maintenance drive)

Type	Approx. weight (kg) for size								
	4	5	6	7	8	9	10	11	12
T3.H	262	377	427	630	710	1015	1135	1595	1860
B3.H	272	392	447	655	740	1055	1185	1665	1940

Type	Approx. weight (kg) for size									
	13	14	15	16	17	18	19	20	21	22
B3.H	2700	3070	4110	4335	5370	5875	6740	7450	9080	9840

Table 3: Weights (approximate values) for gear units including auxiliary drive (load drive)

Type	Approx. weight (kg) for size								
	4	5	6	7	8	9	10	11	12
T3.H	285	432	482	670	750	1090	1210	1775	2040
B3.H	295	447	502	695	780	1130	1260	1845	2120

Type	Approx. weight (kg) for size									
	13	14	15	16	17	18	19	20	21	22
B3.H	2930	3300	4450	4675	5920	6425	7100	8000	9730	10490



All weights are for units without oil charge, however with fitted auxiliary drive, including motor of the auxiliary drive.
For the exact weights, refer to the drawings in the gear-unit documentation.

1.2.3 Measuring-surface sound-pressure level

The gear unit has a measuring-surface sound-pressure level at a distance of 1 m, which can be found in tables 4 to 7.

The measurement is carried out to DIN EN ISO 9614 Part 2, using the sound-intensity method.

The workplace of the operating personnel is defined as the area on the measuring-surface at a distance of 1 metre in the vicinity of which persons may be present.

The sound-pressure level applies to the warmed-up gear unit at input speed n_1 and output power P_2 stated on the rating plate, as measurement obtained on the Siemens test bench. If several figures are given, the highest speed and power values apply.

The measuring-surface sound-pressure level includes add-on lubrication units, if applicable. With outgoing and incoming pipes, the interfaces are the flanges.

The sound-pressure levels stated in the table were obtained by statistical calculation by our Quality Control Dept. The gear unit can be statistically expected to comply with these sound-pressure levels.

1.2.3.1 Measuring-surface sound-pressure level for bevel-helical gear units (B...) with fan

Table 4: Measuring-surface sound-pressure level L_{pA} in dB(A) for bevel-helical gear units with fan

Type	i _N	n ₁ 1/min	Gear-unit size																					
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
B2	5	1500	71	72	73	76	79	81	83	84	85	87	88	89	91	92	94	-	-	-	-	-	-	-
	.	1000	66	66	67	71	73	74	77	78	79	80	82	83	84	85	87	89	90	-	-	-	-	-
	8	750	1)	60	61	64	66	67	70	71	72	73	75	76	77	78	81	82	83	85	-	-	-	-
	9	1500	68	69	70	73	75	76	78	81	82	83	84	85	86	87	88	90	-	-	-	-	-	-
	.	1000	61	62	63	67	68	70	73	74	75	77	79	80	81	82	83	84	86	87	-	-	-	-
	14	750	1)	60	1)	61	62	64	66	67	68	70	72	73	74	75	77	78	79	80	-	-	-	-
	16	1500	65	66	67	71	74	76	78	79	80	81	83	84	87	88	89	90	-	-	-	-	-	-
	.	1000	1)	1)	60	64	67	68	70	72	73	74	78	79	80	81	82	83	84	84	-	-	-	-
	22.4	750	1)	1)	1)	1)	61	63	65	67	68	69	71	72	73	73	74	74	75	76	-	-	-	-
	T3 B3	12.5	1500	-	-	69	72	75	77	79	80	81	82	83	85	88	89	90	91	93	93	93	93	95
.		1000	-	-	62	65	68	69	71	72	73	74	77	78	80	82	83	83	84	85	86	86	88	88
31.5		750	-	-	1)	1)	63	64	66	68	69	70	71	73	74	75	76	77	78	78	79	79	81	81
35.5		1500	-	-	67	69	72	73	74	75	77	79	82	84	86	87	88	89	90	91	92	92	93	93
.		1000	-	-	1)	63	65	66	67	69	71	72	73	75	77	78	79	80	81	82	83	84	85	86
56		750	-	-	1)	1)	1)	1)	62	64	65	67	69	70	71	72	73	74	75	76	77	78	79	79
63		1500	-	-	66	68	70	71	73	74	76	78	81	83	85	86	87	88	89	90	91	91	92	92
.		1000	-	-	1)	61	63	64	66	68	69	71	73	75	77	78	79	80	81	81	82	82	83	84
90		750	-	-	1)	1)	1)	1)	61	63	64	66	67	68	70	71	72	73	74	75	75	76	77	77

¹⁾ $L_{pA} < 60$ dB(A)

1.2.3.2 Measuring-surface sound-pressure level for bevel-helical gear units (B...) without fan

Table 5: Measuring-surface sound-pressure level L_{pA} in dB(A) for bevel-helical gear units without fan

Type	i _N	n ₁ 1/min	Gear-unit size																					
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
B2	5	1500	70	71	72	75	78	80	82	83	84	86	87	88	89	90	93	-	-	-	-	-	-	-
	.	1000	64	65	66	70	72	73	76	77	78	79	81	82	83	84	86	88	89	-	-	-	-	-
	8	750	1)	1)	1)	63	65	66	69	71	72	73	74	75	77	78	80	82	83	84	-	-	-	-
	9	1500	65	66	67	71	74	75	77	79	80	81	83	84	85	86	87	89	-	-	-	-	-	-
	.	1000	59	60	61	65	67	69	72	73	74	76	77	78	80	81	82	83	85	86	-	-	-	-
	14	750	1)	1)	1)	1)	60	63	65	66	67	69	71	72	73	74	76	77	78	79	-	-	-	-
	16	1500	62	65	63	66	69	71	72	74	75	77	78	80	81	82	85	85	-	-	-	-	-	-
	.	1000	1)	1)	1)	61	63	65	67	68	69	71	72	74	75	77	79	80	81	81	-	-	-	-
	22.4	750	1)	1)	1)	1)	1)	1)	60	62	63	64	66	67	68	70	72	73	74	75	-	-	-	-
T3 B3	12.5	1500	-	-	65	68	71	74	75	76	77	79	81	83	84	85	86	87	87	88	89	90	91	92
	.	1000	-	-	1)	63	66	68	69	70	72	73	75	77	78	80	80	81	82	82	84	85	86	86
	31.5	750	-	-	1)	1)	1)	61	62	64	65	66	68	71	71	73	73	74	75	75	77	78	79	79
	35.5	1500	-	-	60	65	67	70	71	71	72	74	77	79	80	81	82	83	83	84	86	86	88	88
	.	1000	-	-	1)	1)	62	65	65	66	66	69	71	73	75	76	76	77	77	78	80	81	82	83
	56	750	-	-	1)	1)	1)	1)	1)	1)	62	65	67	68	69	70	70	71	72	74	74	75	76	
	63	1500	-	-	1)	61	64	70	67	68	68	70	73	75	76	78	78	79	79	80	82	83	84	84
	.	1000	-	-	1)	1)	1)	63	62	62	62	65	68	70	71	72	73	73	74	75	76	77	78	79
	90	750	-	-	1)	1)	1)	1)	1)	1)	1)	61	63	64	65	66	67	67	68	70	70	72	72	
B4	80	1500	-	-	-	-	64	65	67	68	70	72	75	76	77	79	80	81	82	83	84	85	86	86
	.	1000	-	-	-	-	1)	1)	61	63	64	67	69	70	72	73	74	75	76	77	78	79	80	80
	125	750	-	-	-	-	1)	1)	1)	1)	1)	62	64	65	66	68	68	69	71	71	72	73	74	
	140	1500	-	-	-	-	60	61	63	65	66	68	71	72	73	75	76	77	78	79	80	81	82	82
	.	1000	-	-	-	-	1)	1)	1)	1)	61	63	65	67	68	69	71	71	72	74	75	75	76	77
	224	750	-	-	-	-	1)	1)	1)	1)	1)	1)	1)	1)	61	62	64	65	66	67	68	69	69	70
	250	1500	-	-	-	-	1)	1)	1)	62	63	65	67	69	70	71	73	73	75	76	77	77	78	79
	.	1000	-	-	-	-	1)	1)	1)	1)	1)	62	63	64	66	67	68	69	70	71	72	73	73	
	400	750	-	-	-	-	1)	1)	1)	1)	1)	1)	1)	1)	1)	1)	61	62	63	64	65	66	66	

¹⁾ $L_{pA} < 60$ dB(A)

1.2.3.3 Measuring-surface sound-pressure level for helical-gear units (H...) with fan

Table 6: Measuring-surface sound-pressure level L_{pA} in dB(A) for helical-gear units with fan

Type	i_N	n_1 1/min	Gear-unit size																					
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
H1	1.25	1500	-	-	76	-	81	-	84	-	87	-	91	-	-	-	-	-	-	-	-	-	-	-
	.	1000	-	-	71	-	76	-	79	-	81	-	83	-	85	-	-	-	-	-	-	-	-	-
	2	750	-	-	67	-	72	-	75	-	78	-	80	-	82	-	85	-	-	-	-	-	-	-
	2.24	1500	-	-	73	-	79	-	82	-	84	-	89	-	90	-	-	-	-	-	-	-	-	-
	.	1000	-	-	68	-	74	-	77	-	79	-	82	-	84	-	87	-	-	-	-	-	-	-
	3.55	750	-	-	64	-	70	-	72	-	75	-	78	-	80	-	83	-	84	-	-	-	-	-
	4	1500	-	-	70	-	77	-	81	-	83	-	86	-	89	-	93	-	-	-	-	-	-	-
	.	1000	-	-	65	-	71	-	75	-	77	-	80	-	82	-	84	-	85	-	87	-	-	-
	5.6	750	-	-	61	-	68	-	71	-	72	-	75	-	77	-	79	-	81	-	83	-	-	-
H2	6.3	1500	-	-	-	75	76	77	80	81	82	84	85	86	88	90	92	94	96	96	-	-	-	-
	.	1000	-	-	-	69	71	72	74	75	77	79	80	81	83	84	85	86	87	88	88	89	90	-
	10	750	-	-	-	66	68	69	70	72	73	75	76	77	79	80	81	82	83	83	84	84	85	85
	11.2	1500	-	-	-	73	75	77	79	80	81	82	85	88	90	91	92	93	95	95	-	-	-	-
	.	1000	-	-	-	68	69	70	72	73	75	77	79	80	82	83	84	85	85	86	86	87	87	87
	16	750	-	-	-	64	66	67	69	70	71	73	74	76	78	79	79	80	81	81	82	82	83	83
	18	1500	-	-	-	71	73	75	77	78	80	82	84	86	87	90	91	92	93	94	94	95	95	95
	.	1000	-	-	-	65	67	68	71	72	73	75	77	78	80	81	82	83	83	84	85	85	86	86
	28	750	-	-	-	62	64	65	67	68	69	71	73	74	75	77	78	79	79	80	80	81	81	81
H3	22.4	1500	-	-	-	-	71	72	75	75	77	77	80	80	81	81	84	84	84	85	-	-	-	-
	.	1000	-	-	-	-	65	66	69	70	71	72	74	75	75	75	78	78	78	79	-	-	-	-
	35.5	750	-	-	-	-	62	62	66	67	67	68	70	70	71	72	74	74	75	76	-	-	-	-
	40	1500	-	-	-	-	70	71	73	74	76	76	79	79	80	80	83	82	83	83	-	-	-	-
	.	1000	-	-	-	-	64	65	67	68	69	70	73	73	73	74	77	77	77	77	-	-	-	-
	63	750	-	-	-	-	62	62	63	64	65	66	69	69	69	70	72	73	73	73	-	-	-	-
	71	1500	-	-	-	-	70	70	72	72	75	75	78	78	78	78	82	82	82	82	-	-	-	-
	.	1000	-	-	-	-	64	64	65	66	68	69	71	72	72	72	75	75	75	76	-	-	-	-
	112	750	-	-	-	-	61	61	62	62	64	65	67	67	68	68	71	71	71	72	-	-	-	-

1.2.3.4 Measuring-surface sound-pressure level for helical-gear units (H...) without fan

Table 7: Measuring-surface sound-pressure level L_{pA} in dB(A) for helical-gear units without fan

Type	i_N	n_1 1/min	Gear-unit size																					
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
H1	1.25	1500	75	-	73	-	77	-	79	-	81	-	83	-	-	-	-	-	-	-	-	-	-	-
	.	1000	70	-	69	-	72	-	75	-	76	-	78	-	80	-	-	-	-	-	-	-	-	-
	2	750	66	-	65	-	69	-	71	-	73	-	75	-	77	-	79	-	-	-	-	-	-	-
	2.24	1500	72	-	70	-	75	-	77	-	79	-	81	-	83	-	-	-	-	-	-	-	-	-
	.	1000	67	-	66	-	70	-	72	-	74	-	76	-	78	-	80	-	-	-	-	-	-	-
	3.55	750	63	-	62	-	67	-	68	-	71	-	73	-	75	-	77	-	79	-	-	-	-	-
	4	1500	69	-	67	-	72	-	74	-	76	-	78	-	79	-	82	-	-	-	-	-	-	-
	.	1000	64	-	1)	-	67	-	70	-	71	-	73	-	75	-	77	-	79	-	81	-	-	-
	5.6	750	60	-	1)	-	63	-	66	-	67	-	70	-	71	-	74	-	76	-	78	-	-	-
H2	6.3	1500	-	-	-	71	74	75	76	77	79	79	80	81	81	82	84	85	85	86	-	-	-	-
	.	1000	-	-	-	66	69	70	71	72	74	74	75	76	76	77	80	80	80	81	83	83	84	-
	10	750	-	-	-	63	66	67	67	69	70	71	72	73	73	74	76	77	77	78	80	80	81	81
	11.2	1500	-	-	-	69	72	73	74	75	77	77	78	79	79	80	82	83	83	84	-	-	-	-
	.	1000	-	-	-	64	67	68	69	70	72	72	73	74	74	75	77	78	78	79	81	81	82	82
	16	750	-	-	-	61	64	65	66	67	69	69	70	71	71	72	74	75	75	76	77	78	79	79
	18	1500	-	-	-	66	69	70	71	72	74	74	75	76	77	78	80	80	81	82	83	84	84	85
	.	1000	-	-	-	61	64	65	66	68	69	69	70	71	72	73	75	75	76	77	78	79	79	80
	28	750	-	-	-	1)	61	62	63	64	66	66	67	68	69	70	72	72	73	73	75	75	76	76
H3	22.4	1500	-	-	-	-	68	69	73	74	74	75	77	77	78	79	81	81	82	83	83	84	85	86
	.	1000	-	-	-	-	63	65	68	69	69	71	72	73	73	74	76	77	77	78	79	79	81	81
	31.5	750	-	-	-	-	60	61	65	66	65	67	69	69	70	71	73	73	74	75	75	76	77	78
	35.5	1500	-	-	-	-	65	67	70	71	71	73	74	75	76	76	78	79	79	80	81	81	83	83
	.	1000	-	-	-	-	1)	62	65	66	66	68	69	70	71	72	73	74	75	75	76	77	78	78
	63	750	-	-	-	-	1)	1)	62	63	63	65	66	67	67	68	70	71	71	72	73	73	75	75
	71	1500	-	-	-	-	62	64	67	68	68	70	71	72	73	74	76	76	77	78	78	79	80	81
	.	1000	-	-	-	-	1)	1)	62	63	63	65	66	67	68	69	71	71	72	73	73	74	75	76
	112	750	-	-	-	-	1)	1)	1)	1)	1)	62	63	64	65	66	68	68	69	70	70	71	72	72
H4	100	1500	-	-	-	-	-	-	66	67	68	69	70	71	72	73	75	75	76	76	77	78	78	78
	.	1000	-	-	-	-	-	-	62	63	63	64	65	66	67	68	70	70	71	72	72	73	73	74
	140	750	-	-	-	-	-	-	1)	1)	1)	61	62	63	64	64	66	67	68	68	69	69	70	70
	160	1500	-	-	-	-	-	-	64	65	66	66	68	68	69	70	72	73	73	74	74	75	75	76
	.	1000	-	-	-	-	-	-	1)	60	61	62	63	64	64	65	67	68	68	69	70	70	71	71
	250	750	-	-	-	-	-	-	1)	1)	1)	1)	60	61	61	62	64	64	65	66	66	67	67	68
	280	1500	-	-	-	-	-	-	61	62	63	64	65	66	67	67	69	70	70	71	72	72	73	73
	.	1000	-	-	-	-	-	-	1)	1)	1)	1)	60	61	62	63	64	65	66	66	67	68	68	68
	450	750	-	-	-	-	-	-	1)	1)	1)	1)	1)	1)	1)	1)	61	62	62	63	64	64	65	65

¹⁾ $L_{pA} < 60$ dB(A)

2. General notes

2.1 Introduction

These instructions are an integral part of the gear unit supplied and must be kept in its vicinity for reference at all times.



All persons carrying out work on the gear unit must have read and understood these instructions and must adhere to them. Siemens accepts no responsibility for damage or disruption caused by disregard of these instructions.

The "**FLENDER gear unit**" dealt with in these instructions has been developed for driving machines in general engineering applications. Possible applications for gear units of this series are the chemical, rubber, food processing, plastics and other industries.

The gear unit is designed only for the application specified in section 1, "Technical data". Other operating conditions must be contractually agreed.

The gear unit has been manufactured in accordance with the state of the art and is delivered in a condition for safe and reliable use.

The gear unit must be used and operated strictly in accordance with the conditions laid down in the contract governing performance and supply agreed by Siemens and the customer.

The gear unit described in these instructions reflects the state of technical development at the time these instructions went to print.

In the interest of technical progress we reserve the right to make changes to the individual assemblies and accessories which we regard as necessary to preserve their essential characteristics and improve their efficiency and safety.

2.2 Copyright

The copyright to these instructions is held by **Siemens AG**.

These instructions must not be wholly or partly reproduced for competitive purposes, used in any unauthorised way or made available to third parties without our agreement.

Technical enquiries should be addressed to the following works or to one of our customer services:

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Thierbacher Straße 24
09322 Penig

Tel.: +49 (0)37381 / 61-0
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3. Safety instructions



Entry to the gear unit is not permitted during operation!
Entry for maintenance and repair work is only permitted when the gear unit is at a standstill!
Caution, risk of falling!



Any changes on the part of the user are not permitted. This applies equally to safety features designed to prevent accidental contact.

3.1 Obligations of the user

- The operator must ensure that everyone carrying out work on the gear unit has read and understood these instructions and is adhering to them in every point in order to:
 - avoid injury or damage,
 - ensure the safety and reliability of the unit,
 - avoid disruptions and environmental damage through incorrect use.
- During transport, assembly, installation, dismantling, operation and maintenance of the unit, the relevant safety and environmental regulations must be complied with at all times.
- The gear unit must be operated, maintained and/or repaired only by authorised, properly trained and qualified personnel.
- The outside of the gear unit must not be cleaned with high-pressure cleaning equipment.
- All work must be carried out with great care and with due regard to safety.



All work on the gear unit must be carried out only when it is not in operation.
The drive unit must be secured against being switched on accidentally (e.g. by locking the key switch or removing the fuses from the power supply). A notice should be attached to the start switch stating clearly that work is in progress.
At the same time the complete installation must be without load, so that no danger occurs during demounting operations (e.g. change of backstop).

- No electrical welding work must be done at all on the drive.
The drives must not be used as an earthing point for welding operations. Toothed parts and bearings may be irreparably damaged by welding.
- A potential equalisation in accordance with the applying regulations and directives must be carried out! If no threaded holes for earth connection are available on the gear unit, other appropriate measures must be taken. This work must always be done by electrotechnical specialists.



If any inexplicable changes are noticed during operation of the gear unit, such as an important increase in temperature or unusual noises, the drive assembly must be switched off immediately.



Rotating and/or movable drive components must be fitted with suitable safeguards to prevent contact.



When the gear unit is incorporated in plant or machinery, the manufacturer of such plant or machinery must ensure that the contents of these instructions are incorporated in his own instructions.

- When removing the safety equipment the fixation means should be stored for later use. Removed safety equipment must be re-installed prior to starting up.
- Notices attached to the gear unit, e.g. rating plate, direction arrows etc., must always be observed. They must be kept free from dirt and paint at all times. Missing plates must be replaced.
- Screws which have been damaged during assembly or disassembly work must be replaced with new ones of the same strength class and type.
- Spare parts should always be obtained from Siemens (see also section 11).

3.2 Environmental protection

- Dispose of any packing material in accordance with regulations or separate it for recycling.
- When changing oil, the used oil must be collected in suitable containers. Any pools of oil which may have collected should be removed at once with an oil-binding agent.
- Preservative agents should be stored separately from used oil.
- Used oil, preservative agents, oil-binding agents and oil-soaked cloths must be disposed of in accordance with environmental legislation.
- Disposal of the gear unit after its useful life:
 - Drain all the operating oil, preservative agent and/or cooling agent from the gear unit and dispose of in accordance with regulations.
 - Depending on national regulations, gear-unit components and/or add-on parts may have to be disposed of or sent for recycling separately.

3.3 Special dangers and personal protective equipment

- Depending on operating conditions, the surface of the gear unit may heat up or cool down to extreme temperatures.



In the case of hot surfaces (> 55 °C) there is a risk of burns!



In the case of cold surfaces (< 0 °C) there is a risk of frost injury (pain, numbness, frostbite)!



During oil changes there is a risk of scalding from escaping oil!



**Small foreign matter such as sand, dust, etc. can get into the cover plates of the rotating parts and be thrown back by these.
Risk of eye injury!**



In addition to any generally prescribed personal safety equipment (such as safety shoes, safety clothing, helmet) handling the gear unit requires wearing **suitable safety gloves** and **suitable safety glasses**!



The gear unit is not suitable for operation in explosion hazard locations. It must under no circumstances be used in such locations because of the risk to life and limb.

4. Transport and storage

Observe the instructions in section 3., "Safety instructions"!

4.1 Scope of supply

The products supplied are listed in the despatch papers. Check immediately on receipt to ensure that all the products listed have actually been delivered. Parts damaged and/or missing parts must be reported to Siemens in writing immediately.



If there is any visible damage, the gear unit must not be put into operation.

4.2 Transport



**When transporting Siemens products, use only lifting and handling equipment of sufficient load-bearing capacity!
Observe the notes regarding load distribution on the packing.**

The gear unit is delivered in the fully assembled condition. Additional items are delivered separately packaged, if applicable.

Different forms of packaging may be used, depending on the size of the unit and method of transport. Unless otherwise agreed, the packaging complies with the **HPE Packaging Guidelines**.

The symbols marked on the packing must be observed at all times. These have the following meanings:

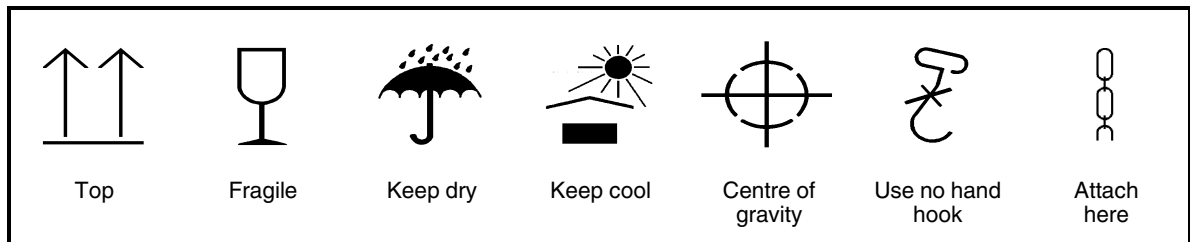


Fig. 2: Transport symbols



**Transport of the gear unit must be carried out so as to avoid personal damage and damage to the gear unit.
If, for example, the free shaft ends are knocked, this may damage the gear unit.**



The gear units must be transported with suitable equipment only.
During transport the gear unit should be left without oil filling.



Exception: In the case of gear units with auxiliary drive, the auxiliary gear unit will be delivered ex works with oil filling.



**Use only the eyes provided to attach lifting equipment to the unit.
Handling of the gear unit by attaching it to the piping is not permitted.
The pipework must not be damaged.
Do not use the front threads at the shaft ends to attach slinging equipment for the transport.
Slinging equipment must be adequate for the weight of the gear unit.**

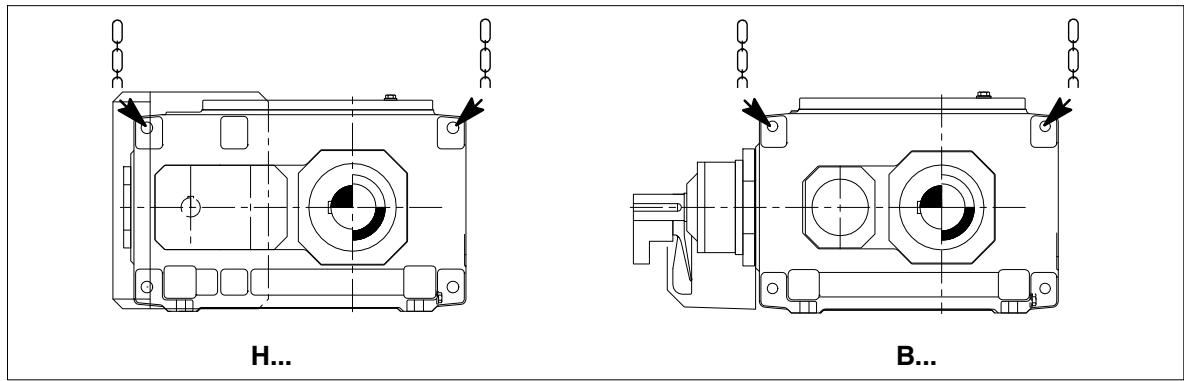


Fig. 3: Attachment points on gear units types H... and B...

For drive units where add-on parts such as motor, add-on coupling etc. are mounted on the gear unit an additional attachment point may be required because of the shift in the centre of gravity.



Units which are slung by eyebolts must not be tilted.

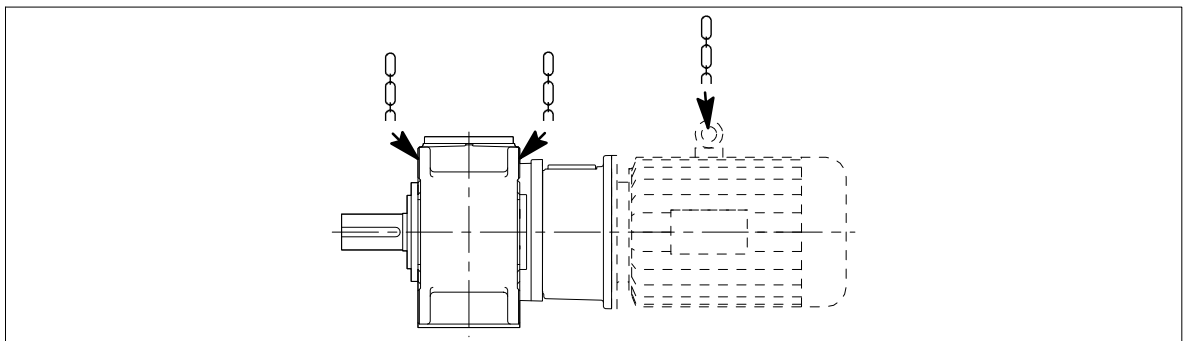


Fig. 4: Attachment points on gear units types H... with motor

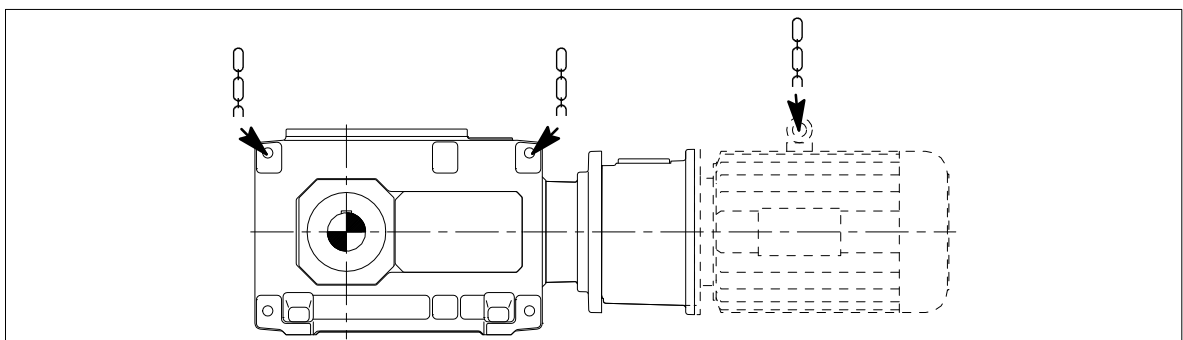


Fig. 5: Attachment points on gear units types B... with motor

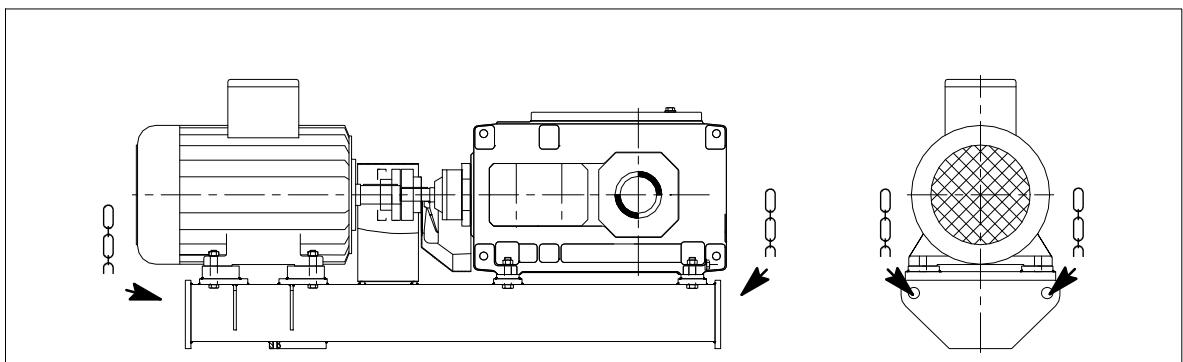


Fig. 6: Attachment points on gear units types B... with gear-unit swing base

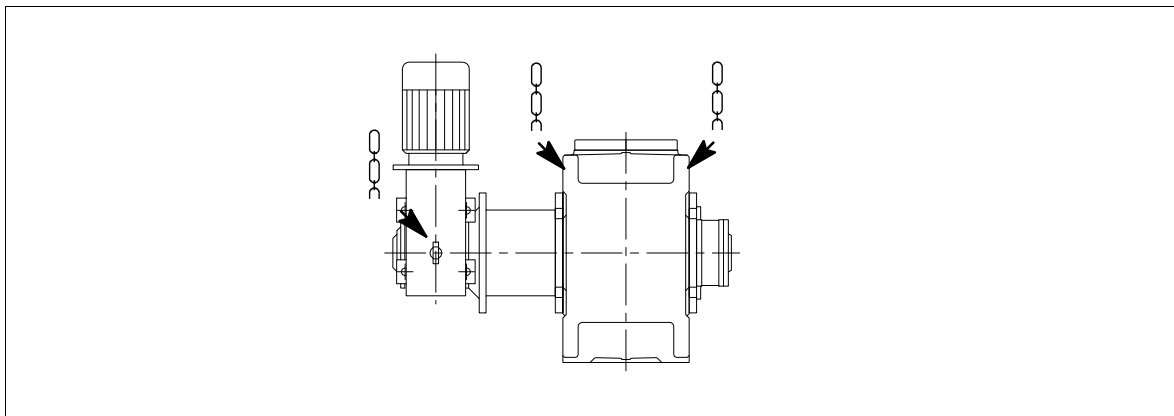


Fig. 7: Attachment points on gear-units types B3.H / T3.H with auxiliary drive

A detailed view of the gear unit can be obtained from the drawings in the gear-unit documentation.

4.3 Storing the gear unit

The gear unit must be stored in the position of use in a sheltered place; it must be placed on a vibration-free, dry base and covered over.



When temporarily storing the gear unit and any single components supplied with it, the preservative agent should be left on them. It must not be damaged, otherwise there is a risk of corrosion.



Do not stack gear units on top of one another.



If the gear unit is being stored out of doors, it must be particularly carefully covered, and care must be taken that neither moisture nor foreign material can collect on the unit. Waterlogging should be avoided.



Unless otherwise agreed by contract, the gear unit must not be exposed to harmful environmental factors such as chemically aggressive products.

Provision for special environmental conditions during transport (e.g. transport by ship) and storage (climate, termites, etc.) must be contractually agreed.

4.4 Standard coating and preservation

The gear unit is provided with an interior preservative agent; the free shaft ends are painted for protection.

The characteristics of the external coat depend on the ambient conditions stipulated in the order relating to method of transport and area of application.



The gear unit is normally delivered completely ready, with a priming and a finish coat.

Where gear units are delivered with a priming coat only, it is necessary to apply a finish coat in accordance with directives applying to the specific application.

The priming coat alone is not suitable to provide a sufficient long-term corrosion protection.



Ensure that the coat is not damaged!

Any damage may cause failure of the external protective coating and corrosion.



Unless otherwise contractually agreed, the interior preservation is guaranteed for 6 months, and the preservation of the free shaft ends for 24 months, provided that storage is in dry, frostfree sheds.

The guarantee period starts on the date of delivery or that of the notice that the item is ready for shipment.

For longer periods of storage (> 6 months) we advise regular checking and, if necessary, renewal of the interior and exterior preservation (see section 7, "Start-up").

The output shaft must then be rotated at least one turn to change the position of the rolling element in the bearings. The input shaft must not be in the same position as before rotation.

This procedure must be repeated and documented every 6 months until start-up.

4.4.1 Interior preservation with preservative agent

Table 8: Durability period and measures for interior preservation when using mineral oil or PAO-based synthetic oil

Duration of protection	Preservative agent	Special measures
up to 6 months	Castrol Alpha SP 220 S	none
up to 24 months		- Close all holes in the gear unit - Replace air filter or breather screw with screw plug. (replace screw plug with air filter or breather screw before start-up)
For storage periods longer than 24 months, renew the preservative agent. For storage periods longer than 36 months, Siemens should be consulted before.		

Table 9: Durability period and measures for interior preservation when using PG-based synthetic oil

Duration of protection	Preservative agent	Special measures
up to 6 months	Special anti-corrosion oil TRIBOL 1390 ¹⁾	none
up to 36 months		- Close all holes in the gear unit - Replace air filter or breather screw with screw plug. (replace screw plug with air filter or breather screw before start-up)
For storage periods longer than 36 months, Siemens should be consulted before.		

¹⁾ Resistant to tropical conditions and sea water; max. ambient temperature 50 °C

4.4.2 Exterior preservation

Table 10: Durability period for exterior preservation of shaft ends and other bright machined surfaces

Duration of protection	Preservative agent	Layer thickness	Remarks
in case of indoor storage up to 36 months ¹⁾	Tectyl 846 K19	approx. 50 µm	Long-term wax-based preservative agent: - resistant to seawater - resistant to tropical conditions - (soluble with CH compounds)
in case of outdoor storage up to 12 months ²⁾			

¹⁾ The gear unit must be stored in the position of use in a sheltered place; it must be placed on a vibration-free, dry wooden base and covered over.

²⁾ If the gear unit is being stored out of doors, it must be particularly carefully covered, and care must be taken that neither moisture nor foreign material can collect on the unit. Waterlogging should be avoided.



The procedure for interior and exterior preservation treatment is described in section 7 (see items 7.3.1.3 and 7.3.2.1)!

5. Technical description

Observe the instructions in section 3. "Safety instructions"!

5.1 General description

The helical gear unit is supplied as a one-, two-, three- or four-stage gear unit. The bevel-helical gear unit is supplied as a two-, three- or four-stage gear unit. The gear unit may also be supplied as a multi-stage bevel-helical gear unit or helical gear unit with fitted auxiliary drive. It is designed for installation in the horizontal mounting position. If necessary, it can also be designed for installation in a different position.



As a principle, the gear unit can be operated in both directions of rotation. The only exceptions are gear types with backstop or overrunning clutch. If rotation reversal is required for these types of unit, Siemens should be consulted.

A number of shaft configurations (types and rotation directions) are possible. These are shown in the following table as solid shafts:

Table 11: Types and rotation directions

Type	Type								
	A	B	C	D	E	F	G	H	I
H1SH									
H2SH H2HM H2HH H2DM H2DH H2KM H2KH H2FM H2FH H2VH H2PH									
H3SH H3HM H3HH H3DM H3DH H3KM H3KH H3FM H3FH H3VH									
H4SH H4HM H4HH H4DM H4DH H4KM H4KH H4FM H4FH H4VH									
B2SH B2HM B2HH B2DM B2DH B2KM B2KH B2FM B2FH B2VH									
B3SH B3HM B3HH B3DM B3DH B3KM B3KH B3FM B3FH B3VH T3SH T3VH T3DH T3HH T3KH T3FH									

Type	A	B	C	D	Type E	F	G	H	I
B4SH B4HM B4HH B4DM B4DH B4KM B4KH B4FM B4FH B4VH									



When mounting the auxiliary drive (as maintenance and/or load drive) the assignment of the direction of rotation to the design is defined in the dimensioned drawing.

The gear units are characterised by a low noise level. This is achieved by helical and bevel-helical gears with a high contact ratio and special sound-damping housings.

The good temperature characteristics of the gear unit are achieved by its high degree of efficiency, large housing surface and performance-related cooling system.

5.2 Output designs

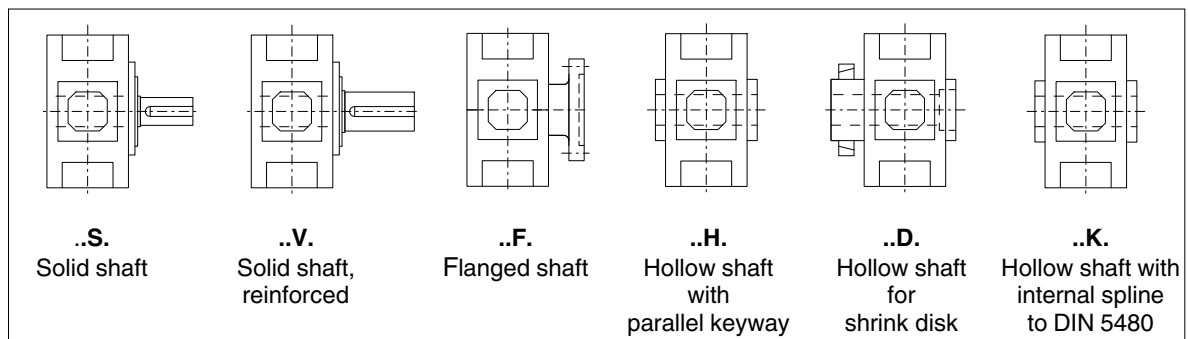


Fig. 8: Output designs

5.3 Housing

The housing is of cast iron; if required, they may also be of steel.

Housings up to size 12 are made in one part. The exception are types H1SH and H2PH, which have a two-part housing similar to those of sizes 13 to 22 of the other types. The housing is rigid in design and due to its form has excellent noise and temperature characteristics.

The gear-unit housing comes with the following equipment:

- Lifting eyes (adequately dimensioned for transport)
- Inspection and/or assembly cover (for oil filling and/or inspection)
- Oil-sight glass or oil dipstick (to check the oil level)
- Oil-drain plug (for oil drain)
- Air filter or venting screw (for aeration and ventilation)

Colour codes for ventilating, oil inlet, oil level and oil drainage:

Air relief point:	yellow		Oil-draining point:	white	
Oil-filling point:	yellow		Lubrication point:	red	
Oil level:	red		Oil level:	red	

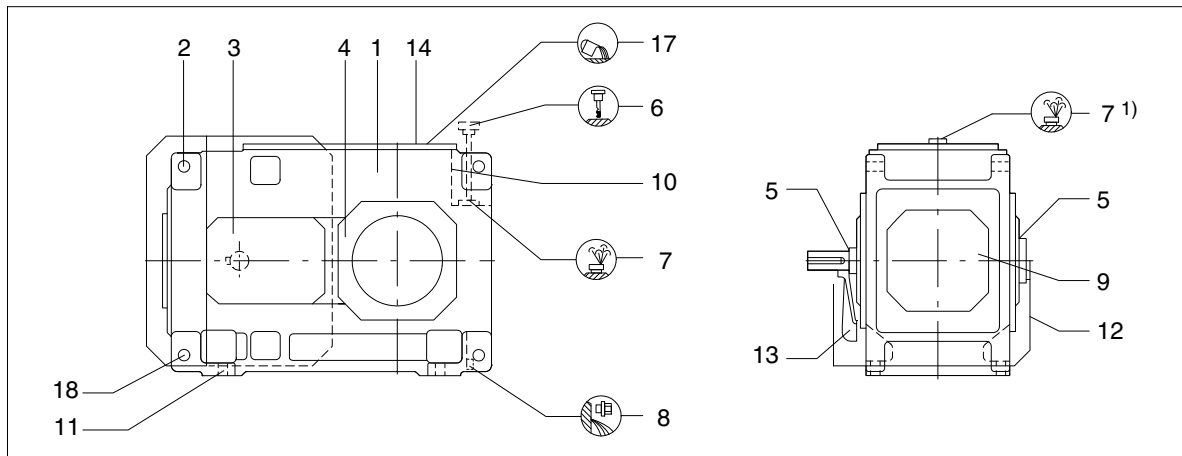


Fig. 9: Gear-unit features on gear units type H.. $H \leq 12$

1) for H1SH only

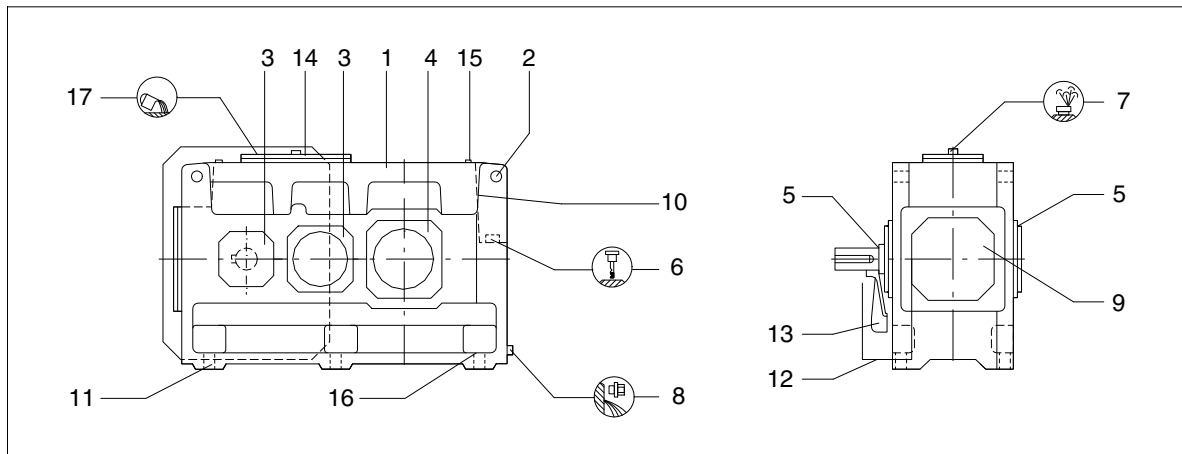


Fig. 10: Gear-unit features on gear units type H.. $H \geq 13$

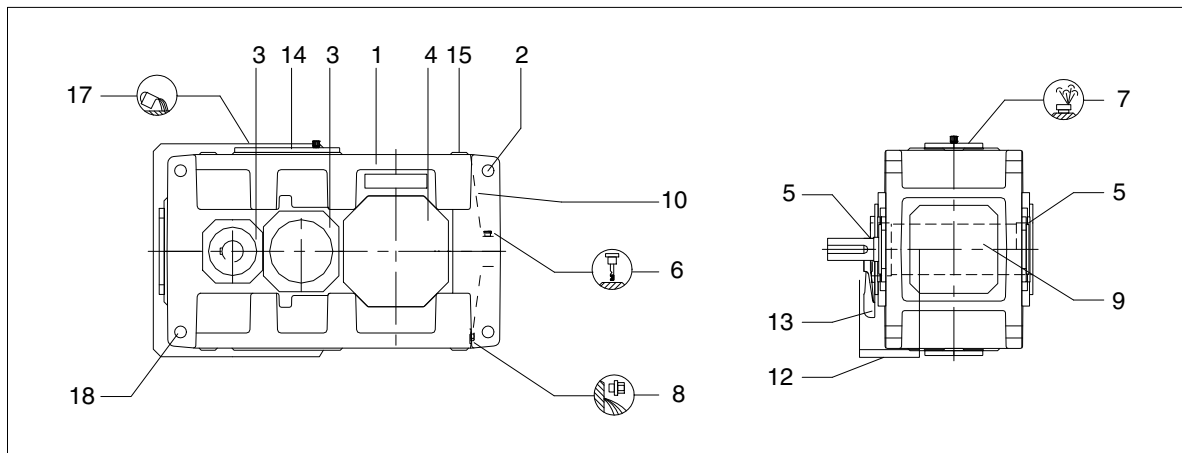


Fig. 11: Gear-unit features on gear units type H.. $M \geq 13$

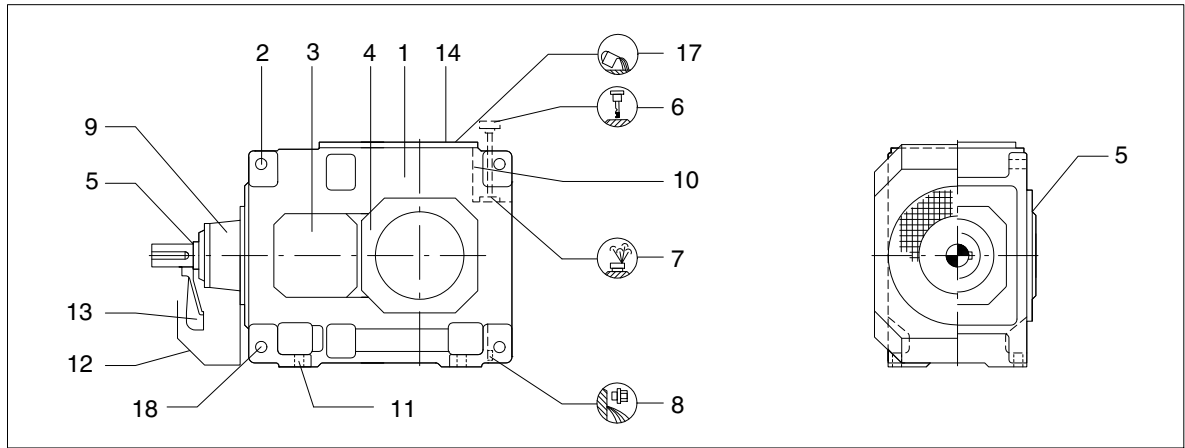


Fig. 12: Gear-unit features on gear units types B..H and T..H ≤ 12

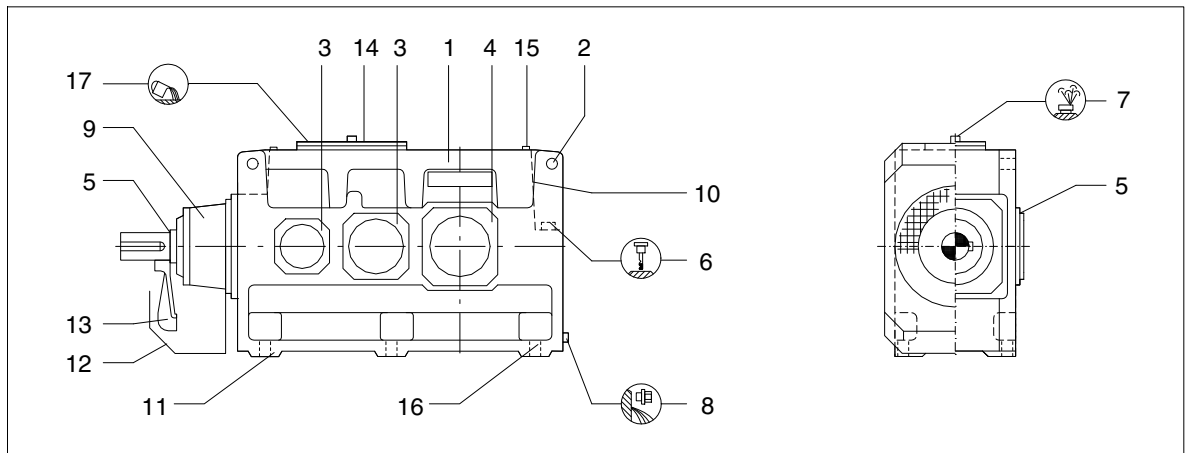


Fig. 13: Gear-unit features on gear units type B..H ≥ 13

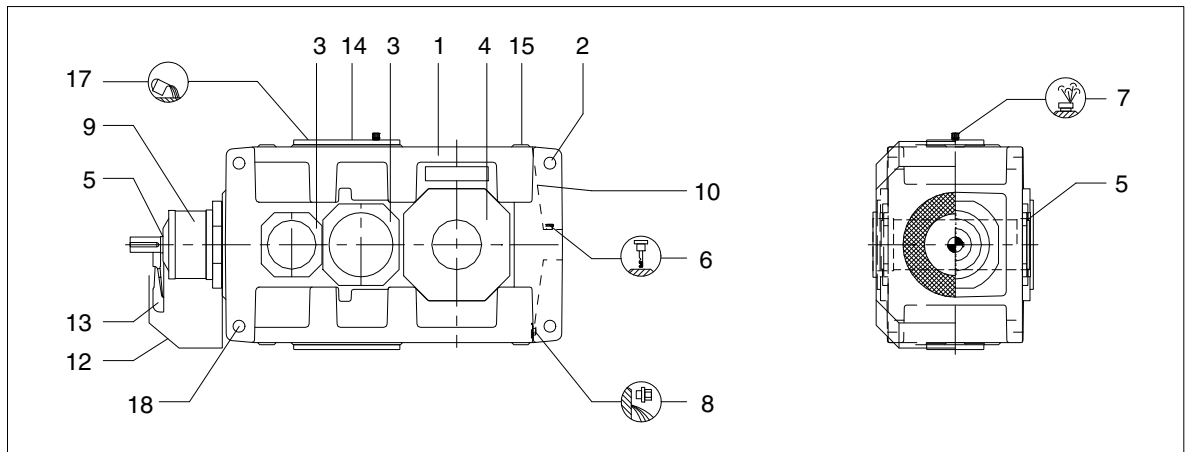


Fig. 14: Gear-unit features on gear units type B..M ≥ 13

- | | |
|--------------------------------|-------------------------------------|
| 1 Housing | 10 Rating plate |
| 2 Lifting eyes | 11 Gear-unit fastening |
| 3 Cover | 12 Fan cowl |
| 4 Cover | 13 Fan |
| 5 Shaft seals | 14 Inspection and/or assembly cover |
| 6 Oil dipstick | 15 Alignment surfaces |
| 7 Housing ventilation | 16 Alignment thread |
| 8 Oil-drain plug | 17 Oil inlet |
| 9 Cover and/or bearing journal | 18 Fastening for torque arm |

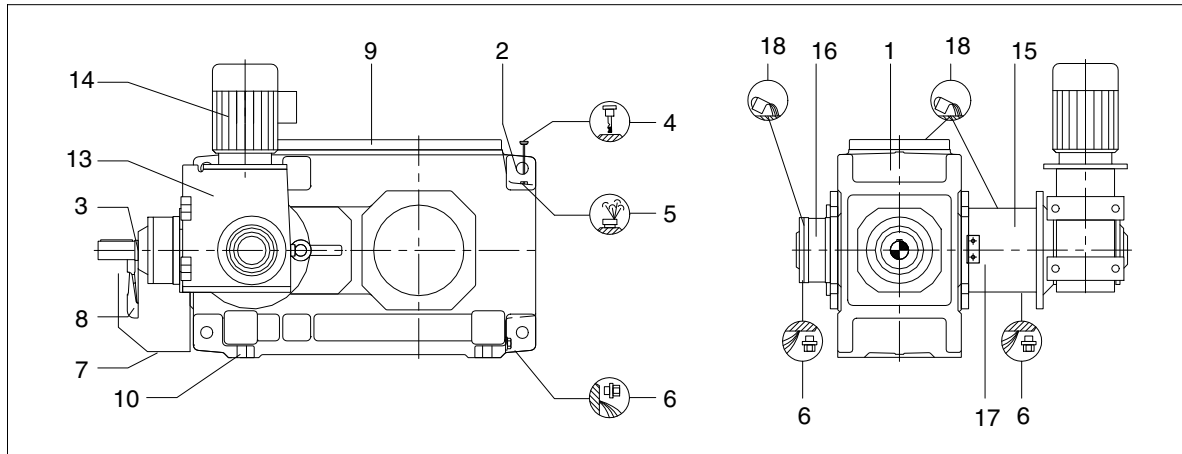


Fig. 15: Gear-unit features on gear units types B3.H / T3.H ≤ 12

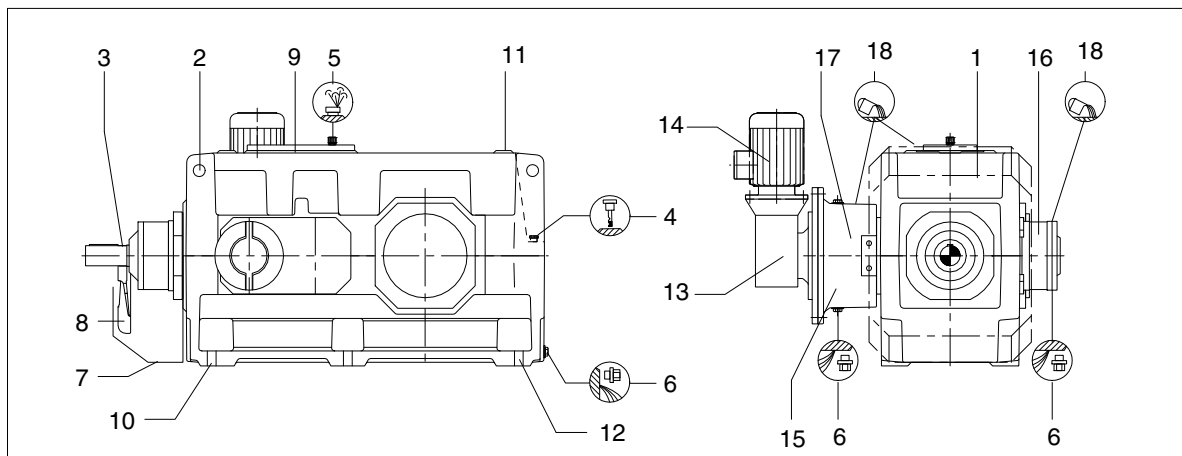


Fig. 16: Gear-unit features on gear units type B3.H ≥ 13

- | | |
|--------------------------------|----------------------------|
| 1 Main gear unit | 10 Gear-unit fastening |
| 2 Lifting eyes | 11 Alignment surfaces |
| 3 Shaft seals | 12 Alignment thread |
| 4 Oil dipstick | 13 Auxiliary gear unit |
| 5 Housing ventilation | 14 Electric motor |
| 6 Oil-drain plug | 15 Overrunning clutch |
| 7 Fan cowl | 16 Backstop |
| 8 Fan | 17 Speed-monitoring device |
| 9 Inspection or assembly cover | 18 Oil-filler plug |

A detailed view of the gear unit can be obtained from the drawings in the gear-unit documentation.

5.4 Toothed components

The externally toothed components of the gear unit are case-hardened. Helical-gear teeth are ground. The high quality of the teeth leads to a significant noise reduction and ensures safe and reliable running.

The gears are connected with the shafts by interference fits and parallel keys or by shrink fits. These types of joints transmit with adequate reliability the torques generated.

5.5 Lubrication

5.5.1 Splash lubrication

Unless otherwise agreed in the order, the teeth and bearings are adequately splash-lubricated with oil. The gear unit thus requires very little maintenance.

5.5.2 Pressure lubrication through add-on oil-supply system

In non-horizontal positions, with high bearing speeds or peripheral velocities on the teeth, the splash lubrication system may be supported and/or replaced with a pressure-lubrication system.

The oil-supply system is permanently attached to the gear unit and consists of a flange pump, a coarse filter, a pressure-monitoring device and pipework. For gear units of sizes 13 to 22, the coarse filter is replaced with a double change-over filter.

The direction of flow from the flange pumps is **independent of the direction of rotation**.

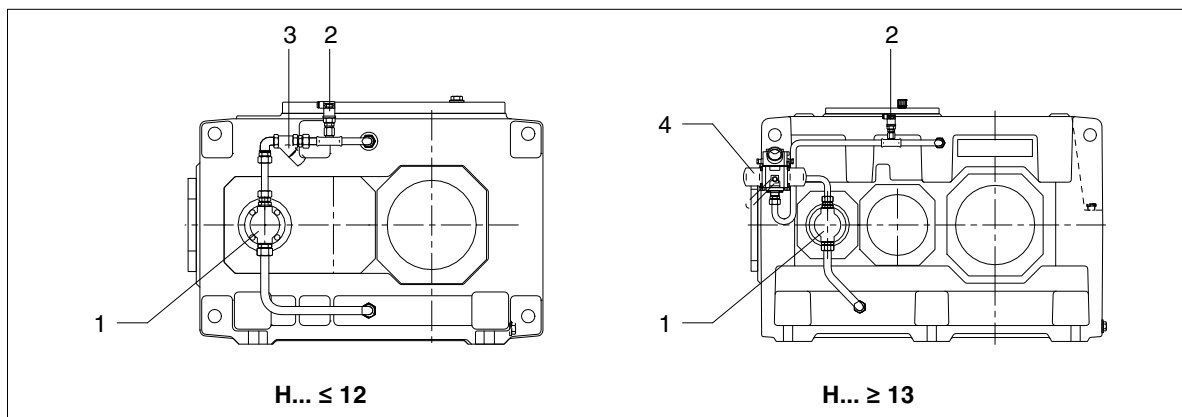


Fig. 17: Add-on oil-supply system on gear units type H...

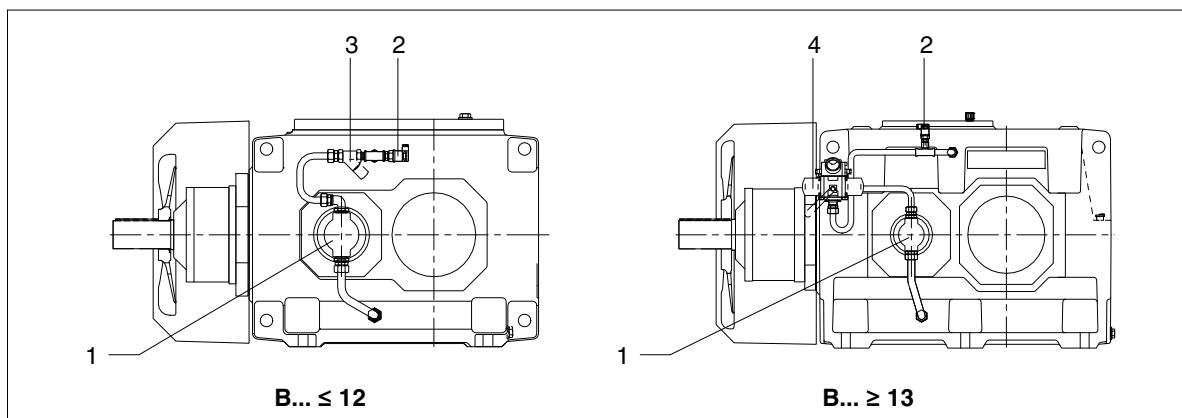


Fig. 18: Add-on oil-supply system on gear units type B...

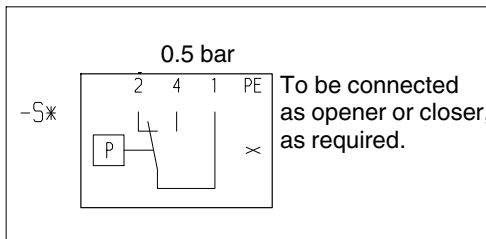
- | | |
|--------------------|-----------------------------|
| 1 Flange pump | 3 Coarse filter |
| 2 Pressure monitor | 4 Double change-over filter |

A detailed view of the gear unit can be obtained from the drawings in the gear-unit documentation.



In the case of gear units with add-on oil-supply system, before starting the unit up the pressure monitor must always be connected so as to be ready for operation.

Depending on the order specification and application, the flange pump may be replaced with a motor pump.



When operating and servicing the components of the oil-supply system, observe the operating instructions of the components.

For technical data, refer to the data sheet and/or the list of equipment.

5.6 Shaft bearings

All shafts are mounted in rolling bearings.

5.7 Shaft seals

Depending on requirements, radial shaft sealing rings, labyrinth seals, or Taconite seals or Tacolab seals are mounted at the shaft exits to prevent oil from leaking from the housing and dirt from entering it.

5.7.1 Radial shaft-sealing rings

Radial shaft-sealing rings are the standard type of seal. They are fitted preferably with an additional dust lip to protect the actual sealing lip from external contamination.



Use in an area with much dust is not possible.

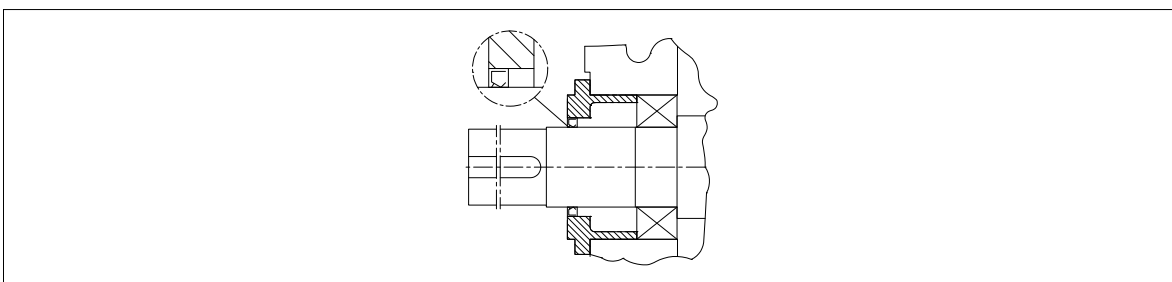


Fig. 19: Radial shaft-sealing ring

5.7.2 Labyrinth seals

Labyrinth seals are non-contacting and avoid wear to the shaft. They therefore require no maintenance and ensure favourable temperature characteristics. They can be used only with certain transmission ratios and minimum speeds.

Check in the spare parts drawing and the spare parts list whether the gear unit is provided with labyrinth seals.

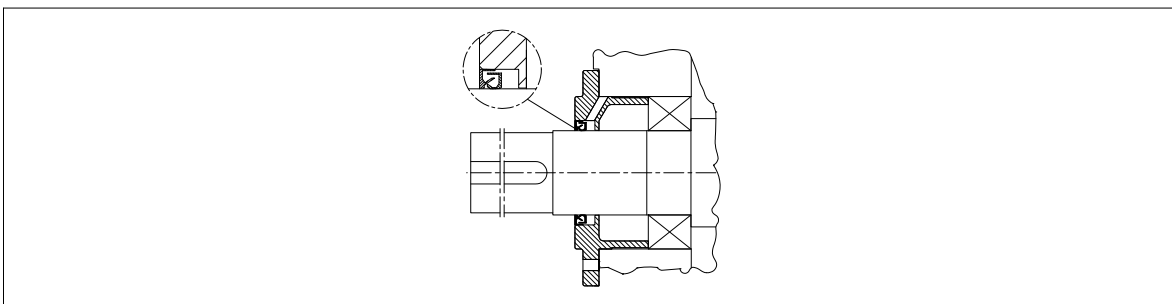


Fig. 20: Labyrinth seal



For reliable operation, this type of seal requires stationary, horizontal positioning in a splash-free and relatively dust-free environment. Overfilling of the gear unit can cause leakage, as can oil with high foam content.

5.7.3 Taconite seals

Taconite seals were specially developed for use in a dusty environment. The penetration of dust is prevented by a combination of three seal elements (radial shaft-sealing ring, lamellar seal and grease-charged labyrinth seal).

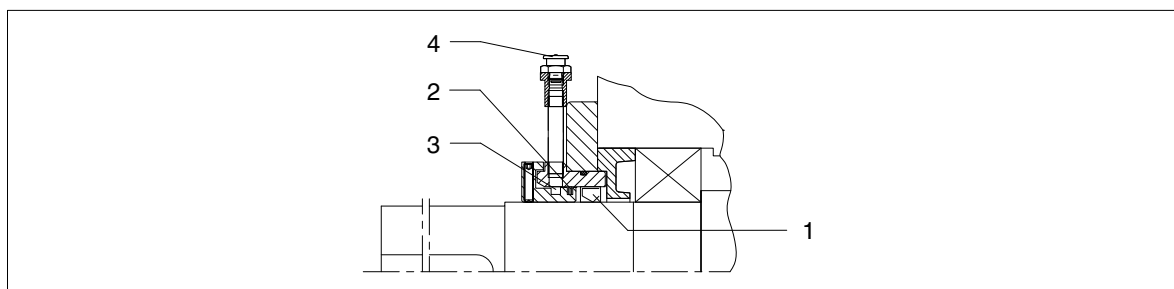


Fig. 21: Taconite seal

- | | | | |
|---|---------------------------|---|---|
| 1 | Radial shaft-sealing ring | 3 | Grease-charged labyrinth seal (re-chargeable) |
| 2 | Lamellar seal | 4 | Flat grease nipple AM10x1 to DIN 3404 |

Taconite seals are divided into the following types:

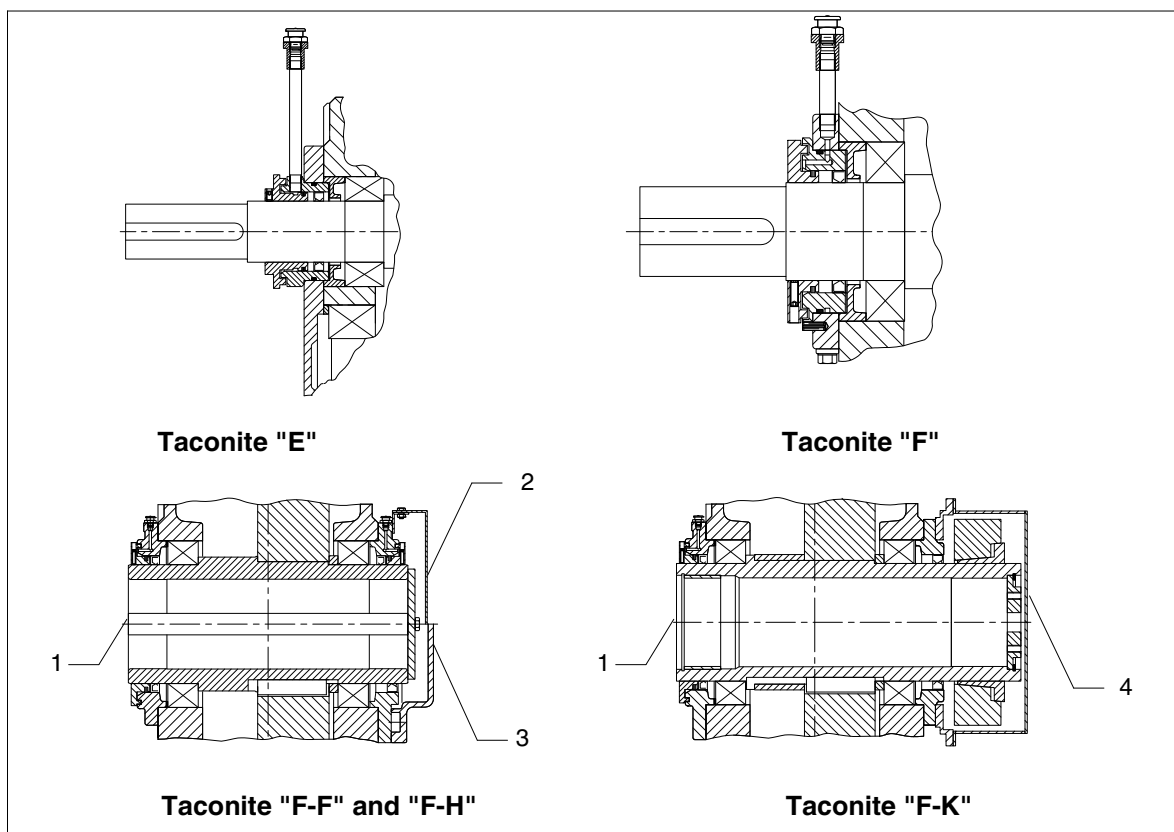


Fig. 22: Taconite seal, variants E, F, F-F, F-H and F-K

- | | | | |
|---|----------------|---|----------------|
| 1 | Output | 3 | Taconite "F-H" |
| 2 | Taconite "F-F" | 4 | Taconite "F-K" |

Table 12: Variant description Taconite seal

Taconite type variant	Application	Remarks
"E"	All input shafts with or without fan	Re-chargeable labyrinth
"F"	Output shaft Type S (Solid shaft) Type V (Solid shaft, reinforced) Type F (Flanged shaft)	
"F-F"	Output shaft Type H (Hollow shaft with parallel keyway) Type K (Hollow shaft with internal spline to DIN 5480)	Labyrinth re-chargeable on both sides, incl. dustproof cowl to prevent contact on gear side facing away from output
"F-H"	Output shaft Type H (Hollow shaft with parallel keyway) Type K (Hollow shaft with internal spline to DIN 5480)	Labyrinth re-chargeable on output side; dustproof cowl on opposite side
"F-K"	Output shaft Type D (Hollow shaft for shrink disk)	



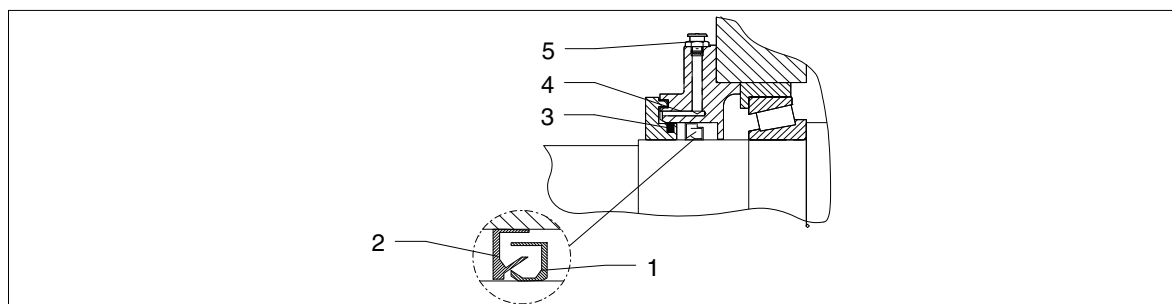
The specified frequencies must be observed (see section 10, "Maintenance and repair") for re-charging the labyrinth seals with grease.

5.7.4 Tacolab seal

Tacolab seals are non-contacting seals, operating wearfreely and requiring very little maintenance and which thus do not cause operating interruptions.

The Tacolab seal is made up of two parts:

- an oil labyrinth preventing lubricating oil from escaping,
- dust seal filled with grease, which permits the use in very dusty environments.

**Fig. 23:** Tacolab seal

- | | | | |
|---|------------------------|---|---|
| 1 | Labyrinth sealing ring | 4 | Grease-charged labyrinth seal (re-chargeable) |
| 2 | Labyrinth sealing ring | 5 | Flat grease nipple AM10x1 to DIN 3404 |
| 3 | Lamellar seal | | |

5.8 Backstop

For certain requirements, the gear unit can be fitted with a mechanical backstop. This backstop permits only the specified direction of rotation during the operation of the unit. The direction of rotation is marked by a corresponding arrow on the input and output side of the gear unit.

The backstop is mounted oiltight on an adapter flange on the gear unit and integrated in its oil-circulation system.

The backstop is fitted with centrifugally operated sprags. If the gear unit rotates in the prescribed direction, the inner ring rotates with the sprag cage in the direction of shaft rotation, while the outer ring remains stationary. From a specific speed up (disengagement speed) the sprags disengage from the outer ring. In this operating condition the backstop operates wearfreely.

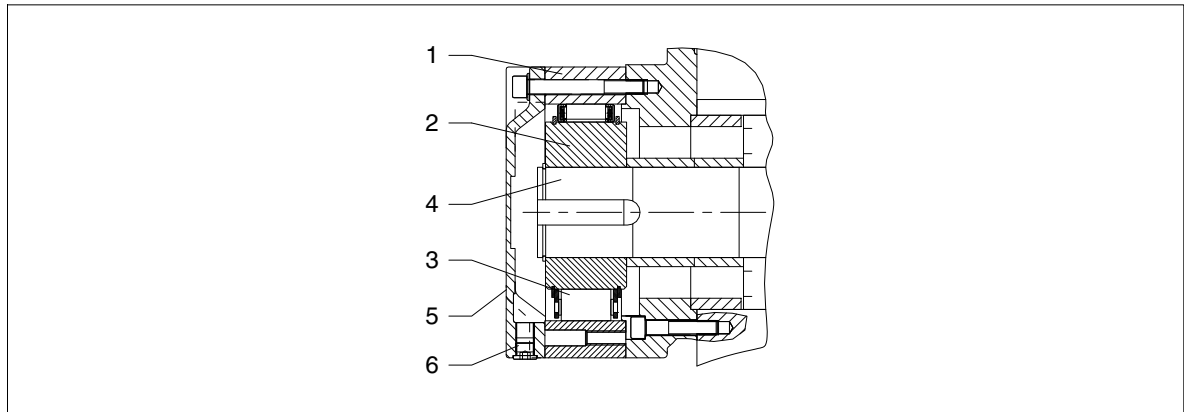


Fig. 24: Backstop

- | | | | |
|---|------------------|---|--------------------|
| 1 | Outer ring | 4 | Shaft |
| 2 | Inner ring | 5 | Cover |
| 3 | Cage with sprags | 6 | Residual-oil drain |



The backstop direction can be changed by turning the cage around. If a change in backstop direction is required, Siemens should be consulted beforehand.



To avoid damaging the backstop or the gear unit, the motor must not be run adversely to the stop direction of the gear unit.

Observe the notice fixed to the gear unit.

The minimum lift-off speeds must not be exceeded during operation.

Before connecting the motor, determine the direction of rotation of the three-phase current supply using a phase-sequence indicator, and connect the motor in accordance with the pre-determined direction of rotation.

5.9 Torque-limiting backstop (special design)

A torque-limiting backstop is available for special uses, e.g. dual drives. The backstop is a combination of a backstop with centrifugally operated sprags and a brake. The slipping torque is set by a number of compression springs.

This "slipping" will protect the gear unit and the sprags of the backstop from inadmissibly high stresses during negative rotation. In addition, a uniform load distribution onto both gear units is achieved during negative rotation when using dual drives.

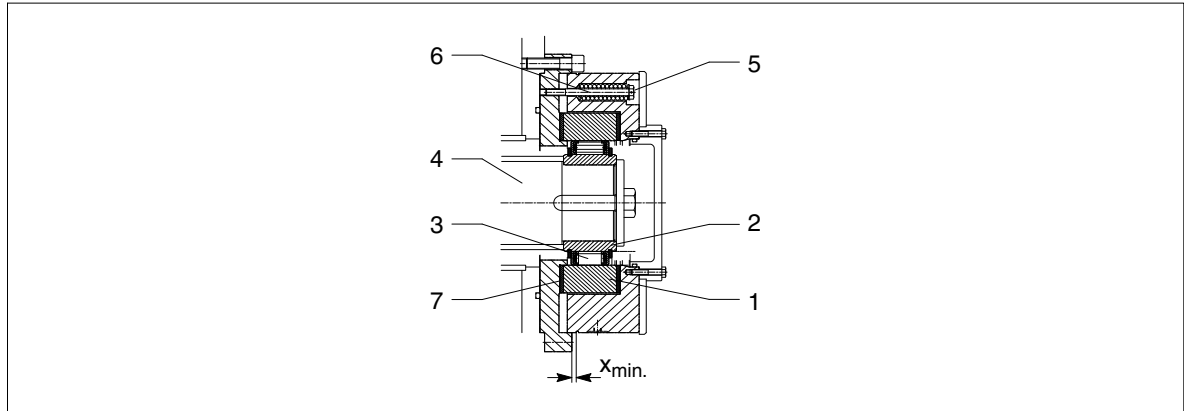


Fig. 25: Torque-limiting backstop

- | | | | |
|---|------------------------|---|------------------------------------|
| 1 | Outer ring | 5 | Locking wire |
| 2 | Inner ring | 6 | Lead screw with compression spring |
| 3 | Cage with sprags | 7 | Friction lining |
| 4 | Shaft (adapter flange) | | |

The torque-limiting backstop is attached to the gear unit by means of an adapter flange to form an oiltight seal and is integrated in its oil circulation system.



The stop direction can be changed by turning the cage around. If a change in stop direction is required, Siemens should be consulted beforehand.



The slipping torque was set at the correct value at the Siemens works; resetting during startup is not permissible.

To safeguard the set slipping torque, the lead screws of the compression springs are secured with locking wire. The warranty will expire if the locking wire for the screws is missing or has been damaged.



For safety reasons, it is absolutely prohibited to change the slipping torque. After having stopped the motor, there is a danger that the load is not safely held in its position and can run in reverse direction at high speed.



As a rule, the backstop operates without wear. As a precaution, the dimension " $x_{min.}$ " must be checked once yearly and after every releasing operation (Type FXRT only).



To avoid damaging the backstop or the gear unit, the motor must not be run adversely to the stop direction of the gear unit. Observe the notice fixed to the gear unit.

5.10 Cooling

Depending on requirements, the gear unit is fitted with a fan, a cooling coil, an added-on oil-supply system with oil cooler or a separately provided oil-supply system. In the case of a separate oil-supply system, the specific instructions for this oil-supply system must be observed.



When installing the gear unit free convection must be ensured on the housing surface, in order to definitely avoid overheating the gear unit.

5.10.1 Fan

As a rule the fan is mounted on the high-speed shaft of the gear unit and is protected from accidental contact by an air guide cover. The fan sucks air through the grid of the air guide cover and blows it along the air ducts on the side of the gear housing. It thereby dissipates a certain amount of heat from the housing.

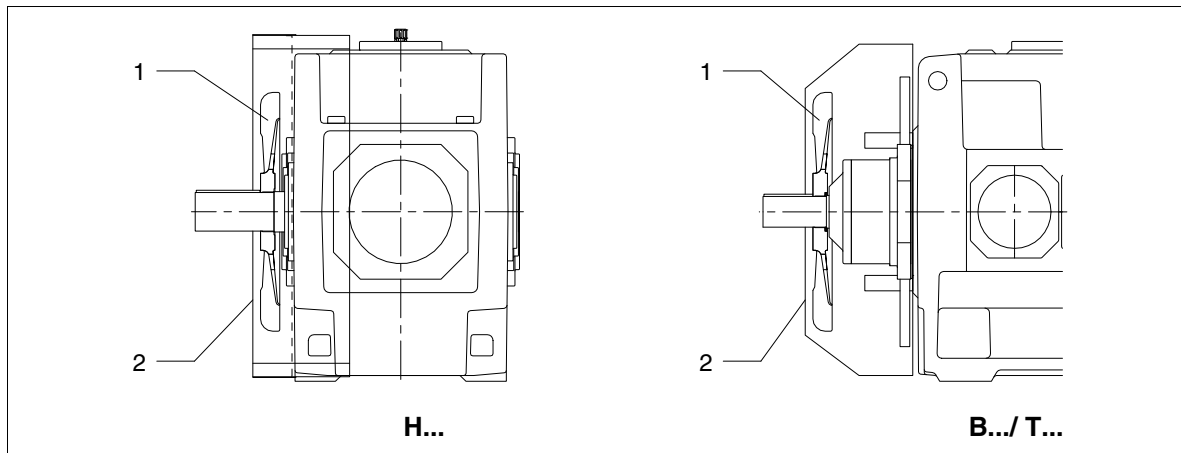


Fig. 26: Fan on gear units types H..., B... and T...

1 Fan

2 Air guide cover

A detailed view of the gear unit can be obtained from the drawings in the gear-unit documentation.



For gear units fitted with a fan, sufficient space must be allowed for air intake when mounting the safety guards for the coupling or other components. The correct distance is given in the dimensioned drawing in the gear-unit documentation.

It must be ensured that the fan cowl is correctly fastened. The fan cowl must be protected against damage from outside. The fan must not come into contact with the fan cowl.



The cooling effect is considerably reduced if the fan or the gear housing are dirty (see section 10, "Maintenance and repair").

5.10.2 Cooling coil

The gear unit can be fitted with a cooling coil in the oil sump. Cooling water is supplied by way of a water connection. The operator must ensure this. Either fresh water, sea water or brackish water can be used for cooling purposes.

When water is flowing through the cooling coil, a certain amount of heat is transferred from the oil to the water and thereby removed from the system.

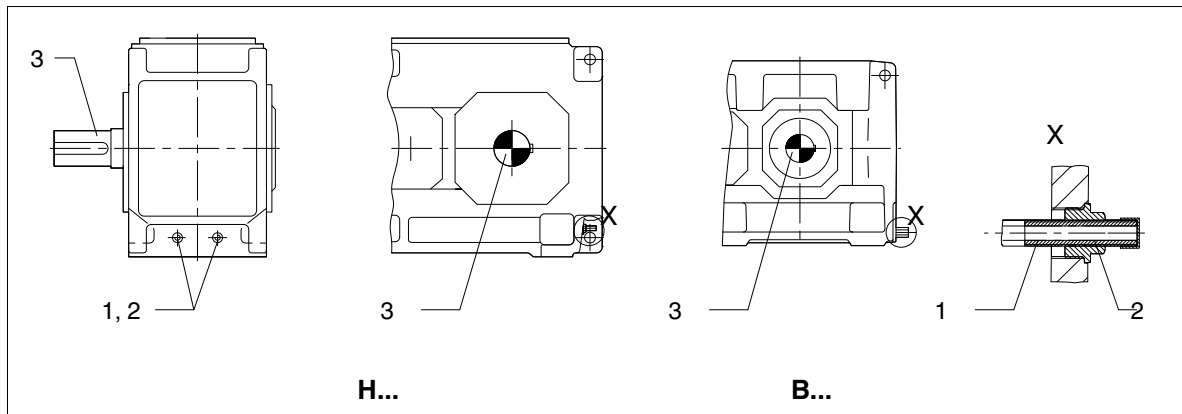


Fig. 27: Cooling coil on gear units types H... and B...

1 Cooling-water connection 2 Reducing screw 3 Output shaft

A detailed view of the gear unit can be obtained from the drawings in the gear-unit documentation.



The water can flow through the gear unit in either direction. The pressure of the cooling water must not exceed 8 bar.

If the gear unit is being withdrawn from service for a longer period and if there is a danger of freezing, the cooling water must be drained off. Remove any remaining water with compressed air.

The ends of the cooling coil must never be twisted because this could destroy the cooling coil.

The reducing bolt must not be tightened or demounted because this may result in damage to the cooling coil.



**Be especially careful when blowing with compressed air.
Wear protective glasses!**



Avoid too high pressure on the cooling-water entry. For this a cooling-water flow control must be used (e.g. a pressure reducer or a suitable valve).



For connecting dimensions, refer to the dimensioned drawing of the gear unit. The required cooling water quantity and the max. permissible inlet temperature are given on the data sheet and/or the list of equipment.

Table 13: Cooling-water quantity required (l/min)

Type	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20 to 22
H1SH	4	-	4	-	4	-	8	-	8	-	8	-	8	-	8	-	1)	-
H2.H	-	4	4	4	4	4	8	8	8	8	8	8	8	8	8	8	1)	1)
H2.M	-	-	-	-	-	-	-	-	-	-	8	8	8	8	8	8	1)	1)
H3.H	-	-	4	4	4	4	4	4	8	8	8	8	8	8	8	8	1)	1)
H3.M	-	-	-	-	-	-	-	-	-	-	8	8	8	8	8	8	1)	1)
B2.H	-	4	8	4	8	4	8	8	8	8	8	8	8	8	8	8	-	-
B3.H	-	4	4	4	4	4	8	8	8	8	8	8	8	8	8	8	1)	1)
B2.M	-	-	-	-	-	-	-	-	-	-	8	8	8	8	8	8	-	-
B3.M	-	-	-	-	-	-	-	-	-	-	8	8	8	8	8	8	1)	1)

1) on request



Refer to the order-specific dimensioned drawing for connecting dimensions.

5.10.3 Add-on oil-supply system with air oil-cooler

For types H1.., H2.. and B2.., an oil-supply system with air oil-cooler may be applied. This oil-cooling system is permanently attached to the gear unit.

Components:

- air oil-cooler
- flange pump
- coarse filter (double change-over filter from size 13)
- pressure-monitoring device
- temperature-control valve
- pipework

The air oil-cooler is designed to cool the gear oil by means of air from the surrounding atmosphere. Depending on the volume flow, the oil passes through the cooler in one or more streams and through the current of air blown in by the fan. For cold starts, a bypass pipe with a temperature-control valve is provided for.



The flow direction of the pump used is **independent of the direction of rotation**, if nothing is specified in the documentation to the contrary.

When connecting the fittings the actual flow direction must however be observed.

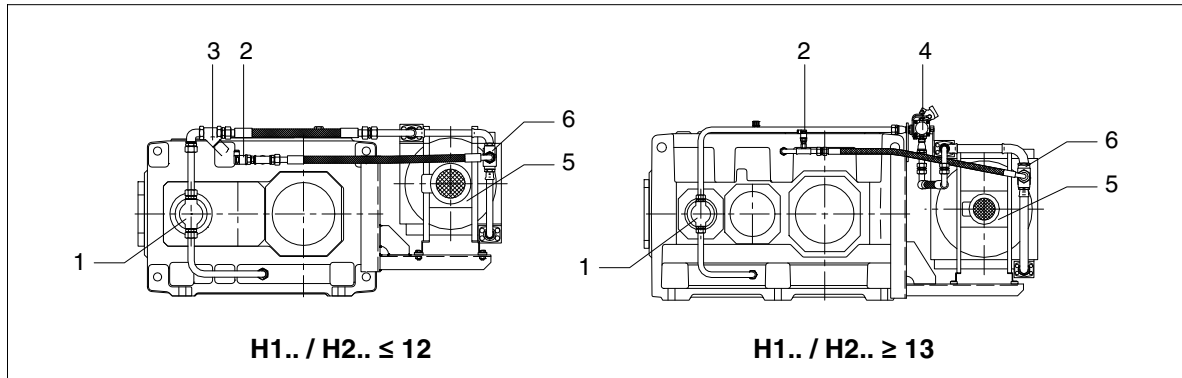


Fig. 28: Air oil-cooling system on gear units types H1.. and H2..

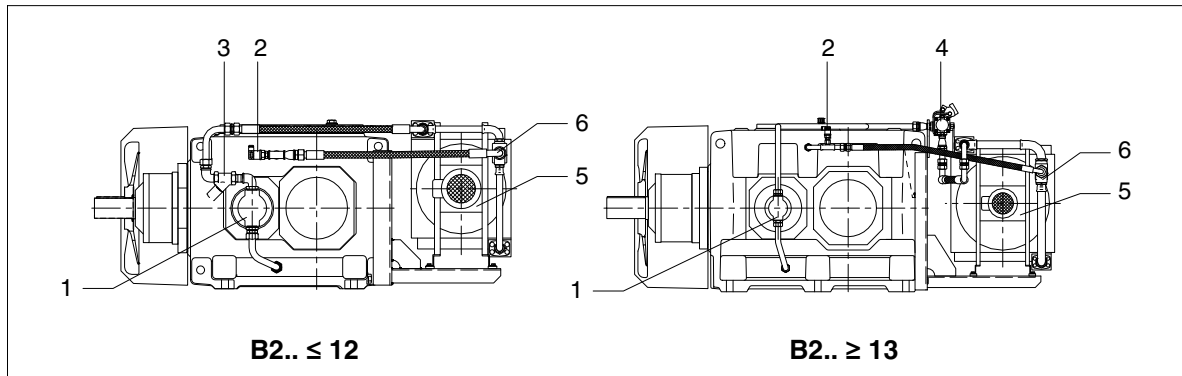


Fig. 29: Air oil-cooling system on gear units types B1.. and B2..

- | | | | |
|---|---|---|---------------------------|
| 1 | Flange pump | 4 | Double change-over filter |
| 2 | Pressure monitor (circuit diagram see item 5.5.2) | 5 | Air oil-cooler |
| 3 | Coarse filter | 6 | Temperature-control valve |

A detailed view of the gear unit can be obtained from the drawings in the gear-unit documentation.



When installing gear units with add-on air oil-cooling units, it must be ensured that the air circulation is not obstructed.

The required minimum distance from adjacent components, walls, etc. is indicated in the drawings in the unit documentation.

Add-on pressure monitors must be connected as shown in item 5.5.2.

Depending on the application, the flange pump may have been replaced with a motor pump.



When operating and servicing the components of the oil-supply system, observe the operating instructions of the components.

For technical data, refer to the data sheet and/or the list of equipment.

The cooling effect is considerably reduced if the cooler or the gear housing are dirty (see section 10., "Maintenance and repair").

5.10.4 Add-on oil-supply unit with water oil-cooler

For types H1.., H2.. and B2.., an oil-supply system with water oil-cooler may be applied, if required in the order. This is permanently attached to the gear unit.

Components:

- pump
- water oil-cooler
- pipework

Depending on size and/or order-specification the oil-supply unit with water oil-cooler may in addition include the following components:

- filter
- monitoring equipment



The flow direction of the pump used is **independent of the direction of rotation**, if nothing is specified in the documentation to the contrary.
When connecting the fittings the actual flow direction must however be observed.



The required water connection must be provided by the user.

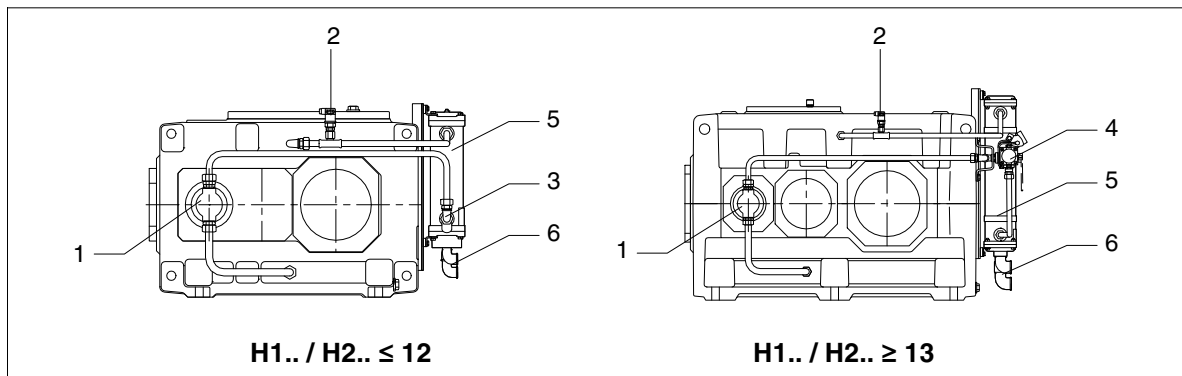


Fig. 30: Water oil-cooling system on gear units types H1.. and H2..

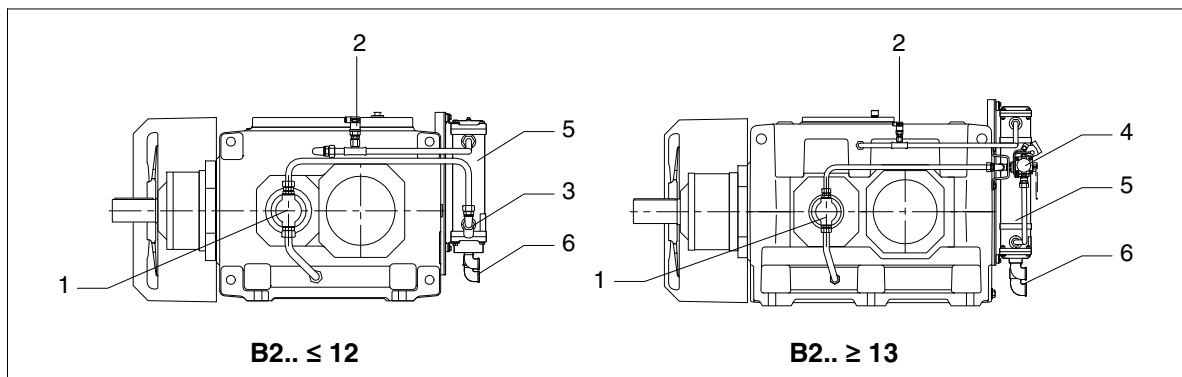


Fig. 31: Water oil-cooling system on gear units type B2..

- | | |
|---|-----------------------------|
| 1 Flange pump | 4 Double change-over filter |
| 2 Pressure monitor (circuit diagram see item 5.5.2) | 5 Water oil-cooler |
| 3 Coarse filter | 6 Water inlet and outlet |

A detailed view of the gear unit can be obtained from the drawings in the gear-unit documentation.



To ensure optimum cooling performance, the specified direction of flow in the water oil-cooler must be observed. The cooling-water inlet and outlet must not be reversed. The pressure of the cooling water must not exceed 8 bar.
If the gear unit is being withdrawn from service for a longer period and if there is a danger of freezing, the cooling water must be drained off. Remove any remaining water with compressed air.
Add-on pressure monitors must be connected as shown in item 5.5.2.



**Be especially careful when blowing with compressed air.
Wear protective glasses!**

Depending on the order specification and application, the flange pump may have been replaced with a motor pump.



For operation and maintenance, always observe the operating instructions indicated in the order-specific appendix.
For technical data, refer to the order-specific list of equipment.

5.10.4.1 Pump

The pumps used are suitable for the delivery of lubricants. The flow medium must not contain abrasive components and must not chemically affect the materials of the pump. A precondition of a proper functioning, high reliability and long service life of the pump is in particular a clean and lubricating delivery medium.

5.10.4.2 Water oil-cooler

Water oil-coolers are suitable for cooling oils. The cooling medium used is water.



For connecting dimensions, refer to the dimensioned drawing of the gear unit. The required cooling water quantity and the max. permissible inlet temperature are given on the data sheet and/or the list of equipment.

5.10.4.3 Filter

The filter protects downstream aggregates, measuring and control devices from contamination. The filter comprises a housing with connections and a sieve. The medium flows through the housing where the dirt particles flowing through the pipe are retained.
Dirty filter elements must be cleaned or replaced.

5.11 Heating

At low temperatures it may be necessary to heat the gear oil before switching on the drive unit or even during operation. In such cases the use of heating elements is possible. These heating elements convert electrical energy into heat which is conducted to the surrounding oil. The heating elements are located in protective tubes inside the housing, thus making it possible to replace them without draining off the oil.

Complete immersion of the heating elements in the oil bath must be guaranteed.

The heating elements can be controlled by a temperature monitor which emits a signal when maximum and minimum temperatures are reached; the signal requires amplification.

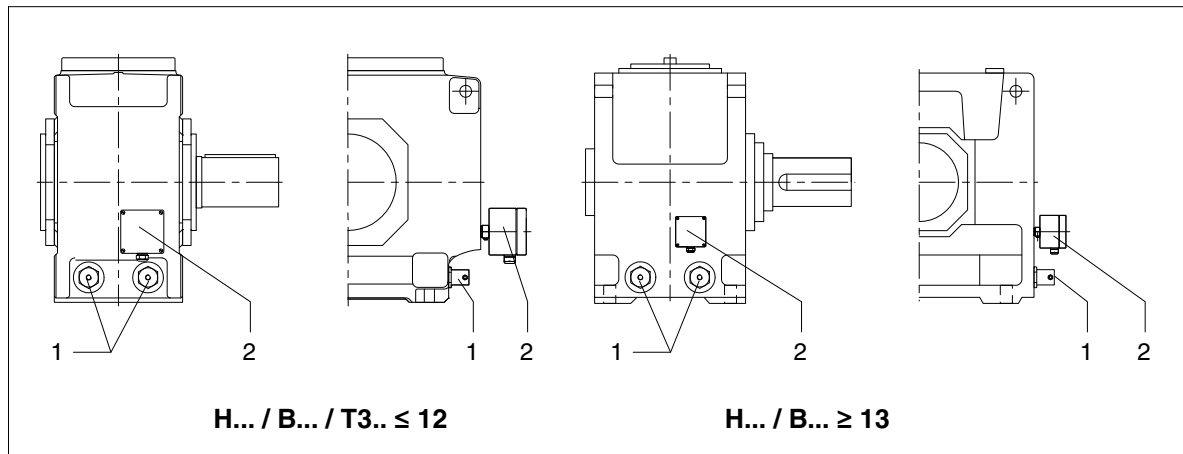


Fig. 32: Heating on gear units types H..., B... and T3..

1 Heating element

2 Temperature monitor

For a detailed illustration of the gear unit and the position of the add-on parts, refer to the drawings in the gear-unit documentation.



Never switch the heating elements on, unless complete immersion of the heating rod in the oil bath is ensured. Fire hazard!
If heating elements are installed afterwards the max. heating capacity (see table 14) on the outer surface of the heating element must not be exceeded.

Table 14: Specific heating output P_{Ho}
as a function of the ambient temperature

P_{Ho} (W/cm ²)	Ambient temperature °C
0.9	+ 10 to 0
0.8	0 to - 25
0.7	- 25 to - 50



Operation and maintenance must be in accordance with the pertinent operating instructions.
For technical data, refer to the list of equipment.

5.12 Oil-temperature monitoring

Depending on the order specification, the gear unit may be fitted with a Pt 100 resistance thermometer for monitoring the oil temperature in the sump. In order to measure the temperatures or temperature differences, the Pt 100 resistance thermometer should be connected to a suitable instrument provided by the customer. The thermometer has a connection head (protection type IP 54) for the wiring. A two-conductor circuit is provided by the manufacturer. However, the customer may fit his own three- or four-conductor circuit if required.



For control information, refer to the list of equipment.
Observe the operating instructions relating to the device in all instances.

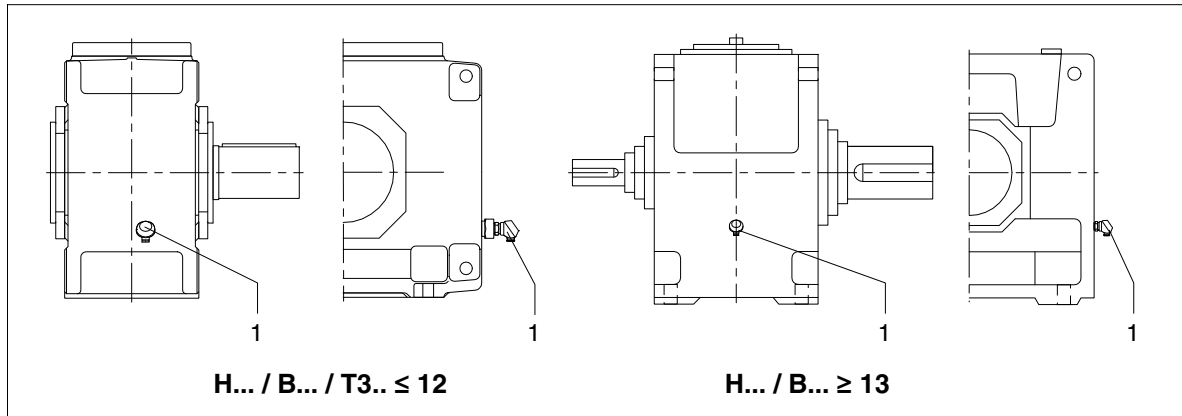


Fig. 33: Oil-temperature measurement on gear units types H..., B... and T3..

1 Pt 100 resistance thermometer

For a detailed illustration of the gear unit and the position of the add-on parts, refer to the drawings in the gear-unit documentation.



When operating and servicing the components, observe the operating instructions relating to the components.

For technical data, refer to the data sheet and/or the list of equipment.

5.13 Oil-level monitoring system

Depending on the order specification, the gear unit can be fitted with an oil-level monitor in the form of a level limit switch. This monitoring is designed as a standstill monitoring (gear unit stop) and checks the level of the oil before the unit is started up. When the signal "oil level too low" is given, it should be wired in such a way that the drive motor cannot start and an alarm is given. During operation, any signal should be bridged.

If an the oil-level monitoring device is in use, it is very important that the unit is in a horizontal position.

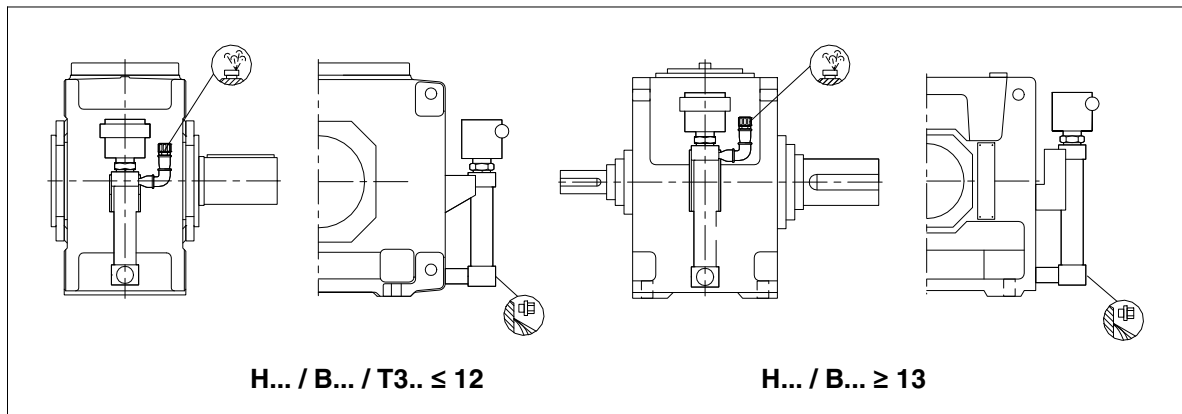


Fig. 34: Oil-level monitoring on gear units types H..., B... and T3..

For a detailed illustration of the gear unit and the position of the add-on parts, refer to the drawings in the gear-unit documentation.



When operating and servicing the components, observe the operating instructions relating to the components.

For technical data, refer to the data sheet and/or the list of equipment.

5.14 Bearing-monitoring system

The gear unit may be fitted with measuring nipples for monitoring the bearings. These nipples are intended for attachment of shock-pulse sensors with rapid-action coupling and are located in the vicinity of the bearings to be monitored.

The gear unit may also be set up for temperature monitoring at the bearing points. The gear unit is then equipped with tapped holes for fitting Pt 100 resistance thermometers. For this version Siemens must be consulted.

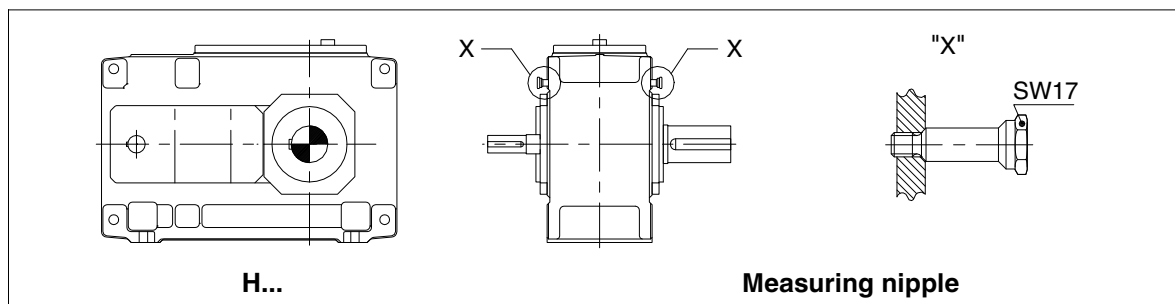


Fig. 35: Bearing monitoring on gear units type H...

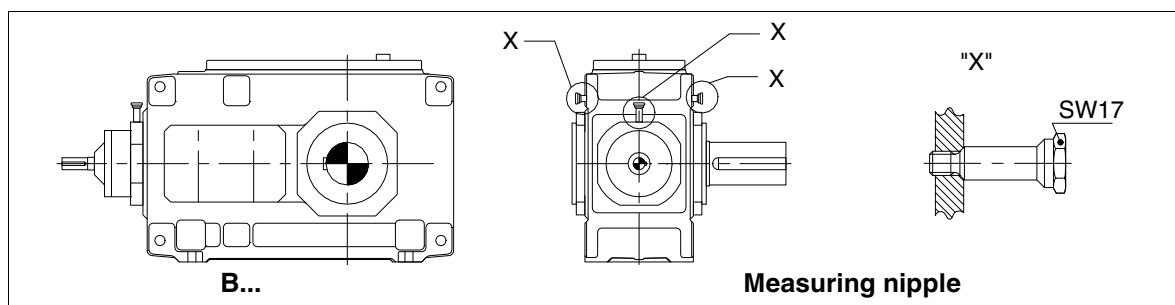


Fig. 36: Bearing monitoring on gear units types B... and T3..

For a detailed illustration of the gear unit and the position of the add-on parts, refer to the drawings in the gear-unit documentation.

5.15 Speed transmitter

An incremental speed transmitter may be mounted. Wiring and evaluation instrument should be provided by the customer.

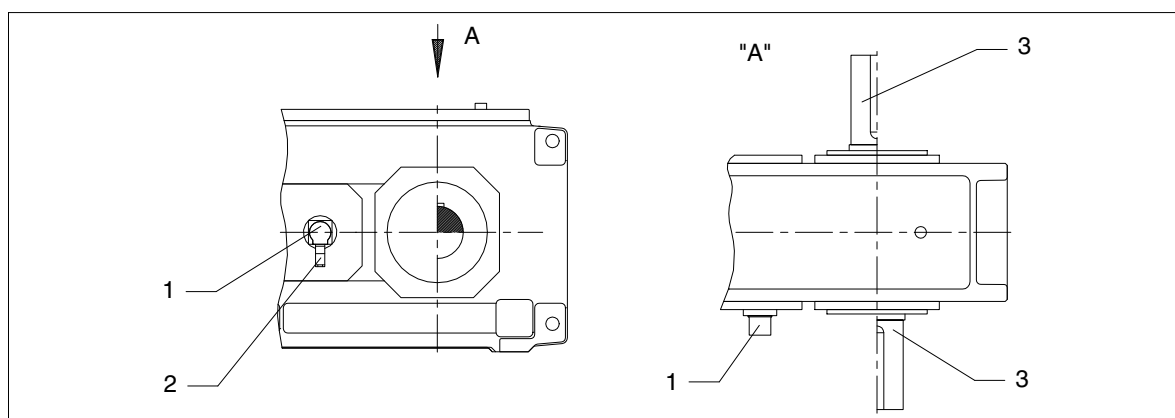


Fig. 37: Speed-monitoring device

- | | | | | | |
|---|-------------------------|---|--------------------|---|--------|
| 1 | Incremental transmitter | 2 | 12-pole brass plug | 3 | Output |
|---|-------------------------|---|--------------------|---|--------|



When operating and servicing the components, observe the operating instructions relating to the components.

For technical data, refer to the data sheet and/or the list of equipment.

A detailed view of the gear unit can be obtained from the drawings in the gear-unit documentation.

5.16 Auxiliary drive

For certain applications the gear unit can, in addition to the main drive unit, be equipped with an auxiliary drive unit. This enables the main gear unit to be operated at a lower output speed in the same direction of rotation. The auxiliary drive is connected with the main gear unit by an overrunning clutch. For the basic drive arrangement, see figure 38.

Basic design of the gear unit with main and auxiliary drives

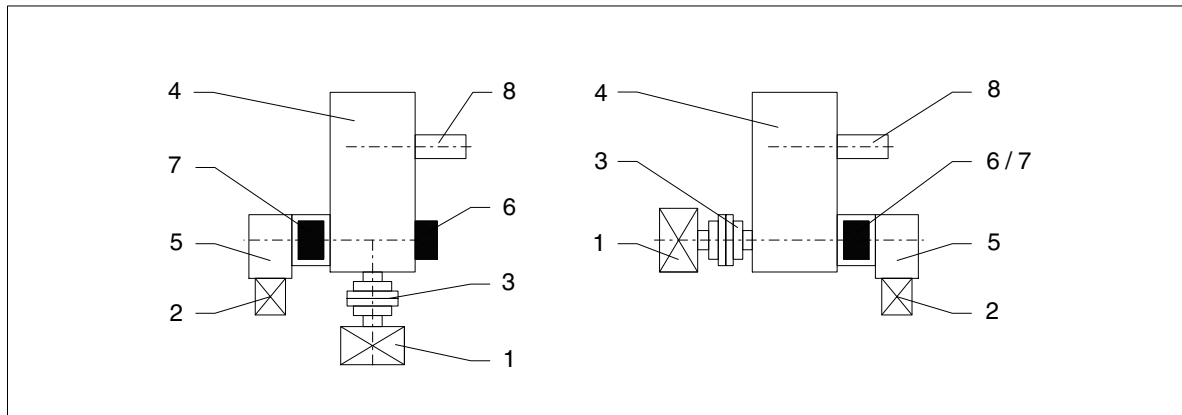


Fig. 38: Design of the gear unit with main and auxiliary drives

- | | | |
|-------------------|-----------------------|--------------------------------------|
| 1 Main motor | 4 Main gear unit | 7 Overrunning clutch |
| 2 Auxiliary motor | 5 Auxiliary gear unit | 8 Output shaft of the main gear unit |
| 3 Coupling | 6 Backstop | |

Depending on use, two auxiliary drives of different capacities are available for each gear-unit size.

5.16.1 Auxiliary drive, designed as a maintenance drive



The auxiliary drive should be protected from overloads.

The drive of the bucket elevator via the auxiliary drive must only be effected during idle running, i.e. without load.

For the exact designation of the geared motor as well as the mounting position, refer to the drawings (see section 1, "Technical Data"). The auxiliary gear unit has its own oil circulation system which is separated from that of the main gear unit. The auxiliary gear unit is already filled with oil when delivered.



Before connecting the motor, determine the direction of rotation of the three-phase current supply using a phase-sequence indicator, and connect the motor in accordance with the pre-determined direction of rotation.
Observe the notice fixed to the gear unit.



The Special operating instructions should be observed for operation of the auxiliary gear unit (Type KF MOTOX bevel-gear motor).

To avoid overspeeds in case of malfunctions of the overrunning clutch, the drive combination must be equipped with a speed-monitoring device by the customer for safety reasons. The speed-monitoring device consists of a pulse generator mounted in the intermediate flange (figure 39) and of an evaluating instrument.



A threaded hole M12x1 for the pulse generator to be made available by the customer is provided at a suitable position in the intermediate flange. The dimension "x" depends on the instructions given by the equipment manufacturer (see figure 39). The pulse generator must satisfy the requirement **"suitable for flush mounting"**.

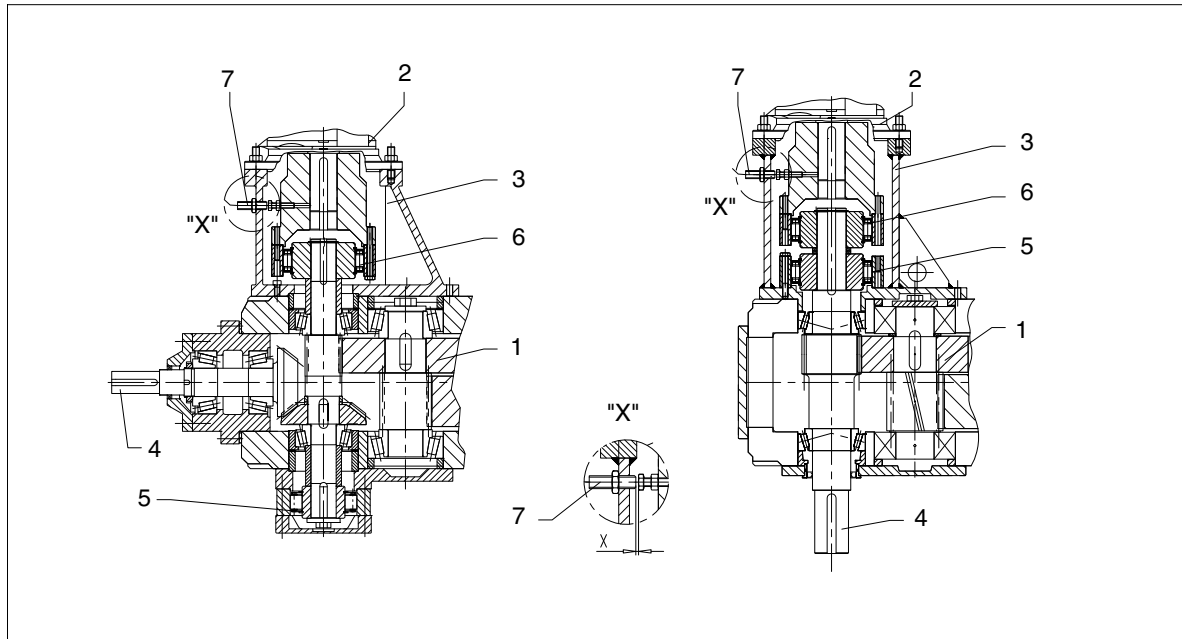


Fig. 39: Design of the gear unit with main and auxiliary drives

- | | | | | | |
|---|---------------------|---|-------------------------------|---|--------------------------------------|
| 1 | Main gear unit | 4 | Drive shaft of main gear unit | 6 | Overrunning clutch |
| 2 | Auxiliary gear unit | 5 | Backstop | 7 | Pulse generator for speed monitoring |
| 3 | Intermediate flange | | | | |

The speed-monitoring device must be connected in such a manner that the main drive is disabled automatically at "> zero" speed at the output shaft of the auxiliary drive. For safety reasons, the disabling function must be checked at regular intervals, i.e. at least quarterly. To check the disabling function, the auxiliary drive is switched on. If the speed-monitoring device connects - which can be verified, for instance, by means of a warning light - the speed-monitoring device is ready for operation.



This speed-monitoring device is an absolute must for safety reasons since in case of a defect in the overrunning system the auxiliary drive can be destroyed with explosive effect due to overspeeds.

5.16.2 Auxiliary drive, designed as a load drive

Unlike the maintenance drive, type KF and KZ bevel-helical MOTOX gear units are used here for the auxiliary drive. The motors of the auxiliary drives are dimensioned in such a manner that a properly loaded conveyor system can be operated at low speed in the same direction of rotation.



Furthermore the conditions set out in item 5.16.1 are applicable.

5.16.3 Overrunning clutch

If the gear unit is fitted with an auxiliary drive in addition to the main drive, coupling is realized by an overrunning clutch. This allows, when driving by the auxiliary drive, a torque transmission in one direction of rotation, while there is **"free-wheeling operation"** when driven by the main drive.

The output shaft of the main drive will rotate in the same direction of rotation both if the drive is effected via the main motor and via the auxiliary drive.

The overrunning clutch is mounted in an adapter flange and integrated in the oil circuit of the gear unit. Maintenance and oil change take place simultaneously with maintenance and oil change of the main drive.

The overrunning clutch is provided with centrifugally operated grippers. If the main gear unit is rotating in the specified direction of rotation, the inner ring will rotate together with the grippers, while the outer ring remains stationary. From a certain speed of rotation, the grippers will lift off and the overrunning clutch will operate without wear. If the drive is effected by the motor of the auxiliary drive via the outer ring, the overrunning clutch will be in **"carrier operation"**, i.e. the main gear unit is turned over slowly in the chosen direction of rotation. At the same time, the drive shaft of the main gear unit and, if a flexible coupling is used between main motor and gear unit, possibly the main motor will rotate slowly along with it.



The main motor and the motor of the auxiliary drive should be interlocked electrically in such a manner that only one of the two motors can be switched on.



When driving via the auxiliary drive, the drive shaft of the main gear unit will rotate along simultaneously. This rotary motion **must not** be impeded. A brake arranged on the drive side in the main drive must be released if the drive is effected via the auxiliary drive.



When filling the main gear unit with oil, start by supplying lubricating point 1 at the intermediate flange with the oil quantity and oil grade specified on the rating plate. Prior to startup, the overrunning clutch should be checked for proper function according to item 7.2.4.

6. Fitting

Observe the instructions in section 3, "Safety instructions"!

6.1 General information on fitting

When transporting the gear unit observe the notes in section 4, "Transport and storage".

Fitting work must be done with great care by authorised, trained and qualified personnel. The manufacturer cannot be held liable for damage caused by incorrect assembly and installation.

During the planning phase sufficient space must be allowed around the gear unit for later care and maintenance work.



Free convection through the surface of the housing must be ensured by suitable measures.

If the gear unit is fitted with a fan, there should be sufficient space for air intake.

Adequate lifting equipment must be available before beginning the fitting work.



During operation the unit must not be allowed to heat up through exposure to heat from external sources such as sunlight, and suitable measures must be taken to prevent this!

Such measures may be:

- fitting a sunshade roof,
or
- fitting an additional cooling unit,
or
- fitting the oil sump with a temperature-monitoring device with a cut-out function.



If a sunshade roof is fitted, heat must be prevented from building up!

If a temperature-monitoring device is fitted, a warning signal must be emitted when the maximum permitted oil-sump temperature is reached. If the maximum permitted oil-sump temperature is exceeded, the drive must be shut off. Such shutting off may cause the operator's system to stop!



The operator should ensure that no foreign bodies affect the proper function of the gear unit (e.g. falling objects or heaping over).

No electrical welding work must be done at all on the drive.

The drives must not be used as an earthing point for welding operations. Toothed parts and bearings may be irreparably damaged by welding.

All the fastening points provided by the design of the unit must be used.

Screws which have been damaged during assembly or disassembly work must be replaced with new ones of the same strength class and type.



To ensure proper lubrication during operation, the mounting position specified on the drawings must always be observed.

6.2 Unpacking

The products supplied are listed in the despatch papers. Check immediately on receipt to ensure that all the products listed have actually been delivered. Parts damaged and/or missing parts must be reported to Siemens in writing immediately.



The packaging must not be opened or damaged, when this is part of the preservation method!

- Remove packaging material and transporting equipment and dispose of in accordance with regulations.
- Perform a visual check for any damage and contamination.



If there is any visible damage, the gear unit must not be put into operation. The instructions in section 4, "Transport and storage", must be observed.

6.3 Installation of gear unit on housing base

6.3.1 Foundation



The foundation must be horizontal and level. The gear unit must not be excessively stressed when tensioning the fastening bolts.

The foundation should be designed in such a way that no resonance vibrations are created and that no vibrations are transmitted from adjacent foundations. The structure on which the unit is to be mounted must be rigid. It must be designed according to the weight and torque, taking into account the forces acting on the gear unit.

Careful alignment with the units on the in- and output sides must be ensured. Any elastic deformation through operating forces must be taken into consideration.



Fastening bolts or nuts must be tightened to the prescribed torque. For the correct torque, refer to item 6.23. Bolts of the minimum strength class 8.8 must be used.

If external forces are acting upon the gear unit, it is advisable to prevent the unit from displacement by means of lateral stops.



For dimensions, space requirement and arrangement of supply connections, refer to the drawings in the gear-unit documentation.

6.3.2 Description of installation work

- Remove the anti-corrosion paint on the shafts with suitable cleaning agent such as benzine.



Do not allow the cleaning agent (e.g. benzine) to contact the shaft-sealing rings.



**Ensure adequate ventilation. Do not smoke!
Danger of explosion!**

- Mount and secure input and output drive elements (e.g. coupling components) on the shafts.
If these are to be heated before mounting, refer to the dimensioned drawings in the coupling documentation for the correct joining temperatures.

Unless otherwise specified, the components may be heated inductively, with a burner, or in a furnace.



Take precautions to avoid burns from hot components!



**Protect shaft sealing rings from damage and heating to over + 100 °C
(use heat-protective screens to protect against radiant heat.)**

The components must be pushed smartly onto the shaft up to the position specified in the order-specific dimensioned drawing.



Fit the coupling with the aid of suitable fitting equipment. The parts must not be driven on by abrupt force, as this may damage the gear unit (see also item 6.8). The shaft-sealing rings and running surfaces of the shaft must not be damaged when pulling in the coupling parts.



When installing the drives, make absolutely certain that the individual components are accurately aligned in relation to each other. Inadmissibly large errors in the alignment of the shaft ends to be connected due to angular and/or axial misalignments result in premature wear and/or material damage. Insufficiently rigid base frames or sub-structures can also during operation cause a radial and/or axial misalignment, which cannot be measured when the unit is at a standstill.



Gear units whose weight requires the use of lifting gear must be attached at the points shown in section 4, "Transport and storage". If the gear unit is to be transported with add-on parts, additional attachment points may be required. The position of these attachment points is shown in the order related dimensioned drawing.

6.3.2.1 Alignment surfaces, alignment thread

Preliminary alignment of the gear units (sizes 3 to 12) in a horizontal direction is done by the surfaces of the inspection and/or assembly cover.

Aligning surface:



For the exact position of the aligning surfaces refer to the drawings in the gear-unit documentation.

The alignment surfaces are for aligning the gear unit horizontally, in order to ensure correct running of the gear unit.



The values punched into the alignment surfaces must always be observed.

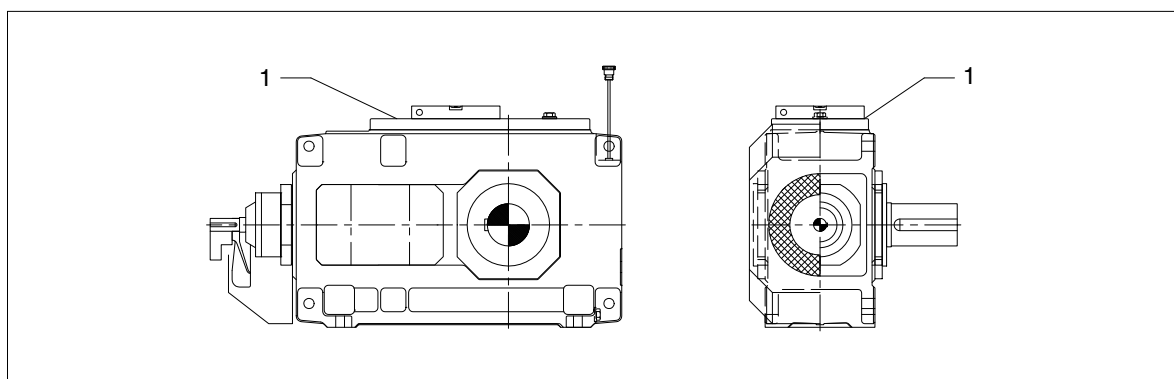


Fig. 40: Alignment surfaces on gear unit up to size 12

Gear units of sizes 13 to 22 have special alignment surfaces on the top of the housing for preliminary alignment of the gear units. To facilitate alignment of gear units of these sizes, alignment threads are provided in the housing base.

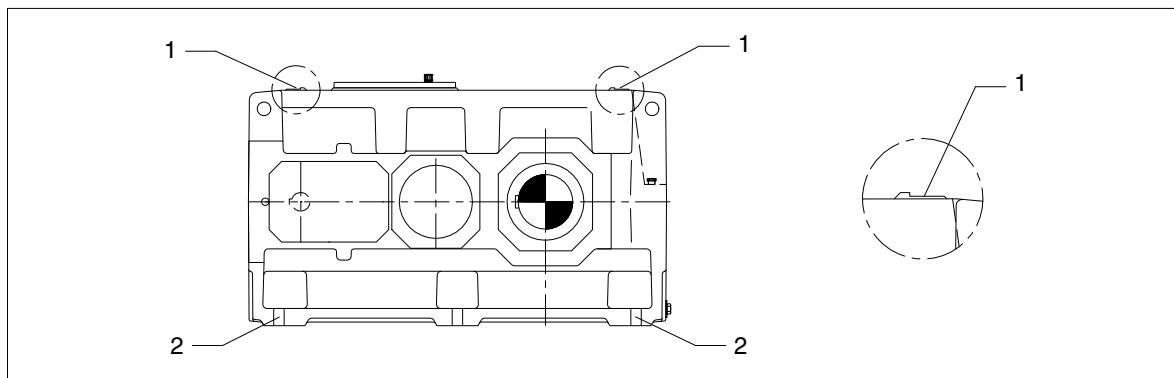


Fig. 41: Alignment surfaces on gear units from size 13

1 Alignment surfaces

2 Alignment thread

The final fine alignment with the assemblies on the in- and output side must be carried out accurately by the shaft axes, using:

- rulers
- spirit level
- dial gauge
- feeler gauge, etc.

Only then should the gear unit be fastened and afterwards the alignment be checked once again.

- Record alignment dimensions.



The report must be kept with these instructions.



The accuracy of shaft axis alignment is an important factor in determining the life span of shafts, bearings and couplings. If possible, the deviation should be zero (exception: ZAPEX couplings). For amongst others the special requirements for the couplings, refer to the specific operating instructions.



Non-observance can cause shaft rupture, resulting in serious injury or danger of life.

6.3.2.2 Mounting on a foundation frame

- Clean the undersurface of the gear-unit base.
- Using suitable lifting gear, place the gear unit on the foundation frame.
- Tighten the foundation bolts to the specified torque (see item 6.23); if necessary, use stops to prevent displacement.



The gear unit must not be excessively stressed when tensioning the fastening bolts.

- Align the gear unit exactly with the input and output units (see item 6.3.2.1).
- Record alignment dimensions.



The report must be kept with these instructions.

6.3.2.3 Mounting on a concrete foundation by means of stone bolts or foundation blocks

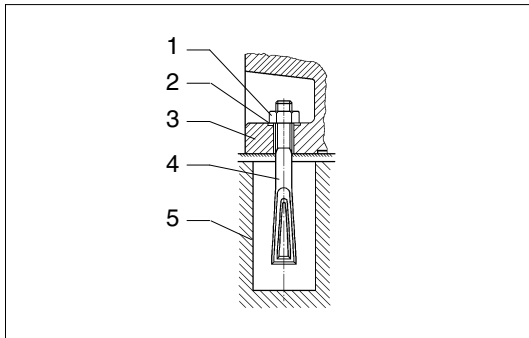
- Clean the undersurface of the gear-unit base.

Placing stone bolts:

- Hook stone bolts with washers and hexagon nuts into the foundation fastening points on the gear-unit housing (see figure 42).



The hexagon nuts must only be tightened when the concrete has set.



- 1 Hexagon nut
- 2 Washer
- 3 Gear-unit base
- 4 Stone bolt
- 5 Foundation

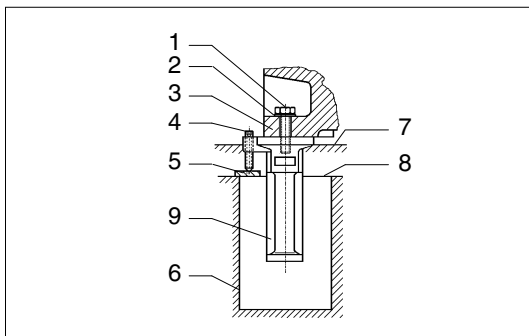
Fig. 42: Stone bolt

Placing the foundation blocks:

- Hook the foundation blocks with washers and fastening bolts into the foundation fastening points on the gear-unit housing (see figure 43).



The fastening bolts must only be tightened when the concrete has set.



- 1 Fastening bolt
- 2 Washer
- 3 Gear-unit base
- 4 Threaded stud
- 5 Flat steel plate
- 6 Foundation
- 7 Final foundation height
- 8 Prepared foundation height
- 9 Foundation block

Fig. 43: Foundation block

- Using suitable lifting gear, place the gear unit on the concrete foundation.
- Align gear unit horizontally by in- and output shafts:
 - if using stone bolts, with shims,
 - if using foundation blocks, with the aid of the set screws (if available).
- If considerable forces may apply, use stops to prevent the unit from displacement.



Before pouring the concrete foundation, fill up the openings in the foundation blocks with adequate material such as polystyrene.

With types **H1** and **H2**, remove the air-conducting cowl before tightening the foundation bolts and then bolt it back into position.

- Pour concrete into the recesses of the stone bolts or foundation blocks.



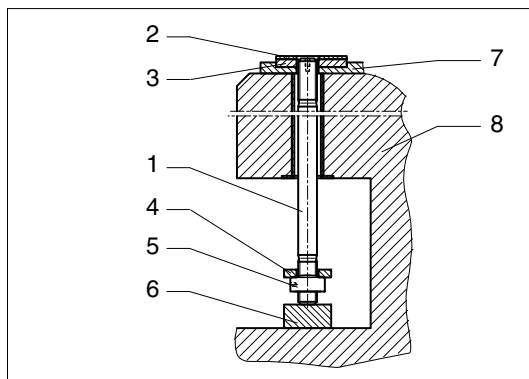
When the concrete has set, tighten the hexagon nuts of the stone bolts or fastening bolts of the foundation blocks to the specified torque (see item 6.23).



The gear unit must not be excessively stressed when tensioning the hexagon nuts or fastening bolts.

6.3.2.4 Mounting on a concrete foundation by means of anchor bolts

- Clean the undersurface of the gear-unit base.
- Place support on the base plate in the fine grout.
- Insert anchor bolts.
- Place pressure plates in position and screw nuts on.
- Pack the anchor bolts with wood so that they are about 10 mm from the upper edge of the support (see figure 44).



- | | |
|---|---------------------|
| 1 | Anchor bolt |
| 2 | Support |
| 3 | Base plate |
| 4 | Pressure plate |
| 5 | Hexagon nut |
| 6 | Wood |
| 7 | Fine-grout concrete |
| 8 | Raw foundation |

Fig. 44: Anchor bolt

- Place gear unit on foundation.



**Use only the eyes provided to attach lifting equipment to the unit.
Do not use the front threads at the shaft ends to attach slinging equipment for the transport.**

- Pull anchor bolts up (for this a bolt or threaded rod can be screwed into the thread on the front face).
- Fit washer.
- Unscrew hexagon nut a few turns by hand.
- Align gear unit with supports (see figure 45).
 - The values punched into the screeds must always be observed.
 - Alignment tolerances in relation to the units on the input and output sides are to be in accordance with the permissible angular and axial displacements of the couplings (see coupling drawings).
 - The alignment values must be recorded.

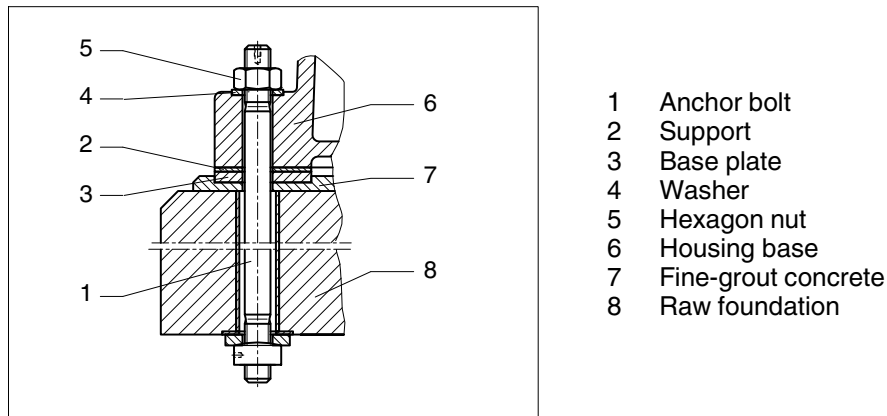


Fig. 45: Anchor bolt



Prior to tensioning the anchor bolts, the fine-grout concrete must have set for at least 28 days.

- Keep anchor bolts in their position by tightening the nut with your fingers.
- Place the protective sleeve.
- Place hydraulic tensioning device in position.
- Initially tension the bolts alternately (for initial tensioning forces, see item 6.23).
- Using a suitable tool, screw hexagon nuts on as far as the stop.



To ensure correct handling and adjustment of the pretensioning tool, the manufacturer's operating instructions must be adhered to.

The tensioning pressures and/or the initial tensioning forces should be recorded (see also item 7.2.9).

6.4 Assembly of a shaft-mounting gear unit with hollow shaft and parallel keyway

The end of the driven-machine shaft (material C60+N or higher strength) must be provided with a parallel key to DIN 6885 Part 1 Form A. Furthermore, a centring hole to DIN 332 Form DS (tapped) should be provided (for the connection dimensions of the driven machine shaft, see dimensioned drawing in the gear unit documentation).

6.4.1 Preparatory work

To facilitate demounting (see also item 6.4.3), we recommend providing a connection for pressure oil on the end of the driven machine shaft. For this a hole must be drilled through to the hollow shaft bore (see figure 46). This connection may also be used for supplying rust-releasing agent.

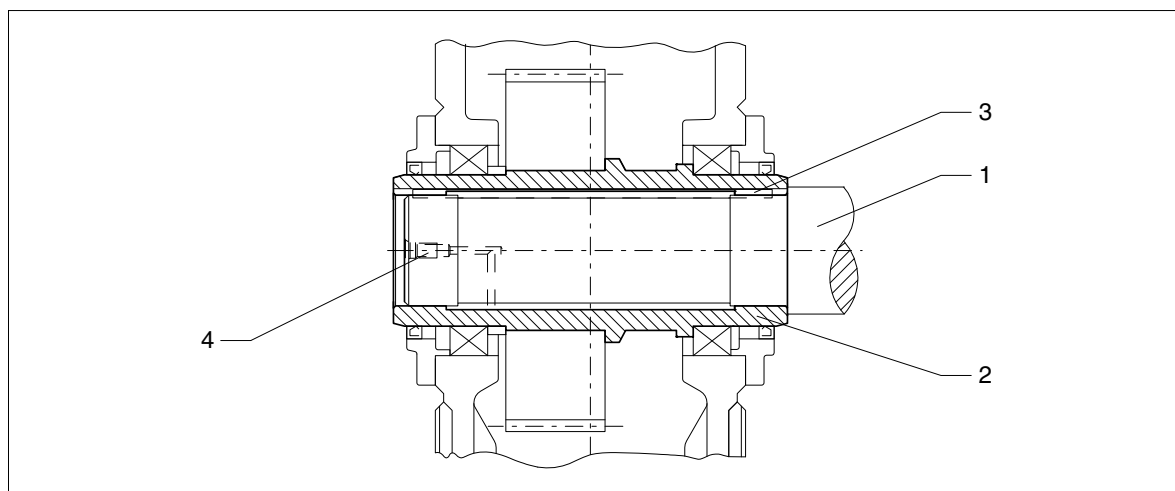


Fig. 46: Hollow shaft with parallel keyway, preparation

- | | | | |
|---|---------------|---|-------------------------|
| 1 | Machine shaft | 3 | Parallel key |
| 2 | Hollow shaft | 4 | Pressure oil connection |

6.4.2 Fitting

- Remove the preservative agent from the hollow shaft and the machine shaft with a suitable cleaning agent (such as benzine).



Do not allow the cleaning agent (e.g. benzine) to contact the shaft-sealing rings.



**Ensure adequate ventilation. Do not smoke!
Danger of explosion!**

- Check the hollow and machine shafts to ensure that seats and edges are not damaged. If necessary, rework the parts with a suitable tool and clean them again.



Coat with a suitable lubricant to prevent frictional corrosion of the contact surfaces.

6.4.2.1 Fitting

- Fit the gear unit by means of nut and threaded spindle. The counterforce is provided by the hollow shaft.



The hollow shaft must be exactly aligned with the machine shaft to avoid canting.

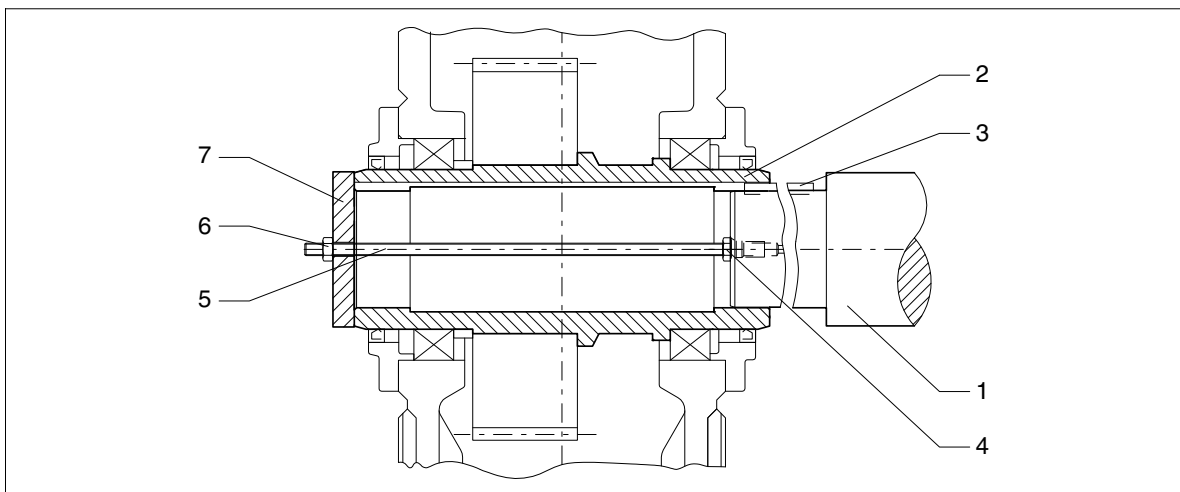


Fig. 47: Hollow shaft with parallel keyway, mounting with threaded spindle

1	Machine shaft	4	Nut	7	End plate
2	Hollow shaft	5	Threaded spindle		
3	Parallel key	6	Nut		

Instead of the nut and threaded spindle shown in the diagram, other types of equipment such as a hydraulic lifting equipment (type "Lukas") may be used.



The hollow shaft may be tightened against a machine-shaft collar only if the gear-unit configuration is one of the following:

- Torque arm
- Support with gear-unit swing base

With a different arrangement the bearings may be excessively stressed during the mounting procedure.

6.4.2.2 Axial fastening

Depending on type, secure the hollow shaft axially on the machine shaft (e.g. with locking ring, end plate, set screw).

6.4.3 Demounting

- Remove the axial securing device from the hollow shaft.
- If frictional corrosion has occurred on the seating surfaces, rust-releasing agent may be used in order to facilitate forcing off the gear unit. The rust releaser can be injected through the pressure-oil connection (see figure 46), e.g. using a pump.
- When the rust-releasing agent has taken effect, pull the gear unit off with the device (see figures 48 and 49).
- Removing the gear unit from the driven-machine shaft can be done locally as follows:
 - using forcing screws in an end plate (see figure 49) or
 - using a central threaded spindle or
 - preferably using a hydraulic lifting unit ("Lukas").



The end plate and/or the auxiliary plate for forcing off the gear unit are not included in our delivery.
Each of the two end faces of the hollow shaft is provided with 2 threaded holes (for dimensions, see figure 50) to receive bolts for fastening the end plate to the hollow shaft.

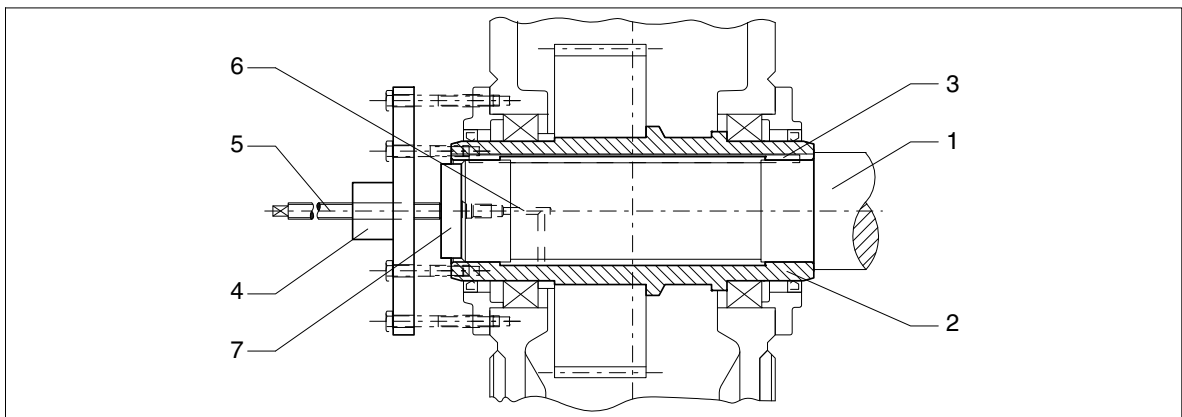


Fig. 48: Hollow shaft with parallel keyway, demounting with hydraulic lifting equipment ("Lukas")

- | | |
|------------------------------------|---------------------------|
| 1 Machine shaft | 5 Threaded spindle |
| 2 Hollow shaft | 6 Pressure oil connection |
| 3 Parallel key | 7 Plate for forcing out |
| 4 Hydraulic lifting unit ("Lukas") | |

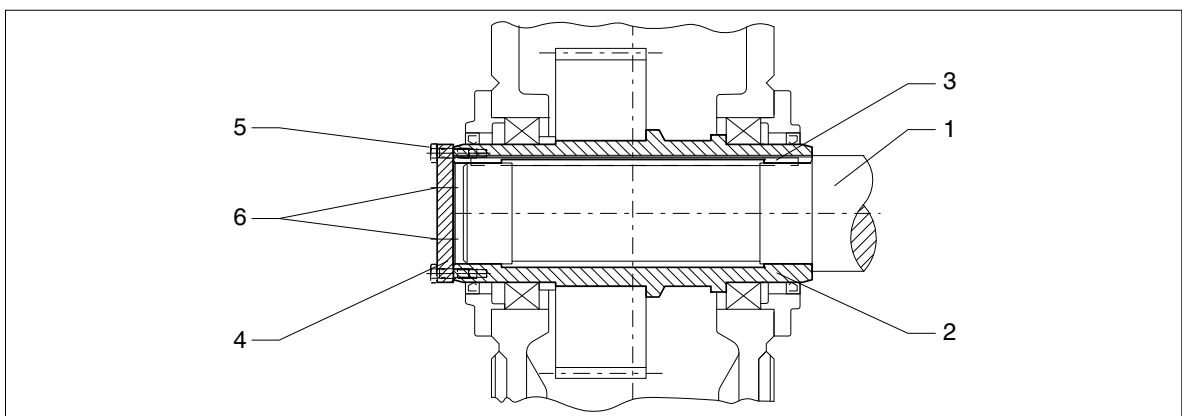


Fig. 49: Hollow shaft with parallel keyway, demounting with end plate

- | | |
|-----------------|-----------------------------|
| 1 Machine shaft | 4 End plate for forcing out |
| 2 Hollow shaft | 5 Screws |
| 3 Parallel key | 6 Forcing screws |



Avoid canting when pulling the unit off.



The plate for forcing-out is not included in our delivery.

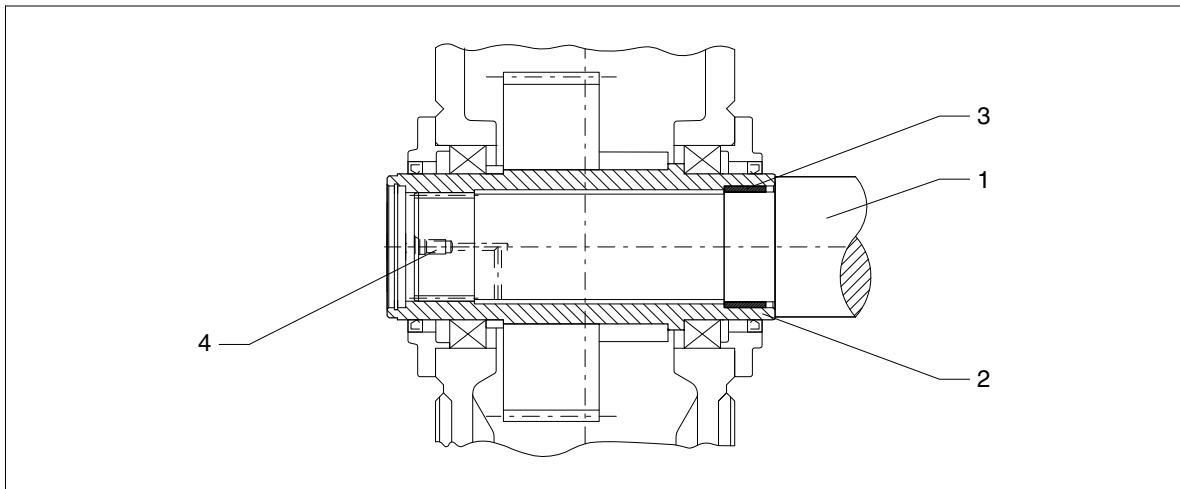


Fig. 50: Hollow shaft with parallel keyway

*) 2 threads offset 180°

Table 15: Threaded holes on the end faces of the gear-unit hollow shafts

Gear-unit size	m mm	s	t mm	Gear-unit size	m mm	s	t mm
4	95	M 8	14.5	12	215	M 12	19.5
5	115	M 8	14.5	13	230	M 12	19.5
6	125	M 8	14.5	14	250	M 12	19.5
7	140	M 10	17	15	270	M 16	24
8	150	M 10	17	16	280	M 16	24
9	160	M 10	17	17	300	M 16	24
10	180	M 12	19.5	18	320	M 16	24
11	195	M 12	19.5	19 ... 22	on request		



If the counterforce is provided not only by the hollow shaft, but also by the housing, as shown in figure 48, the forces used must not exceed the values given in the following table 16.

Table 16: Maximum forcing pressures

Gear-unit size	Maximum forcing pressure N	Gear-unit size	Maximum forcing pressure N
4	22600	12	113600
5	33000	13	140000
6	37500	14	160000
7	50000	15	193000
8	56000	16	215000
9	65000	17	240000
10	82000	18	266000
11	97200	19 ... 22	on request



If the above values are exceeded, the housing, the hollow-shaft bearings or other gear components may be irreparably damaged. Before replacing the gear unit on the machine shaft, always check the bearings for any signs of damage.



When using forcing screws or threaded spindles, the head of the thread pressing against the driven machine should be rounded and well greased to reduce the risk of seizing at this point.

6.5 Shaft-mounting gear unit with hollow shaft and internal spline to DIN 5480

The shaft end of the driven machine must be designed with internal splines to DIN 5480. Furthermore, a centring hole to DIN 332 Form DS (tapped) should be provided (for the connection dimensions of the driven machine shaft, see dimensioned drawing in the gear unit documentation).

6.5.1 Preparatory work

To facilitate demounting (see also item 6.4.3), we recommend providing a connection for pressure oil on the end of the driven machine shaft. For this a hole must be drilled through to the hollow shaft bore (see figure 51). This connection may also be used for supplying rust-releasing agent.

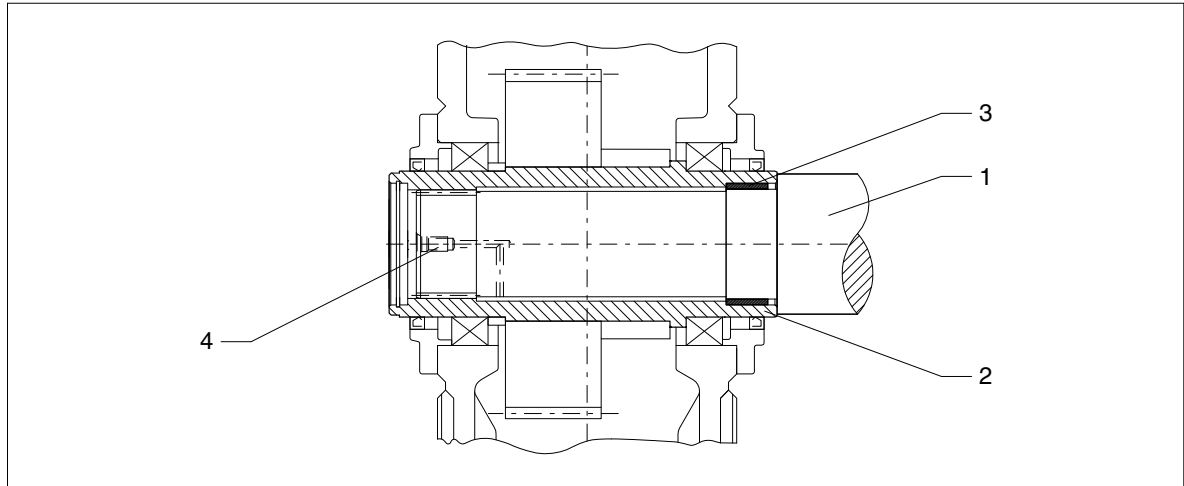


Fig. 51: Hollow shaft with internal spline, preparation

- | | | | |
|---|---------------|---|-------------------------|
| 1 | Machine shaft | 3 | DU bush |
| 2 | Hollow shaft | 4 | Pressure oil connection |

6.5.2 Fitting

- Remove the preservative agent from the hollow shaft and the machine shaft with a suitable cleaning agent (such as benzine).



Do not allow the cleaning agent (e.g. benzine) to contact the shaft sealing rings.



**Ensure adequate ventilation. Do not smoke!
Danger of explosion!**

- Check the hollow and machine shafts to ensure that seats, teeth or edges are not damaged. If necessary, rework the parts with a suitable tool and clean them again.



Coat with a suitable lubricant to prevent frictional corrosion of the contact surfaces.

6.5.2.1 Fitting with integrated DU bush

- Fit the gear unit by means of threaded spindle and nut. The counterforce is provided by the hollow shaft.



The hollow shaft must be exactly aligned with the machine shaft to avoid canting. When fitting, ensure that the position of the teeth between the machine shaft and hollow shaft is correct. The correct position can be determined by turning the input shaft and/or by swivelling the gear unit lightly around the hollow shaft.

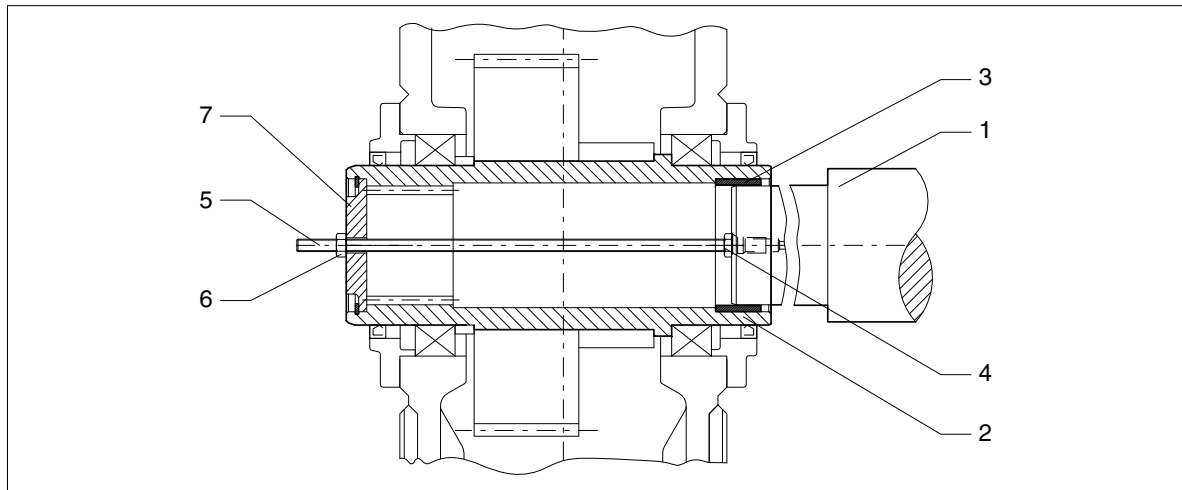


Fig. 52: Hollow shaft with internal spline, mounting with DU bush

- | | | | | | |
|---|---------------|---|------------------|---|-----------|
| 1 | Machine shaft | 4 | Nut | 7 | End plate |
| 2 | Hollow shaft | 5 | Threaded spindle | | |
| 3 | DU bush | 6 | Nut | | |

6.5.2.2 Fitting with loose DU bush

The loose DU bush is pushed onto the machine shaft, fixed rigidly in position with a locating tie and then pulled into the hollow shaft along with the machine shaft (see figure 52).



The hollow shaft must be exactly aligned with the machine shaft to avoid canting. When fitting, ensure that the position of the teeth between the machine shaft and hollow shaft is correct. The correct position can be determined by turning the input shaft and/or by swivelling the gear unit lightly around the hollow shaft.

Instead of the nut and threaded spindle shown in the diagram, other types of equipment such as a hydraulic lifting equipment (type "Lukas") may be used.



The hollow shaft may be tightened against a machine-shaft collar only if the gear-unit configuration is one of the following:

- Torque arm
- Support with gear-unit swing base

With a different arrangement the bearings may be excessively stressed.

6.5.2.3 Axial fastening

Depending on type, secure the hollow shaft axially on the machine shaft (e.g. with locking ring, end plate, set screw).

6.5.3 Demounting

- Remove the axial securing device from the hollow shaft.
- If frictional corrosion has occurred on the seating surfaces, rust-releasing agent may be used in order to facilitate forcing off the gear unit. The rust releaser can be injected through the pressure-oil connection (see figure 53), e.g. using a pump.
- The end plate and the locking ring must first be removed.
- When the rust-releasing agent has taken effect, pull the gear unit off with the device (see figures 53 and/or 54).
- Removing the gear unit from the driven-machine shaft can be done locally as follows:
 - using forcing screws in an end plate (see figure 54) or
 - using a central threaded spindle or
 - preferably using a hydraulic lifting unit ("Lukas").

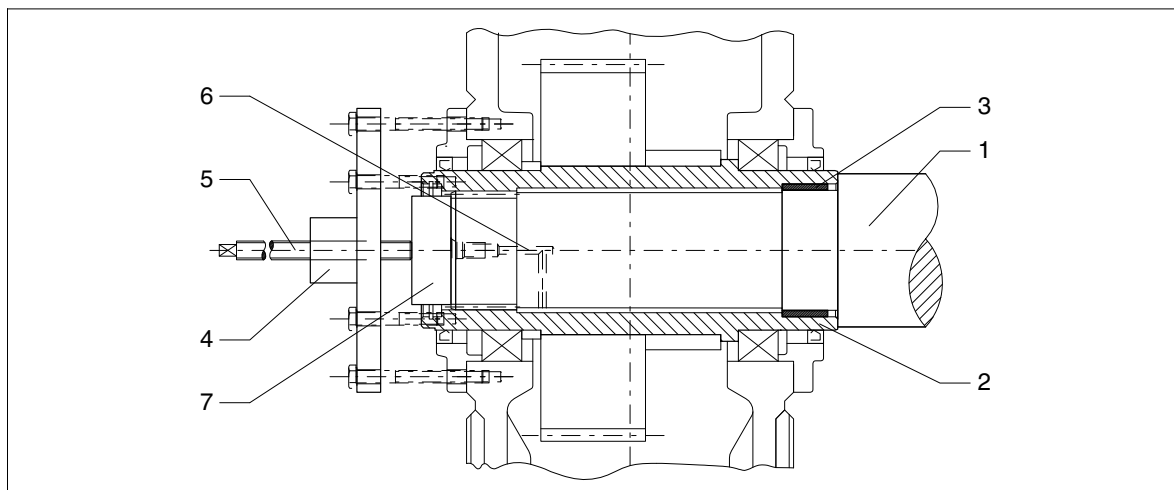


Fig. 53: Hollow shaft with internal spline, demounting with hydraulic lifting equipment ("Lukas")

- | | |
|---|---------------------------|
| 1 Machine shaft | 5 Threaded spindle |
| 2 Hollow shaft | 6 Pressure oil connection |
| 3 DU bush | 7 Plate for forcing out |
| 4 Hydraulic lifting equipment ("Lukas") | |

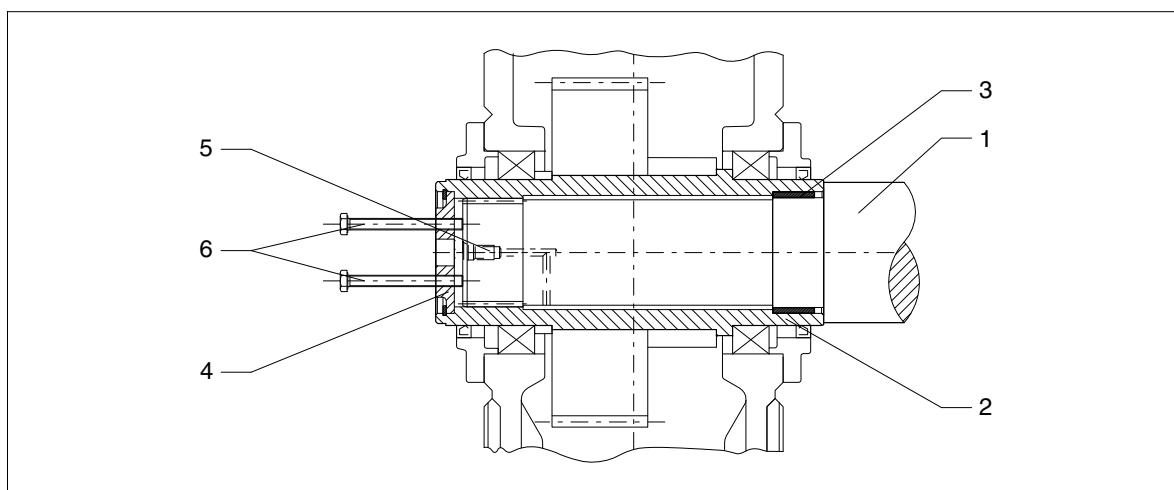


Fig. 54: Hollow shaft with internal spline, demounting with end plate

- | | |
|-----------------|---------------------------|
| 1 Machine shaft | 4 End plate |
| 2 Hollow shaft | 5 Pressure oil connection |
| 3 DU bush | 6 Forcing screws |



Avoid canting when pulling the unit off.



The plate for forcing-out is not included in our delivery.



If the counterforce is provided not only by the hollow shaft, but also by the housing, as shown in figure 53, the forces used must not exceed the values given in the following table 17.

Table 17: Maximum forcing pressures

Gear-unit size	Maximum forcing pressure N	Gear-unit size	Maximum forcing pressure N
4	22600	12	113600
5	33000	13	140000
6	37500	14	160000
7	50000	15	193000
8	56000	16	215000
9	65000	17	240000
10	82000	18	266000
11	97200	19 ... 22	on request



If the above values are exceeded, the housing, the hollow-shaft bearings or other gear components may be irreparably damaged. Before replacing the gear unit on the machine shaft, always check the bearings for any signs of damage.



When using forcing screws or threaded spindles, the head of the thread pressing against the driven machine should be rounded and well greased to reduce the risk of seizing at this point.

6.6 Shaft-mounting gear unit with hollow shaft and shrink disk

The end of the driven machine shaft (material C60+N or higher strength) should have a centring means to DIN 332 Form DS (with thread) in its end face (for connecting dimensions of the driven machine shaft, see dimensioned drawing in the gear-unit documentation).

6.6.1 Fitting

- Remove the preservative agent from the hollow shaft and the machine shaft with a suitable cleaning agent (such as benzine).



Do not allow the cleaning agent (e.g. benzine) to contact the shaft sealing rings.



**Ensure adequate ventilation. Do not smoke.
Danger of explosion!**

- Check the hollow and machine shafts to ensure that seats and edges are not damaged. If necessary, rework the parts with a suitable tool and clean them again.



**The bore of the hollow shaft and the machine shaft must be absolutely free of grease in the area of the shrink disk seat.
This is essential for safe and reliable torque transmission.
Do not use contaminated solvents or dirty cloths for removing grease.**

6.6.1.1 Fitting with integrated DU bush

- Fit the gear unit by means of nut and threaded spindle. The counterforce is provided by the hollow shaft.



The hollow shaft must be exactly aligned with the machine shaft to avoid canting.

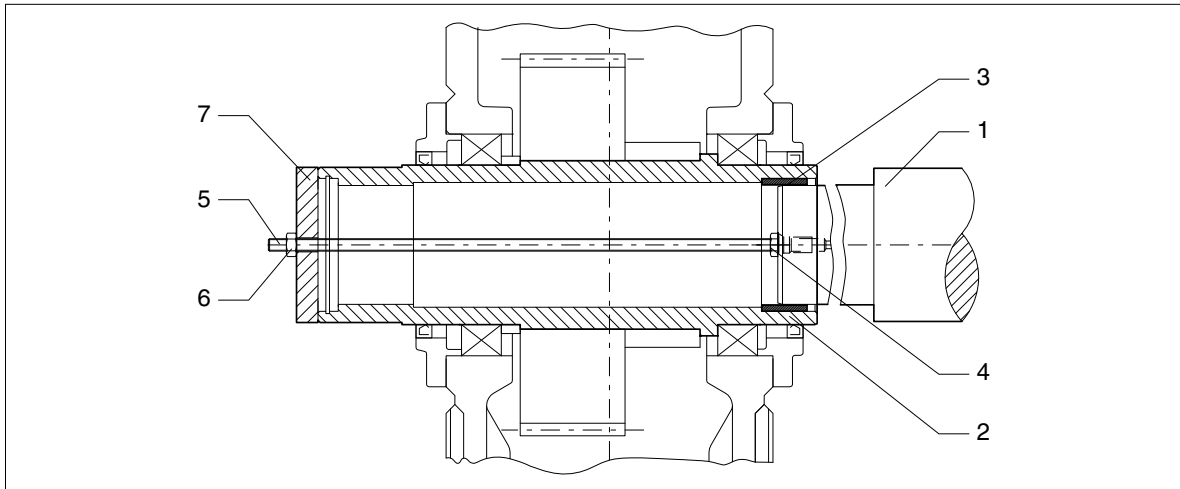


Fig. 55: Hollow shaft in shrink-disk design, mounting with DU bush

1	Machine shaft	4	Nut	7	End plate
2	Hollow shaft	5	Threaded spindle		
3	DU bush	6	Nut		

6.6.1.2 Fitting with loose DU bush

The loose DU bush is pushed onto the machine shaft, fixed rigidly in position with a locating tie and then pulled into the hollow shaft along with the machine shaft (see figure 55).



The hollow shaft must be exactly aligned with the machine shaft to avoid canting.

Instead of the nut and threaded spindle shown in the diagram, other types of equipment such as a hydraulic lifting equipment (type "Lukas") may be used.



The hollow shaft may be tightened against a machine-shaft collar only if the gear-unit configuration is one of the following:

- Torque arm
- Support with gear-unit swing base

With a different arrangement the bearings may be excessively stressed.

6.6.1.3 Axial fastening

If the shrink disk is fitted according to instructions (see item 6.6), the gear unit is fixed securely in the axial direction. Additional fastening is not required.

6.7 Shrink disk

The shrink disk realizes a press fit connection between a hollow shaft and a stub/machine shaft (in the following called "stub shaft"). The interference fit can transfer torques, bending moments and forces. The jointing pressure between the hollow and stub shafts generated by the shrink disk is essential for the torque and force transmission.

The shrink disk is delivered ready for installation.



The shrink disk must not be dismantled before mounting for the first time.

Mounting and start-up must be carried out by properly trained specialist personnel. Prior to start-up these instructions must be read, understood and adhered to. We accept no liability for personal injury or damage due to non-observance.

6.7.1 Fitting the shrink disk

- Before beginning installation, the hollow shaft and the stub shaft must be carefully cleaned.



Observe manufacturer's instructions for handling lubricants and solvents.



Do not allow cleansing agent or solvent to affect surfaces with paint coating.



The bore of the hollow shaft and the stub shaft must be absolutely clean, free of grease and oil in the area of the shrink disk seat.

This is essential for safe and reliable torque transmission.

Do not use contaminated solvents or dirty cloths nor cleansing agents containing oil (such as paraffin or terpentine) for removing grease.

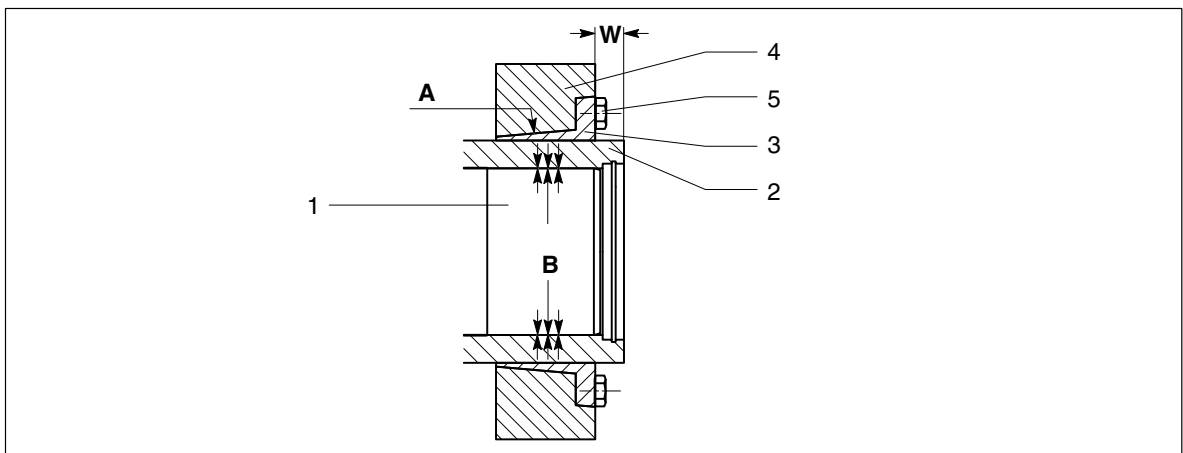


Fig. 56: Fitting the shrink disk

A	Greased	B	Absolutely grease- and oil-free	W	Installation height
1	Stub shaft	3	Inner ring	5	Tensioning bolt
2	Hollow shaft	4	Outer ring		



The outer surface of the hollow shaft must be lightly greased in the area of the shrink disk seat.

For a detailed view, refer to the dimensioned drawing in the gear-unit documentation.

- Place the shrink disk on the hollow shaft and fasten, if required. For the exact installation height (W) of the shrink disk, refer to the dimensioned drawing.



For transporting and lifting the shrink disk it may be required to use a suitable lifting device!

Make sure that the shrink disk cannot slip off the hollow shaft.



Do not tighten the tensioning bolts (5) until the stub shaft is installed too.

- Tighten the tensioning bolts (5) gradually one after the other, working round several times by quarter turns.
- Tighten all tensioning bolts (5) until the end faces of the inner ring (3) and the outer ring (4) are flush and the maximum tightening torque of the tensioning bolts has been achieved. The correct alignment is to be checked using a ruler. The max. tolerance is ± 0.2 mm.



The correct clamping condition can thus be checked visually.



To avoid overloading the individual bolts, the maximum tightening torque (see table 18) must not be exceeded. If, when tightening the clamping bolts at max. tightening torque, the inner and outer ring are not aligned, Siemens must be consulted.

Table 18: Maximum torques for tensioning bolts

Tensioning-bolt thread	max. tightening torque per bolt Strength class 12.9 Nm	Tensioning-bolt thread	max. tightening torque per bolt Strength class 12.9 Nm
M 8	35	M 20	570
M 10	70	M 24	980
M 12	120	M 27	1450
M 14	193	M 30	1970
M 16	295	M 33	2650



The shrink disk has been identity-marked on the outer ring (4). In case of contacting Siemens this identification must be referred to.



For safety reasons, a protective cover should be mounted to prevent contact! This cover must be applied after completion of all works on the shrink disk.



Only the complete shrink disks supplied by the manufacturer may be used. Combining of components from different shrink disks is not permitted.



Tightening the fastening bolts using an impact screwdriver is not permitted!

6.7.2 Demounting the shrink disk

- Remove the protective cover.
- Remove any rust deposits from the shaft and the hollow shaft.



Under no circumstances must the tensioning bolts be unscrewed one after the other.

- Undo all tensioning bolts one after the other by approx. 1/4 turn.



The stored energy of the outer ring is slowly loosened during disassembly via the bolts to be loosened. In order that this is carried out correctly, the procedure described here must be carefully adhered to!

- All tensioning bolts should now be further loosened one after the other by approx. 1 turn.



The outer ring should now release of its own accord from the inner ring. If this is not the case, the outer ring can be detensioned with the forcing threads. To this purpose screw some of the adjacent fastening bolts into the forcing threads. The now loosening outer ring is braced against the remaining bolts. This operation must be carried out until the outer ring completely releases of its own accord.

- The shrink disk is to be secured against axial shifting.
- Draw the stub shaft out of the hollow shaft.
- Pull the shrink disk off the hollow shaft.



For transporting and lifting the shrink disk it may be required to use a suitable lifting device!

6.7.3 Cleaning and greasing the shrink disk



Only dirty shrink disks must be disassembled and cleaned.

- Inspection of all parts for any damage.



Damaged parts must be replaced with new ones! The use of damaged parts is not permissible!



Only the complete shrink disks supplied by the manufacturer may be used. Combining of components from different shrink disks is not permitted.

- Thoroughly clean all parts.



Do not use contaminated solvents or dirty cloths nor cleansing agents containing oil (such as paraffin or terpentine) for removing grease.

- The conical surfaces of the inner and outer rings (3 and 4, see figure 56) must be free of grease and oil.
 - A thin layer of grease must be applied evenly to the conical surfaces of the inner and outer rings (3 and 4, see figure 56).
 - Provide the tensioning bolts (5, see figure 56) on the contact surface and on the thread with lubricant.

- Use a solid lubricant paste with a **high MoS₂-based molybdenum disulphide content** which will not slide during fitting work and which shows the following characteristics:
 - friction coefficient “ μ ” = 0.04
 - resistant to pressure up to a maximum pressure of 300 N/mm²
 - ageing-resistant

Table 19: Recommended lubricants for shrink disks after their cleaning ¹⁾

Lubricant	Form	Manufacturer
Molykote G Rapid	Spray or paste	DOW Corning
Aemasol MO 19 P	Spray or paste	A. C. Matthes
Unimoly P 5	Powder	Klüber Lubrication
gleitmo 100	Spray or paste	Fuchs Lubritec

¹⁾ Other lubricants may be used if they have the same characteristics.

- Join inner ring (3) and outer ring (4).
- Place the tensioning bolts and screw in some threads by your fingers.



Observe the manufacturer's instructions for handling lubricants!

Mounting and start-up must be carried out by properly trained specialist personnel.

6.7.4 Re-mounting the shrink disk



For re-mounting the shrink disk the procedure described in item 6.7.1 must be adhered to.

6.7.5 Inspection of the shrink disk



In all cases the inspection relating to the shrink disk should be carried out simultaneously with the examination of the gear unit, **however at least every 12 months**.

Inspection of the shrink disk is limited to a visual assessment of its condition. The following must be observed when carrying out this work:

- loose screws
- damage caused by force
- flush position of the inner ring (3) in relation to outer ring (4).

6.8 Couplings, clutches

As a rule, flexible couplings or safety slip clutches are provided for the drive of the gear unit.

If rigid couplings or other in- and/or output elements, which create additional radial and/or axial forces, (e.g. gear wheels, belt pulleys, disk flywheels, hydraulic couplings) are to be used, this must be agreed by contract.



Couplings must be balanced in accordance with the specifications in the pertinent instructions manual!



For maintenance and operation of the couplings, refer to the specific operating instructions for the couplings.



When installing the drives, make absolutely certain that the individual components are accurately aligned in relation to each other. Inadmissibly large errors in the alignment of the shaft ends to be connected due to angular and/or axial misalignments result in premature wear and/or material damage.

Insufficiently rigid base frames or sub-structures can also during operation cause a radial and/or axial misalignment, which cannot be measured when the unit is at a standstill.



For permissible alignment errors in the case of couplings supplied by Siemens, refer to the operating instruction manuals for the couplings.

If you use couplings manufactured by other manufacturers, ask these manufacturers which alignment errors are permissible, stating the radial loads occurring.



Increased system service life and reliability and reduced running noise can be achieved through the least possible radial and angular misalignment.

The coupling parts may get out of alignment

- by imprecise alignment during assembly or installation
- during operation of the system by:
 - heat expansion,
 - shaft flexure,
 - too weak machine frames and the like.

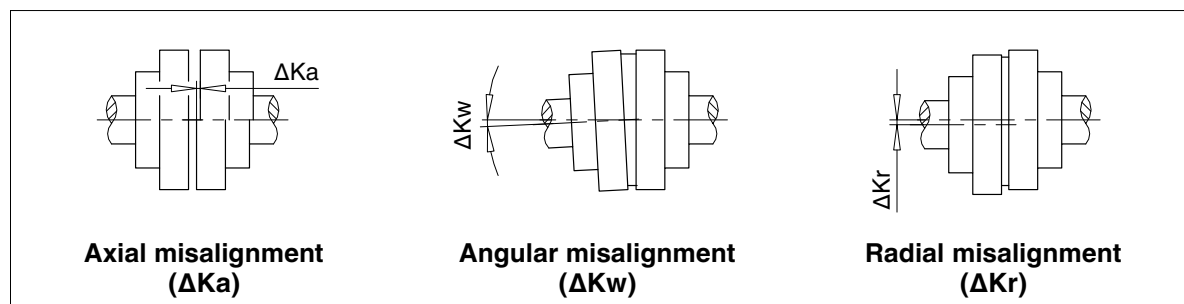


Fig. 57: Possible misalignments

Alignment has to be done in two axial planes arranged perpendicularly to each other. This can be done by means of a ruler (radial misalignment) and feeler gauge (angular misalignment), as shown in the illustration. The aligning accuracy can be increased by using a dial gauge or a laser alignment system.

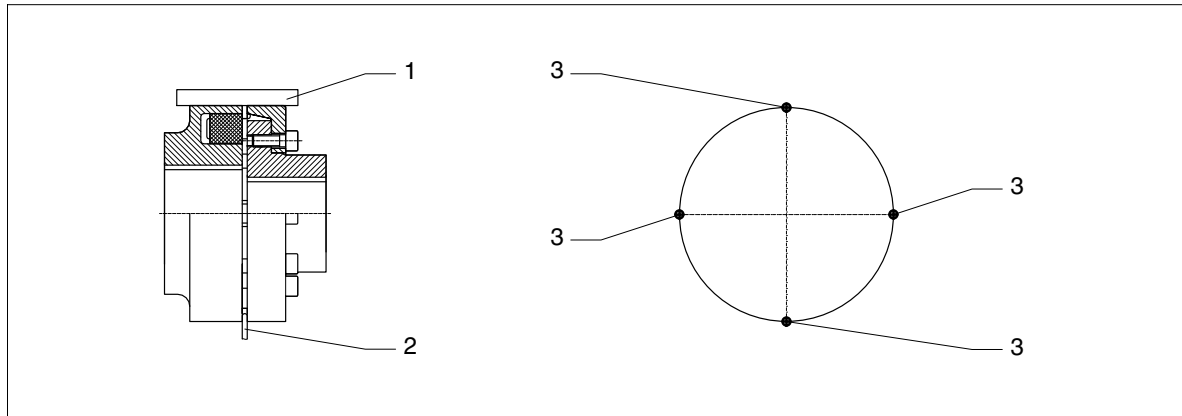


Fig. 58: Example of alignment on a flexible coupling

1 Ruler

2 Feeler gauge

3 Measuring points



The maximum permissible misalignment values are specified in the operating instructions for the coupling; they must under no circumstances be exceeded during operation.

Angular and radial misalignments may occur at the same time. The sum of both misalignments must not exceed the maximum permissible value of the angular or radial misalignment.

If you use couplings manufactured by other manufacturers, ask these manufacturers which alignment errors are permissible, stating the radial loads occurring.



For alignment of the drive components (vertical direction), it is recommended to use packing or foil plates underneath the mounting feet. The use of claws with set screws on the foundation for lateral adjustment of the drive components is also advantageous.

In the case of gear units with hollow output shafts or flange output shafts, the coupling on the output side is not required. Gear units with hollow output shafts must be mounted on the shafts of the customer's machinery. Gear units with flanged output shafts must be mounted on the customer's shaft via a counterflange.

6.9 Shaft-mounting gear unit with flanged shaft



The front area of the flanged shaft must be absolutely free of grease.

This is essential for safe and reliable torque transmission. Do not use contaminated solvents or dirty cloths for removing grease.



Before tightening the tensioning bolts it must be ensured that the flange centring means are inserted one inside the other.

Then tighten diametrically opposed tensioning bolts to full torque.

Tightening torques of flange bolts for gear units:

Table 20: Tightening torques for flange connections

Gear-unit size	Strength class		Tightening torque
	Bolt DIN 931	Nut DIN 934	
5 ... 6	10.9	10	610 Nm
7 ... 10	10.9	10	1050 Nm
11 ... 16	10.9	10	2100 Nm
17 ... 20	10.9	10	3560 Nm
21 ... 22	10.9	10	5720 Nm

6.10 Shaft mounting gear unit with block flange



The front area of the block flange must be absolutely free of grease. This is essential for safe and reliable torque transmission. Do not use contaminated solvents or dirty cloths for removing grease.



Tighten diametrically opposed tensioning bolts to full torque.

The joint bolts must be tightened to the prescribed torque. For the correct torque, refer to item 6.23. Bolts of the minimum strength class 8.8 must be used. The transmittable gear-unit torque is limited by the bolted joint on bolt circle K_1 .

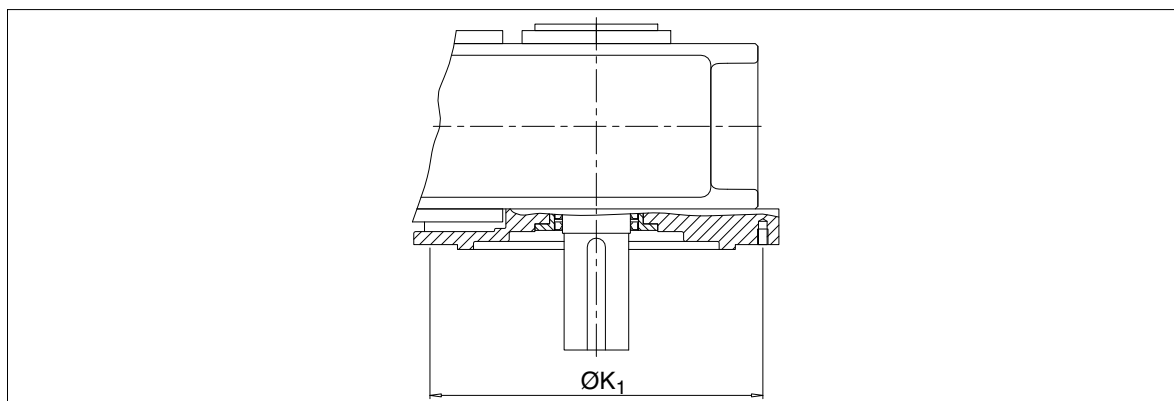


Fig. 59: Illustration with block flange

Table 21: Types and rotation directions

Type	Type ¹⁾	
	B	C
H2..		
H3..		
H4..		
B2..		
B3..		
B4..		

¹⁾ Types with hollow output shaft on request

6.11 Mounting the torque arm for the gear-unit housing

6.11.1 Attaching the torque arm



The torque arm must be mounted stress-free on the machine side

On helical gear units with a motor bell housing the torque arm is located opposite the motor bell housing.

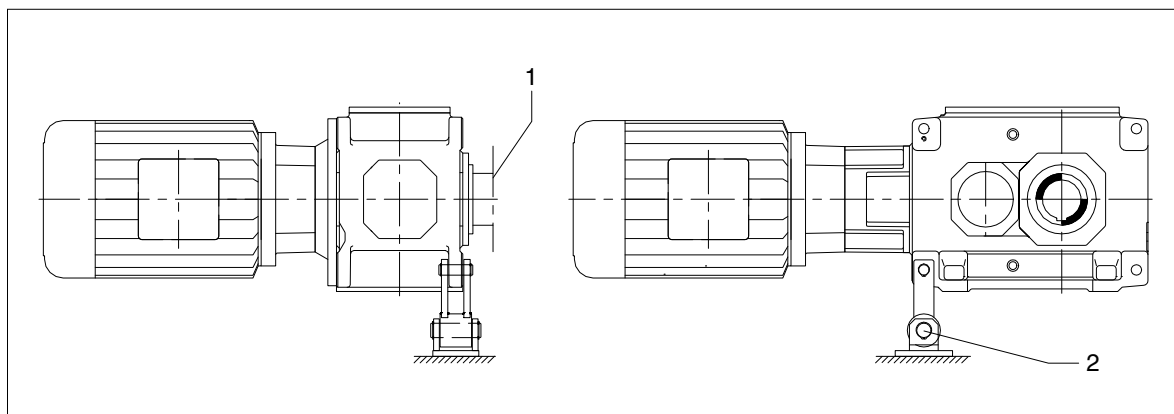


Fig. 60: Torque arm for gear-unit housing

1 Machine side

2 Flexible support block

Table 22: Motor types and torque arms

Gear-unit size	Max. perm. standard-motor size					
	Gear type			Gear type		
	H2	H3	H4	B2	B3	B4
4	200	-	-	200	200	-
5 ... 6	225	225	-	225	225	160
7 ... 8	280	280	180	280	280	200
9 ... 10	280	280	225	280	280	225
11 ... 12	315M	315M	250	315M	315M	280
13 ... 14	-	355	315M	355	355	315M
15 ... 16	-	355	315	-	355	355M
17 ... 18	-	355	355M	-	355	355
19 ... 22	on request					



Larger motors should be used only with the approval of Siemens.

- Foundation type for fastening the torque arm, see item 6.3.1, "Foundation".
- If the customer fits a torque arm, connection to the foundation must be by means of a flexible element.

6.12 Mounting supports for gear-unit swing bases

6.12.1 Attaching the support



The support for the gear-unit swing base must be mounted free of stress.

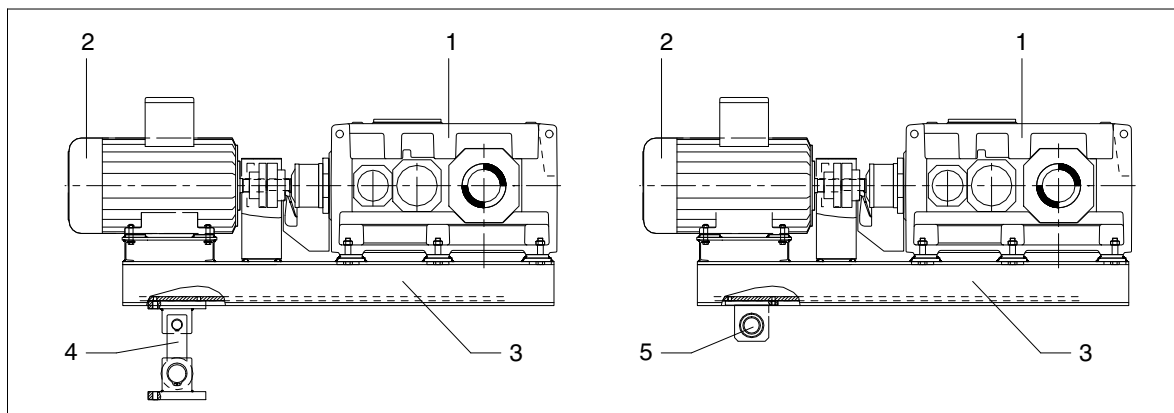


Fig. 61: Support for gear-unit swing base

- | | | | |
|---|----------------------|---|------------------------|
| 1 | Gear unit | 4 | Torque arm |
| 2 | Motor | 5 | Flexible support block |
| 3 | Gear-unit swing base | | |

Table 23: Motor assignment gear-unit swing base

Gear-unit size	Max. perm. standard-motor size		
	Gear type		
	B2	B3	B4
4	on request	200	-
5 ... 6		225M	160
7 ... 8		280M	200
9 ... 10		315	225M
11 ... 12		355	280S
13 ... 14		400M	315M
15 ... 16		400M	315
17 ... 18		400M	355L
19 ... 22	on request		



Larger motors should be used only with the approval of Siemens.

- Foundation type for fastening the torque arm, see item 6.3.1, "Foundation".
- If the customer fits a torque arm, connection to the foundation must be by means of a flexible element.

6.13 Gear units with cooling coil

- Before connecting the cooling coil the screw plugs must be removed from the connecting bushes.
- Flush the cooling coil (in order to remove any contamination).
- Connect the cooling-water in- and outflow pipes (for exact position of the connections, see dimensioned drawing).



Observe also item 5.10.2.

6.14 Gear unit with add-on components

- For the technical data to the add-on components, as stated in item 6.15 to 6.21, refer to the order-specific list of equipment.
- The electrical equipment for regulation and control must be wired in accordance with the equipment suppliers' instructions.
- For operation and maintenance the operating instructions provided specifically for the order and the specifications in items 5.10.2 to 5.15 must be observed.

6.15 Gear units with air oil-cooler

- Connect the contamination indicator for switchover filter (for gear sizes ≥ 13) and pressure monitor electrically.
- Connect the fan motor electrically.

6.16 Gear units with fitted water oil-cooler

- Before connecting the water oil-cooler remove the plugs from the cooling-water connections.
- Flush the water oil-cooler (in order to remove any contamination).
- Install the cooling-water in- and outflow pipes (for flow direction and exact position of connections, see dimensioned drawing).



Make sure when installing the piping that no forces, moments or vibrations act upon the connections of the water oil-cooler.

- Wire the pressure monitor electrically (in case of gear units with corresponding equipment only).



Observe also item 5.10.4.

6.17 Gear unit with heating element

- Wire heating elements electrically.

6.18 Gear unit with oil-temperature monitoring system

- Wire resistance thermometer with evaluating instrument (to be provided by customer) electrically.

6.19 Gear unit with oil-level monitoring

- Wire the level-limit switch electrically.
- Wire oil-level monitor electrically.

6.20 Bearing-monitoring system

- The bearing monitoring device must be fitted by the customer.

6.21 Gear unit with speed transmitter

- Wire speed transmitter electrically.

6.22 Final work

- After installation of the gear unit check all screw connections for tight fit.
- Check the alignment after tightening the fastening elements (the alignment must not have been changed).
- Check that all the devices which have been demounted for transport reasons have been refitted. For this refer to the details in the data sheet, the list of equipment and the associated drawings.
- Any oil-drain cocks must be secured against accidental opening.
- The gear unit must be protected against falling objects.
- Protective devices for rotating parts must be checked for correct seating. Contact with rotating parts is not permitted.
- A potential equalisation in accordance with the applying regulations and directives must be carried out! If no threaded holes for earth connection are available on the gear unit, other appropriate measures must be taken. This work must always be done by electrotechnical specialists.
- Cable entries should be protected against moisture.
- Check that protective measures have been taken!

6.23 Screw-connection classes, tightening torques and initial tensioning forces

6.23.1 Screw-connection classes

The specified screw connections are to be fastened applying the tightening torques specified in the table below.

Table 24: Screw-connection classes

Screw-connection class	Distribution of emitted torque on the tool	Tightening procedure (Usually the tightening processes lie within the stated tool distribution)
C	$\pm 5 \%$ up to $\pm 10 \%$	- Hydraulic tightening with mechanical screwdriver - Torque-controlled tightening with torque wrench, signal-emitting torque wrench - Tightening with precision mechanical screwdriver with dynamic torque measuring
D	$\pm 10 \%$ up to $\pm 20 \%$	- Torque-controlled tightening with mechanical screwdriver
E	$\pm 20 \%$ up to $\pm 50 \%$	- Tightening with pulse screwdriver or impact wrench without adjustment checking device - Tightening by hand, using a spanner without torque measuring device

6.23.2 Tightening torques and initial tensioning forces



The tightening torques apply to friction coefficients of $\mu_{\text{total}} = 0.14$.

The friction coefficient $\mu_{\text{total}} = 0.14$ applies here to lightly oiled steel bolts, black-annealed or phosphatised and dry, cut mating threads in steel or cast iron. Lubricants which alter the friction coefficient must not be used and may overload the screw connection.

Table 25: Initial tensioning forces and tightening torques for screw connections of strength classes **8.8; 10.9; 12.9** with a common friction coefficient of $\mu_{\text{total}} = 0.14$

Nominal thread diameter d mm	Strength class of the screw	Initial tensioning force for screw-connection classes from table 22			Tightening torque for screw-connection classes from table 22		
		C	D $F_{M \text{ min.}}$ N	E	C	D M_A Nm	E
M10	8.8	18000	11500	7200	44.6	38.4	34.3
	10.9	26400	16900	10600	65.4	56.4	50.4
	12.9	30900	19800	12400	76.5	66.0	58.9
M12	8.8	26300	16800	10500	76.7	66.1	59.0
	10.9	38600	24700	15400	113	97.1	86.6
	12.9	45100	28900	18100	132	114	101
M16	8.8	49300	31600	19800	186	160	143
	10.9	72500	46400	29000	273	235	210
	12.9	85000	54400	34000	320	276	246
M20	8.8	77000	49200	30800	364	313	280
	10.9	110000	70400	44000	520	450	400
	12.9	129000	82400	51500	609	525	468
M24	8.8	109000	69600	43500	614	530	470
	10.9	155000	99200	62000	875	755	675
	12.9	181000	116000	72500	1020	880	790
M30	8.8	170000	109000	68000	1210	1040	930
	10.9	243000	155000	97000	1720	1480	1330
	12.9	284000	182000	114000	2010	1740	1550
M36	8.8	246000	157000	98300	2080	1790	1600
	10.9	350000	224000	140000	2960	2550	2280
	12.9	409000	262000	164000	3460	2980	2670
M42	8.8	331000	212000	132000	3260	2810	2510
	10.9	471000	301000	188000	4640	4000	3750
	12.9	551000	352000	220000	5430	4680	4180
M48	8.8	421000	269000	168000	4750	4090	3650
	10.9	599000	383000	240000	6760	5820	5200
	12.9	700000	448000	280000	7900	6810	6080
M56	8.8	568000	363000	227000	7430	6400	5710
	10.9	806000	516000	323000	10500	9090	8120
	12.9	944000	604000	378000	12300	10600	9500
M64	8.8	744000	476000	298000	11000	9480	8460
	10.9	1060000	676000	423000	15600	13500	12000
	12.9	1240000	792000	495000	18300	15800	14100
M72x6	8.8	944000	604000	378000	15500	13400	11900
	10.9	1340000	856000	535000	22000	18900	16900
	12.9	1570000	1000000	628000	25800	22200	19800

Nominal thread diameter d mm	Strength class of the screw	Initial tensioning force for screw-connection classes from table 22			Tightening torque for screw-connection classes from table 22		
		C	D	E	C	D	E
		$F_{M \min.}$ N			M_A Nm		
M80x6	8.8	1190000	760000	475000	21500	18500	16500
	10.9	1690000	1100000	675000	30500	26400	23400
	12.9	1980000	1360000	790000	35700	31400	27400
M90x6	8.8	1510000	968000	605000	30600	26300	23500
	10.9	2150000	1380000	860000	43500	37500	33400
	12.9	2520000	1600000	1010000	51000	43800	39200
M100x6	8.8	1880000	1200000	750000	42100	36200	32300
	10.9	2670000	1710000	1070000	60000	51600	46100
	12.9	3130000	2000000	1250000	70000	60400	53900



Damaged bolts must be replaced with new bolts of the same type and strength class.

7. Start-up

Observe the instructions in section 3., "Safety instructions"!



The gear unit must not be started up if the required instructions are not to hand.

7.1 Procedure before start-up

7.1.1 Removal of preservative agent

The location of the oil-draining points is marked by an appropriate symbol in the dimensioned drawing in the gear-unit documentation.

Oil-draining point:



- Place suitable containers under the oil-draining points.
- Unscrew the oil-drain plug or open the oil-drain cock.
- Remove remaining preservative agent and/or running-in oil from the gear unit using a suitable container, unscrew any existing residual-oil drain plugs, to do so.
- Dispose of remaining preservative agent and/or running-in oil in accordance with regulations.



**Remove any oil spillage immediately with an oil-binding agent.
The oil must not come into contact with the skin (e.g. the operator's hands).
The safety notes on the data sheets for the oil used must be observed here!**

- Screw in oil-drain plug or reclose oil-drain cock.
- Screw in any unscrewed residual-oil drain plugs again.

A detailed view of the gear unit can be obtained from the drawings in the gear-unit documentation.

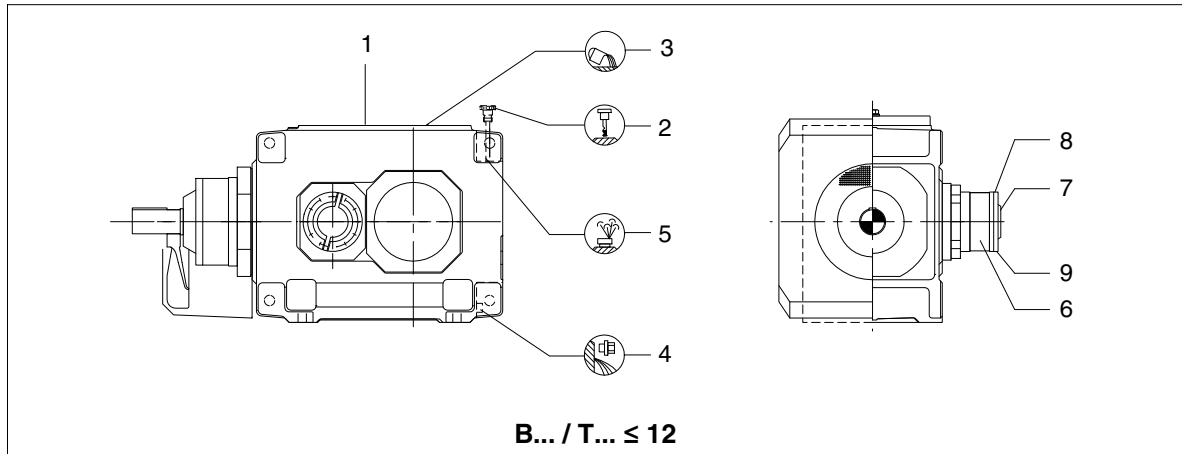


Fig. 62: Oil inlet / oil drain on gear unit

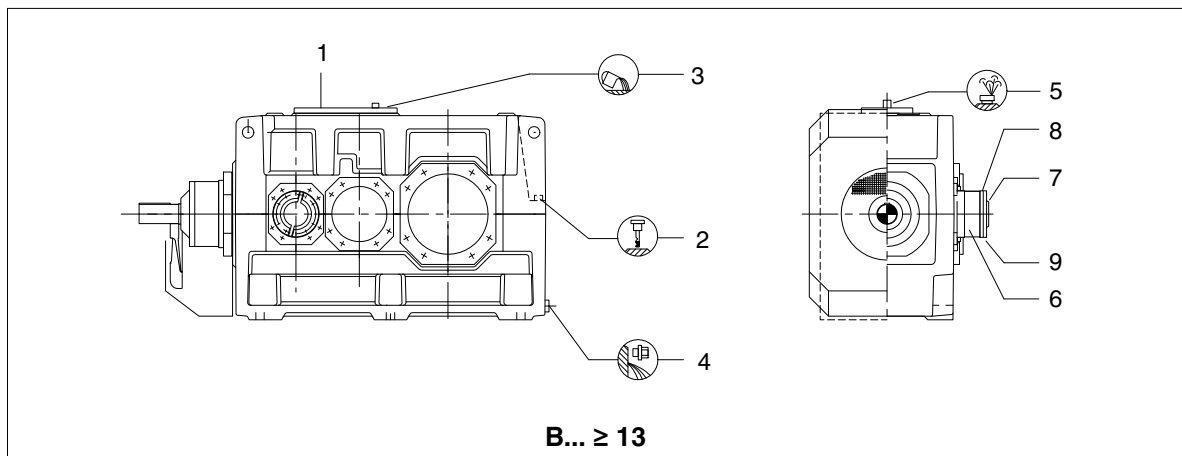


Fig. 63: Oil inlet / oil drain on gear unit

- | | | | |
|---|----------------------------------|---|---|
| 1 | Inspection and/or assembly cover | 6 | Backstop |
| 2 | Oil dipstick | 7 | Cover for backstop |
| 3 | Oil inlet | 8 | Screw plug for oil inlet backstop |
| 4 | Oil-drain plug | 9 | Screw plug for residual-oil drainage backstop |
| 5 | Breather screw or screw plug | | |

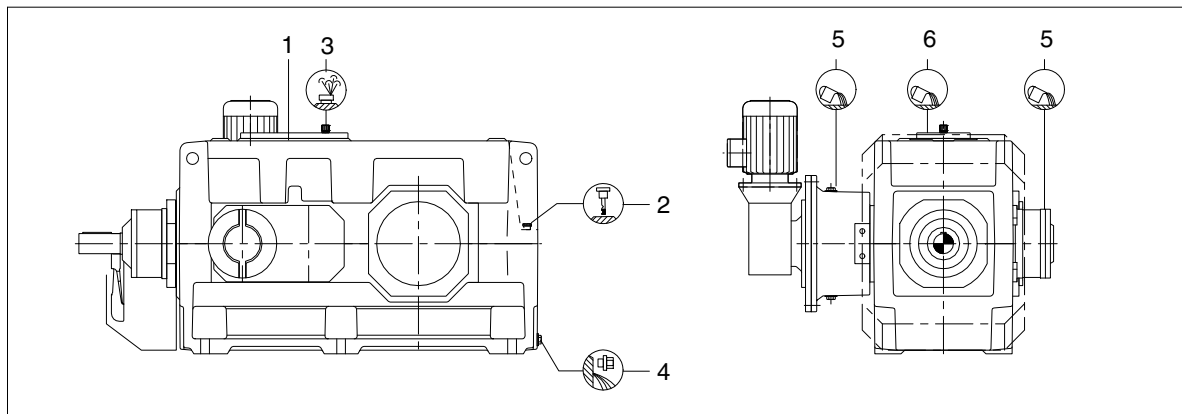


Fig. 64: Oil inlet / oil drain on gear units type B3.. ≥ 13 with auxiliary drive

- | | | | |
|---|----------------------------------|---|-----------------|
| 1 | Inspection and/or assembly cover | 4 | Oil-drain plug |
| 2 | Oil dipstick | 5 | Oil-filler plug |
| 3 | Breather screw or screw plug | 6 | Oil inlet |

A detailed view of the gear unit can be obtained from the drawings in the gear-unit documentation.

7.1.2 Filling with lubricant

- Undo and remove fixing screws on the inspection and/or assembly cover.
- Remove cover with seal from housing (seal will be used again).



Using a filter (max. mesh 25 µm), fill the gear unit with fresh oil up to the MAX mark on the oil dipstick or the middle of the oil-sight glass.

Remember to fill the oil pockets above the bearings and (with bevel-gear units) at the input shaft on the inside.

- In the case of gear units with add-on backstop, release the screw plug on the backstop cover and pour in approx. 0.5 l fresh oil of the total oil quantity via a filter (max. filter mesh 25 µm).
- Screw in the screw plug.



The sealing surfaces must not be contaminated nor damaged.



For the correct type of oil (of various brands) to be used, refer to the BA 7300 EN operating instructions supplied separately.

Information on the type, quantity and viscosity of the oil is given on the rating plate on the gear unit.

The quantity of oil indicated on the rating plate is an approximation only. The marks on the dipstick or oil sight glass are decisive for the amount of oil to be filled in.



In the case of gear units fitted with pressure lubrication or an oil-cooling system, the oil circuit must also be charged with oil. To do this, briefly start up the gear unit with add-on pump (observing the description in section 8).

- Check the oil level in the gear-unit housing with the oil dipstick or by means of the oil-sight glass.



The oil level must be at the upper mark on the oil dipstick or the middle of the oil-sight glass.



Remove any oil spillage immediately with an oil-binding agent.

- Place inspection or assembly cover including seal on the housing.
- Place the fastening bolts of the cover and tighten them to the specified torque (see item 6.23).

7.1.2.1 Oil quantities

Table 26: Approximate figures for required oil quantities in horizontal gear units **with radial shaft seals and Taconite seals**

Type	Oil quantity (approximate value) in litres for size										
	1	2	3	4	5	6	7	8	9	10	11
H1SH	2.5	-	7	-	22	-	42	-	68	-	120
H2.H	-	-	-	10	15	16	27	30	42	45	71
H2.M	-	-	-	-	-	-	-	-	-	-	-
H3.H	-	-	-	-	15	17	28	30	45	46	85
H3.M	-	-	-	-	-	-	-	-	-	-	-
H4.H	-	-	-	-	-	-	25	27	48	50	80
H4.M	-	-	-	-	-	-	-	-	-	-	-
B2.H	3.5	8	-	10	16	19	31	34	48	50	80
B2.M	-	-	-	-	-	-	-	-	-	-	-
B3.H	-	-	-	9	14	15	25	28	40	42	66
B3.M	-	-	-	-	-	-	-	-	-	-	-
B4.H	-	-	-	-	16	18	30	33	48	50	80
B4.M	-	-	-	-	-	-	-	-	-	-	-

Type	Oil quantity (approximate value) in litres for size										
	12	13	14	15	16	17	18	19	20	21	22
H1SH	-	175	-	190	-	270	-	390	-	-	-
H2.H	76	135	140	210	215	290	300	320	340	320	340
H2.M	-	110	115	160	165	230	240	300	320	350	370
H3.H	90	160	165	235	245	305	315	420	450	470	490
H3.M	-	125	130	190	195	240	250	390	415	515	540
H4.H	87	130	140	230	235	290	305	360	380	395	420
H4.M	-	120	125	170	175	225	230	310	330	430	450
B2.H	95	140	155	220	230	320	335	-	-	-	-
B2.M	-	120	130	180	190	260	275	-	-	-	-
B3.H	72	130	140	210	220	290	300	380	440	370	430
B3.M	-	110	115	160	165	230	235	360	420	420	490
B4.H	90	145	150	230	235	295	305	480	550	540	620
B4.M	-	120	125	170	175	230	235	440	510	590	680

Table 27: Approximate values for required oil quantities in horizontal gear units **with labyrinth seals**

Type	Oil quantity (approximate value) in litres for size																			
	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
H1SH	5.5	-	19	-	36	-	60	-	106	-	155	-	156	-	225	-	-	-	-	-
H2SH	4.5	7	11	12	21	23	33	34	58	60	120	130	190	200	260	270	-	-	-	-
H2PH	-	-	25	-	45	-	72	-	110	-	-	-	-	-	-	-	-	-	-	-

Table 28: Approximate values for additional oil quantities for the **intermediate flange** when adding on the auxiliary drive to the main gear unit

Type	Additional oil quantity (approximate value) in litres for size								
	4	5	6	7	8	9	10	11	12
B3.H T3.H	1	2	2	5	5	5	6	12	12

Type	Additional oil quantity (approximate value) in litres for size									
	13	14	15	16	17	18	19	20	21	22
B3.H	15	15	20	20	25	25	40	40	60	60



For details regarding the auxiliary gear unit, see the Special operating instructions.
(The auxiliary gear unit is delivered ex works with oil filling.)

7.2 Start-up



Before start-up, replace the yellow plastic screw plug with the air filter (see notice on gear unit).

7.2.1 Oil level

The oil level must be monitored by means of the existing oil-level monitoring equipment. To do so, the gear unit must be shut down.

When the oil is cool, the level should be at the upper mark on the oil dipstick or the middle of oil-sight glass. When the oil is warm it may slightly exceed these marks.



It must in no case be allowed to fall below the mark. If necessary, top up to the correct level.

7.2.2 Gear unit with cooling coil or external oil-supply system



The permissible pressure and temperature values specified in the data sheet and/or list of equipment must not be exceeded.
This is to be checked before the start-up.

- Fully open the stop valves in the coolant in- and outflow pipes of the cooling system.
- Check that connecting lines are correctly fastened and tight.



For connecting dimensions, refer to the dimensioned drawing of the gear unit. The required cooling water quantity and the max. permissible inlet temperature are given on the data sheet and/or the list of equipment.

7.2.3 Gear unit with backstop



Observe details at item 5.8, "Backstop"!

Before start-up, check whether the backstop can be turned manually in the free-wheeling direction without exerting undue force. Observe the direction-of-rotation arrows on the housing.



**To avoid damaging the backstop or the gear unit, the motor must not be run adversely to the stop direction of the gear unit.
Observe the notice fixed to the gear unit.
The minimum lift-off speeds must not be exceeded during operation.**



**If a backstop with release mechanism is used, the operating instructions for this backstop must be followed.
Moreover, the dimension " $x_{min.}$ " must be checked; " $x_{min.}$ " must not be smaller than that stated on the backstop rating plate.**

Before connecting the motor, determine the direction of rotation of the three-phase current supply using a phase-sequence indicator, and connect the motor in accordance with the pre-determined direction of rotation.



The unit can be started up once the amount of oil indicated on the notice has been poured in through the oil filler plug screw of the backstop. Always use oil of the same type and viscosity as for the gear unit.

7.2.4 Gear unit with overrunning clutch



Observe details at item 5.16.3, "Overrunning clutch"!

Before start-up, check whether the overrunning clutch can be turned manually in the free-wheeling direction without exerting undue force. Observe the direction-of-rotation arrows on the housing.



The overrunning clutch is in free-wheeling operation if the motor shaft of the auxiliary drive is rotated in opposed operating direction of rotation.
When rotating in the operating direction of rotation, the blocking action of the overrunning clutch (carrier operation) becomes effective. Coupling and, thus, rotation of the output shaft of the main gear unit in operating direction of rotation takes place.
In the case of auxiliary drives designed for load operation, the brake at the auxiliary motor must first be lifted in order to check for proper functioning of the overrunning clutch.

The overrunning clutch is accommodated within the intermediate flange and is supplied with oil from the main gear unit.

7.2.5 Temperature measurement



During first start-up and after maintenance work, the oil sump temperature must be measured during correct use (maximum machine performance) after appropriate running in.

The maximally permissible oil-sump temperature is 90 °C (for mineral oil) or 100 °C (for synthetic oil).

At higher temperatures the gear unit must be shut down immediately and Siemens customer service consulted.

7.2.6 Oil-level monitoring system



This monitoring is designed as a standstill monitoring (gear-unit stop) and checks the level of the oil before the unit is started up. When the signal "oil level too low" is given, it should be wired in such a way that the drive motor cannot start and an alarm is given. During operation, any active signal should be bridged.

7.2.7 Bearing monitoring (vibration measurement)



If the necessary measures have been made for vibration measurement as bearing monitoring (see item 5.14), vibration measurements must be taken in order to obtain initial values and/or standard values for the diagnosis. These measurements must be recorded and filed.

7.2.8 Heating



Never switch the heating on, unless complete immersion of the rod heater in the oil bath is ensured. Fire hazard!
If heating elements are installed afterwards the maximum heating capacity (see table 14 in item 5.11) on the outer surface of the heating element must not be exceeded.



The correct setting of the switch points must be checked!

7.2.9 Checking procedure

The following visual checks must be conducted and recorded when starting up:

- ☐ Oil level
- ☐ Leaktightness of the oil-cooling or oil-supply lines
- ☐ Opening condition of the shut-off valves
- ☐ Effectiveness of the shaft seals
- ☐ Freedom of the rotating parts from contact

The tension pressures and/or pretensioning forces in accordance with item 6.3.2.4 must also be recorded in this document.



The document must be kept with the instructions.

7.3 Removal from service

- To take the gear unit out of service, first switch off the drive unit.



Secure the drive unit to prevent it from being started up unintentionally. Attach a warning notice to the start switch!

- In the case of gear units fitted with cooling coil or water oil-coolers, close the stop valves on the water in- and outflow pipes. To prevent freezing, drain the water from the cooling coil or the water oil-cooler.
- Start the gear unit and allow it to run briefly (5 to 10 minutes) approx. every 3 weeks (during a shut-down period no longer than 6 months).
- Treat the gear unit with preservative, see items 7.3.1 and 7.3.2 (before a shut-down period exceeding 6 months).

7.3.1 Interior preservation during longer disuse

Depending on the type of lubrication and/or shaft sealing, the following types of interior preservation can be applied.

7.3.1.1 Interior preservation with gear oil

Gear units with splash-lubrication systems and contacting shaft seals can be filled with the correct type of service oil up to a point just below the air filter.

7.3.1.2 Interior preservation with preservative agent

Before longer shut-down periods gear units with pressure lubrication systems, oil circulation cooling and/or non-contacting shaft seals should be filled with preservative agent and run without load.

7.3.1.3 Interior-preservation procedure

- Stop the gear unit.
- Drain oil into a suitable container (see section 10., "Maintenance and Repair").
- Unscrew the air filter including the reducing screw.
- Pour in the preservative agent through the hole of the reducing screw up to the top mark on the oil-sight glass.



For preservative agent see table 8 or 9 in item 4.4.1!

- Screw in air filter including reducing screw.
- Start the gear unit and allow it to idle briefly.
- Unscrew the oil-drain plug.
- Drain preservative agent into a suitable container.
- Dispose of preservative agent in accordance with regulations.



There is a risk of scalding from the hot preservative agent draining from the gear unit. Wear protective gloves!

- Screw in the oil-drain plug.
- Replace air filter with screw plug.



Before re-starting the gear unit, replace the screw plug with the air filter. Observe the instructions in item 7.1.1.

7.3.2 Exterior preservation

7.3.2.1 Exterior-preservation procedure

- Clean the surfaces.



For separation between the sealing lip of the shaft-sealing ring and the preservative agent, the shaft should be brushed with grease in way of the sealing lip.

- Apply preservative agent.



For preservative agent see table 10 in item 4.4.2!

8. Operation

Observe the instructions in section 3., "Safety instructions", in section 9, "Faults, causes and remedy", and in section 10., "Maintenance and repair"!

8.1 General

To achieve a satisfactory and trouble-free operation of the equipment, be certain to observe the operating values specified in section 1, "Technical Data", as well as the information given in the operating instructions of the oil-supply system, if applicable.

During operation the gear unit must be monitored for:

- Operating temperature The gear unit is designed for an operating temperature of:
90 °C (mineral oil only)
The maximum permitted temperature is:
100 °C (synthetic oil only)
- Oil pressure of the oil-supply system (min. 0.5 bar)
- Changes in gear noise
- Possible oil leakage at the housing and shaft seals

8.2 Oil level



To check the oil level, stop operation of the gear unit.
When the oil is warm, the oil level may slightly exceed the upper mark of the oil dipstick or the middle of the oil-sight glass. It must not be allowed to fall below the lower mark. If necessary, top up to the correct level.



The oil level in the oil-supply system must be checked.
For this, the operating instructions of the oil-supply system must be observed!

8.3 Irregularities



The drive unit must be switched off at once,

- **if irregularities are found during the operation**

or

- **if the pressure-monitoring device in the oil-cooling system triggers alarm (only with correspondingly equipped gear units).**

Determine the cause of the fault, using table 29, "Faults, causes and remedy" (see item 9.2).

Table 29, "Faults, causes and remedies", contains a list of possible faults, their causes and suggested remedies.

If the cause cannot be found, a specialist from one of our customer-service centres should be called in (see section 2).

9. Faults, causes and remedy

Observe the instructions in section 3., "Safety instructions", and in section 10., "Maintenance and repair"!

9.1 General information on faults and malfunctions



Faults and malfunctions occurring during the guarantee period and requiring repair work on the gear unit must be carried out only by Siemens customer service.

In the case of faults and malfunctions occurring after the guarantee period and whose cause cannot be precisely identified, we advise our customers to contact our customer service.



Siemens will not be bound by the terms of the guarantee or otherwise be responsible in cases of improper use of the gear unit, modifications carried out without the agreement of Siemens or use of spare parts not supplied by Siemens.



To remedy faults and malfunctions, the gear unit must always be taken out of service. Secure the drive unit to prevent it from being started up unintentionally.

Attach a warning notice to the start switch!

9.2 Possible faults

Table 29: Faults, causes and remedy

Faults	Causes	Remedy
Changes in gear-unit noise.	Damage to gear teeth.	Contact Customer Service. Check all toothed components and replace any damaged parts.
	Excessive bearing play.	Contact Customer Service. Adjust bearing backlash.
	Bearing defective.	Contact Customer Service. Replace defective bearings.
Loud noises in the area of the gear-unit fastening.	Gear-unit fastening has worked loose.	Tighten bolts / nuts to specified torque. Replace damaged bolts / nuts.
Increased temperature at the bearing points.	Oil level in gear-unit housing too low or too high.	Check oil level at room temperature and, if necessary, top up oil.
	Oil too old.	Check date of last oil change and, if necessary, change oil. See section 10.
	Oil-supply system defective.	Check the oil-supply system, replace any defective parts. Consult operating instructions for oil-supply system.
	Bearing defective.	Contact Customer Service. Check and, if necessary, replace bearings.
Exterior of gear unit is oiled up.	Inadequate sealing of housing covers and/or joints.	Seal joints.

Faults	Causes	Remedy
Oil leakage from the gear unit.	Inadequate sealing of housing covers and/or joints. Radial shaft-sealing rings defective.	Check and, if necessary, replace sealings. Seal joints. Check radial shaft-sealing rings and, if necessary, replace.
Oil foaming in the gear unit.	Preservation agent not completely drained. Oil-supply system has been operated too long at low temperatures. Gear unit too cold in operation. Water in oil. Oil too old (defoaming agent used up). Unsuitable oils mixed up	Oil change. Stop oil-supply system. Allow the oil to degas. Shut down gear unit and have oil degassed. Restart without cooling water. Test the oil, change oil if necessary. Test the oil, change oil if necessary. Test the oil, change oil, if necessary.
Water in oil.	Oil foams in sump. Defective oil-supply unit or cooling coil. Gear unit exposed to cold air from machine-room ventilator: Water condensing. Climatic conditions.	Check state of oil by the test-tube method for water contamination. Have oil analysed by laboratory. Check the oil-supply system or cooling coil, replace any defective parts. Consult operating instructions for oil-supply system. Protect gear unit with suitable heat insulation. Close air outlet or alter its direction by structural measures. Contact Customer Service. If necessary, fit wet-air filter.
Increased operating temperature.	Oil level in housing too high. Oil too old. Oil badly contaminated. Defective oil-supply unit or cooling coil.	Check oil level and, if necessary, adjust. Check date of last oil change and, if necessary, change oil. See section 10. Change oil. See section 10. Check the oil-supply system or cooling coil, replace any defective parts. Consult operating instructions for oil-supply system.
Fault in oil-supply system.		Consult operating instructions for oil-supply system.

10. Maintenance and repair

Observe the instructions in section 3., "Safety instructions", and in section 9, "Faults, causes and remedy"!

10.1 General notes on maintenance

All maintenance and repair work must be done with care and by duly trained and qualified personnel only.

The following applies to all work in item 10.2:



Switch the gear unit and add-on components off.

**Secure the drive unit to prevent it from being started up unintentionally.
Attach a warning notice to the start switch!**



The periods indicated in table 30 depend on the conditions under which the gear unit is operated. Only average periods can therefore be stated here. These refer to:

a daily operating time of	24	h
a duty factor "ED" of	100	%
an input-drive speed of	1500	1/min
a maximum oil temperature of	90	°C (mineral oil only)
	100	°C (synthetic oil only)

**The operator must ensure that the intervals stated in table 30 are adhered to.
This also applies if the maintenance work is included in the operator's internal maintenance schedules.**

Table 30: Maintenance and repair work

Measures	Periods	Remarks
Check oil temperature	Daily	
Check for unusual gear-unit noise	Daily	
Check oil level	Monthly	
Check gear unit for leaks	Monthly	
Test the water content of the oil	Approx. 400 operating hours, at least once per year	see item 10.2.1
Perform the first oil change	Approx. 400 operating hours after start-up	see item 10.2.2
Perform subsequent oil changes	Every 24 months or 10 000 operating hours	see item 10.2.2
Clean air filter	Every 3 months	see item 10.2.3
Clean fan and gear unit	Depending on requirements, at least every 2 years	See item 10.2.4
Refill Taconite seals with grease	Every 3000 operating hours or at least every 6 months	see item 10.2.5
Refill Tacolab seals with grease	Every 3000 operating hours or at least every 6 months	see item 10.2.6
Check hose lines	Yearly	see item 10.2.10

Measures	Periods	Remarks
Change the hose lines	6 years from the manufacturing date impressed	see item 10.2.10
Check cooling coil	Every 2 years	see item 10.2.7
Check friction linings of torque-limiting backstop	Once per year at least	see item 5.9
Check auxiliary drive		see item 5.16
Check tightness of fastening bolts	After first oil change, then every 2 years	see item 10.2.14
Check shrink disk	Every 12 months	see item 6.7.5
Inspection of the gear unit	Approx. every 2 years	see item 10.4

10.1.1 General oil-service lives

According to the manufacturers, the following are the expected periods during which the oils can be used without undergoing any significant change in quality. They are calculated on the basis of an average oil temperature of 80 °C:

- for mineral oils, biologically degradable oils and physiologically safe (synthetic esters) oils 2 years or 10 000 operating hours (**does not apply to natural esters such as rape seed oils**).
- for poly- α -olefins and polyglycols: 4 years or 20 000 operating hours.



The actual service lives may differ. The general rule is that an increase in temperature of 10 K will halve the service life and a temperature decrease of 10 K will approximately double the service life.

10.2 Description of maintenance and repair work

10.2.1 Test water content of oil

More information about examining the oil for water content or conducting oil analyses is obtainable from your lubricant manufacturer or our customer service.

- For reference purposes, a fresh sample of the operating lubricating oil used must be sent with the used oil sample to the analysing institute for analysis.
- The oil sample must be taken downstream of the filter of the oil-supply system while the gear unit is running. A suitable connection point is normally located upstream of the gear unit input (e.g. oil-drain cock in the pressure line).
- A special sample container should be filled with the specified quantity of oil.
If there is no such sample container available, at least one litre of oil must be put in a **clean**, transportworthy, sealable vessel.

10.2.2 Change oil

As an alternative to the oil-change intervals indicated in table 30 (see item 10.1) it is possible to have the oil sample tested at regular intervals by the Technical Service of the relevant oil company and to have it released for further use.

If re-usability has been confirmed, no oil change will be necessary.



Please observe the separately attached operating instructions BA 7300 EN.

- The instructions in item 7.1 must be observed!
- Drain the oil while the gear unit is still warm, i.e. immediately after shutting down the machinery.



When changing the oil, always re-fill the gear unit with the same type of oil. Never mix different types of oil and/or oils made by different manufacturers. Polyglycol-based synthetic oils in particular must not be mixed with PAO-based synthetic or mineral oils. If changing to a different grade or make of oil, the gear unit must, if necessary, be flushed out with the new oil grade. Flushing is not necessary, if the new service oil is fully compatible with the old service oil in all respects. Compatibility must be confirmed by the oil supplier. If there is a change to another oil grade or make, Siemens recommends flushing out the gear unit with the new grade of service oil.



When changing the oil, the housing and the oil-supply system, if available, must be flushed with oil to remove sludge, metal particles and oil residue. Use the same type of oil as is used for normal operation. High-viscosity oils must be heated beforehand using suitable means. Ensure that all residues have been removed before filling with fresh oil.

- Place a suitable container under the oil-draining point of the gear-unit housing.
- Unscrew the air filter including reducing screw at the housing top.
- Unscrew oil-drain plug or open oil-drain cock and drain the oil into the collecting container.
- Drain the oil from the oil-supply system (see operating instructions to the oil-supply system).



**There is a danger of scalding from the hot oil emerging from the housing.
Wear protective gloves!
Remove any oil spillage immediately with an oil-binding agent.**



Check the condition of the sealing ring (the sealing ring is vulcanised onto the oil-drain plug). If necessary, use a new oil-drain plug.

- Screw in the oil-drain plug or close oil-drain cock.
- Clean the oil filter in the oil-cooling system (see operating instructions of the oil-supply system).
- Clean the air filter (see item 10.2.3).
- Screw in air filter including reducing screw.
- Fill fresh oil into the gear unit (see item 7.1.2).

10.2.3 Clean the air filter



If a layer of dust has built up, the air filter must be cleaned, whether or not the minimum period of 3 months has expired.

- Unscrew the air filter including the reducing screw.
- Clean the air filter using benzine or a similar cleanser.
- Dry the air filter and/or blow with compressed air.



**Be especially careful when blowing with compressed air.
Wear protective glasses!**



Foreign bodies must be prevented from entering the gear unit.

10.2.4 Clean the fan and gear unit

- The instructions in item 5.10.1 must be observed!
- Remove the fan cowl.
- Using a stiff brush, remove any dirt adhering to the fan wheel, fan cowl and safety grid.
- Remove any corrosion.
- Screw safety grid back onto the fan cowl.



The gear unit must not be cleaned with high-pressure cleaning equipment.



It must be ensured that the fan cowl is correctly fastened. The fan must not come into contact with the fan cowl.

10.2.5 Refill Taconite seals with grease

- Inject approx. 30 g lithium-based bearing grease into each of the lubrication points of the Taconite seal. The lubrication points are fitted with flat grease nipples type AM10x1 to DIN 3404.



Remove and dispose of any old grease escaping.

10.2.6 Refill Tacolab seals with grease

- Inject approx. 30 g lithium-based bearing grease into each of the lubrication points of the Tacolab seal. The lubrication points are fitted with flat grease nipples type AM10x1 to DIN 3404.



Remove and dispose of any old grease escaping.

10.2.7 Check cooling coil

- Shut off the cooling-water supply.
- Disconnect the cooling-water in- and outflow pipes from the cooling coil.
- Check the inside walls of the cooling coil for deposits.



If the cooling coil is dirty, heat is no longer withdrawn effectively from the gear unit. Any dirt adhering to the inside of the coil should be removed by chemical cleaning or the cooling coil should be replaced with a new one.

- If thick deposits have formed on the inside of the cooling coil, the cooling water and/or the deposits themselves should be chemically analysed. These analyses are carried out by companies which specialise in chemical cleaning. They also supply the special cleaning agents required.
- Before using these cleaning agents, ensure that they will not damage the coil materials (contact Siemens). Observe the manufacturer's instructions at all times when using different cleaning agents by several manufacturers.



Avoid burns when working with corrosive cleaning agents. Always observe the manufacturers' instructions for safety and use.

Wear personal protective equipment (gloves, safety glasses)!

- Seriously contaminated cooling coils must be replaced. Consult our Customer Service.
- Re-connect the water in- and outflow pipes.

10.2.8 Check air oil-cooler

- The instructions in items 5.10.3, 7.1.2 and 10.1 must be observed!
- Close the stop valves in the coolant in- and outflow pipes.
- Remove dirt from the cooler block.
- Check the condition of screw connections and, if necessary, replace.

10.2.9 Check water oil-cooler

- The instructions in items 5.10.4, 7.1.2 and 10.1 must be observed!
- Close the stop valves in the coolant in- and outflow pipes.
- Inspect cooler for leaks in the piping.
- Check the condition of screw connections and, if necessary, replace.

10.2.10 Check hose lines

Even when adequately stored and subjected to permissible loads, hoses and hose lines are subject to a natural ageing process. This limits their period of use.



The period of use of the hose lines must not exceed 6 years from the manufacturing date stamped on them.

The period of use can be determined using available test and empirical values, taking into account the conditions of use.



The operator of the system must ensure that hose lines are replaced at suitable intervals of time, even if no defects which may affect their safe operation are identifiable on them.

Hose lines must be inspected for safe working condition by an expert before the plant is first put into operation and thereafter at least once a year.



If during inspections faults are found, these must be rectified immediately or suitable countermeasures taken.

10.2.11 Top up oil

- The instructions in item 7.1.2 must be observed!
- Always top up with the same type of oil as already used in the unit (see also item 10.2.2).

10.2.12 Checking friction linings of torque-limiting backstop

- The instructions in item 5.9 must be observed!

10.2.13 Checking auxiliary drive

- The instructions in item 5.16 must be observed!
- Be sure to observe the supplied operating instructions relating to the auxiliary gear unit for operation and maintenance.

10.2.14 Check tightness of fastening bolts

- The instructions in item 10.1 must be observed!
- Close the stop valves in the coolant in- and outflow pipes (gear units with cooling coil or water oil-cooling system).
- Check tightness of all fastening bolts.



Damaged bolts must be replaced with new bolts of the same type and strength class.

10.3 Final work



For operating and servicing the components, the pertinent instruction manuals and the specifications in sections 5 and 7 must be observed.
For technical data, refer to the data sheet and/or the list of equipment.



Observe also item 6.22.



Damaged bolts must be replaced with new bolts of the same type and strength class.

10.4 General inspection of the gear unit

The general inspection of the gear unit should be carried out by the Siemens Customer Service, as our engineers have the experience and training necessary to identify any components requiring replacement.

10.5 Lubricants

The quality of the oil used must meet the requirements of the separately supplied BA 7300 EN operating instructions, otherwise the guarantee given by Siemens will lapse. We urgently recommend using one of the oils listed in BA 7300 EN, because they have been tested and meet the requirements.



To avoid misunderstandings, we should like to point out that this recommendation is in no way intended as a guarantee of the quality of the lubricant supplied. Each lubricant manufacturer is responsible for the quality of his own product.

Information on the type, quantity and viscosity of the oil is given on the rating plate on the gear unit or in the supplied documentation.

The quantity of oil indicated on the rating plate is an approximation only. The marks on the dipstick or oil-sight glass are decisive for the amount of oil to be filled in.

The manual containing the current lubricants recommended by Siemens can also be consulted on the Internet (see back cover).

The oils listed there are subjected to continuous tests. Under certain circumstances the oils recommended there may therefore later be removed from the range or replaced with further developed oils.

We recommend regularly checking whether the selected lubricating oil is still recommended by Siemens. If it is not, the brand of oil should be changed.

11. Spare parts, customer service

11.1 Stocking spare parts

By stocking the most important spare and wearing parts on site you can ensure that the gear unit is ready for use at any time.

To order spare parts, refer to the spare-parts list.

For further information refer to the spare-parts drawing stated in the spare parts list.



We guarantee only the original spare parts supplied by us. Non-original spare parts have not been tested or approved by us. They may alter technical characteristics of the gear unit, thereby posing an active or passive risk to safety. Siemens will assume no liability or guarantee for damage caused by spare parts not supplied by Siemens. The same applies to any accessories not supplied by Siemens.

Please note that certain components often have special production and supply specifications and that we supply you with spare parts which comply fully with the current state of technical development as well as current legislation.

When ordering spare parts, always state the following:

Order number, item	Type, size	Part number	Quantity
--------------------	------------	-------------	----------

11.2 Spare parts and customer-service addresses

When ordering spare parts or requesting a service specialist, please contact Siemens first (see section 2).

12. Declarations

12.1 Declaration of incorporation

Declaration of Incorporation

in accordance with Directive 2006/42/EC, Annex II 1 B

The manufacturer, Siemens AG, 46393 Bocholt, declares with regard to the partly completed machinery,

Gear Unit

**H.SH, H.VH, H.HH, H.DH, H.KH, H.FH, H.HM,
H.DM, H.KM, H.FM, H.PH, B.SH, B.VH, B.HH,
B.DH, B.KH, B.FH, B.HM, B.DM, B.KM, B.FM,
T.SH, T.HH, T.KH, T.DH, T.FH**

Sizes 1 to 22

developed for driven machines in general engineering applications:

- The special technical documents described in Annex VII B have been prepared.
- The following basic health and safety requirements set out in Directive 2006/42/EC, Annex I, are applied and are satisfied:
1.1, 1.1.2, 1.1.3, 1.1.5; 1.2.6; 1.3.1 - 1.3.4, 1.3.6 - 1.3.8.1; 1.4.1, 1.4.2.1;
1.5.1, 1.5.2, 1.5.4 - 1.5.11, 1.5.13; 1.6.1, 1.6.2; 1.7.1 - 1.7.2, 1.7.4 - 1.7.4.3
- The partly completed machinery must not be put into service until it has been established that the machinery into which the partly completed machinery is to be incorporated has been declared in conformity with the provisions of Directive 2006/42/EC, as appropriate.
- The manufacturer undertakes, in response to a reasoned request by the national authorities, to transmit in electronic form relevant information about the partly completed machinery.
- The person authorised to compile the relevant technical documentation is:
Friedheim Schreier (Director ZGE Engineering)



Penig, 2010-06-01

Friedheim Schreier
(Director ZGE Engineering)



Penig, 2010-06-01

Hans-Hermann Olm
(Director ZG Business Subsegment)

Gear-unit lubrication

for
helical gear units, bevel-gear units,
bevel-helical gear-units, planetary-gear units
and geared motors
(with the exception of worm-geared motors)

Assembly and operating instructions
BA 7300 en 01/2012

FLENDER supplies

SIEMENS

Gear-unit lubrication

for
helical gear units, bevel-gear units,
bevel-helical gear-units,
planetary-gear units and geared motors
(with the exception of worm-geared motors)

Assembly and operating instructions

Translation of the original assembly and operating instructions

Lubricating oils

1

Lubricating greases

2

Check list for
oil filling and oil change

3

Documentation sheet
for oil change

4

Notes and symbols in these assembly and operating instructions

Note: The term "Assembly and operating instructions" will in the following also be shortened to "instructions" or "manual".

Legal notes

Warning note concept

This manual comprises notes which must be observed for your personal safety and for preventing material damage. Notes for your personal safety are marked with a warning triangle or an "Ex" symbol (when applying Directive 94/9/EC), those only for preventing material damage with a "STOP" sign.



WARNING! Imminent explosion!

The notes indicated by this symbol are given to prevent **explosion damage**.
Disregarding these notes may result in serious injury or death.



WARNING! Imminent personal injury!

The notes indicated by this symbol are given to prevent **personal injury**.
Disregarding these notes may result in serious injury or death.



WARNING! Imminent damage to the product!

The notes indicated by this symbol are given to prevent **damage to the product**.
Disregarding these notes may result in material damage.



NOTE!

The notes indicated by this symbol must be treated as general **operating information**.
Disregarding these notes may result in undesirable results or conditions.



WARNING! Hot surfaces!

The notes indicated by this symbol are made to prevent **risk of burns due to hot surfaces** and must always be observed.
Disregarding these notes may result in light or serious injury.

Where there is more than one hazard, the warning note for whichever hazard is the most serious is always used. If in a warning note a warning triangle is used to warn of possible personal injury, a warning of material damage may be added to the same warning note.

Qualified personnel

The product or system to which these instructions relate may be handled only by persons qualified for the work concerned and in accordance with the instructions relating to the work concerned, particularly the safety and warning notes contained in those instructions. Qualified personnel must be specially trained and have the experience necessary to recognise risks associated with these products or systems and to avoid possible hazards.

Intended use of Siemens products

Observe also the following:



Siemens products must be used only for the applications provided for in the catalogue and the relevant technical documentation. If products and components of other makes are used, they must be recommended or approved by Siemens. The faultfree, safe operation of the products calls for proper transport, proper storage, erection, assembly, installation, start-up, operation and maintenance. The permissible ambient conditions must be adhered to. Notes in the relevant documentations must be observed.

Trademarks

All designations indicated with the registered industrial property mark ® are registered trademarks of Siemens AG. Other designations used in these instructions may be trademarks the use of which by third parties for their own purposes may infringe holders' rights.

Exclusion of liability

We have checked the content of the instructions for compliance with the hard- and software described. Nevertheless, variances may occur, and so we can offer no warranty for complete agreement. The information given in these instructions is regularly checked, and any necessary corrections are included in subsequent editions.

Symbols

Earth connection point		Air relief point		yellow	
Oil-filling point		yellow	Oil-draining point		white
Oil level		red	Oil level		red
Oil level		red	Connection for vibration-monitoring device		
Lubrication point		red	Apply grease		
Lifting eye			Eye bolt		
Do not unscrew					
Alignment surface, horizontal			Alignment surface, vertical		



These symbols indicate the oil-level checking procedure using the oil dipstick.



These symbols indicate that the oil dipstick must always be firmly screwed in.

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1. Lubricating oils for helical gear units, bevel-gear units, bevel-helical gear-units, planetary-gear units and geared motors (with the exception of worm-geared motors)



These assembly and operating instructions do not apply to gear units made by WINERGY AG and Flender Graffenstaden S.A.S. These plants provide different instructions.



For "FLENDER gear units" not referred to in these assembly and operating instructions the following manuals will apply:

Ship's gearboxes:	BA 7301
Worm-gear unit:	BA 7303

1.1 Oil groups and base oils

Siemens Mechanical Drives (MD), Bocholt, distinguishes between the following oil groups in regard to gear oils for FLENDER gears:

- Standard oils
- Biologically degradable oils (also called "BIO" for short)
- Physiologically safe oils approved in accordance with NSF-H1 (also called "PHY" for short)

The tested and recommended gear oils can be manufactured on the basis of the following four base oils:

- Mineral oils
- Poly- α -olefins (also called "PAO" for short)
- Polyglykole (also called "PG" or "PAG" for short)
- synthetic esters



Theoretically all four base oil types may occur in each oil group.

1.2 Quality, code numbers, approval lists

1.2.1 Required quality of the gear oils to be used

For "FLENDER gear units", Siemens approves only CLP quality oils which contain constituents to DIN 51517-3 for improvement of corrosion prevention and resistance to ageing and which reduce wear in mixed-friction areas. The oils must have passed the following tests (see also - Specification of gear oil approvals for FLENDER gear units):

- | | |
|---|---|
| • Scuffing resistance in the FZG-test to DIN ISO 14635-1 under the A/8.3/90 test conditions | Fail stage > 12 |
| • FE-8 rolling-bearing test to DIN 51819-3 under the D-7.5/80-80 test conditions | Rolling-element wear < 30 mg
Cage wear: state value |
| • Resistance to grey staining to FVA 54 I-IV | GF fail stage ≥ 10
Resistance to grey staining
GFT = high |
| • Flender foam test | – Increase in total volume
1 minute after switching off $\leq 15\%$

– Increase in oil-air dispersion
5 minutes after switching off $\leq 10\%$ |
| • Static and dynamic "FLENDER-Freudenberg test" for shaft-sealing rings | Approval by Messrs. Freudenberg |
| • Compatibility with liquid seals | Approval by Messrs. Loctite |
| • "FLENDER-Mäder-Farbstest" (colour test) | Approval by Messrs. Mäder |



Oil group, base oil and viscosity class as stated in the specifications on the rating plates and information in the gear unit documentation must be adhered to! Failure to do so will result in invalidation of the guarantee.

If oil filters are used in the system the oil supplier should be requested to confirm that the filters do not negatively affect the characteristics of the oil.

The unapproved use of oils which do not comply with the above quality requirements will invalidate the Siemens product guarantee obligation. In addition adherence to the instructions given in these operating instructions is conditional for any claims under warranty.

Deviations are permitted only after consultation with Siemens!

If the operating conditions have been subsequently modified and differ from those stated in the customer's order, the gear oil to be used must be approved by Siemens in writing.

1.2.2 Code numbers, approval lists

To help its customers Siemens MD, Bocholt, has grouped oils in approval lists. The above-mentioned test results have been verified to Siemens for these oils by the oil manufacturers. Also worldwide adherence to the characteristics, features and minimum requirements required by Siemens has been guaranteed by the oil manufacturers.



Assignment of the approved oils to the oil groups, the base oils and the viscosities is indicated by code numbers in the approval lists.

Code number A13 indicates e.g. all standard mineral oils with the viscosity ISO VG 460.

Table 1: Code numbers of lubricant assignment, **standard oils**
CLP oils for "FLENDER" helical-, bevel- and planetary-gear units and geared motors

ISO VG:	100	150	220	320	460	680	1000
Mineral oils	A 17	A 16	A 15	A 14	A 13	A 12	A 11
Polyglycols	A 27	A 26	A 25	A 24	A 23	A 22	A 21
Poly- α -olefins	A 37	A 36	A 35	A 34	A 33	A 32	A 31

Table 2: Code numbers of lubricant assignment, **biologically degradable oils**
CLP oils for "FLENDER" helical-, bevel- and planetary-gear units and geared motors

ISO VG:	100	150	220	320	460	680	1000
Synthetic esters	A 47	A 46	A 45	A 44	A 43	A 42	A 41

Table 3: Code numbers of lubricant assignment, **physiologically safe oils**
CLP oils for "FLENDER" helical-, bevel- and planetary-gear units and geared motors

ISO VG:	100	150	220	320	460	680	1000
Polyglycols	A 57	A 56	A 55	A 54	A 53	A 52	A 51
Poly- α -olefins	A 67	A 66	A 65	A 64	A 63	A 62	A 61

Table 4: Code numbers of lubricant assignment
Rolling-bearing greases for use in "FLENDER gear units"

NLGI class:	2	3
Mineral oil base	H 14	H 13



The approval lists "T 7300" with the current gear-oil recommendations for "FLENDER gear units" are available as "FAQ" on the Internet under the following link:

<http://support.automation.siemens.com/WW/view/en/44231658>

We recommend checking before any oil change whether the chosen lubricant is still approved by Siemens.

The approval of these gear oils for use in FLENDER gear units and their consequent inclusion in the relevant lists does not mean that Siemens is liable for the suitability and quality of the oils. Nor is Siemens liable for any damage caused to FLENDER gear units by the use of these gear oils. Each gear-oil manufacturer is always responsible for the suitability and quality of his own product.

1.3 Oil temperatures

Synthetic gear oils have a wider temperature range and a higher viscosity index (i.e. a flatter viscosity-temperature gradient) than mineral oils.

The operating temperature range of the base oils most frequently used can be approximately specified as follows:

- | | |
|---|---|
| – Mineral oils | approx. - 10 °C to + 90 °C (short term + 100 °C) |
| – Polyglycols and poly- α -olefins | approx. - 20 °C to + 100 °C (short term + 110 °C) |
| – Synthetic esters | approx. - 20 °C to + 90 °C |



The upper and lower operating temperatures of certain gear oils may deviate widely from the values indicated. For operating conditions outside the above-mentioned temperature ranges the flash point and pour point provide information which must normally not be understood to represent operating limit values. For data on and characteristics of the gear oils please see the oil manufacturers' technical data sheets and safety data sheets.



The operating temperatures of the gear unit and the type of base oil must not be changed without consulting Siemens!

1.4 General service lives of oils

The service lives set out below are guaranteed by the oil manufacturers for the gear oils approved by Siemens (see also item 1.2) without substantial changes in oil qualities.

These specifications apply to an average oil temperature of 80 °C in the oil sump of the gear unit.

- | | |
|---|-----------------------------------|
| – Mineral oils and synthetic esters | 2 years or 10 000 operating hours |
| – Poly- α -olefins and polyglycols | 4 years or 20 000 operating hours |



Actual service lives may deviate from these. Here the rule applies that a temperature rise of 10 K will roughly halve and a temperature reduction of 10 K will roughly double the service life.

1.5 Information about initial filling and oil change

The purity level of the gear oil affects the operational reliability and service life of the oil and the gear unit. Accordingly, care should be taken to use only clean oil in the gear unit. For initial filling and oil changes the supplementary specifications, if any, in the gear unit instructions must also be observed.

Careful initial filling and careful oil changing contribute considerably to an increase in the operational reliability and service life of both the gear unit and the oil.



When putting in and changing oil no impurities such as foreign bodies, water or other liquids must be allowed to get into the interior of the gear unit.



In the case of large quantities of oil it is recommended to carry out an oil analysis to determine whether the oil needs to be cleaned or changed.



For recording purposes it is recommended to use table 7 (see section 4).

1.6 Safety at work, environmental protection and general information

All work carried out on the gear unit must be carried out carefully and by qualified personnel only (see also "Qualified Personnel" on page 3 of these instructions).

The following regulations and documents must be adhered to:

- Regulations for safety at work and environmental protection.
- Regulations for the disposal of used oil.



Gear oils are recyclable materials. Local legal requirements must be observed when disposing of them.

- Product data sheets for the oils used, including flushing oil, if necessary
- Safety data sheets for the oils and aids used
- Operating and maintenance instructions for the gear units scheduled for initial filling or oil change, including the oil supply system
- Documentation sheet for oil change (see table 7)

The following preparations must be made on the gear unit:

- Suitable body protection must be used (skin care media, gloves, work shoes, safety glasses, ear protectors, etc.).
- Sufficient space and cleanliness in the work place must be ensured.
- Provision of following aids and parts:
 - suitable oil-collecting and/or extraction equipment of a sufficient size
 - suitable, cleaned equipment and containers for oil sampling (see item 1.11).
 - suitable equipment for marking the oil sample
 - sufficient quantity of clean flushing oil (if necessary)
 - correct type and sufficient quantity of fresh oil for refilling the gear unit
 - cleaned filling system and further aids, including filling filters
 - fresh seals (if necessary)
 - oil binding and cleaning media
 - lintfree, clean cleaning cloths in sufficient quantity



For initial filling the procedure set out in item 1.8 must be adhered to.

1.7 Procedure for emptying the gear unit



It is recommended to use the check list in table 6 as an aid.

Before the oil is changed, a sample of the oil still warm from operation should be taken from the gear unit in an adequate manner and documented and kept for later reference (see item 1.11).

1.7.1 Draining off the used oil filling

- Note information given in the operating instructions for draining off the used oil. This also applies particularly to central lubricating systems and other oil supply equipment. If no relevant information is given, the lowest possible point must be used to drain off or suck out the oil.
- Shut down gear unit and ensure that it cannot be started up.
- Check and record oil level.



An oil level above the max mark may indicate that a foreign liquid (e.g. water) has got in. An oil level below the min mark may indicate that there is a leak. Both conditions must be rated as a fault and may result in damage to the gear unit. The cause must be found and, if necessary, rectified before refilling!



After shutting off the gear unit the oil should be changed as soon as possible to prevent any solid matter from settling. Should this not be possible, the gear unit must be run again for a short while before draining off the oil. The oil should as far as possible be drained off while it is still hot (approx. 50 °C).



Risk of scalding from escaping hot oil!

- Visual check for possible leakage. If there is leakage, the cause must be found and rectified before refilling.
- Before draining off or sucking out the oil open all vent holes provided.
- Position the oil collector under the oil drain.
- Drain off or suck out oil and, if necessary, also empty all oil pockets.



Risk of scalding from escaping hot oil! Open drain carefully.

- The oil draining off must be collected in accordance with regulations. If necessary, ancillary units (such as oil supply systems, filters) and pipework must likewise be emptied.
- On gear units with an oil circulation system and on oil supply systems the oil-conducting system must be emptied in accordance with the manufacturer's instructions in the operating and maintenance instructions.



Any spilled oil must be immediately bound with oil binding material and disposed of in accordance with regulations!

1.7.2 Checks and work to be done before refilling the gear unit

- The quantity of oil drained off must be compared with the manufacturer's rated quantity. If despite a correct oil level the quantity is considerably less than the rated quantity, it must be assumed that there are considerable quantities of used oil remaining in the gear unit and/or the ancillary units.



In this case the flushing procedure set out in item 1.9 must be followed.

- The oil drained off must be checked visually (appearance, colour, impurities, etc).
- If there is serious contamination (e.g. by water, metal particles, sludge), a specialist (preferably a Siemens specialist) must be called in to trace the cause.



The lubricating oil system must then be thoroughly flushed out before refilling (see item 1.9).

- Oil drain plugs with a magnet or magnetic extractors provided must be inspected for and freed of abraded metal. If a large amount of abraded metal is suspected, a specialist (preferably a Siemens specialist) must be called in to trace the cause.
- On gears with inspection holes the interior of the gear unit must be checked for deposits and the teeth and bearings checked for damage. Any damage must be repaired. If there are heavy deposits or other impurities, the flushing procedure set out in item 1.9 must be followed. If it is evident that the impurities cannot be removed by flushing alone, they must be removed mechanically. Residues may be removed from storage containers by using a rubber scraper.



Before opening inspection covers on gear units the area around the inspection hole must be thoroughly cleaned to prevent dirt from getting into the interior of the gear unit from outside when the cover is opened.

- Oil filters and seals provided must be cleaned or changed in accordance with regulations.
- The used oil must be disposed of in accordance with regulations.



Certain synthetic oils or impurities may have to be disposed of by different means from that for normal used oils. The oil data sheets must be observed.

1.8 Putting in the fresh gear oil



For initial filling it is recommended to use table 6, "Oil-change check list", from point 27 on as an aid.



For oil changing the gear unit must be filled only with fresh oil of the type and make previously used. If necessary, the specifications for the required purity class in the operating instructions must be observed. The make of oil or even the type of base oil must not be changed without very good reasons. If the oils previously used and those subsequently used are incompatible, a flushing procedure must be followed (see item 1.9).



It is recommended to keep a sample of the fresh oil before filling as a reference sample for later comparison.

The following steps must be carried out:

- Clean the seats of the vent covers, drain plugs, etc).
- If necessary, fit previously new seals on previously opened vent covers, drain plugs, etc., and close.
- If necessary, a flushing procedure must be followed at this point. Unless otherwise prescribed by the gear unit and/or oil manufacturer, the procedure as set out in item 1.9 is recommended.
- Check fresh oil visually for the following parameters:
 - Homogeneity
 - Freedom from solid impurities and water (clouding)
 - Colour and appearance in compliance with the oil manufacturer's product-specific data
- Put in fresh oil up to the prescribed fill level.



The oil quantity specified on the rating plates is only a guide value. Decisive for the quantity of oil to be put in are the marks on the oil dipstick or the other oil level indication systems.

- Fresh oil may be contaminated. The oil should therefore be put in through a filter as set out in the operating and maintenance instructions. If no specifications are available for this, filling filters with a filter mesh of 10 µm have proved highly effective. It is recommended to preheat the oil to between 35 and 40 °C. It sometimes also suffices to store the oil to be used to fill the gear unit with for approx. 2 to 3 days in a room at a temperature of approx. 25 °C before putting it in.
- If necessary, also fill ancillary units (oil systems, filters, etc.) and pipework.
- After a short operating time (approx. 10 minutes) and a rest time of at least 15 minutes the oil level must be rechecked and, if necessary, more oil put in or oil drained off.



It is recommended to take an oil sample approx. 5 operating hours after changing the oil and to record and keep it for later checking.

- Fill out and keep the documentation sheet for the oil change (see table 7).
- After a week at most check the oil filter (if provided) for deposits and, if necessary, clean or replace it. Impurities still remaining in the system may have dissolved and may be blocking the filters.

1.9 Flushing procedure



When changing the oil make, particularly the base oil type, if the gear oil put in is badly contaminated or major repairs have been carried out on the gear unit, a flushing procedure is necessary before finally filling the system with the operating oil.

If necessary, specifications for flushing included in the gear unit operating instructions must be adhered to. The following procedure is given only as an example and must be adjusted to suit local conditions.



The described procedure must be scheduled **in addition** to the activities specified for oil changing and must be integrated at the appropriate point in the procedure plan.

- Ensure that gear unit cannot be started up.
- To prevent incompatibilities the same type of lubricating oil as that to be used later should be used for flushing. To achieve a better flushing and dissolving effect a lower viscosity level of the same type of oil may have to be selected. Here the lubrication requirements of all component assemblies, including the pumps, must be taken into consideration.



The use of special cleaning or flushing oils is possible and is sometimes unavoidable.



If special cleaning or flushing oils are used, prior agreement with the oil supplier and Siemens is necessary.

- As with oil changing for the fresh oil as set out in item 1.8, the flushing oil must be visually checked before use.
- The system must be filled with the oil designated for flushing until a short run without load or in the part-load range is possible without disruptions



Heated flushing oil improves the flushing action and shortens flushing time. Because of the risk of scalding 50 °C must not be exceeded.



Risk of scalding from escaping hot oil!

- When filling with flushing oil any ancillary units provided must be taken into consideration. If necessary, these should be filled separately and emptied again after the flushing run.
- The gear unit must be operated for a short time with the flushing oil filling to ensure that it is mixed intensively with the residual oil remaining in the system and to as far as possible dissolve impurities. If no flushing time is specified in the operating and maintenance instructions, at least 10 to 60 minutes are recommended. The greater the operating oil quantity and the worse the contamination, the longer the time taken for flushing should be.



It is recommended to clean the flushing oil in the bypass flow during the flushing operation.

- The flushing oil must then be drained off. The procedure is identical to that followed for draining off the used oil (see item 1.7.1).
- The flushing oil must be visually checked. If there are still unacceptable impurities in the oil and/or the gear unit, this flushing procedure must be repeated until the desired result is achieved.

Depending on its condition, the flushing oil may be cleaned and re-used as flushing oil or must be disposed of.



The flushing oil must not be used as operating oil.

1.10 Procedure when changing the type of oil



The type of oil should be changed in exceptional cases only for important reasons. When changing to another base oil type or to an oil with very different additive technology, incompatibilities may result.

Ultimately only oil manufacturers can judge the compatibility of different gear oils. It is therefore urgently advisable to involve them in the decision process.



When changing to a lubricating oil with another base oil type or with different additive technology, the entire system must be thoroughly cleaned and flushed after draining off the used oil (see item 1.9). In this case the flushing oil may be used at most a second time to flush out the same gear unit. The flushing oil must then be disposed of.



When changing to a new type of oil of similar composition a flushing procedure as set out in item 1.9 is likewise to be recommended.

The following points must be **additionally** observed when changing the type of base oil:

- The compatibility of the new oil with other gear unit components, e.g. sealing materials or varnishes and paints, must be checked. To be recommended are the oils approved by Siemens (see item 1.2).
- Cavitation problems may arise with oils of a higher density. Larger suction line cross-sections may be required.

1.11 Oil samples

1.11.1 Taking an oil sample

- For reference purposes a fresh oil sample of the operating lubricating oil must, before the gear unit is filled, be analysed by the laboratory, in which the samples of the used oil are also to be inspected later. The test results must be recorded and kept for later comparisons.
- The oil sample must be taken so that it is representative. A precondition for this is that the oil filling is as homogeneous as possible. For this reason the oil sample should be taken, if possible, while the gear unit is running, when the oil is still warm from operation, from the same point and always by the same method and by the same specialist. If the gear unit has to be stationary, the sample must be taken no later than 10 minutes after the gear unit is taken out of operation. In this way demixture or deposition is largely avoided.
- When choosing the sampling point, it is important that the agitated oil is accessible. Places where the oil has not been agitated or samples of the immediate oil surface or from the bottom must be avoided. If there is a filter in the oil circuit, the sample must be taken from a point upstream of the filter.
- The sampling point must be cleaned before the oil sample is taken. While the sample is being taken, moisture (water) and contaminants (e.g. dust) from the environment must be prevented from entering.
- The required size of the sample is determined by the required scope of the analysis. In case of doubt it is recommended that the analysing laboratory be consulted. 250 ml are normally sufficient for routine analyses. The sample containers must be suitable, clean, dry vessels with tight closures (see standard ISO 3722: Hydraulic fluid power – Fluid sample containers – Qualifying and controlling cleaning methods <4>). They must be resistant to the sample material up to operating oil temperature. Transparent and tightly sealable wide-neck bottles of glass or HD-PE have proved suitable. If it should be necessary to clean the sample containers, it is recommended that they be thoroughly flushed out several times on site with the operating oil. The flushing oil must then be disposed of.



Milk, wine, beer, mineral-water bottles or other containers used to store foodstuffs must not be used as sample containers, even if they have been properly cleaned and marked.

- The sample container must be marked with an indelible pencil or unlosable label. It is recommended that the following data be given:
 - Company / system operator
 - Designation and, if applicable, serial number of the system
 - If applicable the point at which the sample has been taken
 - Oil type, commercial designation
 - Date at which sample has been taken

In addition to the data on the sample container a sample-consignment note which can be clearly assigned to the sample must be completed. A copy of the sample-consignment note should be kept with the gear-unit documentation.

- After the sample has been taken, the oil level must be checked and, if necessary, the oil topped up by adding fresh oil.

1.11.2 Assessing the test results

The water content of the oil must be assessed as follows:

- Mineral oils, PAOs, non-water-dissolving polyglycols, esters:

< 300 ppm	= 0.03 %	⇒ In order
300 ppm – 600 ppm	= 0.03 % – 0.06 %	⇒ Observation, check trend behaviour; if necessary, consult Siemens.
> 600 ppm	= 0.06 %	⇒ Not in order; determine and rectify cause; if necessary, change oil.

- Water-dissolving polyglycols:

< 15 000 ppm	= 1.5 %	⇒ In order
15 000 ppm – 20 000 ppm	= 1.5 % – 2.0 %	⇒ Observation, check trend behaviour; if necessary, consult Siemens.
> 20 000 ppm	= 2.0 %	⇒ Not in order; determine and rectify cause; if necessary, change oil.

- The deviation in the viscosity of the oil sample from the viscosity of the oil specified in the technical data should not exceed 10 %.
- The purity of the oil should not be less than specified in the following:
–/20/15 acc. to ISO 4406
The test laboratory must be informed that for measurement the oil must be thinned with **toluol**.
- Trend analyses of the measured element contents are mostly more important than the magnitude of the absolute values. Without filtration higher contamination levels are naturally to be expected than are found in the case of systems with filtration.

Table 5: Possible cause of changes in test results (trend analysis)

Change		Possible causes
Viscosity ↑	–	Short-chain constituents volatilised
	–	High-viscosity foreign fluids / greases got in
Viscosity ↓	–	Long-chain constituents cracked
	–	Viscosity-index improvers cut or used up
	–	Low-viscosity foreign fluids got in
TAN (NZ) ↑	–	Oil aged, acids produced
PQ-Index ↑	–	Wear caused by magnetisable substances (particularly Fe)
Fe ↑	–	Wear and oxidation on gears and rolling bearings, corrosion on housing
Cr ↑	–	Wear on rolling bearings and gears
Ni ↑	–	Wear on rolling bearings and gears
Al ↑	–	Wear on oil-pump bearings
Cu ↑	–	Wear on rolling-bearing cages and slide bearings
Zn ↑	–	Washed out of priming coat
Zn ↓	–	Additive with Zn content used up
Sn ↑	–	Wear on slide bearings
Si ↑	–	Contamination from outside (dust)
Si ↓	–	Anti-foaming additive used up
Ca ↑	–	Contamination from outside (e.g. chalk dust)
Oil clouded	–	Contamination from water and/or other foreign fluids (including oils)
Oil darker	–	Oil aged



The oil supplier must decide about the further usability of the oil. He must guarantee that the properties of the oil conform to the Siemens specification.

2. Lubricating greases for gear units and rolling bearings

For special gear-unit applications, grease lubrication of the bearings may be necessary.



Greases may be used only if their use is specified in the operating instructions for the gear unit. The relubrication periods must be adhered to.

As well as lubrication, rolling bearing greases are used for the special sealing of bearing points, e.g. with vertical gear-unit connection shafts or against environmental action such as dust or water spray.



In closed gear units with internal oil lubrication the gear oil must not be allowed to mix with bearing grease.



The approval lists "T 7300" with the current grease recommendations for "FLENDER gear units" are available as "FAQ" on the Internet under the following link:

<http://support.automation.siemens.com/WW/view/en/44231658>

When using greases, we recommend checking every year whether the chosen lubricant is still approved by Siemens.

3. Check list for oil filling and oil change

Table 6: Oil-change check list

1	Oil-change record ready for completion?	☐	21	Gear unit internal parts inspected for damage, repaired or replaced (if necessary)?	☐
2	Sufficient fresh oil available?	☐	22	Gear unit and oil supply system with pipework flushed out (if necessary)?	☐
3	Sufficiently large oil collector and used oil receptacle(s) available?	☐	23	Oil filter cleaned or replaced (if necessary)?	☐
4	Sufficient oil binding and cleaning materials available?	☐	24	Seals changed?	☐
5	Sufficient room available for oil change?	☐	25	Opened vent and drain points closed again?	☐
6	Oil samples (used and fresh oil) taken and stored?	☐	26	Cause(s) of leaks removed (see item 10)?	☐
7	Protective clothing (glasses, gloves, etc.) available and used?	☐	27	Fresh oil visually checked?	☐
8	Oil change by specialist personnel?	☐	28	Oil filling hole(s) opened?	☐
9	Gear unit and/or oil supply system manufacturers' maintenance instructions read?	☐	29	Fresh oil put in through filter up to mark on the oil-level check point?	☐
10	Leaks in the drive and/or output recorded after visual check?	☐	30	Oil-filling hole closed?	☐
11	Used oil at a temperature of between 30 and 50 °C and drive started?	☐	31	Drive started and run for a short while (approx. 10 minutes)?	☐
12	Drive shut down and prevented from restarting?	☐	32	Oil level checked (after min. 15 minutes rest time)?	☐
13	Vent hole opened?	☐	33	Oil topped up or drained off until the oil level is correct (if necessary)?	☐
14	Oil collector placed under the oil drain?	☐	34	Oil-change record completed?	☐
15	Drain opened carefully? Caution - risk of scalding!	☐	35	Sample of used oil checked?	☐
16	Used oil completely drained off?	☐	36	Used oil correctly disposed of?	☐
17	Ancillary units, oil pockets and pipework emptied (if any)?	☐	37	Sample of new operating oil taken after approx. 5 hours and stored (for checking when changing type of oil)?	☐
18	Spilled oil immediately bound with suitable material?	☐	38	Binding material and cleaning cloths correctly disposed of?	☐
19	Used oil assessed and recorded?	☐	39	Drive re-started?	☐
20	Gear unit and other units thoroughly cleaned?	☐	40	Filter checked and cleaned or replaced (after one week's operation at most)?	☐

4. Documentation sheet for oil change

Table 7: Oil-change oil-filling record

Oil-change oil-filling record			
System:		Gear unit:	
Date:	Operating hours:		Last oil change
Oil to be used according to rating plate:		Quantity according to rating plate:	
Condition before oil change			
Designation of oil:		Oil level:	
Gear unit/oil supply system leaktight? yes ☐ no ☐		Filter dirty? yes ☐ no ☐	
Operating temperature: °C			
Other noticeable features:			
.....			
Used oil			
Identification mark of used-oil sample:		Laboratory number:	
Quantity:			
Result of visual check of oil:			
Gear unit / oil supply system			
Deposits? yes ☐ no ☐			
Result of visual check of oil:			
Fresh oil			
Designation of oil:			
Designation of fresh-oil sample:		Laboratory number:	
Filtered during filling? yes ☐ no ☐		Filter mesh: µm	Quantity put in:
Result of visual check of oil:			
Was/were the gear unit and/or the oil-supply system flushed before being filled with fresh oil?			
yes ☐ with no ☐			
Noticeable features (if applicable):			
.....			
Designation of 5-hour operating oil sample:		Laboratory number:	
Name of recorder:		Department:	
Date:		Signature:	

Further Information:

"FLENDER gear units" on the Internet

www.siemens.com/gearunits

"FLENDER couplings" on the Internet

www.siemens.com/couplings

Service & Support:

<http://support.automation.siemens.com/WW/view/en/10803928/133300>

Lubricants:

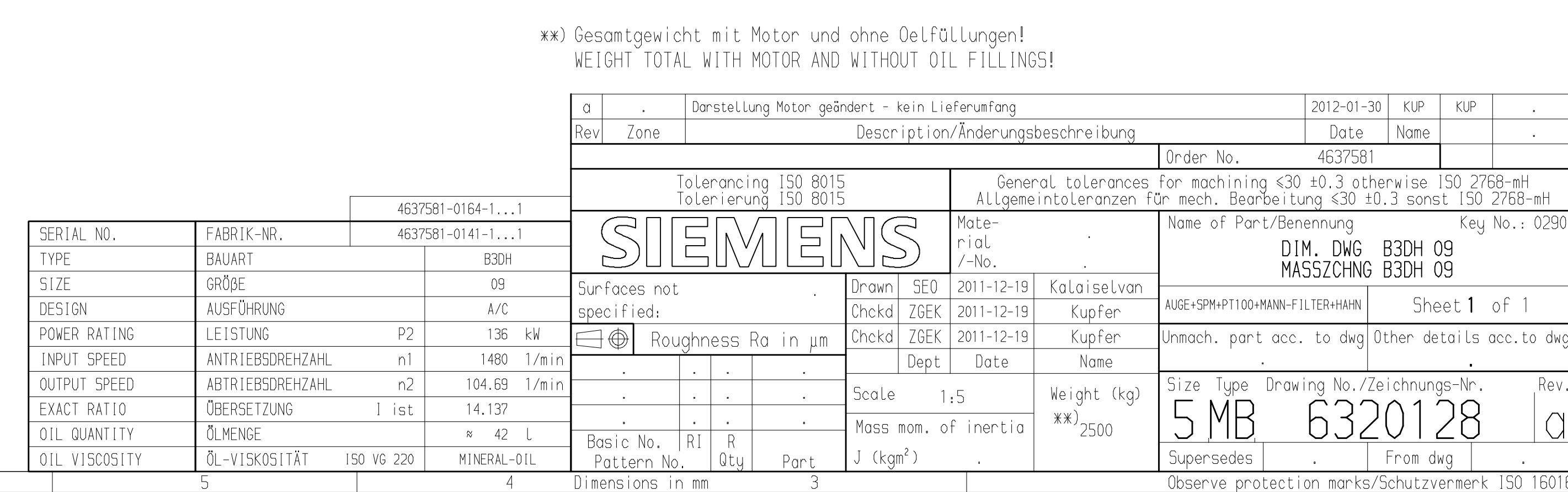
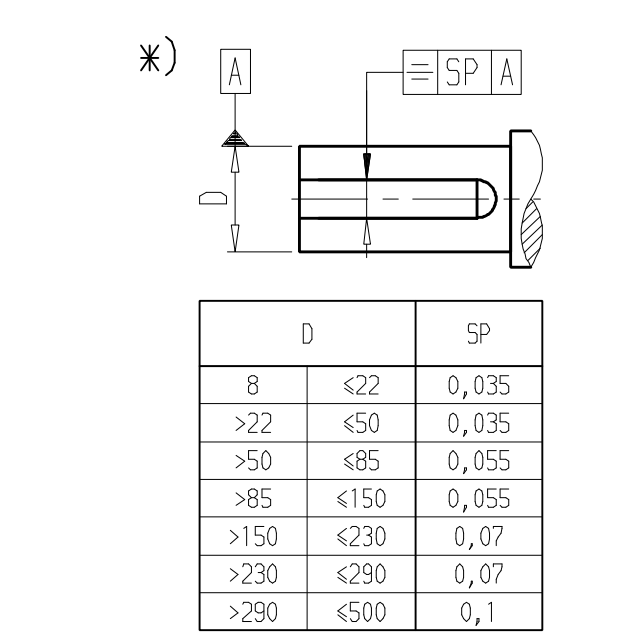
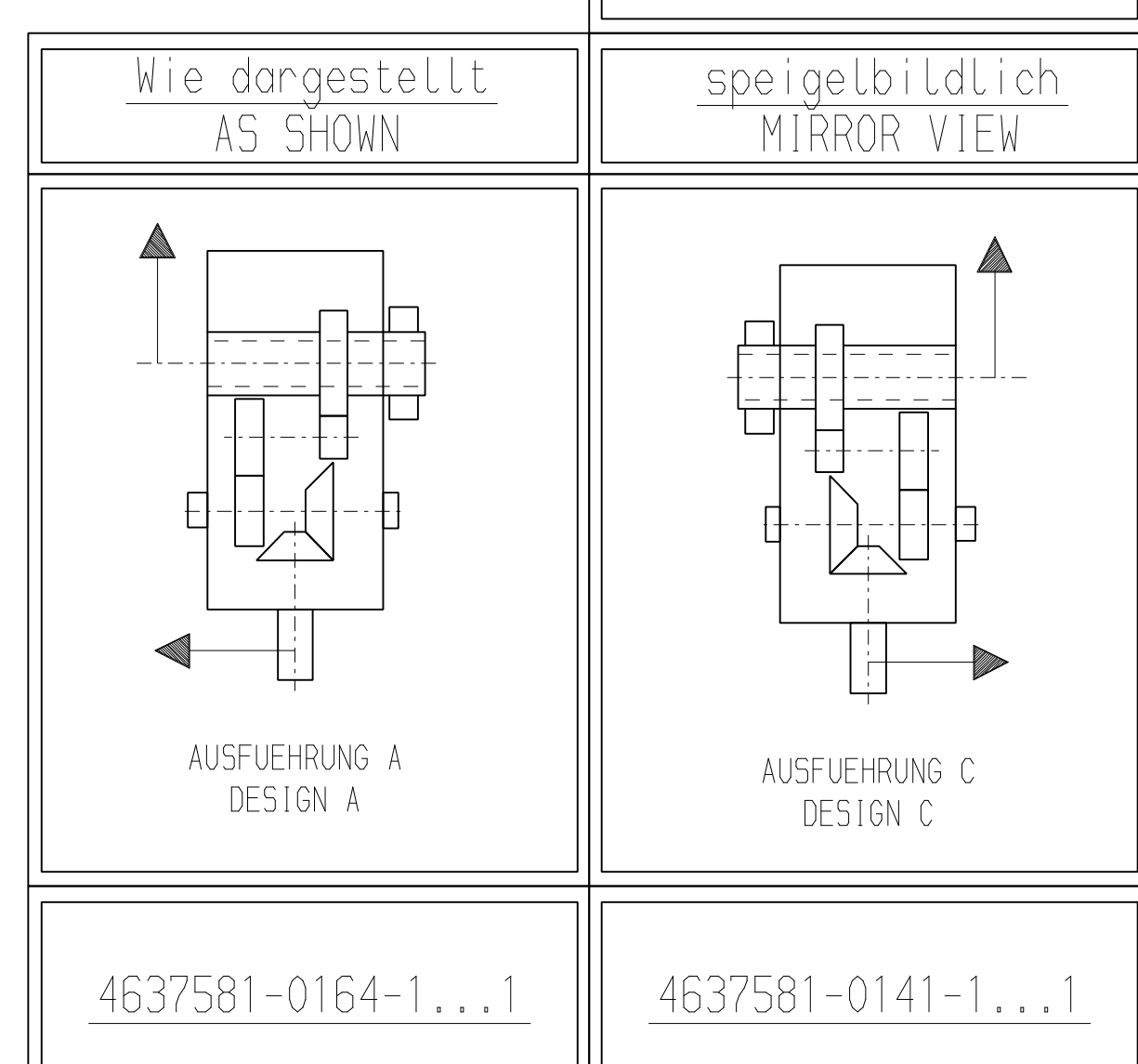
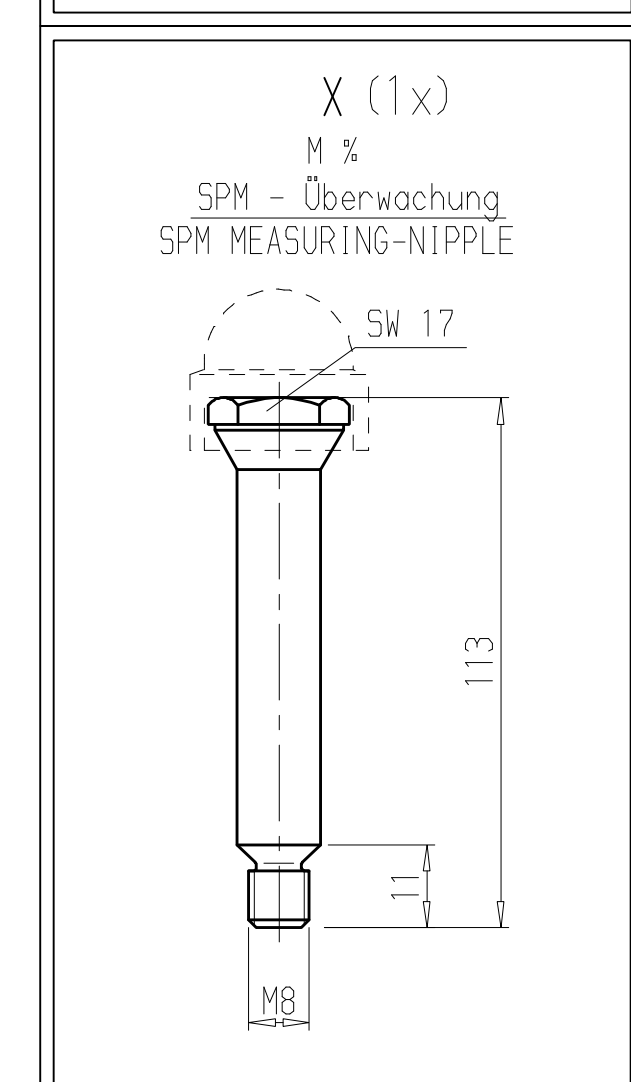
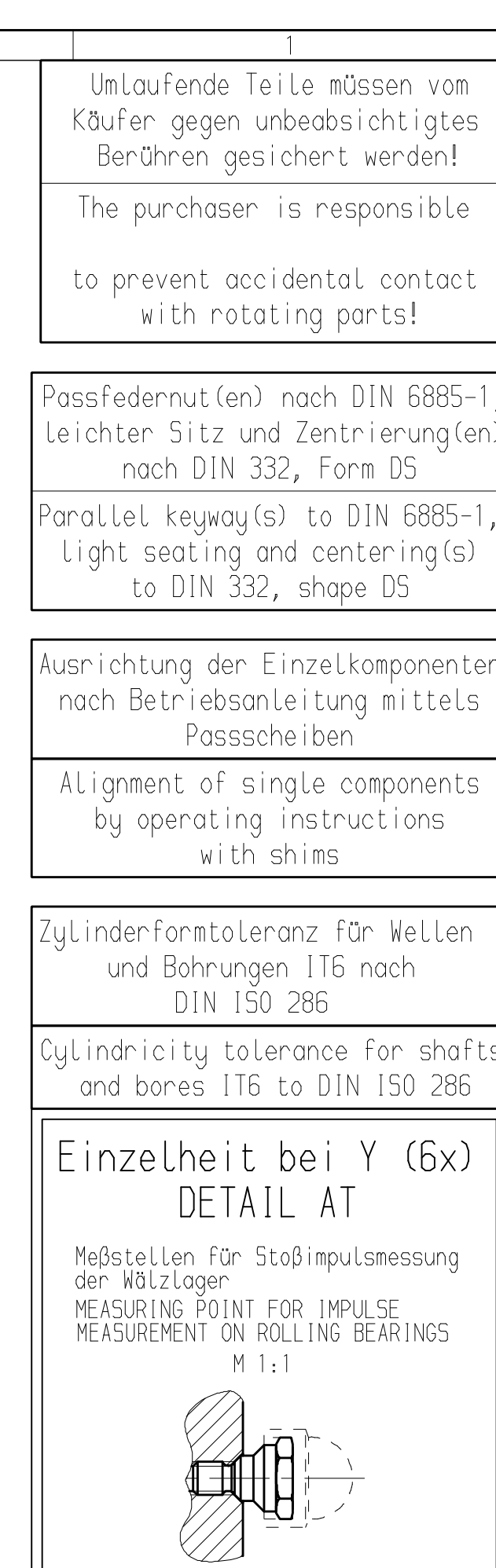
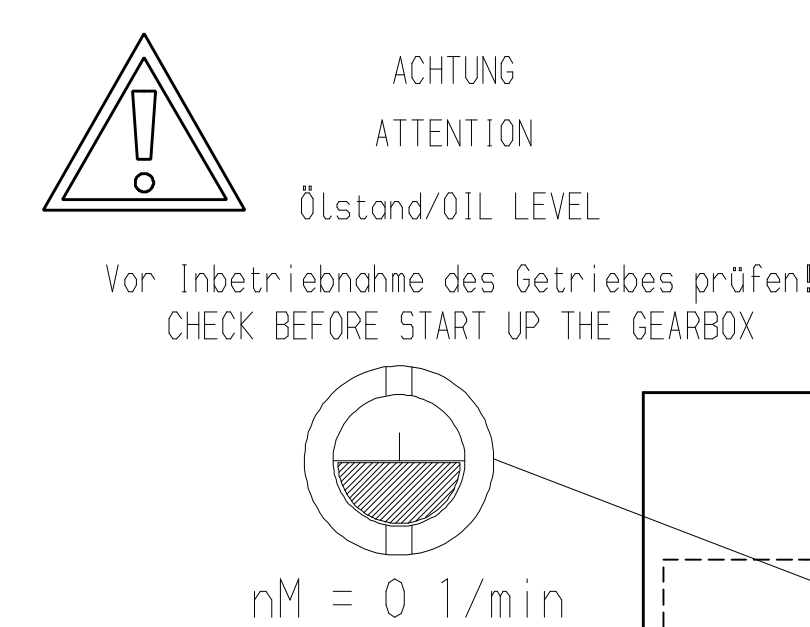
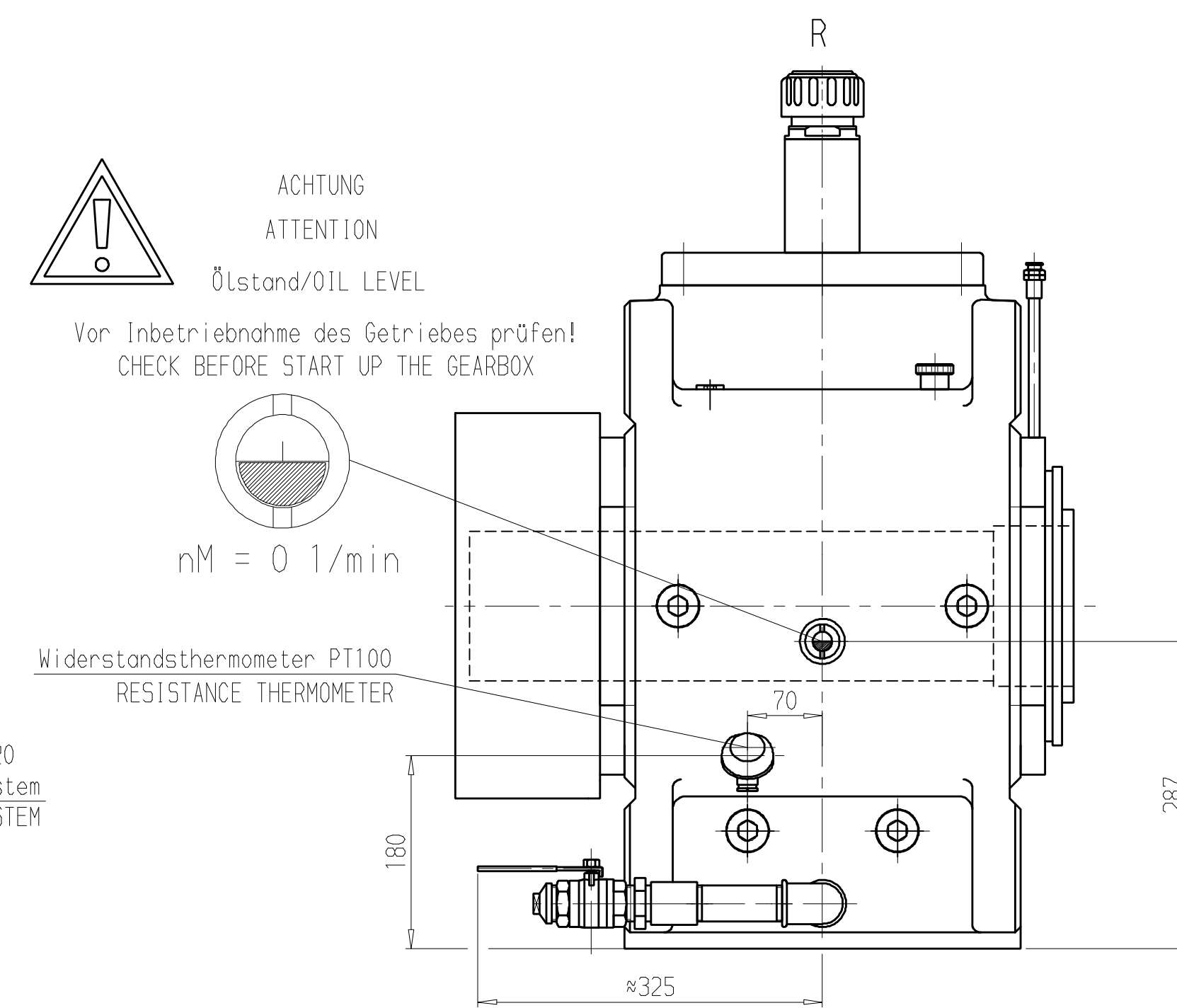
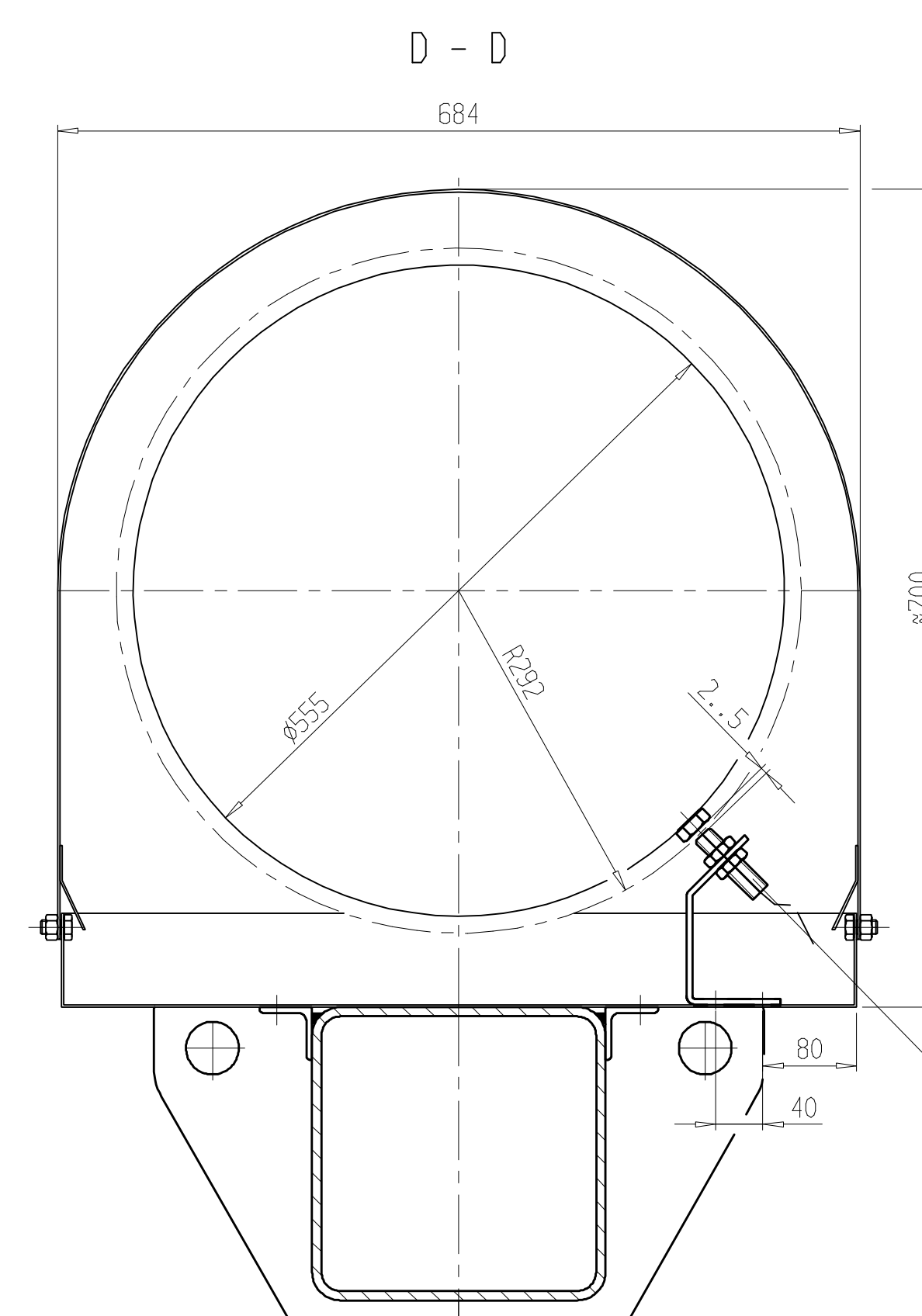
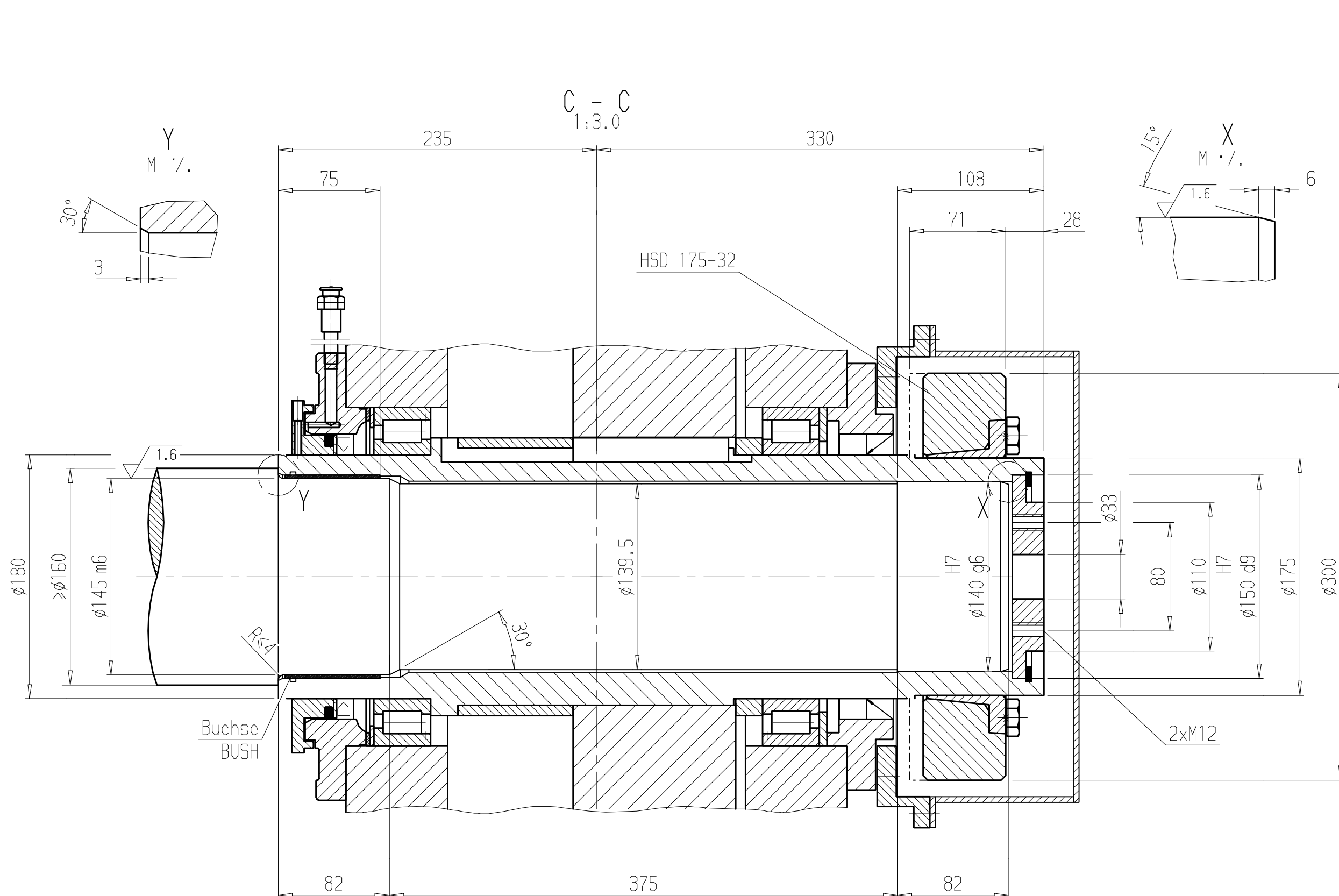
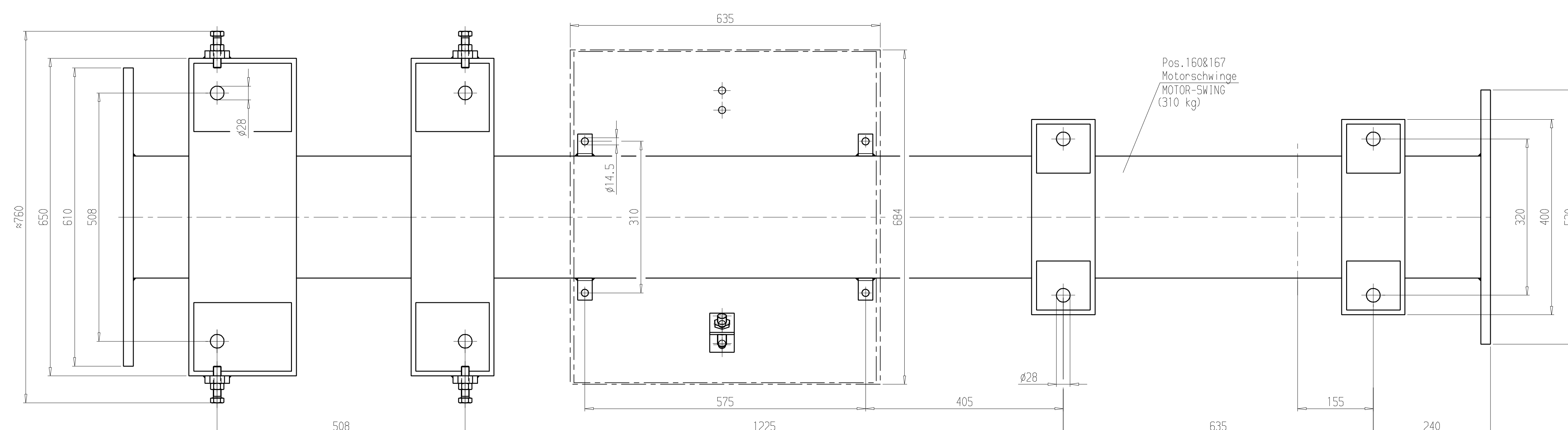
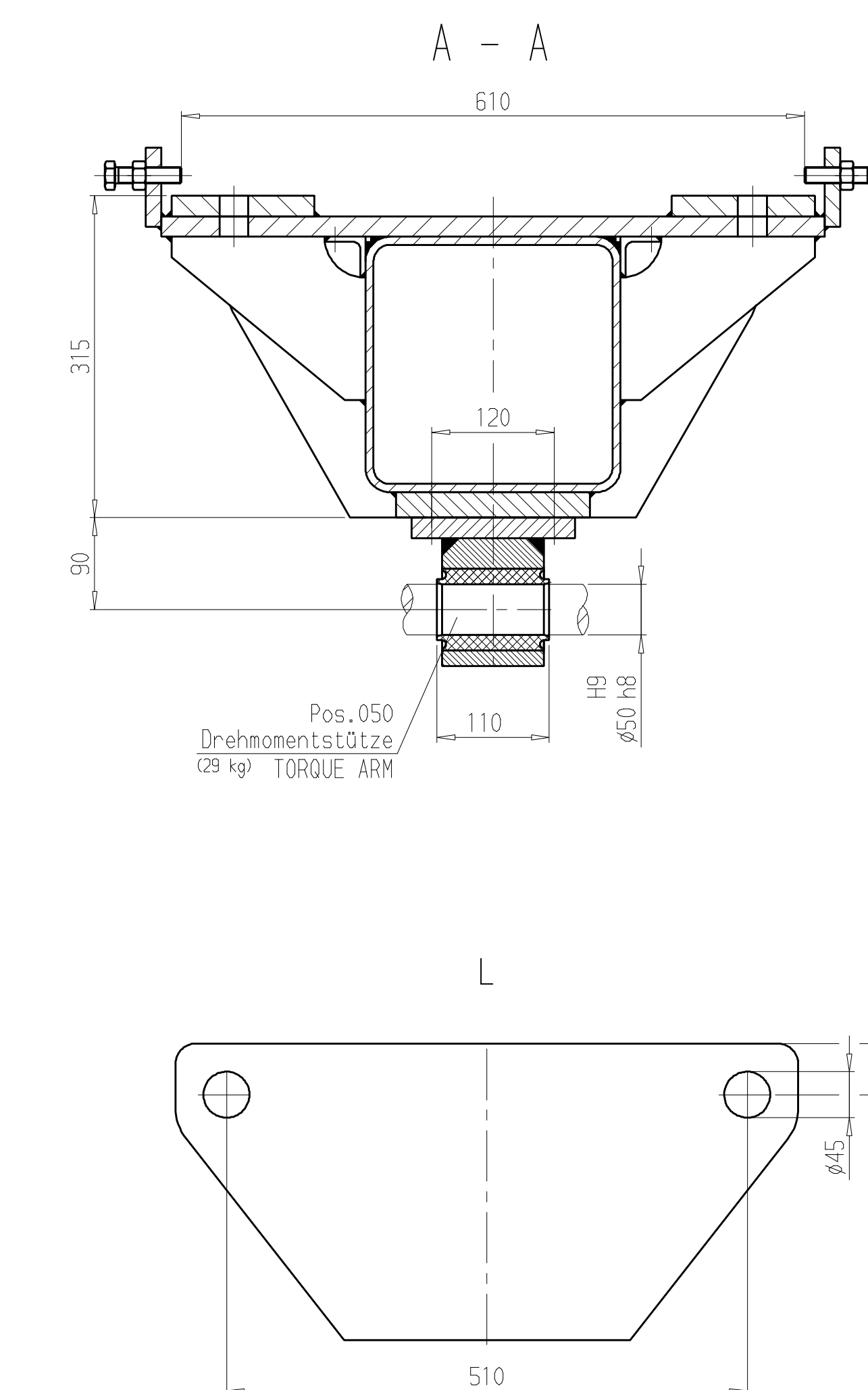
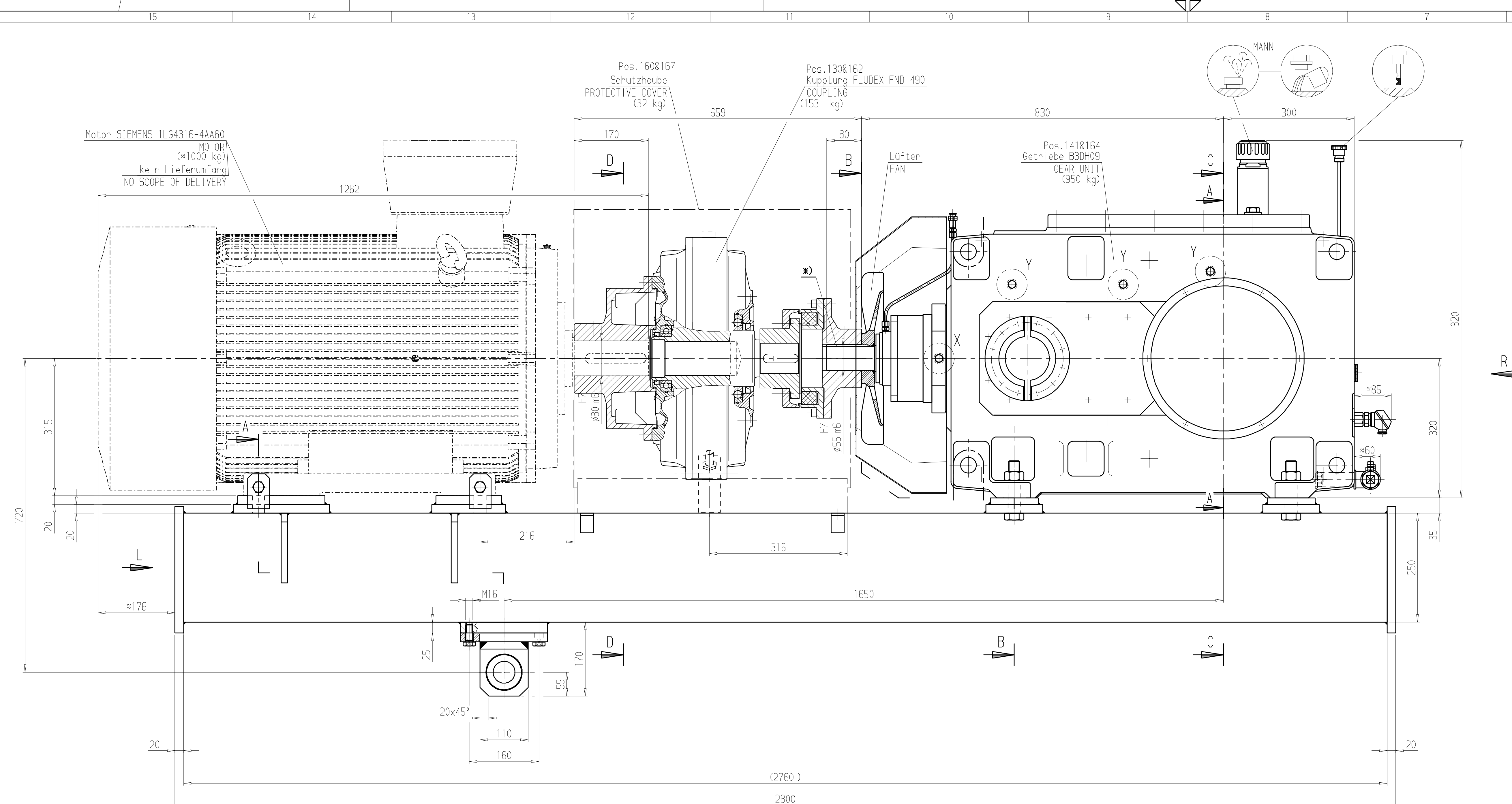
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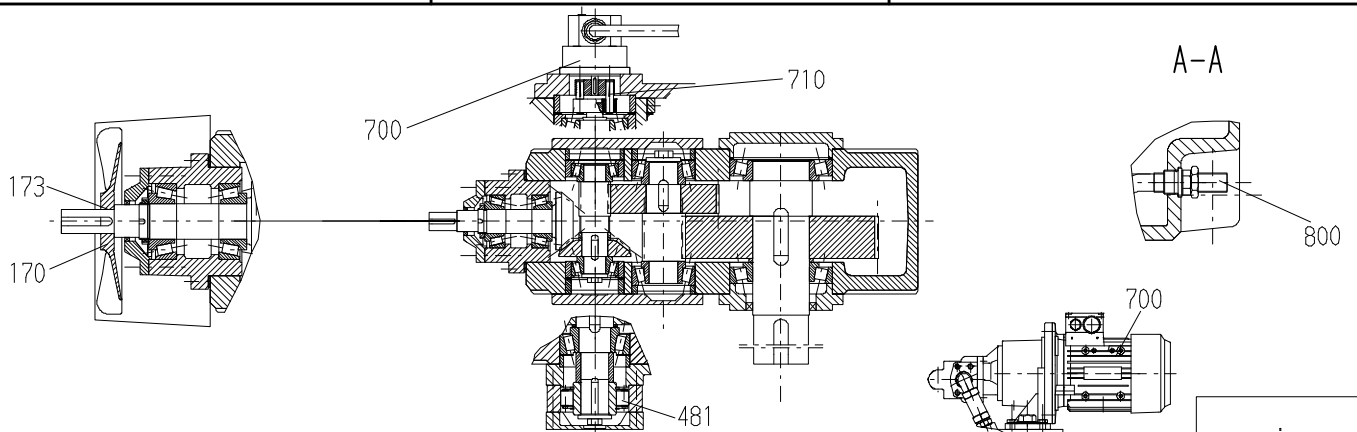
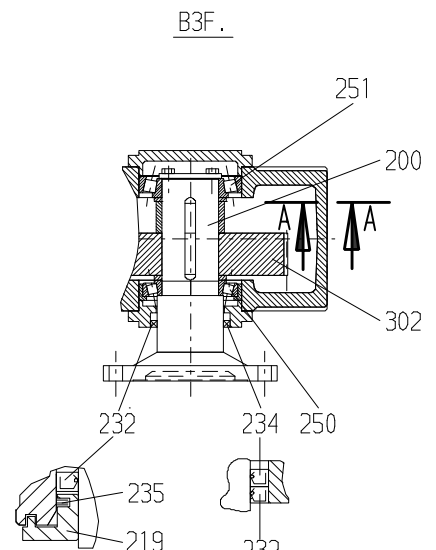
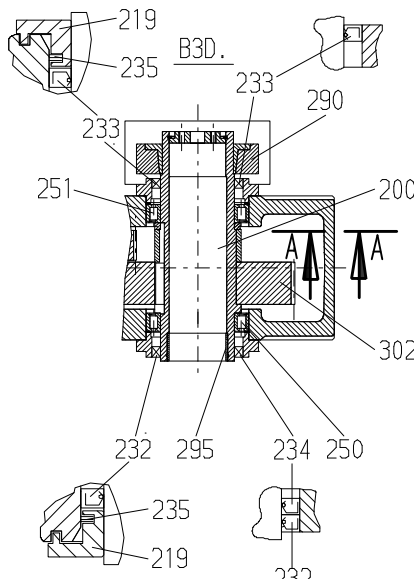
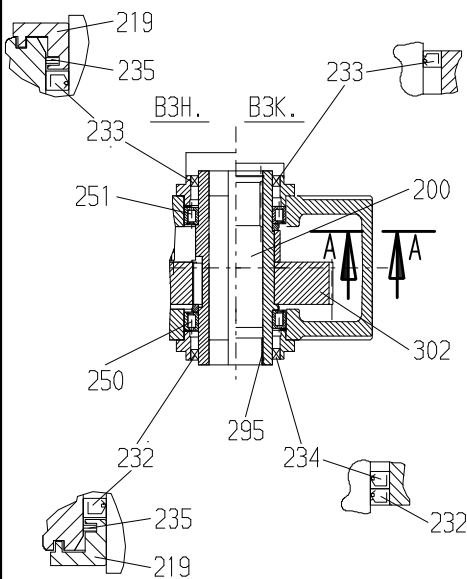
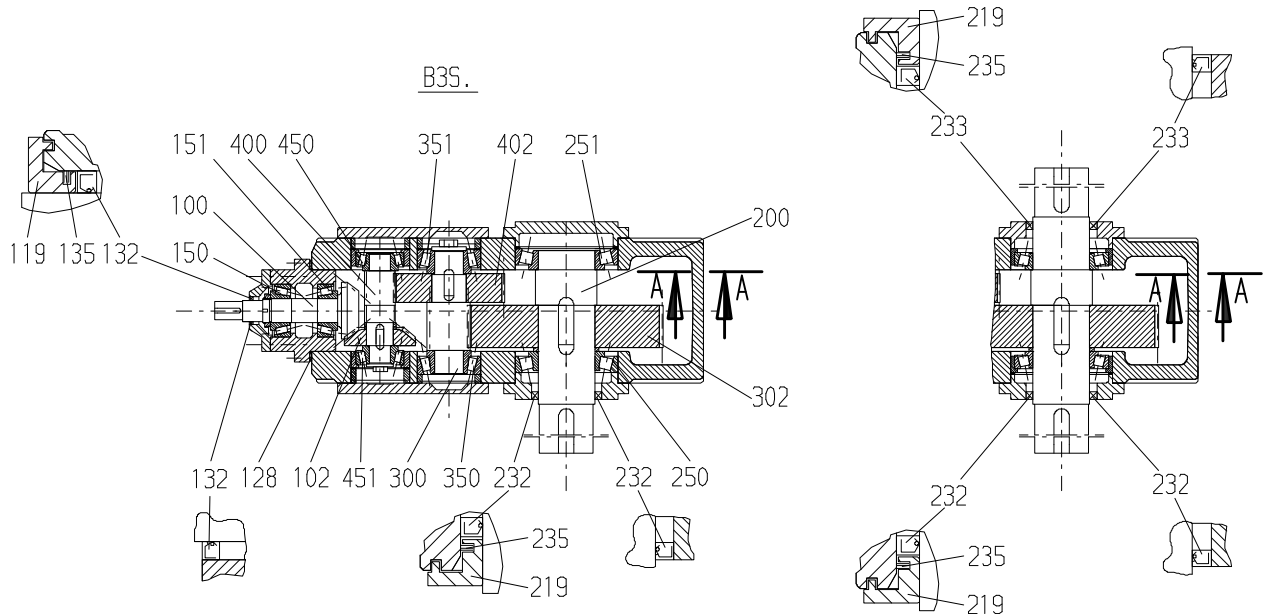
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Ausführung
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Sheet 1 of 1

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09322 Penig, Tel.+49(0)37381-61-0, Fax+49(0)37381-80286

Datum/ Date
2010-09-14

Name	Kästner	ZGEK
Y	6153212	g

Diese technische Unterlage hat gesetzlichen Schutz (DIN 34)

Rev. ID 008
Medusa

SIEMENS		Ersatzteilliste (EL) Spare parts list		Bauart B3DH Type Größe 9,0 Size Übersetzung 25,843 ratio		Seite Page 1/2	
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Hierzu gehört Zeichnungs-Nr. Please refer to DWG No.				6153212			
Teil-Nr.		Menge	Benennung		Zeichnungs-Nr.	Material-Nr.	Gw(kg)
Part No.		No. off	Description		Drawings No.	Ident no.	Weight
G	0099	1	ST KEGELRADSATZ ZY224/1,62 1.6587		9508191/-	000.000.564.945	22,0
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			ADAPTER RING 201X276X05 E ST				
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E	0300	1	ST STRDWELLE LP 265,0/3,94 1.6587		5876797/B	000.000.676.797	20,5
			PINION SHAFT LP 265,0/3,94 1.6587				
E	0302	1	ST STIRNRAD RP 265,0/3,94 1.6587		6115338/-	000.001.325.944	102,0
			GEAR WHEEL RP 265,0/3,94 1.6587				
E	0350	1	ST WAELZL 32317 D 720 ST-KAEF			000.000.380.777	6,9
			ROLLG BRG 32317 D 720 ST-KAEF				
Die mit * gekennzeichneten Teile gehören zu einer Baugruppe (G). Die Baugruppe ist nur komplett auszutauschen. The parts marked with * belong to a subassembly (G). The subassembly must be replaced complete.							
Siemens Industriegetriebe GmbH, Thierbacher Str.24, 09322 Penig Tel. +49(0)37381-61-0, Fax +49(0)37381-80286					Datum Date KALAISELVAN,J.,MR. CHENN-7858		

SIEMENS	Ersatzteilliste (EL) Spare parts list		Bauart B3DH Type Größe 9,0 Size Übersetzung 25,843 ratio		Seite Page 2/2																																																																																				
	Bei Korrespondenz bitte angeben Please quote in correspondence Hierzu gehört Zeichnungs-Nr. Please refer to DWG No. 4503614363 EL 4637581-031 DE/EN 6153212																																																																																								
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Siemens Industriegetriebe GmbH, Thierbacher Str.24, 09322 Penig Tel. +49(0)37381-61-0, Fax +49(0)37381-80286			Datum Date KALAISELVAN,J.,MR. CHENN-7858																																																																																						

SIEMENS	Ersatzteilliste (EL) Spare parts list		Bauart B3DH Type Größe 13,0 Size Übersetzung 13,721 ratio		Seite Page 1/2
Bei Korrespondenz bitte angeben Please quote in correspondence			4503614363 EL 4637581-081 DE/EN		
Hierzu gehört Zeichnungs-Nr. Please refer to DWG No.			6153212		
Teil-Nr.	Menge	Benennung	Zeichnungs-Nr.	Material-Nr.	Gw(kg)
Part No.	No. off	Description	Drawings No.	Ident no.	Weight
E 0100	1	ST KEGELRADW R ZS 285/1,27 1.6587 BEVEL PINION SHAFT R ZS 285/1,27 1.6587	5881186/B	000.000.561.186	39,1
E 0102	1	ST KEGELRAD L ZS 285/1,27 1.6587 BEVEL GEAR L ZS 285/1,27 1.6587	5881351/B	000.000.561.351	20,9
E 0119	1	ST RINGTRAEGER085X150,0X21 ST/GJL RING CARRIER085X150,0X21 ST/GJL	6144020/-	000.001.352.730	1,2
E 0128	1	ST RING-PASS 271X381X06 G S355J2G3 ADAPTER RING 271X381X06 G S355J2G3	5881756/C	000.000.561.756	3,2
E 0132	1	ST WDR 085X110X12 BAUMSL W 5710 75FKM585 SHAFT SEAL 085X110X12 BAUMSL W 5710		000.000.306.042	0,1
E 0135	3	ST RING-LAM AS110/04,6X0,98 W5730 FED-ST RING-LAM AS110/04,6X0,98 W5730 FED-ST		000.000.347.205	0,0
E 0150	1	ST WAELZL 32318 D 720 ST-KAEF ROLLG BRG 32318 D 720 ST-KAEF		000.000.380.778	8,4
E 0151	1	ST WAELZL 32318 D 720 ST-KAEF ROLLG BRG 32318 D 720 ST-KAEF		000.000.380.778	8,4
E 0170	1	ST LUEFTER 620/ 80 AC45000 FAN 620/ 80 AC45000	5882406/E	000.000.562.406	8,8
E 0173	1	ST RING-TOL BN0820-075-05 W 3372 FED-ST DISTANCE RING BN0820-075-05 W 3372		000.000.346.009	0,0
E 0200	1	ST WELLE-HOHL 295X190X0815 C60+N HOLLOW SHAFT 295X190X0815 C60+N	5877461/C	000.000.677.461	166,0
E 0219	1	ST RINGTRAEGER260X358,0X26 ST/GJL RING CARRIER260X358,0X26 ST/GJL	5877867/-	000.000.677.867	6,0
E 0232	1	ST WDR 260X300X20 BAUMSL W 5710 75FKM585 SHAFT SEAL 260X300X20 BAUMSL W 5710		000.000.307.019	0,5
E 0233	1	ST WDR 260X300X20 BAUMSL W 5710 75FKM585 SHAFT SEAL 260X300X20 BAUMSL W 5710		000.000.307.019	0,5
E 0235	3	ST RING-LAM AS300/09,0X1,5 W5730 FED-ST RING-LAM AS300/09,0X1,5 W5730 FED-ST		000.000.347.220	0,1
E 0250	1	ST WAELZL182952 SL ROLLG BRG 182952 SL		000.000.380.192	18,5
E 0251	1	ST WAELZL182952 SL ROLLG BRG 182952 SL		000.000.380.192	18,5
E 0290	1	ST SCHB-SCHR GR.260 F 3233 ST SHRINK DISC GR.260 F 3233 ST		000.000.309.471	82,0
E 0295	1	ST BUCHSE PAP 195X200X080 P10 INA BUSH PAP 195X200X080 P10 INA		000.000.375.903	1,0
E 0300	1	ST STRDWELLE RP 370,0/3,94 1.6587 PINION SHAFT RP 370,0/3,94 1.6587	5877097/D	000.000.677.097	53,3
E 0302	1	ST STIRNRAD LP 370,0/3,94 1.6587 GEAR WHEEL LP 370,0/3,94 1.6587	5877110/H	000.000.677.110	271,0
E 0350	1	ST WAELZL 22322 E D 635 SKF/FAG ROLLG BRG 22322 E D 635 SKF/FAG		000.000.380.878	17,7
E 0351	1	ST WAELZL 22322 E D 635 SKF/FAG ROLLG BRG 22322 E D 635 SKF/FAG		000.000.380.878	17,7
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Siemens Industriegetriebe GmbH, Thierbacher Str.24, 09322 Penig Tel. +49(0)37381-61-0, Fax +49(0)37381-80286			Datum Date	KALAISELVAN,J.,MR. CHENN-7858	

SIEMENS	Ersatzteilliste (EL) Spare parts list		Bauart B3DH Type Größe 13,0 Size Übersetzung 13,721 ratio		Seite Page 2/2																																																																		
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Bei Korrespondenz bitte angeben Please quote in correspondence						4503614363 EL 4637581-141 DE/EN		
Hierzu gehört Zeichnungs-Nr. Please refer to DWG No.						6153212		
Teil-Nr. Part No.		Menge No. off	Benennung Description		Zeichnungs-Nr. Drawings No.	Material-Nr. Ident no.	Gw(kg) Weight	
G	0099	1	ST KEGELRADSATZ ZY200/1,27 1.6587 PAIR OF MATING BEVEL WHEELS ZY200/1,27		9508189/-	000.000.564.943	21,1	
E*	0100	1	ST KEGELRADW L ZY 200/1,27 1.6587 BEVEL PINION SHAFT L ZY 200/1,27 1.6587		5877021/B	000.000.677.021	14,0	
E*	0102	1	ST KEGELRAD R ZY 200/1,27 1.6587 BEVEL GEAR R ZY 200/1,27 1.6587		5877051/C	000.000.677.051	7,1	
E	0119	1	ST RINGTRAEGER060X112,0X19 ST/GJL RING CARRIER060X112,0X19 ST/GJL		5877842/-	000.000.677.842	0,6	
E	0128	1	ST RING-PASS 201X276X05 E ST ADAPTER RING 201X276X05 E ST		5881754/D	000.000.561.754	1,2	
E	0132	1	ST WDR 060X075X08 BAUMSL W 5710 75FKM585 SHAFT SEAL 060X075X08 BAUMSL W 5710			000.000.307.230	0,0	
E	0135	3	ST RING-LAM AS075/03,3X0,82 W5730 FED-ST RING-LAM AS075/03,3X0,82 W5730 FED-ST			000.000.347.200	0,0	
E	0150	1	ST WAELZL 32313 B D 720 ST-KAEF ROLLG BRG 32313 B D 720 ST-KAEF			000.000.380.566	3,7	
E	0151	1	ST WAELZL 32313 B D 720 ST-KAEF ROLLG BRG 32313 B D 720 ST-KAEF			000.000.380.566	3,7	
E	0170	1	ST LUEFTER 395/ 55 AC45000 FAN 395/ 55 AC45000		5882391/C	000.000.562.391	3,4	
E	0173	1	ST RING-TOL BN0820-055-01 W 3372 FED-ST DISTANCE RING BN0820-055-01 W 3372			000.000.346.101	0,0	
E	0200	1	ST WELLE-HOHL 205X140X0565 16MnCrS5 HOLLOW SHAFT 205X140X0565 16MnCrS5		5877301/D	000.000.677.301	47,3	
E	0219	1	ST RINGTRAEGER180X260,0X26 ST/GJL RING CARRIER180X260,0X26 ST/GJL		5877861/B	000.000.677.861	3,1	
E	0232	1	ST WDR 180X210X15 BAUMSL W 5710 75FKM585 SHAFT SEAL 180X210X15 BAUMSL W 5710			000.000.306.062	0,3	
E	0233	1	ST WDR 180X210X15 BAUMSL W 5710 75FKM585 SHAFT SEAL 180X210X15 BAUMSL W 5710			000.000.306.062	0,3	
E	0235	3	ST RING-LAM AS210/08,0X1,5 W5730 FED-ST RING-LAM AS210/08,0X1,5 W5730 FED-ST			000.000.347.182	0,1	
E	0250	1	ST WAELZL182936 SL ROLLG BRG 182936 SL			000.000.379.147	6,2	
E	0251	1	ST WAELZL182936 SL ROLLG BRG 182936 SL			000.000.379.147	6,2	
E	0290	1	ST SCHB-SCHR GR.175 F 3233 ST SHRINK DISC GR.175 F 3233 ST			000.000.309.467	22,5	
E	0295	1	ST BUCHSE PAP 145X150X070 P10 INA BUSH PAP 145X150X070 P10 INA			000.000.375.899	0,6	
E	0300	1	ST STRDWELLE LP 265,0/3,94 1.6587 PINION SHAFT LP 265,0/3,94 1.6587		5876797/B	000.000.676.797	20,5	
E	0302	1	ST STIRNRAD RP 265,0/3,94 1.6587 GEAR WHEEL RP 265,0/3,94 1.6587		6115338/-	000.001.325.944	102,0	
E	0350	1	ST WAELZL 32317 D 720 ST-KAEF ROLLG BRG 32317 D 720 ST-KAEF			000.000.380.777	6,9	
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SIEMENS	Ersatzteilliste (EL) Spare parts list		Bauart B3DH Type Größe 9,0 Size Übersetzung 14,137 ratio		Seite Page 2/2																																																																																				
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Siemens Industriegetriebe GmbH, Thierbacher Str.24, 09322 Penig Tel. +49(0)37381-61-0, Fax +49(0)37381-80286			Datum Date KALAISELVAN,J.,MR. CHENN-7858																																																																																						

SIEMENS			Ersatzteilliste (EL) Spare parts list			Bauart B3DH Type Größe 9,0 Size Übersetzung 14,137 ratio			Seite Page 1/2		
Bei Korrespondenz bitte angeben						4503614363					
Please quote in correspondence						EL 4637581-164 DE/EN					
Hierzu gehört Zeichnungs-Nr.											
Please refer to DWG No.						6153212					
Teil-Nr.		Menge	Benennung			Zeichnungs-Nr.		Material-Nr.		Gw(kg)	
Part No.		No. off	Description			Drawings No.		Ident no.		Weight	
G	0099	1	ST KEGELRADSATZ ZY200/1,27 1.6587			9508189/-		000.000.564.943		21,1	
			PAIR OF MATING BEVEL WHEELS ZY200/1,27								
E*	0100	1	ST KEGELRADW L ZY 200/1,27 1.6587			5877021/B		000.000.677.021		14,0	
			BEVEL PINION SHAFT L ZY 200/1,27 1.6587								
E*	0102	1	ST KEGELRAD R ZY 200/1,27 1.6587			5877051/C		000.000.677.051		7,1	
			BEVEL GEAR R ZY 200/1,27 1.6587								
E	0119	1	ST RINGTRAEGER060X112,0X19 ST/GJL			5877842/-		000.000.677.842		0,6	
			RING CARRIER060X112,0X19 ST/GJL								
E	0128	1	ST RING-PASS 201X276X05 E ST			5881754/D		000.000.561.754		1,2	
			ADAPTER RING 201X276X05 E ST								
E	0132	1	ST WDR 060X075X08 BAUMSL W 5710 75FKM585					000.000.307.230		0,0	
			SHAFT SEAL 060X075X08 BAUMSL W 5710								
E	0135	3	ST RING-LAM AS075/03,3X0,82 W5730 FED-ST					000.000.347.200		0,0	
			RING-LAM AS075/03,3X0,82 W5730 FED-ST								
E	0150	1	ST WAELZL 32313 B D 720 ST-KAEF					000.000.380.566		3,7	
			ROLLG BRG 32313 B D 720 ST-KAEF								
E	0151	1	ST WAELZL 32313 B D 720 ST-KAEF					000.000.380.566		3,7	
			ROLLG BRG 32313 B D 720 ST-KAEF								
E	0170	1	ST LUEFTER 395/ 55 AC45000			5882391/C		000.000.562.391		3,4	
			FAN 395/ 55 AC45000								
E	0173	1	ST RING-TOL BN0820-055-01 W 3372 FED-ST					000.000.346.101		0,0	
			DISTANCE RING BN0820-055-01 W 3372								
E	0200	1	ST WELLE-HOHL 205X140X0565 16MnCrS5			5877301/D		000.000.677.301		47,3	
			HOLLOW SHAFT 205X140X0565 16MnCrS5								
E	0219	1	ST RINGTRAEGER180X260,0X26 ST/GJL			5877861/B		000.000.677.861		3,1	
			RING CARRIER180X260,0X26 ST/GJL								
E	0232	1	ST WDR 180X210X15 BAUMSL W 5710 75FKM585					000.000.306.062		0,3	
			SHAFT SEAL 180X210X15 BAUMSL W 5710								
E	0233	1	ST WDR 180X210X15 BAUMSL W 5710 75FKM585					000.000.306.062		0,3	
			SHAFT SEAL 180X210X15 BAUMSL W 5710								
E	0235	3	ST RING-LAM AS210/08,0X1,5 W5730 FED-ST					000.000.347.182		0,1	
			RING-LAM AS210/08,0X1,5 W5730 FED-ST								
E	0250	1	ST WAELZL182936 SL					000.000.379.147		6,2	
			ROLLG BRG 182936 SL								
E	0251	1	ST WAELZL182936 SL					000.000.379.147		6,2	
			ROLLG BRG 182936 SL								
E	0290	1	ST SCHB-SCHR GR.175 F 3233 ST					000.000.309.467		22,5	
			SHRINK DISC GR.175 F 3233 ST								
E	0295	1	ST BUCHSE PAP 145X150X070 P10 INA					000.000.375.899		0,6	
			BUSH PAP 145X150X070 P10 INA								
E	0300	1	ST STRDWELLE LP 265,0/3,94 1.6587			5876797/B		000.000.676.797		20,5	
			PINION SHAFT LP 265,0/3,94 1.6587								
E	0302	1	ST STIRNRAD RP 265,0/3,94 1.6587			6115338/-		000.001.325.944		102,0	
			GEAR WHEEL RP 265,0/3,94 1.6587								
E	0350	1	ST WAELZL 32317 D 720 ST-KAEF					000.000.380.777		6,9	
			ROLLG BRG 32317 D 720 ST-KAEF								
Die mit * gekennzeichneten Teile gehören zu einer Baugruppe (G). Die Baugruppe ist nur komplett auszutauschen.											
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Tel. +49(0)37381-61-0, Fax +49(0)37381-80286											

SIEMENS	Ersatzteilliste (EL) Spare parts list	Bauart B3DH Type Größe 9,0 Size Übersetzung 14,137 ratio	Seite Page 2/2
Bei Korrespondenz bitte angeben Please quote in correspondence		4503614363 EL 4637581-164 DE/EN	
Hierzu gehört Zeichnungs-Nr. Please refer to DWG No.		6153212	
Teil-Nr. Menge Part No. No. off		Benennung Description	Zeichnungs-Nr. Material-Nr. Gw(kg) Drawings No. Ident no. Weight
E 0351	1	ST WAE LZL 32317 D 720 ST-KAEF ROLLG BRG 32317 D 720 ST-KAEF	000.000.380.777 6,9
E 0400	1	ST STRDWELLE R 185,0/2,83 1.6587 PINION SHAFT R 185,0/2,83 1.6587	5877350/A 000.000.677.350 13,9
E 0402	1	ST STIRNRAD L 185,0/2,83 1.6587 GEAR WHEEL L 185,0/2,83 1.6587	5880873/C 000.000.560.873 32,1
E 0450	1	ST WAE LZL 32312 B D 720 ST-KAEF ROLLG BRG 32312 B D 720 ST-KAEF	000.000.380.565 3,2
E 0451	1	ST WAE LZL 32312 B D 720 ST-KAEF ROLLG BRG 32312 B D 720 ST-KAEF	000.000.380.565 3,2
E 0481	1	ST RS FXM 100- 50SX/H 55 W 5813 ST BACKSTOP FXM 100- 50SX/H 55 W 5813 ST	000.000.389.012 12,2
Siemens Industriegetriebe GmbH, Thierbacher Str.24, 09322 Penig Tel. +49(0)37381-61-0, Fax +49(0)37381-80286		Datum Date	KALAISELVAN,J.,MR. CHENN-7858

SIEMENS	Ersatzteilliste (EL) Spare parts list		Bauart B3DH Type Größe 15,0 Size Übersetzung 12,172 ratio		Seite Page 1/2
	Bei Korrespondenz bitte angeben Please quote in correspondence Hierzu gehört Zeichnungs-Nr. Please refer to DWG No.		4503614363 EL 4637581-191 DE/EN 6153212		
Teil-Nr. Part No.	Menge No. off	Benennung Description	Zeichnungs-Nr. Drawings No.	Material-Nr. Ident no.	Gw(kg) Weight
E 0100	1	ST KEGELRADW L ZS 350/1,26 1.6587 BEVEL PINION SHAFT L ZS 350/1,26 1.6587	5881201/D	000.000.561.201	74,9
E 0102	1	ST KEGELRAD R ZS 350/1,26 1.6587 BEVEL GEAR R ZS 350/1,26 1.6587	5881366/B	000.000.561.366	42,8
E 0119	1	ST RINGTRAEGER100X160,0X24 ST/GJL RING CARRIER100X160,0X24 ST/GJL	5889335/C	000.000.669.335	1,6
E 0128	1	ST RING-PASS 311X431X6,5 H S355J2G3 ADAPTER RING 311X431X6,5 H S355J2G3	5881757/A	000.000.561.757	3,3
E 0132	1	ST WDR 100X125X12 BAUMSL W 5710 75FKM585 SHAFT SEAL 100X125X12 BAUMSL W 5710		000.000.306.057	0,1
E 0135	3	ST RING-LAM AS125/05,0X0,98 W5730 FED-ST RING-LAM AS125/05,0X0,98 W5730 FED-ST		000.000.347.206	0,0
E 0150	1	ST EINBAUS 32222/DF ST-KAEF PAIRED TAPER ROLLER BEARINGS 32222/DF		000.000.379.978	15,6
E 0151	1	ST WAE LZL 2226 NJ EC/C3 D 5412 SKF/FAG ROLLG BRG 2226 NJ EC/C3 D 5412 SKF/FAG		000.000.770.357	11,7
E 0170	1	ST LUEFTER 620/ 90 AC45000 FAN 620/ 90 AC45000	5882408/D	000.000.562.408	8,7
E 0173	1	ST RING-TOL BN0820-085-01 W 3372 FED-ST DISTANCE RING BN0820-085-01 W 3372		000.000.346.011	0,0
E 0200	1	ST WELLE-HOHL 335X230X0930 C60+N HOLLOW SHAFT 335X230X0930 C60+N	5877463/E	000.000.677.463	223,0
E 0219	1	ST RINGTRAEGER300X405,0X29 ST/GJL RING CARRIER300X405,0X29 ST/GJL	5877869/-	000.000.677.869	8,5
E 0232	1	ST WDR 300X340X20 BAUMSL W 5710 75FKM585 SHAFT SEAL 300X340X20 BAUMSL W 5710		000.000.307.027	1,0
E 0233	1	ST WDR 300X340X20 BAUMSL W 5710 75FKM585 SHAFT SEAL 300X340X20 BAUMSL W 5710		000.000.307.027	1,0
E 0235	3	ST RING-LAM AS340/10,0X1,5 W5730 FED-ST RING-LAM AS340/10,0X1,5 W5730 FED-ST		000.000.347.221	0,1
E 0250	1	ST WAE LZL182960 SL ROLLG BRG 182960 SL		000.000.378.060	31,2
E 0251	1	ST WAE LZL182960 SL ROLLG BRG 182960 SL		000.000.378.060	31,2
E 0290	1	ST SCHB-SCHR GR.300 F 3233 ST SHRINK DISC GR.300 F 3233 ST		000.000.309.473	120,0
E 0295	1	ST BUCHSE PAP 235X240X090 P10 INA BUSH PAP 235X240X090 P10 INA		000.000.375.905	1,3
E 0300	1	ST STRDWELLE LP 442,0/3,47 1.6587 PINION SHAFT LP 442,0/3,47 1.6587	5877492/A	000.000.677.492	103,0
E 0302	1	ST STIRNRAD RP 442,0/3,47 1.6587 GEAR WHEEL RP 442,0/3,47 1.6587	5877499/A	000.000.677.499	429,0
E 0350	1	ST WAE LZL 22328 E D 635 MET-KAEF ROLLG BRG 22328 E D 635 MET-KAEF		000.000.380.879	35,1
E 0351	1	ST WAE LZL 22328 E D 635 MET-KAEF ROLLG BRG 22328 E D 635 MET-KAEF		000.000.380.879	35,1
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Siemens Industriegetriebe GmbH, Thierbacher Str.24, 09322 Penig Tel. +49(0)37381-61-0, Fax +49(0)37381-80286			Datum Date KALAISELVAN,J.,MR. CHENN-7858		

SIEMENS	Ersatzteilliste (EL) Spare parts list		Bauart B3DH Type Größe 15,0 Size Übersetzung 12,172 ratio		Seite Page 2/2																																																																								
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Bei Korrespondenz bitte angeben						4503614363					
Please quote in correspondence						EL 4637581-231 DE/EN					
Hierzu gehört Zeichnungs-Nr.											
Please refer to DWG No.						6153212					
Teil-Nr.		Menge	Benennung			Zeichnungs-Nr.		Material-Nr.		Gw(kg)	
Part No.		No. off	Description			Drawings No.		Ident no.		Weight	
E	0100	1	ST KEGELRADW L ZS 350/1,26 1.6587			5881201/D		000.000.561.201		74,9	
			BEVEL PINION SHAFT L ZS 350/1,26 1.6587								
E	0102	1	ST KEGELRAD R ZS 350/1,26 1.6587			5881366/B		000.000.561.366		42,8	
			BEVEL GEAR R ZS 350/1,26 1.6587								
E	0119	1	ST RINGTRAEGER100X160,0X24 ST/GJL			5889335/C		000.000.669.335		1,6	
			RING CARRIER100X160,0X24 ST/GJL								
E	0128	1	ST RING-PASS 311X431X6,5 H S355J2G3			5881757/A		000.000.561.757		3,3	
			ADAPTER RING 311X431X6,5 H S355J2G3								
E	0132	1	ST WDR 100X125X12 BAUMSL W 5710 75FKM585					000.000.306.057		0,1	
			SHAFT SEAL 100X125X12 BAUMSL W 5710								
E	0135	3	ST RING-LAM AS125/05,0X0,98 W5730 FED-ST					000.000.347.206		0,0	
			RING-LAM AS125/05,0X0,98 W5730 FED-ST								
E	0150	1	ST EINBAUS 32222/DF ST-KAEF					000.000.379.978		15,6	
			PAIRED TAPER ROLLER BEARINGS 32222/DF								
E	0151	1	ST WAE LZL 2226 NJ EC/C3 D 5412 SKF/FAG					000.000.770.357		11,7	
			ROLLG BRG 2226 NJ EC/C3 D 5412 SKF/FAG								
E	0170	1	ST LUEFTER 620/ 90 AC45000			5882408/D		000.000.562.408		8,7	
			FAN 620/ 90 AC45000								
E	0173	1	ST RING-TOL BN0820-085-01 W 3372 FED-ST					000.000.346.011		0,0	
			DISTANCE RING BN0820-085-01 W 3372								
E	0200	1	ST WELLE-HOHL 335X230X0930 C60+N			5877463/E		000.000.677.463		223,0	
			HOLLOW SHAFT 335X230X0930 C60+N								
E	0219	1	ST RINGTRAEGER300X405,0X29 ST/GJL			5877869/-		000.000.677.869		8,5	
			RING CARRIER300X405,0X29 ST/GJL								
E	0232	1	ST WDR 300X340X20 BAUMSL W 5710 75FKM585					000.000.307.027		1,0	
			SHAFT SEAL 300X340X20 BAUMSL W 5710								
E	0233	1	ST WDR 300X340X20 BAUMSL W 5710 75FKM585					000.000.307.027		1,0	
			SHAFT SEAL 300X340X20 BAUMSL W 5710								
E	0235	3	ST RING-LAM AS340/10,0X1,5 W5730 FED-ST					000.000.347.221		0,1	
			RING-LAM AS340/10,0X1,5 W5730 FED-ST								
E	0250	1	ST WAE LZL182960 SL					000.000.378.060		31,2	
			ROLLG BRG 182960 SL								
E	0251	1	ST WAE LZL182960 SL					000.000.378.060		31,2	
			ROLLG BRG 182960 SL								
E	0290	1	ST SCHB-SCHR GR.300 F 3233 ST					000.000.309.473		120,0	
			SHRINK DISC GR.300 F 3233 ST								
E	0295	1	ST BUCHSE PAP 235X240X090 P10 INA					000.000.375.905		1,3	
			BUSH PAP 235X240X090 P10 INA								
E	0300	1	ST STRDWELLE LP 442,0/3,47 1.6587			5877492/A		000.000.677.492		103,0	
			PINION SHAFT LP 442,0/3,47 1.6587								
E	0302	1	ST STIRNRAD RP 442,0/3,47 1.6587			5877499/A		000.000.677.499		429,0	
			GEAR WHEEL RP 442,0/3,47 1.6587								
E	0350	1	ST WAE LZL 22328 E D 635 MET-KAEF					000.000.380.879		35,1	
			ROLLG BRG 22328 E D 635 MET-KAEF								
E	0351	1	ST WAE LZL 22328 E D 635 MET-KAEF					000.000.380.879		35,1	
			ROLLG BRG 22328 E D 635 MET-KAEF								
Die mit * gekennzeichneten Teile gehören zu einer Baugruppe (G). Die Baugruppe ist nur komplett auszutauschen.											
The parts marked with * belong to a subassembly (G). The subassembly must be replaced complete.											
Siemens Industriegetriebe GmbH, Thierbacher Str.24, 09322 Penig						Datum Date		ALBERT JOSEPH,ALWIN MOSE.CHENN			
Tel. +49(0)37381-61-0, Fax +49(0)37381-80286											

SIEMENS	Ersatzteilliste (EL) Spare parts list		Bauart B3DH Type Größe 15,0 Size Übersetzung 12,172 ratio		Seite Page 2/2																																																																								
	Bei Korrespondenz bitte angeben Please quote in correspondence Hierzu gehört Zeichnungs-Nr. Please refer to DWG No. 4503614363 EL 4637581-231 DE/EN 6153212																																																																												
<table border="1"> <thead> <tr> <th>Teil-Nr.</th> <th>Menge</th> <th>Benennung</th> <th>Zeichnungs-Nr.</th> <th>Material-Nr.</th> <th>Gw(kg)</th> </tr> <tr> <th>Part No.</th> <th>No. off</th> <th>Description</th> <th>Drawings No.</th> <th>Ident no.</th> <th>Weight</th> </tr> </thead> <tbody> <tr> <td>E 0400</td> <td>1</td> <td>ST STRDWELLE RP 320,0/2,78 1.6587</td> <td>5877359/A</td> <td>000.000.677.359</td> <td>69,5</td> </tr> <tr> <td></td> <td></td> <td>PINION SHAFT RP 320,0/2,78 1.6587</td> <td></td> <td></td> <td></td> </tr> <tr> <td>E 0402</td> <td>1</td> <td>ST STIRNRAD LP 320,0/2,78 17CRNIM06 5880918/E</td> <td></td> <td>000.000.560.918</td> <td>174,0</td> </tr> <tr> <td></td> <td></td> <td>GEAR WHEEL LP 320,0/2,78 17CRNIM06</td> <td></td> <td></td> <td></td> </tr> <tr> <td>E 0450</td> <td>1</td> <td>ST WAE LZL 32320 D 720 ST-KAEF</td> <td></td> <td>000.000.380.540</td> <td>12,9</td> </tr> <tr> <td></td> <td></td> <td>ROLLG BRG 32320 D 720 ST-KAEF</td> <td></td> <td></td> <td></td> </tr> <tr> <td>E 0451</td> <td>1</td> <td>ST WAE LZL 32320 D 720 ST-KAEF</td> <td></td> <td>000.000.380.540</td> <td>12,9</td> </tr> <tr> <td></td> <td></td> <td>ROLLG BRG 32320 D 720 ST-KAEF</td> <td></td> <td></td> <td></td> </tr> <tr> <td>E 0481</td> <td>1</td> <td>ST RS FXM 200- 63SX/H100 W 5813 ST</td> <td></td> <td>000.000.389.016</td> <td>52,0</td> </tr> <tr> <td></td> <td></td> <td>BACKSTOP FXM 200- 63SX/H100 W 5813 ST</td> <td></td> <td></td> <td></td> </tr> </tbody> </table>						Teil-Nr.	Menge	Benennung	Zeichnungs-Nr.	Material-Nr.	Gw(kg)	Part No.	No. off	Description	Drawings No.	Ident no.	Weight	E 0400	1	ST STRDWELLE RP 320,0/2,78 1.6587	5877359/A	000.000.677.359	69,5			PINION SHAFT RP 320,0/2,78 1.6587				E 0402	1	ST STIRNRAD LP 320,0/2,78 17CRNIM06 5880918/E		000.000.560.918	174,0			GEAR WHEEL LP 320,0/2,78 17CRNIM06				E 0450	1	ST WAE LZL 32320 D 720 ST-KAEF		000.000.380.540	12,9			ROLLG BRG 32320 D 720 ST-KAEF				E 0451	1	ST WAE LZL 32320 D 720 ST-KAEF		000.000.380.540	12,9			ROLLG BRG 32320 D 720 ST-KAEF				E 0481	1	ST RS FXM 200- 63SX/H100 W 5813 ST		000.000.389.016	52,0			BACKSTOP FXM 200- 63SX/H100 W 5813 ST			
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Siemens Industriegetriebe GmbH, Thierbacher Str.24, 09322 Penig Tel. +49(0)37381-61-0, Fax +49(0)37381-80286			Datum Date ALBERT JOSEPH,ALWIN MOSE.CHENN																																																																										

SIEMENS**List of equipment**Type **B3DH**Size **09**Page
1 of 1

Code:

Please quote in correspondence

GL 463 7581-31/141/164 EN

Part No.	Qty.	Description	Manufacturer
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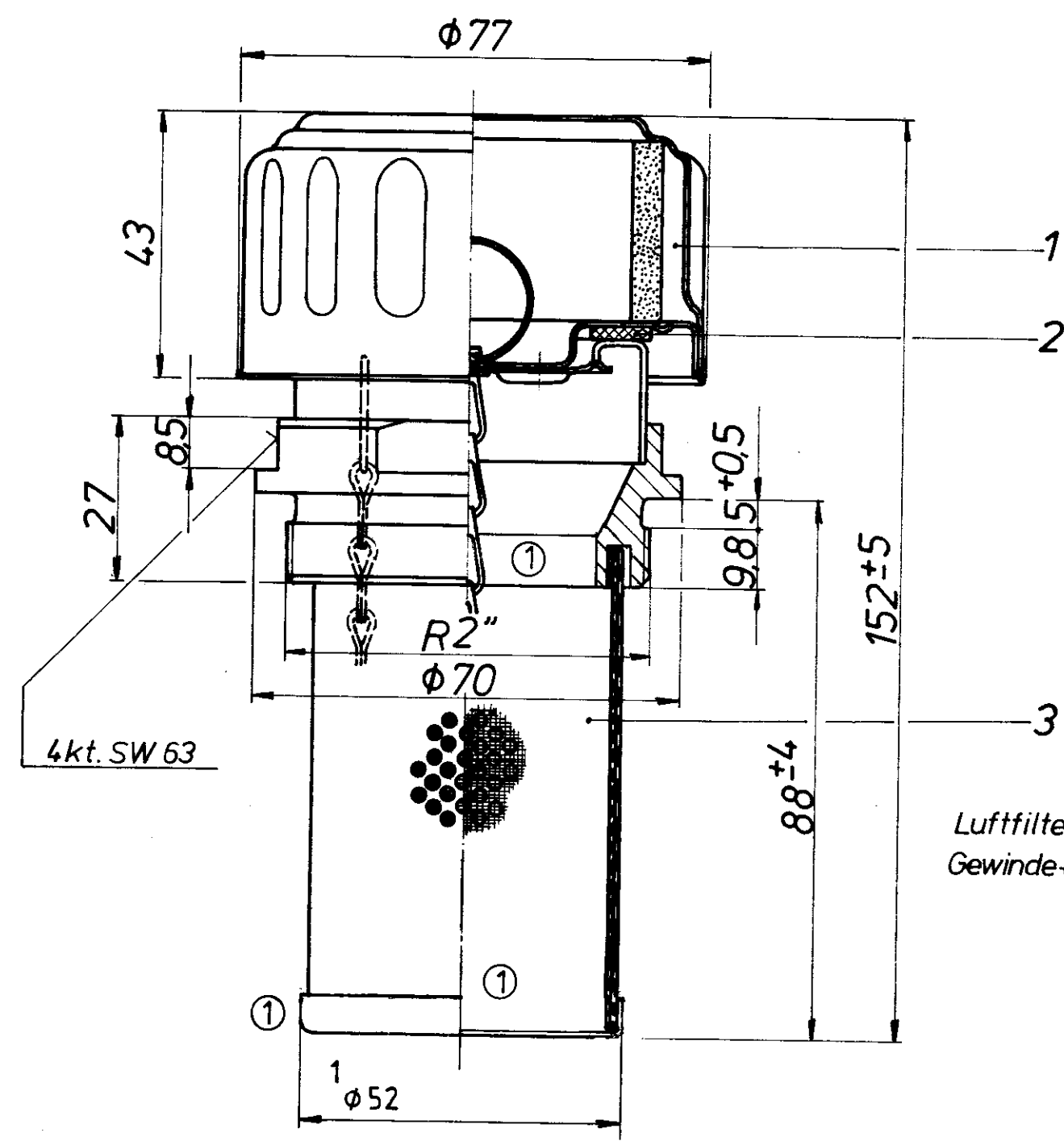
054	1	Breather filter Type 2 Screw connection G 2	MANN
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925	1	Resistance thermometer F 6100-3 Mounting length EL 100 mm Precision resistor 2 x PT 100 DIN IEC Cl. B Two-wire circuit Immersion tube length 100 mm Connection immersion tube G 1/2 thread Degree of protection: IP 65	SIEMENS
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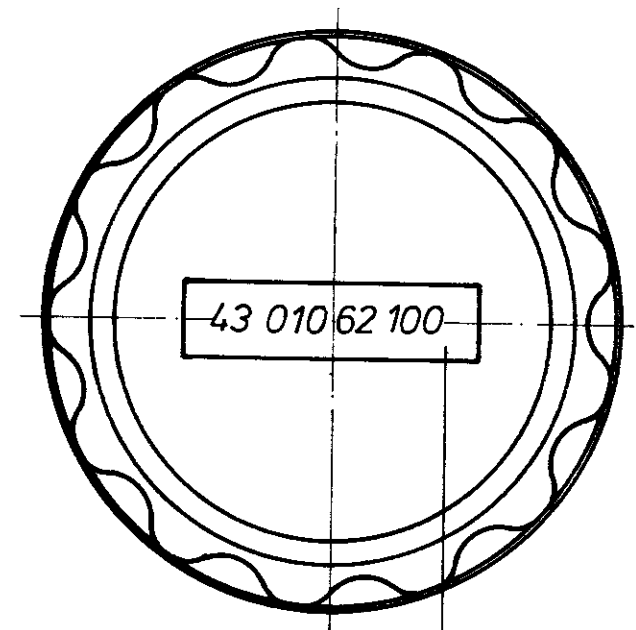
SIEMENS	List of equipment		Type B3DH	Page 1 of 1
			Size 013	
Code:				
Please quote in correspondence			GL 463 7581-081 EN	
Part No.	Qty.	Description	Manufacturer	
054	1	Breather filter Type 2 Screw connection G 2	MANN	
732	1	Filter Type 3/4 1NI PN25 F5911 MS Mesh width 250 µm Operating pressure up to 25 bar	SAMSON	
738	1	Pressure monitor as per Flender Works Standard W 5925 Screw connection G 1/4 Adjustment range p = 0.3-1.5 bar Quantity of switching contacts: 1 Contact loading: 4 A / 250 V AC / 250 VA 2 A / 24 V DC / 1 A / 50 V DC Degree of protection: IP 65 adjusted at p < 0.5 bar	SUCO	
925	1	Resistance thermometer F 6100-3 Mounting length EL 100 mm Precision resistor 2 x PT 100 DIN IEC Cl. B Two-wire circuit Immersion tube length 100 mm Connection immersion tube G 1/2 thread Degree of protection: IP 65	SIEMENS	
Siemens AG, Mechanical Drives, 46395 Bocholt, Tel. 02871 / 92-0, Telefax 02871 / 922596, http://www.siemens.de/antriebstechnik			Datum 2012-02-10	Name: Kup/Lob Rev.:)
			OE01	

SIEMENS	List of equipment Code:		Type B3DH	Page 1 of 1
			Size 015	
Please quote in correspondence			GL 463 7581-191/231 EN	
Part No.	Qty.	Description	Manufacturer	
054	1	Breather filter Type 2 Screw connection G 2	MANN	
732	1	Filter Type 3/4 1NI PN25 F5911 MS Mesh width 250 µm Operating pressure up to 25 bar	SAMSON	
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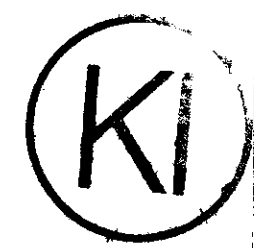
WZ



Draufsicht



Klebeschild



Oberfläche
Luftfilterkappe: Ni Cr
Gewinde+Sieb : ZN ①

Nr.	Ä.-M.	Änderung	Datum	Bearb.	Geprüft
①		8150 OBERFLÄCHE: ZN WAR SN; POS.3 BODEN U. FLANSCH FORMGEÄE.	30.5.97	Mann	97

Pos.	Benennung	Bestell-Nr.	Fremd-Nr.
3	GEWINDE-SIEB VST	21 060 15 981	
2	DECKELDICHUNG	02 200 21 005	
1	LUFTFILTERKAPPE	43 010 60 100	

Ersatzteilliste

Kundenvermerke				Angebotszeichnung			
				Für diese technische Unterlage behalten wir uns alle Rechte vor (vgl. DIN 34). Bei Mißbrauch erfolgt Berufung auf diesen Vorbehalt.			
Freimaßtoleranzen einschl. freier Form- u. Lagerl.		Maßstab 1:1		Datum	Name	Benennung MANN- Luftfilter + Sieb	
zul. Abweichungen			Bearb.	14.12.77	K. H. H.		
Längen			Geprüft	2.2.78			
± 1			Gesehen				
Radial. Fasen	± 1	 FILTERWERK MANN & HUMMEL GMBH D 7140 LUDWIGSBURG			Zeichnungs-Nr.		
Winkel	± 1°				43 010 62 100		
Oberflächen Reihe 2 DIN 3141		ähnlich 43 010 63 100			Ersatz für 41 004 60 352		

SIEMENS

SITRANS TS

Temperature sensors

Resistance thermometer (F6100)/Thermocouple (F6101)

Compact Operating Instructions

General safety information

General information

This device left the factory free from safety problems. In order to maintain this status and to ensure safe operation of the device, please observe the safety information and warnings contained in these instructions.

Correct usage

The device may only be used for the purposes specified in these instructions.

Insofar as they are not expressly stated in these instructions, all changes to the device are the sole responsibility of the user.

Laws and directives

Observe the test certification, provisions and laws applicable in your country during connection, assembly and operation.

For hazardous areas, these are for example:

- IEC 60079-14 (international)
- EN 60079-14 (formerly VDE 0165, T1) (EU, Germany)
- The working reliability regulation (Germany)

Measures

In the interests of safety, the following precautions must be observed:

 WARNING "Intrinsic safety" type of protection Connect the device only to certified, intrinsically safe circuits. These circuits must comply with the technical data specified on the nameplate or in the certificates and approvals. Should these circuits not match the details given in the certificates and approvals, then the safety required for the approval can no longer be guaranteed. The device's protection level "ia" is lowered to protection level "ib" if fail-safe circuits are connected with protection level "ib".
 WARNING Handling aggressive and dangerous media The device can be operated under high pressure as well as aggressive and dangerous media. This means incorrect operation of these devices may result in severe injuries and/or serious damage. This is particularly important when the device was in operation and is being replaced.
 CAUTION Modules susceptible to electrical discharge The device contains modules susceptible to electrical discharge. Modules susceptible to electrical discharge can be destroyed by voltages that fall far below the limits of human perception. These voltages even occur if you touch a component part or electrical connections of a module without being electrostatically discharged. The damage to a module caused by overvoltage cannot normally be detected immediately, it only becomes apparent after a longer period of operating time has elapsed.

Qualified Personnel

Qualified personnel are people who are familiar with the installation, mounting, commissioning, and operation of the product. These people have the following qualifications:

- They are authorized, trained or instructed in operating and maintaining devices and systems according to the safety regulations for electrical circuits, high pressures and aggressive as well as hazardous media.
- For explosion-proof devices: They are authorized, trained, or instructed in carrying out work on electrical circuits for hazardous systems.
- They are trained or instructed in maintenance and use of appropriate safety equipment according to the safety regulations.

Description

Structure of the nameplate

All F610x devices are at least identified with the information of the manufacturer (SIEMENS) and on the device type. The nameplate of the version F610x-6 (explosion protection) is located on the connection head and includes the following information:

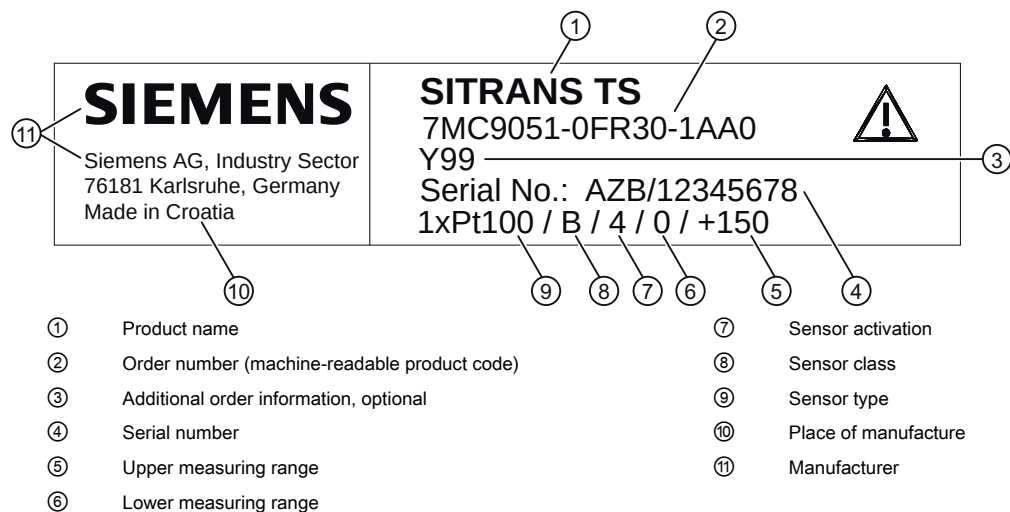


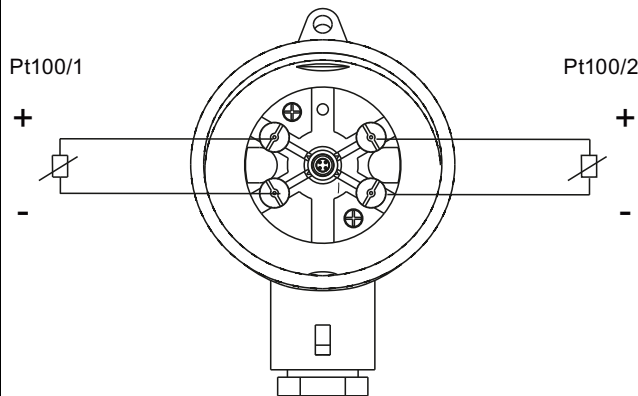
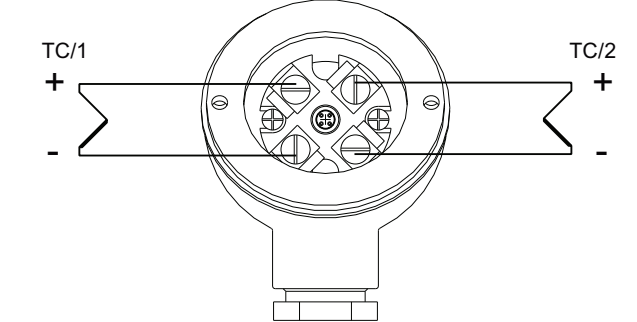
Figure 1 Example for nameplate version F6100-6

Connection without transducer

Connection on ceramic base

Use the following illustrations when connecting the device: The name of the cable colors fulfills IEC 60757.

Use the following illustrations when connecting the device. The name of the cable colors follows IEC 60757.

Connection for type F6100-x thermometer (resistance thermometer)	Connection for type F6101-x thermometer (thermocouple)																																										
 <table><tr><th colspan="2">Pt100/1</th><th colspan="2">Pt100/2</th></tr><tr><td>+</td><td>-</td><td>+</td><td>-</td></tr><tr><td>RD</td><td>WH</td><td>RD/WH</td><td>WH</td></tr></table>	Pt100/1		Pt100/2		+	-	+	-	RD	WH	RD/WH	WH	 <table><tr><th></th><th colspan="2">TC/1</th><th colspan="2">TC/2</th></tr><tr><th>3 Type</th><th>+</th><th>-</th><th>+</th><th>-</th></tr><tr><td>E</td><td>VT</td><td>WH</td><td>VT/WH</td><td>WH</td></tr><tr><td>J</td><td>BK</td><td>WH</td><td>BK/WH</td><td>WH</td></tr><tr><td>K</td><td>GN</td><td>WH</td><td>GN/WH</td><td>WH</td></tr><tr><td>T</td><td>BN</td><td>WH</td><td>BN/WH</td><td>WH</td></tr></table>		TC/1		TC/2		3 Type	+	-	+	-	E	VT	WH	VT/WH	WH	J	BK	WH	BK/WH	WH	K	GN	WH	GN/WH	WH	T	BN	WH	BN/WH	WH
Pt100/1		Pt100/2																																									
+	-	+	-																																								
RD	WH	RD/WH	WH																																								
	TC/1		TC/2																																								
3 Type	+	-	+	-																																							
E	VT	WH	VT/WH	WH																																							
J	BK	WH	BK/WH	WH																																							
K	GN	WH	GN/WH	WH																																							
T	BN	WH	BN/WH	WH																																							

Overview of displacement technologies

You minimize incorrect measurements by using multiple connector technology.

- Usually the connection with three or four conductors will be in place as of the additionally installed terminal box (① and ②)
- Upon customer request, the 3/4 conductor technology can be used as of the connection head (④ and ⑤). The limited space permits a single connection only in case of four-conductor technology (⑤)

Displacement technologies		
① three-conductor technology as of terminal box	② four-conductor technology as of terminal box	
③ two-conductor input as of connection head	④ three-conductor input as of connection head	⑤ four-conductor input as of connection head

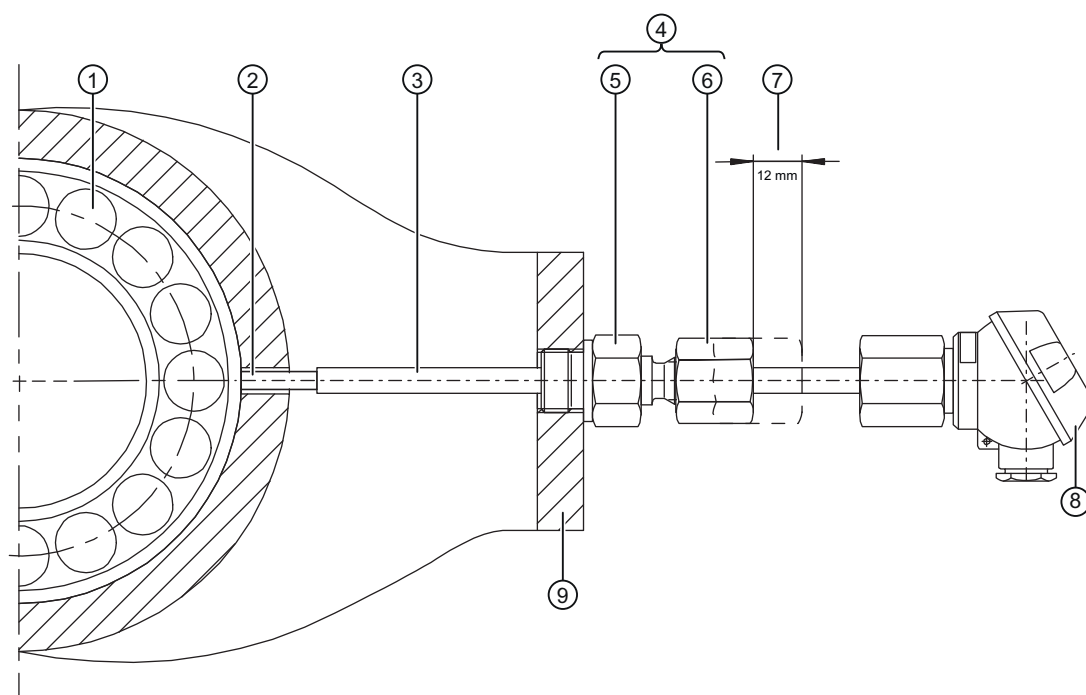
Connection with transducer

Use the following illustrations when connecting the device: The name of the cable colors fulfills IEC 60757.

Connection for type F6100-1, 2, 3, 4, 5 and 7 thermometers	Connection for type F6100-6 thermometer
Transducer type T03 J; 4 ... 20 mA	Transducer type SITRANS TH100; 4 ... 20 mA SITRANS TH300; 4 ... 20 mA, HART SITRANS TH400; PROFIBUS PA SITRANS TH400; FOUNDATION Fieldbus

Adjusting the spring clamp screw

Procedure



- | | |
|--------------------------------|-------------------|
| ① Rolling bearing | ② Probe |
| ③ Guide tube | ④ Screw, finished |
| ⑤ Screw | ⑥ Nut |
| ⑦ Spring excursion about 12 mm | ⑧ Connection head |
| ⑨ Enclosure | |

1. Push the guide tube ③ through the enclosure bore hole until the probe ② contacts the resistance thermometer.
2. Tighten the screw ④ until contact in the enclosure ⑨.
3. Pull out the screw ④ in the direction of the connection head ⑧ approx. 5 mm.
4. Tighten the nut ⑥.

The spring is prestressed. This guarantees permanent contact between the probe and the measured part, e.g. ①.

Technical data

Temperature sensor		
Connection head		
Degree of protection	IP65	
Type	Form J	
Material	Aluminum	
Cable gland		
Type F610x-6	M20x1.5	
Types F610x-1/2/3/4/5/7	M16x1.5	
Wetted parts	Stainless steel	
Sensor type	Pt100; Class B, DIN EN 60751	
	Types E, J, K and T; Class 2, DIN EN 60548	
Temperature range		
Configured measurement range/ Maximum values for measuring temperature	-50° C to +150° C (-122 to +302° F)	
Maximum values for hazardous applications (F610x-6)	• Temperature class T4 = 126° C (258° F) • Temperature class T6 = 76° C (168° F)	
Permissible ambient temperature at connection head		
with built-in transducer	-40° C to +85° C (-40° F to +185° F)	
without transducer	-40° C to +100° C (-40° F to +212° F)	
Power supply		
Type F610x-6	U _b = 7.5 to 15 V DC	
Types F610x-1/2/3/4/5/7	U _b = 7.5 to 30 V DC	
Explosion protection for type F610x-6 only	Zones 1 and 21 Read manufacturer declaration!	
Maximum values of the electrical connection parameters for hazardous applications (F610x-6) ¹⁾		
	• U_i = 15 V • I_i = 100 mA • C_i = C_{i sensor} (700 pF/m) + (C_{i cable}/m) ¹⁾ • L_i = L_{i sensor} (15 µH/m) + (L_{i cable}/m) ¹⁾ • P_i = 100 mW	

¹⁾ The capacity and inductance of the cable connected to the temperature sensor must be taken into consideration prior to installation:

- $C_i = (C_{i \text{ sensor}} + C_{i \text{ cable}})$
- $L_i = (L_{i \text{ sensor}} + L_{i \text{ cable}})$

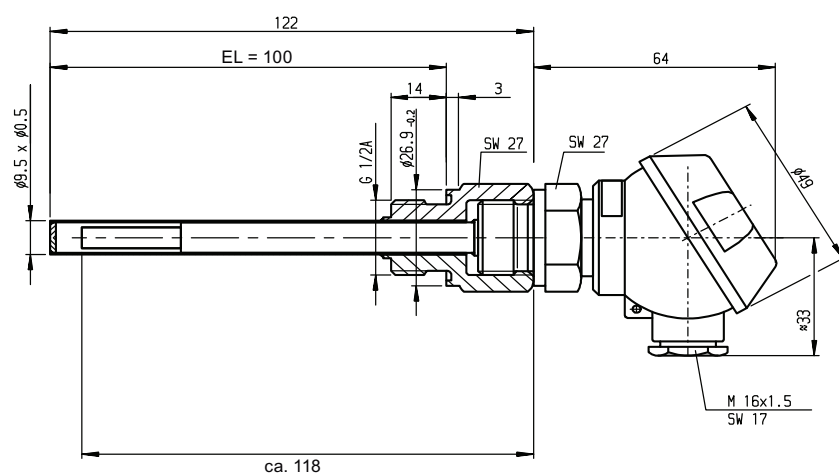
Temperature transducer		
Detail information on temperature transducers		
• T03 J	For types F6100-3, 5, 7	See: Jumo (http://www.Jumo.de/)
• SITRANS TH100	For type F6100-6, 4 ... 20 mA interface	See:
• SITRANS TH300	For type F6100-6, HART interface	Instructions and Manuals
• SITRANS TH400	For type F6100-6, PROFIBUS PA interface	(http://www.siemens.com/processinstrumentation/documentation)
	For type F6100-6, FOUNDATION Fieldbus interface	

Dimensional drawings F610x-x

Type F6100-3 resistance thermometer

Type F6101-3 thermocouple

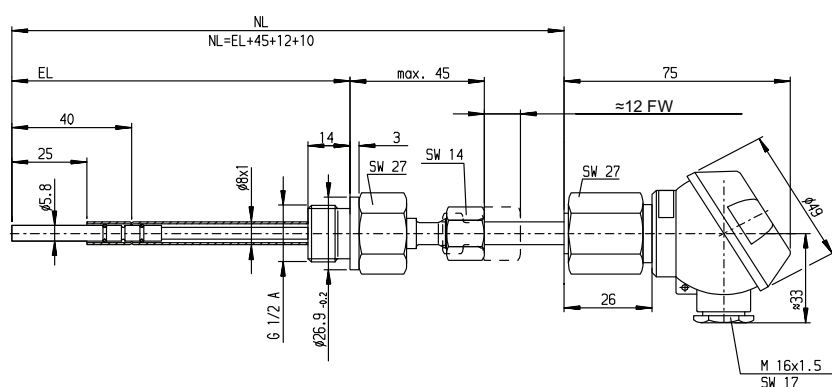
- Oil-tight up to 2 bar.
- Ensure non-critical venting when opening.



Type F6100-4 resistance thermometer

Type F6101-4 thermocouple

- Oil-tight up to 2 bar.
- Ensure non-critical venting when opening.

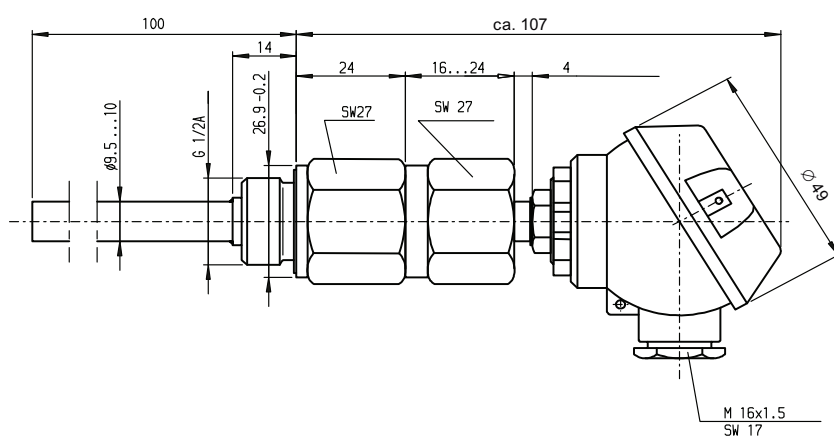


Dimensional drawings F610x-x

Type F6100-7 resistance thermometer

Type F6101-7 thermocouple

- Oil-tight up to 2 bar.
- Ensure non-critical venting when opening.



Manufacturer declaration

SIEMENS

Industry

Manufacturer Declaration Herstellereklärung

No. A5E02171080-02

Manufacturer: Hersteller:	Siemens AG Industry Sector Industry Automation Division
Address: Anschrift:	76181 Karlsruhe, Germany
Product description: Produktbezeichnung	Temperature Sensor / Temperatur Sensor SITRANS TS Type / Typ TMC9051-0Fxxx-xxxx (Typ F6100-6; Typ F6101-6)

The temperature sensor listed above is considered simple electrical equipment according to EN 60079-11 that does not represent a potential ignition hazard when the following connection parameters are observed.

Der oben aufgeführte Temperatur Sensor ist nach EN 60079-11 als einfaches elektrisches Betriebsmittel zu betrachten, von dem unter Einhaltung der nachfolgend genannten Anschlussparameter keine potenzielle Zündgefahr im Sinne der Europäischen Richtlinie 94/9/EG ausgeht.

Limit values of connection parameters: / Höchstwerte der Anschlussparameter:

$U_i = 15 \text{ V}$
 $I_i = 100 \text{ mA}$
 $C_i = C_{i_sensor} (700\text{pF/m}) + (C_{i_cable}/\text{m})^1$
 $L_i = L_{i_sensor} (15\mu\text{H/m}) + (L_{i_cable}/\text{m})^1$
 $P_i = 100 \text{ mW}$

¹The capacity as well as inductance of the connected cable at the temperature sensor will have to be taken into consideration prior to installation: $C_i = (C_{i_sensor} + C_{i_cable})$; $L_i = (L_{i_sensor} + L_{i_cable})$.

¹Die Kapazität sowie Induktivität der angeschlossenen Leitung an den Temperatursensor muss vor der Installation berücksichtigt werden $C_i = (C_{i_sensor} + C_{i_cable})$; $L_i = (L_{i_sensor} + L_{i_cable})$.

Temperature class / Temperaturklasse

Use of the temperature sensor has been approved for temperature class T4 / T6.

Die Verwendung des Temperatursensors ist für die Temperaturklasse T4 / T6 zulässig.

Maximum value measured temperature / Höchstwerte der Messtemperatur:

Temperature class / Temperaturklasse T4: $T^{\text{max}} = 126^\circ\text{C}$
 Temperature class / Temperaturklasse T6: $T^{\text{max}} = 76^\circ\text{C}$

The specific values apply to operation as class 2 device in potentially explosive gas and dust environments (zone 1, zone 21)

Observe the safety instructions in the operating instructions.

Die spezifizierten Werte gelten für den Einsatz als Kategorie 2-Gerät in explosionsfähigen Gas- oder Staubgemischen (Zone 1, Zone 21)

Die Sicherheitshinweise der Betriebsanleitung sind zu beachten.

Karlsruhe, 15.12.2009
Siemens AG

i.v.

Guido Stephan,
Research & Development / Entwicklung
(Name, function / Funktion)

i.v.

Frank Krahl,
Quality / Qualität
(Name, function / Funktion)

Siemens Aktiengesellschaft: Chairman of the Supervisory Board: Gerhard Cromme;
Managing Board: Peter Loescher, Chairman, President and Chief Executive Officer;
Wolfgang Dehen, Heinrich Hiesinger, Joe Kaeser, Barbara Kux, Hermann Requardt, Siegfried Russwurm, Peter Y. Solmsen
Registered offices: Berlin and Munich, Germany; Commercial registries: Berlin Charlottenburg, HRB 12300, Munich, HRB 6684
WEEE-Reg.-Nr. DE 23691322

Page 1 / 1

Siemens AG
Industry Sector
Postfach 48 48
90026 NÜRNBERG

Resistance thermometer (F6100)/Thermocouple (F6101)
A5E02228650, 07/2011



Strainers with threaded end connections

Type 1 N · With standard strainer insert

Type 1 NI · With dual strainer insert

Type 1 FN · With standard strainer insert · Version for district heating systems

Type 1 FNI · With dual strainer insert · Version for district heating systems

Application

For protecting downstream plants, aggregates and measuring and control devices against impurities. Straining and collecting dirt particles carried along by the medium.

With body **G 3/8** to **G 2** · **PN 25** · Suitable for liquids, steam and non-flammable gases up to **max. 300 °C**

Conversion of valve sizing coefficients:

$$C_v \text{ (in U.S. gallons/min)} = 1.17 \cdot K_{vs} \text{ (in m}^3/\text{h)}$$

$$K_{vs} \text{ (in m}^3/\text{h)} = 0.86 \cdot C_v \text{ (in U.S. gallons/min)}$$

The strainers consist of a Y-style body with threaded end connections and a wide-meshed filter element (standard strainer insert) or a wide-meshed filter element with an additional fine-meshed filter element (dual strainer insert).

Special features

- Compact design with lengths according to DIN
- Easy removal of the collected dirt particles
- Easy replacement of the strainer insert

Versions

Inlet/outlet: female thread 3/8" to 2" · Connection DN 15 to DN 50 · PN 25

Type 1 N · With standard strainer insert

Type 1 NI · With dual strainer insert

– Body of brass · Connections G 3/8 to G 2

– Body of malleable iron · Connections G 1/2 to G 2

Type 1 FN · Version for district heating · With standard strainer insert

Type 1 FNI · Same as Type 1 FN, but with dual strainer insert
– Body of red brass · With/without connection nuts and welding ends · Connections G 1/2 to G 2 · Especially suitable for district heating systems

Principle of operation

The medium flows through the strainer in the direction indicated by the arrow on the body. The uncleaned medium first contacts the inner side of the strainer insert (filter element). While it passes through the filter element, the dirt particles carried along by the medium are collected in the filter element. The dirt particles can be removed after unscrewing the screw cap.

Installation

- Install the strainer in the direction of flow indicated by the arrow on the body. Make sure ample space is available for removing and cleaning the strainer insert (see dimensions).
- In vertical pipelines—when the medium flows upward—install the strainer with the screw cap facing upward. In this case, however, the dirt particles will not be collected, but retained.
- Install swing check valves or similar devices to prevent back-flow.
- For further information on installation, refer to the Mounting and Operating Instructions EB 1010 EN.

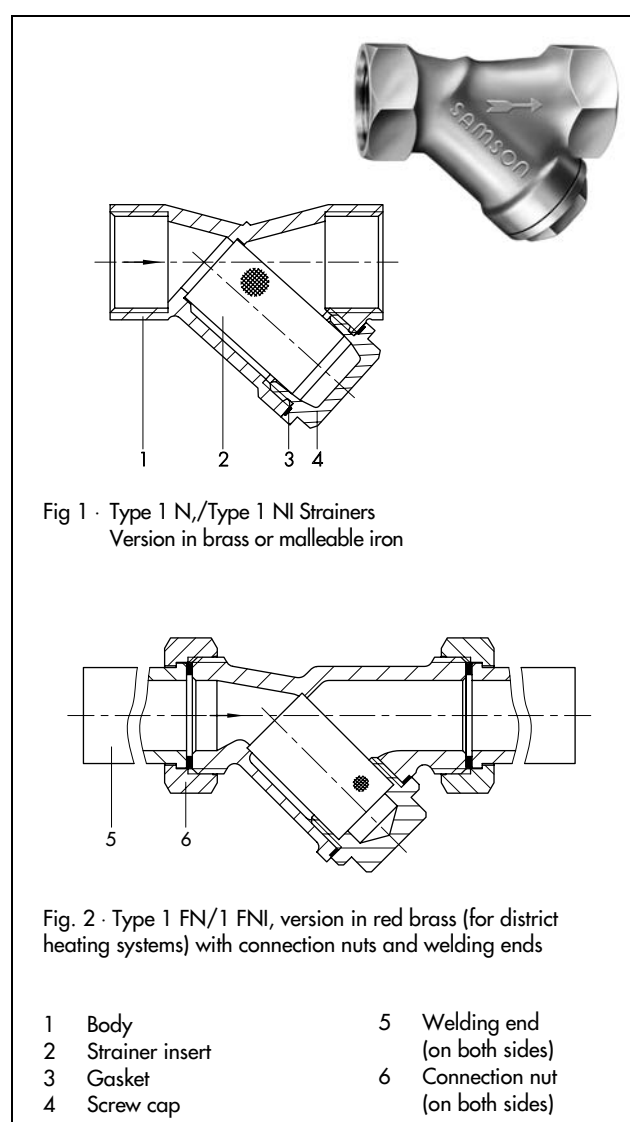


Fig 1 · Type 1 N, Type 1 NI Strainers
Version in brass or malleable iron

Fig. 2 · Type 1 FN/1 FNI, version in red brass (for district heating systems) with connection nuts and welding ends

1	Body	5	Welding end (on both sides)
2	Strainer insert	6	Connection nut (on both sides)
3	Gasket		
4	Screw cap		

Ordering text

Strainer Type 1 N/1 NI, connection G ...

Body material ...

Type 1 FN/1 FNI, connection G ...

With/without connection nuts and welding ends

Table 1 · Technical data

Table 1 - Technical data

Version		PN 25 · Body made of brass, malleable iron, or red brass						
Type 1 N or 1 FN								
Connection	G	3/8 ¹⁾	1/2	3/4	1	1 1/4	1 1/2	2
Kvs		3.7	5.6	10.0	15.6	25.5	40	63
Mesh size	mm	0.5				0.75		
Number of meshes per	cm ²	150				64		
Flow resistance coefficient	ζ	2.5						
Free filter area		Approx. 3 x cross-sectional area of the pipe						
Type 1 NI or 1 FNI								
Kvs	m ³ /h	3.3 ¹⁾	5.1	9.1	14.3	23.0	36.6	57
Mesh size	mm	0.25						
Number of meshes per	cm ²	625						
Flow resistance coefficient	ζ	3						
Free filter area		Approx. 2.5 x cross-sectional area of the pipe						

¹⁾ Only Types 1 N / 1 NI with body of brass

Table 2 · Materials · Material number according to DIN EN

Type	1 FN/1 FNI	1N/1 NI (brass)	1 N/1 NI (malleable iron)
Body	Red brass Rg 5, CC491K (G-CuSn5ZnPb)	Brass CW604N (GK-CuZn37Pb)	Malleable iron EN-JM-1030 (GTW-40-05)
Screw cap	Brass CW610N (CuZn39Pb)		Up to G 1: 9S20K; > G 1: GTW-40-05
Strainer insert (filter element)	Stainless steel 1.4401		
Gasket	Novatec Premium		

Table 3 · Dimensions in mm and weights in kg

Body of brass, malleable iron, or red brass · PN 25								
Connections	G	3/8 ¹⁾	1/2	3/4	1	1 1/4	1 1/2	2
Length L		65	65	75	90	110	120	150
Connection thread R		—	3/4"	1"	1 1/4"	1 3/4"	2"	2 1/2"
Width across flats SW		—	30	36	46	59	65	82
Length L1		—	80	85	100	125	135	160
Length L2		—	225	250	265	293	320	360
Height H1		40	45	56	73	84	84	108
Height H2 (Strainer insert pulled out)		61	75	90	115	134	134	158
Weight, approx. in kg								
Types 1N/1NI (brass)		0.2	0.3	0.47	0.77	1.35	1.35	1.9
Types 1N/1NI (GTW)		—	0.3	0.5	0.6	0.9	1.6	2.4
Types 1 FN/1 FNI (red br.)		—	0.55	0.65	0.8	1.1	1.85	2.6

¹⁾ Only Types 1N/1NI with body made of brass

Dimensions

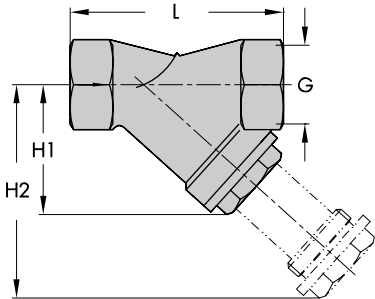
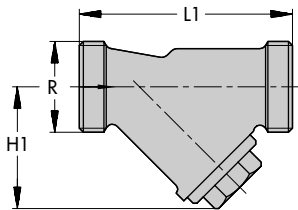
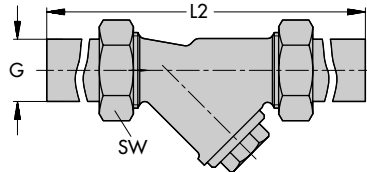
Types 1 N/1 NI – Body of either malleable iron or brass –	Type 1 FN/1 FNI – Body of red brass –
	
	
... with connection nuts and welding ends	

Fig. 4 · Dimensions

Specifications subject to change without notice.



Bedienungsanleitung

Für künftige Verwendung aufbewahren

**Membran-/
Kolbendruckschalter
Baureihe 0180 / 0181
Baureihe 0186 / 0187****D**

Einbau und Inbetriebnahme sind nach dieser Bedienungsanleitung und nur von autorisiertem Fachpersonal vorzunehmen



Robert-Scheuffele GmbH & Co. KG
Keplerstraße 12-14
D-74321 Bietigheim-Bissingen
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Telefax (07142) 597-19
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E-Mail: info@suco.de

**Funktion und Anwendung**

Die Baureihe 0180 / 0181 und 0186 / 0187 öffnet oder schließt einen elektrischen Stromkreis beim Erreichen eines einstellbaren Druckwerts. Durch das Ansteigen des Drucks wird eine Membrane bzw. ein Kolben bewegt. Die Auslenkung Membrane bzw. der Hub des Kolbens hängt von der Druckkraft und der einstellbaren Federvorspannung ab. Bei einer definierten Auslenkung der Membrane bzw. einem definierten Hub des Kolbens wird ein Mikroschalter betätigt, der die elektrischen Kontakte öffnet bzw. schließt (Wechsler).



Der Druckschalter überwacht einen eingestellten Druckwert.

Vorraussetzungen für den Produkteinsatz

Allgemeine, stets zu beachtende Hinweise für den ordnungsgemäßen und sicheren Einsatz des Druckschalters:



- Halten Sie die angegebenen Grenzwerte wie z.B. Drücke, Kräfte, Momente und Temperaturen ein.
- Berücksichtigen Sie die vorherrschenden Umgebungsbedingungen (Temperatur, Luftfeuchte, Luftdruck etc.).
- Beachten Sie die Vorschriften der Berufsgenossenschaften, des Technischen Überwachungsvereins (TÜV) oder die entsprechenden nationalen Bestimmungen.
- Beachten Sie unbedingt die Warnungen und Hinweise in der Bedienungsanleitung.
- Setzen Sie den Druckschalter niemals starken Stößen oder Vibrationen aus.
- Verwenden Sie das Produkt nur im Originalzustand. Nehmen Sie keine eigenmächtige Veränderungen vor.
- Entfernen Sie die alle Transportvorkehrungen wie Schutzfolien, Kappen oder Kartonagen.
- Die Entsorgung der einzelnen Werkstoffe in Recycling-Sammelbehältern ist möglich.

**Operating instructions**

Please keep carefully for future use

**Diaphragm-/
Piston Pressure Switch
Series 0180 / 0181
Series 0186 / 0187****GB**

Installation and commissioning must be carried out in accordance with these operating Instructions and by authorized, qualified personnel only.



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**Operating and use**

The series 0180 / 0181 and 0186 / 0187 switch opens or closes an electrical circuit when a certain (adjustable) pressure is reached. A diaphragm or piston is moved by the increase in pressure. The amount of the diaphragm deflection or piston travel depends on the force of the pressure applied and the (adjustable) spring tension. At a predetermined deflection of the diaphragm or movement of the piston, a micro switch is actuated which opens or closes the electrical contacts (changeover).



The pressure switch monitors a preset pressure.

Conditions governing the use of the product

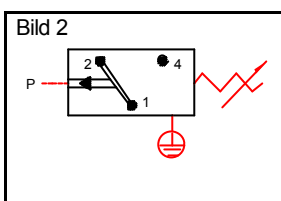
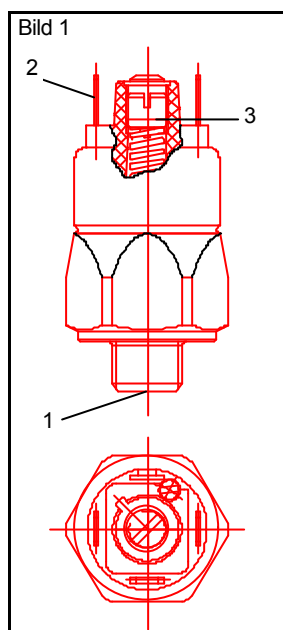
The following general instructions are to be observed at all times to ensure the correct, safe use of the pressure switch:



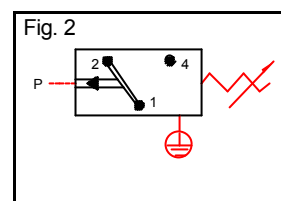
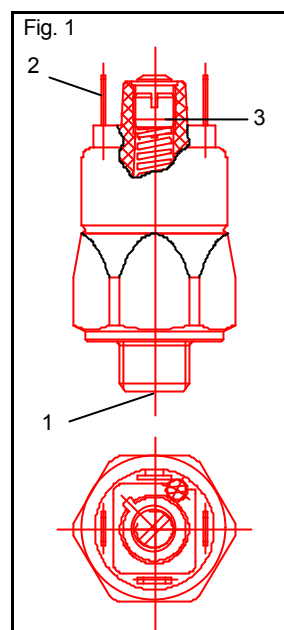
- Do not exceed the specified limits for e.g. pressures, forces, moments or temperatures under any circumstances.
- Give due consideration to the prevailing ambient conditions (temperatures, atmospheric humidity, atmospheric pressure, etc.).
- Observe the applicable safety regulations laid down by the regulatory bodies in the country of use.
- Observe without fail the warning notices and other instructions laid down in the operating instructions.
- Never subject pressure switch to intense blows or high vibrations.
- Never expose the pressure switch to severe side impacts or vibrations.
- Use the product only in its original condition. Do not carry out any unauthorized modifications.
- Remove all items providing protection in transit such as foils, caps or cartons.
- Disposal of the above-named materials in recycling containers is permitted.



Betriebsbedingungen	Operating conditions																																																																																																																																																																												
Bei Medientemperaturen außerhalb der Raumtemperatur (20°C): Extreme Temperatureinflüsse (abweichend von der Raumtemperatur) können zu einer starken Schaltpunktabweichung oder zum Ausfall des Druckschalters führen <u>Schutzart IP65:</u> Die Typenprüfung ist nicht uneingeschränkt auf alle Umweltbedingungen übertragbar. Die Überprüfung, ob die Steckverbindung anderen als den angegebenen Bestimmungen und Vorschriften entspricht bzw. ob diese in speziellen, von uns nicht vorgesehenen Anwendungen eingesetzt werden kann, obliegt dem Anwender. <u>Sauerstoffeinsatz:</u> Membrandruckschalter: Beim Einsatz von Sauerstoff sind die einschlägigen Unfallverhütungsvorschriften zu beachten. Außerdem empfehlen wir, einen maximalen Betriebsdruck von 10 bar nicht zu überschreiten. Kolbendruckschalter: Kolbendruckschalter sind für gasförmige Medien, insbesondere für Sauerstoff nicht geeignet. <u>Überdrucksicherheit:</u> In den Technischen Daten ist die statische Überdrucksicherheit angegeben. Sie bezieht sich auf den hydraulischen bzw. pneumatischen Teil des Druckschalters. Der dynamische Wert ist 30 bis 50 % niedriger.	Media temperatures other than room temperature (20°C): The effects of extreme temperatures (relative to the room temperature) can lead to pronounced variations in the switching point or failure of the pressure switch. <u>Type of protection IP65:</u> Type testing does not apply to all ambient conditions without limitations. The user is responsible for verifying that the plug-and-socket connection complies with the specified rules and regulations of CE, or whether it may be used for specialized purposes other than those intended by us. <u>Use with oxygen:</u> Diaphragm pressure switch: If oxygen is used, the applicable accident prevention regulations must be observed. In addition, we recommend a maximum operating pressure of 10 bars, which should not be exceeded. Piston pressure switch: Piston pressure switches are not suitable for gaseous media, particularly oxygen. <u>Protection against overpressure:</u> The static overpressure safety is included in the technical data. The overpressure safety corresponds to the hydraulic, pneumatic part of the pressure switch. The dynamic rating of the overpressure safety is smaller than 30 to 50%.																																																																																																																																																																												
Technische Daten	Technical data																																																																																																																																																																												
<table><tr><th>Bemessungs- triebsspannung U_e</th><th>Bemessungs- triebsstrom I_e</th><th>Gebrauchs-kategorie</th></tr><tr><td>250 Volt AC 50/60 Hz</td><td>4 Ampere</td><td>AC12</td></tr><tr><td>250 Volt AC 50/60 Hz</td><td>1 Ampere</td><td>AC14</td></tr><tr><td>30 Volt DC</td><td>4 / 4 Ampere</td><td>DC12 / DC13</td></tr><tr><td>50 Volt DC</td><td>2 / 1 Ampere</td><td>DC12 / DC13</td></tr><tr><td>75 Volt DC</td><td>1 / 0,5 Ampere</td><td>DC12 / DC13</td></tr><tr><td>125 Volt DC</td><td>0,3 / 0,2 Ampere</td><td>DC12 / DC13</td></tr><tr><td>250 Volt DC</td><td>0,25 / 0,2 Ampere</td><td>DC12 / DC13</td></tr><tr><td colspan="2">Bemessungsisolationsspannung U_i:</td><td>300 Volt</td></tr><tr><td colspan="2">Bemessungsstoßspannungsfestigkeit U_{imp}:</td><td>2,5 kV</td></tr><tr><td colspan="2">konventioneller thermischer Strom I_{the}:</td><td>5 Ampere</td></tr><tr><td colspan="2">Schaltüberspannung:</td><td>< 2,5 kV</td></tr><tr><td colspan="2">Bemessungsfrequenz:</td><td>DC und 50 / 60 Hz</td></tr><tr><td colspan="2">Nennstrom der Kurzschlusseinrichtung:</td><td>bis 5 Ampere</td></tr><tr><td colspan="2">Bedingter Kurzschlussstrom:</td><td>< 350 Ampere</td></tr><tr><td colspan="2">IP-Schutzart nach EN 60 529:1991:</td><td>IP65 mit Stecker</td></tr><tr><td colspan="2">Anzugsdrehmoment der Anschlussschrauben:</td><td>< 0,35 Nm</td></tr><tr><td colspan="2">Anschlussquerschnitt:</td><td>0,5 bis 1,5 mm²</td></tr><tr><td colspan="2">Schalthäufigkeit:</td><td>< 200 min⁻¹</td></tr><tr><td colspan="2">Schalthysterese:</td><td>10-30%, im Werk einstellbar</td></tr><tr><td colspan="3">mechanische Lebensdauer:</td></tr><tr><td colspan="2">Membranausführung:</td><td>10⁶ Schaltspiele (bei Schaltdrücken bis 40 bar)</td></tr><tr><td colspan="2">Kolbenausführung:</td><td>10⁶ Schaltspiele</td></tr><tr><td colspan="3">Gehäusewerkstoff:</td></tr><tr><td colspan="2" rowspan="3">Temperaturbeständigkeit:</td><td>NBR -30°C bis +100°C</td></tr><tr><td>EPDM -30°C bis +120°C</td></tr><tr><td>FKM -5°C bis +120°C</td></tr><tr><td colspan="3">Überdrucksicherheit:</td></tr><tr><td colspan="2">Membrandruckschalter:</td><td>100 bar (0,3...1,5 bar, 1...10 bar) 300 bar (1...10 bar m. Endnummer 040, 041, 042, 340, 341, 342 und restliche Druckbereiche)</td></tr><tr><td colspan="2">Kolbendruckschalter:</td><td>600 bar</td></tr></table>	Bemessungs- triebsspannung U_e	Bemessungs- triebsstrom I_e	Gebrauchs-kategorie	250 Volt AC 50/60 Hz	4 Ampere	AC12	250 Volt AC 50/60 Hz	1 Ampere	AC14	30 Volt DC	4 / 4 Ampere	DC12 / DC13	50 Volt DC	2 / 1 Ampere	DC12 / DC13	75 Volt DC	1 / 0,5 Ampere	DC12 / DC13	125 Volt DC	0,3 / 0,2 Ampere	DC12 / DC13	250 Volt DC	0,25 / 0,2 Ampere	DC12 / DC13	Bemessungsisolationsspannung U_i :		300 Volt	Bemessungsstoßspannungsfestigkeit U_{imp} :		2,5 kV	konventioneller thermischer Strom I_{the} :		5 Ampere	Schaltüberspannung:		< 2,5 kV	Bemessungsfrequenz:		DC und 50 / 60 Hz	Nennstrom der Kurzschlusseinrichtung:		bis 5 Ampere	Bedingter Kurzschlussstrom:		< 350 Ampere	IP-Schutzart nach EN 60 529:1991:		IP65 mit Stecker	Anzugsdrehmoment der Anschlussschrauben:		< 0,35 Nm	Anschlussquerschnitt:		0,5 bis 1,5 mm ²	Schalthäufigkeit:		< 200 min ⁻¹	Schalthysterese:		10-30%, im Werk einstellbar	mechanische Lebensdauer:			Membranausführung:		10 ⁶ Schaltspiele (bei Schaltdrücken bis 40 bar)	Kolbenausführung:		10 ⁶ Schaltspiele	Gehäusewerkstoff:			Temperaturbeständigkeit:		NBR -30°C bis +100°C	EPDM -30°C bis +120°C	FKM -5°C bis +120°C	Überdrucksicherheit:			Membrandruckschalter:		100 bar (0,3...1,5 bar, 1...10 bar) 300 bar (1...10 bar m. 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Bedienteile und Anschlüsse

- (1) Hydraulischer / Pneumatischer Anschluss
(2) Elektrischer Anschluss (AMP 6,3x0,8)
(3) Einstellschraube für den Schaltpunkt

Operating controls and connections

- (1) Hydraulic / Pneumatic connection
(2) Electrical connection (AMP 6,3x0,8)
(3) Switching point adjusting screw

EinbauMechanisch, pneumatisch, hydraulisch:

Drehen Sie den Druckschalter an dem sechskantförmigen Ansatz mit einem Maulschlüssel der Schlüsselweite 27 (nach DIN 894 o.ä.) in den vorgesehenen Druckanschluss (Anzugsdrehmomente siehe nachfolgende Tabelle).



Zum Abdichten des Systems verwenden Sie einen Standard-Kupferdichtring mit den entsprechenden Abmessungen.

Anschlussgewinde	Drehmoment
M10x1keg. und NPT1/8"	Einschrauben bis System abgedichtet ist
M10x1zyl.	35 Nm
Restliche	50 Nm

Elektrisch:

Verkabeln Sie den Druckschalter gemäß dem Schaltbild (Bild 2).



Verwenden Sie die Gerätesteckdose 1-1-80-652-002 (nicht im Lieferumfang enthalten!).

InstallationMechanical / pneumatic / hydraulic:

With a size 27 open-ended wrench (to DIN 894 or similar), install the pressure switch, by means of the hexagon connector, in the corresponding pressure socket (tightening torque G1/4": 50 Nm).



For sealing the system use a standard copper gasket of the appropriate dimensions.

Connecting thread	Torque
M10x1keg. and NPT1/8"	Tighten until system is hermetically sealed
M10x1 straight	35 Nm
Others	50 Nm

Electrical:

Connect up the pressure switch in accordance with the circuit diagram (Fig. 2).



Use a connector type 1-1-80-652-002 (not include in the delivery specification).

Inbetriebnahme	Entry into service
1. Verkabeln Sie die elektrischen Anschlüsse 1 und 4 mit einem Durchgangsprüfer (Bild 2).  Bei Verwendung einer Prüflampe als Durchgangsprüfer: Achten Sie auf die max. zulässige Schaltleistung (siehe technische Daten). 2. Drehen Sie die Einstellschraube (3) zunächst ganz ein. Verwenden Sie zum Einstellen des Druckschalters einen Schraubendreher mit 6,3 mm Klingenbreite.  Beachten Sie bitte, dass die Einstellschraube (3) nur beim Eindrehen einen Anschlag besitzt. 3. Beaufschlagen Sie den Druckschalter mit dem gewünschten Schaltdruck (Kontrollmanometer erforderlich). 4. Drehen Sie die Einstellschraube (3) so weit heraus, bis der Druckschalter umschaltet (Durchgangsprüfer reagiert). 5. Korrigieren Sie gegebenenfalls den Schaltdruck durch Verdrehen der Einstellschraube (3).  Bei der Inbetriebnahme des Druckschalters beachten Sie bitte die entsprechenden Sicherheitsvorschriften der Berufsgenossenschaften oder die entsprechenden nationalen Bestimmungen.  Die Einstellung der Hysterese ist nur werkseitig durchführbar. Bei unsachgemäßer Vorgehensweise kann der Druckschalter beschädigt werden.	1. Using a continuity tester, wire up the electrical connections 1 and 4 (Fig. 2).  If using a testing lamp as a continuity tester, observe the maximum permissible switching capacity (see technical data) 2. First, screw in the adjusting screw (3) as far as it will go. To adjust the pressure switch use a screwdriver with a 6,3 mm wide blade.  Take care to ensure that the adjusting screw (3) does not seize at any point other than when it is fully tightened down. 3. Adjust the pressure switch to the desired actuating pressure (a test pressure gauge is required). 4. Ease off the adjusting screw (3) to a sufficient extent to cause the pressure switch to trip (continuity tester reacts). 5. If necessary, adjust the trip pressure setting by turning the adjusting screw (3).  When putting the pressure switch into service, please observe the applicable safety regulations laid down by the governing bodies in the country of use.  The adjustment of Hysteresis can only be carried out in the factory. If this is inexpertly undertaken, damage may be caused to the pressure switch.
Ausbau	Removing the pressure switch
 Beachten Sie folgende wichtige Punkte beim Ausbau des Druckschalters: • Das Drucksystem, aus dem der Druckschalter ausgebaut werden soll, muss sich in drucklosem Zustand befinden. • Es müssen alle relevanten Sicherheitsbestimmungen beachtet werden. • Drehen Sie den Druckschalter mit einem Maulschlüssel der Schlüsselweite 27 (nach DIN 894 o.ä.) aus dem Druckanschluss.	 When removing the pressure switch, observe the following important instructions: • The pressurized system from which the pressure switch is intended to be removed must be entirely of pressure. • All the relevant safety regulations must be observed. • Use a size 27 open-ended wrench (to DIN 894 or similar), to remove the pressure switch.
Zeichenerklärung	Key to drawings:
 Achtung  Hinweis, Bemerkung  Recycling  Gefahr	 Caution  Note  Recycling  Danger
Art.-Nr.: 1-1-80-628-014_2 06 / 02	Art.-No.: 1-1-80-628-014_2 06 / 02

Bedienungsanleitung

Für künftige Verwendung aufbewahren

Gerätesteckdose
1-1-80-652-002**D**

Einbau und Inbetriebnahme sind nach dieser Bedienungsanleitung und nur von autorisiertem Fachpersonal vorzunehmen



Robert-Scheuffele GmbH & Co. KG
Keplerstraße 12-14
D-74321 Bietigheim-Bissingen
Telefon (07142) 597-0
Telefax (07142) 597-19
www.suco.de
E-Mail: info@suco.de

**Operating instructions**

Please keep carefully for future use

Connector
1-1-80-652-002**GB**

Installation and commissioning must be carried out in accordance with these operating Instructions and by authorized, qualified personnel only.



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**Vorraussetzungen für den Produkteinsatz**

Allgemeine, stets zu beachtende Hinweise für den ordnungsgemäßen und sicheren Einsatz der Gerätesteckdose:



- Beachten Sie die Vorschriften der Berufsgenossenschaften, des Technischen Überwachungsvereins (TÜV) oder die entsprechenden nationalen Bestimmungen.
- Beachten Sie unbedingt die Warnungen und Hinweise in der Bedienungsanleitung.
- Entfernen Sie die alle Transportvorkehrungen wie Schutzfolien, Kappen oder Kartonagen.



- Die Entsorgung der einzelnen Werkstoffe in Recycling-Sammelbehältern ist möglich.

Conditions governing the use of the product

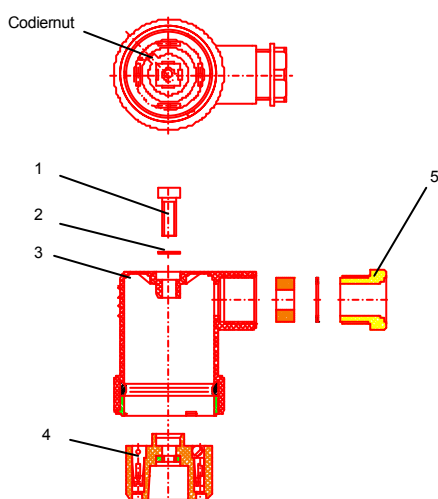
The following general instructions are to be observed at all times to ensure the correct, safe use of the connector:



- Observe the applicable safety regulations laid down by the regulatory bodies in the country of use.
- Observe without fail the warning notices and other instructions laid down in the operating instructions.
- Remove all items providing protection in transit such as foils, caps or cartons.
- Disposal of the above-named materials in recycling containers is permitted.

**Einzelteile und Anschlüsse**

Bild 1



- (1) Befestigungsschraube
- (2) Dichtung
- (3) Steckergehäuse
- (4) Anschlussplatte
- (5) PG-Verschraubung

Bild 2

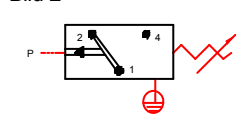
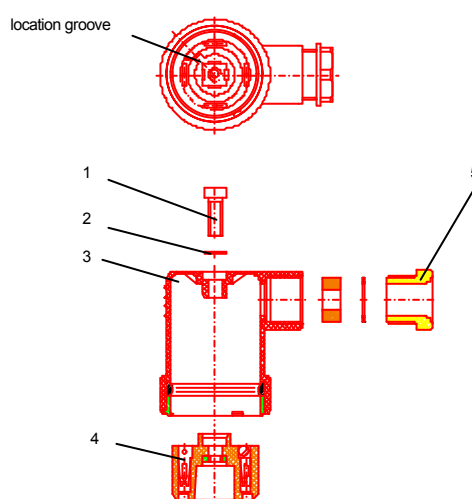
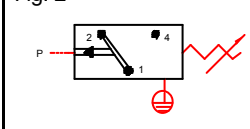
**Operating controls and connections**

Fig. 1



- (1) Mounting screw
- (2) Seal
- (3) Plug housing
- (4) Terminal board
- (5) Pg screw coupling

Fig. 2

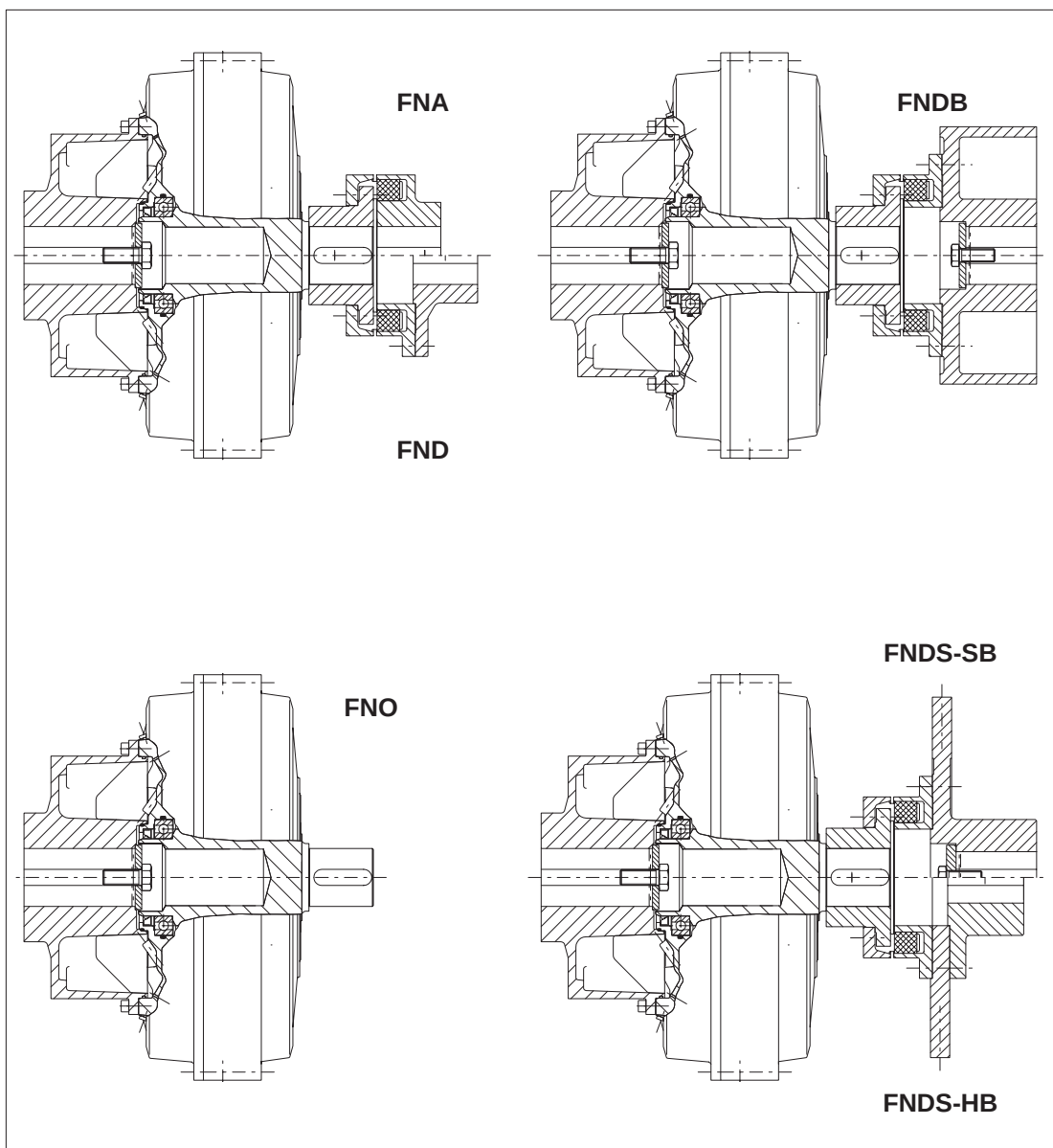


Verkabelung der Gerätesteckdose	Wiring up the connector																
Stellen Sie sicher, dass das Kabel – quetschfrei, – knickfrei, – dehnungsfrei verlegt ist. Verkabelung der Gerätesteckdose: 1. Lösen Sie die Befestigungsschraube (1) und ziehen diese am Kopfende ganz heraus, achten Sie dabei auf den Fiberglas-Dichtring (2) (gut aufbewahren !) 2. Demontieren Sie die nun gelöste Anschlussplatte (4), in den Sie diese mit einem Schraubendreher von oben (Bohrung der Befestigungsschraube (1)) her heraus drücken, beim Herausdrücken ist der Widerstand des Dichtringes zu überwinden. 3. Schließen Sie die Kabel (max. Leitungsquerschnitt 1,5 mm²) an den dafür vorgesehenen Schraubklemmen an (Bild 2). 4. Anschlussplatte (4) wieder in das Steckergehäuse (3) schieben und Befestigungsschraube (1) mit Fiberglas-Dichtring (2) montieren und festschrauben.  Achten Sie bei der Montage der Gerätesteckdose auf den ordnungsgemäßen Sitz der Fiberglas-Dichtung (2) und auf eine sachgemäße Montage der Pg-Verschraubung (5), da sonst die Schutzart IP65 nicht erreicht werden kann.	Take care to ensure that the cable is laid in such a way that it is not: - pinched - kinked - under tension. Wiring up the connector: 1. Remove the mounting screw (1) from the head end, taking care not to damage the seal (2) (set aside carefully for future use). 2. Now remove the loose terminal board (4) by pushing out with a screwdriver inserted from above through the mounting screw (1) hole. In this process it is necessary to overcome the resistance exerted by the seal. 3. Connect the cable (max. lead cross-section 1,5 mm²) to the screw terminal provided (Fig. 2). 4. Reinstall the terminal board (4) in the plug housing (3), install the mounting screw (1) with seal (2) and tighten securely.  Take care to ensure that the profiled seal (2) and Pg screw coupling (5) are correctly installed, otherwise the conditions specified for protection category IP 65 will not be met.																
Montage der Gerätesteckdose	Installing the connector																
 Bei der Montage beachten Sie bitte, dass die Codiernut mit der Codiernase des Druckschalters übereinstimmt. Schieben Sie die Gerätesteckdose soweit auf den Druckschalter, bis diese im Druckschalter einrastet.	 During installation, take care that the locating groove engages the locating lug on the pressure switch. Push the connector onto the pressure switch until it locks in place.																
Technische Daten	Technical data																
<table border="1"> <tr> <td>Bemessungsisolationsspannung U_i:</td> <td>300 Volt</td> </tr> <tr> <td>IP-Schutzart nach EN 60529:1991:</td> <td>IP 65</td> </tr> <tr> <td>Anzugsdrehmoment der Anschlussschrauben:</td> <td>< 0,35 Nm</td> </tr> <tr> <td>Anschlussquerschnitt:</td> <td>0,5 bis 1,5 mm²</td> </tr> </table>	Bemessungsisolationsspannung U _i :	300 Volt	IP-Schutzart nach EN 60529:1991:	IP 65	Anzugsdrehmoment der Anschlussschrauben:	< 0,35 Nm	Anschlussquerschnitt:	0,5 bis 1,5 mm ²	<table border="1"> <tr> <td>Rated insulation voltage U_i:</td> <td>300 volts</td> </tr> <tr> <td>IP protection to EN 60529:1991:</td> <td>IP 65</td> </tr> <tr> <td>Tightening torque for terminal screws:</td> <td>< 0,35 Nm</td> </tr> <tr> <td>Connection size:</td> <td>0,5 to 1,5 mm²</td> </tr> </table>	Rated insulation voltage U _i :	300 volts	IP protection to EN 60529:1991:	IP 65	Tightening torque for terminal screws:	< 0,35 Nm	Connection size:	0,5 to 1,5 mm ²
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Zeichenerklärung	Key to drawings:																
 Achtung  Hinweis, Bemerkung  Recycling	 Caution  Note  Recycling																
Art.-Nr.: 1-1-80-628-015_2 9 / 97	Art.-No.: 1-1-80-628-015_2 9 / 97																

Operating Instructions

BA 4602 EN 09.04

FLUDEX Couplings Types
FNO, FNA, FND, FNDB and FNDS
including design in accordance with 94/9/EC



FLENDER

A. Friedr. Flender AG · 46393 Bocholt · Tel. 02871/92-0 · Telefax 02871/92-2596 · www.flender.com

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1. Technical data

1.1 General data

These Operating Instructions (BA) apply generally to FLUDEX couplings of the FN series.

1.2 Data relating specifically to the coupling

The order-specific data relating to a delivery are, if known in case of an order, listed on the title page. In this case this title page diverges from that published on the Internet. The contents of the operating instructions are, however, in this case identical.

The supply order number serves as the coupling's ident number. This number appears on the coupling in light raised relief.

2. General notes

2.1 Introduction

These Operating Instructions (BA) are an integral part of the coupling delivery and must be kept in its vicinity for reference at all times.

Caution!

All persons involved in the installation, operation, maintenance and repair of the coupling must have read and understood these Operating Instructions and must comply with them at all times. We accept no responsibility for damage or disruption caused by disregard of these Instructions.

The "**Coupling**" described in these operating instructions (BA) has been developed for stationary use in general engineering applications.

The coupling is designed only for the application described in section 1, "Technical data". Other operating conditions must be contractually agreed.

The coupling described in these Operating Instructions (BA) reflects the state of technical development at the time these Instructions went to print.

In the interest of technical progress we reserve the right to make changes to the individual assemblies and accessories which we regard as necessary to preserve their essential characteristics and improve their efficiency and safety.

2.2 Copyright

The copyright to these Operating Instructions (BA) is held by **FLENDER AG**.

These Operating Instructions (BA) must not be wholly or partly reproduced for competitive purposes, used in any unauthorised way or made available to third parties without our agreement.

Technical enquiries should be addressed to the following works

FLENDER AG
D 46393 Bocholt

Telefon: 02871/92-2868
Telefax: 02871/92-2579

or to one of our customer-service addresses. A list of our customer-service addresses is given in section 11, "Spare parts, customer-service addresses".

3. Safety notes

3.1 Proper use

- The coupling has been manufactured in accordance with the state of the art and is delivered in a condition for safe and reliable use. Any changes on the part of the user which may affect safety and reliability are prohibited. This applies equally to safety features designed to prevent accidental contact.
- The coupling must be used and operated strictly in accordance with the conditions laid down in the contract governing performance and supply.

3.2 Obligations of the user

- The operator must ensure that all persons involved in installation, operation, maintenance and repair have read and understood these Operating Instructions (BA) and comply with them at all times in order to:
 - avoid injury or damage,
 - ensure the safety and reliability of the coupling,

and

- avoid disruptions and environmental damage through incorrect use.
- During transport, assembly, installation, dismantling, operation and maintenance of the unit, the relevant safety and environmental regulations must be complied with at all times.
- The housing of the FLUDEX coupling is made of aluminium (AC-Al-Si10Mg). Depending on the individual case, if necessary, further requirements must be taken into account when handling aluminium.
- The coupling must be operated, maintained or repaired only by authorised, duly trained and qualified personnel.
- All work must be carried out with great care and with due regard to safety.
- All work on the gear unit must be carried out only when it is at a standstill.
The drive unit must be secured against being switched on accidentally (e.g. by locking the key switch or removing the fuses from the power supply). A notice should be attached to the ON switch stating clearly that work is in progress.
- The coupling must be fitted with suitable safeguards to prevent accidental contact. This fixture must also offer protection against spurting hot operating fluid without substantially impairing the coupling ventilation. The fusible safety plugs and filling plugs should remain accessible.
- The use of suitable operating fluids (usually oil) is necessary for the operation of the coupling. The instructions for use enclosed by the operating fluid manufacturer must be observed.
- The drive unit must be shut down as soon as changes to the coupling are detected during operation.
- If the coupling is intended for installation in plant or equipment, the manufacturer of such plant or equipment must ensure that the contents of the present Operating Instructions are incorporated in his own instructions.
- All spare parts must be obtained from FLENDER.

3.3 Warnings and symbols used in these Operating Instructions



This symbol indicates safety measures which must be observed with regard to **explosion protection**.



This symbol indicates safety measures which must be observed to avoid **personal injury**.

Caution!

This symbol indicates safety measures which must be observed to avoid **damaging the coupling**.

Note:

This symbol indicates general **operating instructions** which are of particular importance.

4. Handling and storage

4.1 Scope of supply

The products supplied are listed in the despatch papers. Check immediately on receipt to ensure that all the products listed have actually been delivered. Parts damaged during transport or missing parts must be reported in writing immediately.

The equipment is delivered ready for operation, depending upon an order with or without fluid filling. Couplings filled with operating medium are marked by a corresponding label ("...L, HLP 32 DIN 51524/2, operating oil already put in") in the vicinity of the filling hole. Delivered separately with every coupling is a fusible safety plug (depending on design also a thermal circuit breaker) with sealing rings, a locking plate and a retaining screw, if provided for.

The coupling in design in accordance with the guideline 94/9/EC is provided with the CE identification mark as described in section 5.

4.2 Handling

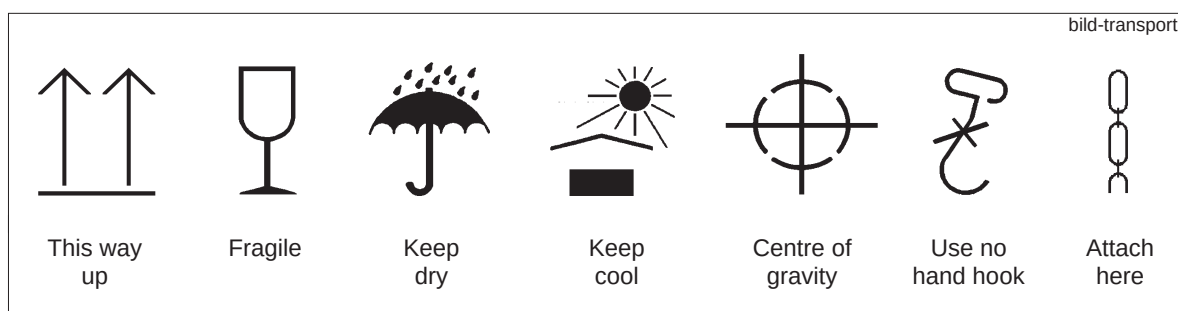


When handling FLENDER products, use only lifting and handling equipment of sufficient load-bearing capacity!

Note: The coupling must be transported using suitable transport equipment only.

Different forms of packaging may be used depending on the size of the coupling and method of transport. Unless otherwise agreed, the packaging complies with the **HPE Packaging Guidelines**.

The symbols marked on the packaging must be observed at all times. These have the following meanings:



4.3 Storage of the coupling

Unless otherwise expressly agreed, the coupling is delivered in a preserved condition and can be stored in a covered, dry place for up to 3 months. If storage is provided for a protracted period, the unit should be treated with a long-term preservative agent on all outer surfaces with the exception of the aluminium housing (FLENDER must be consulted).

Caution!

Before cleaning the coupling and applying the long-term preservative agent, the flexible elements of the N-EUPEX add-on coupling must be removed. The shaft sealing rings must not come into contact with solvents.

Correctly stored couplings retain their properties unchanged for up to five years. Unfavourable storage conditions and improper treatment of the flexible elements and/or seals will negatively affect their physical properties. Such negative effects may be caused by e.g. the action of oxygen, ozone, extreme temperatures, light, moisture, or solvents.

4.4 Storage area for a protracted period

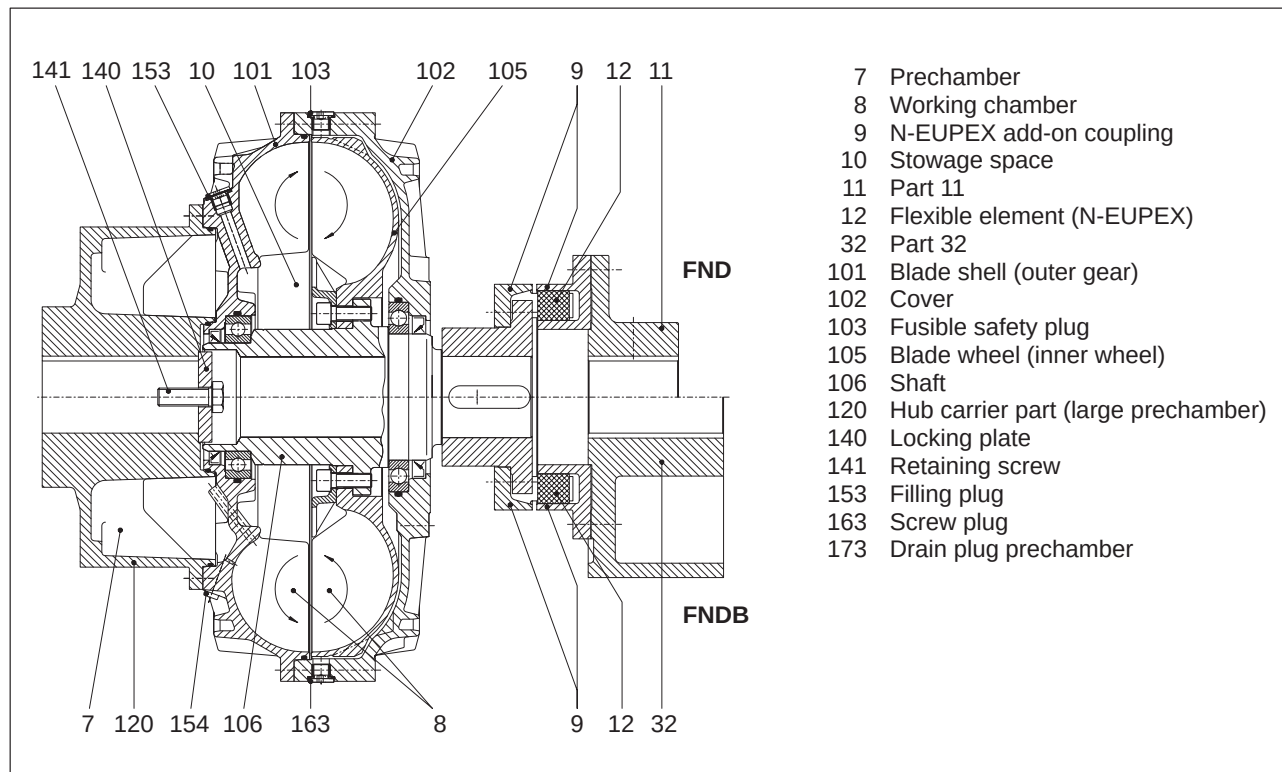
The storage area must be dry and free from dust. The couplings must not be stored with chemicals, solvents, motor fuels, acids, etc. Furthermore, the flexible elements and seals should be protected against light, in particular direct sunlight and bright artificial light with a high ultraviolet content.

Caution!

The storage areas must not contain any ozone-generating equipment, e.g. fluorescent light sources, mercury vapour lamps, high-voltage electrical equipment. Damp storage areas are unsuitable. Ensure that no condensation occurs. The most favourable atmospheric humidity is below 65 %.

5. Technical description

5.1 General description FN series



The FLUDEX coupling is a hydrodynamic fluid coupling working on the Föttinger principle. The coupling parts on the in- and output side are not connected to one other mechanically. The torque is transmitted by the fluid flowing in the coupling and conducted via radial blades. During continuous operation very low rotary speed slip occurs.

FLUDEX couplings of the FN series are suitable for both directions of rotation. They can be mounted in a horizontal, angled or vertical position. In the case of couplings mounted in a sharply angled or vertical position the hub carrier part (120) (large prechamber) must be located at the bottom. The drive should preferably be directed via the hub carrier part (120) to the outer wheel (blade shell 101) to enable the advantages of the prechamber (7) and the working chamber geometry to be utilised.

If locked or overloaded by the driven machine, the coupling heats up until the fusing temperature of the safety fuse is reached.

When the safety fuse fuses, the operating fluid escapes and the drive motor is disconnected from the driven machine. To prevent the operating fluid spurting out, electronically or mechanically controlled thermal monitoring equipment (see separate operating instructions) can be used.

5.2 Construction of couplings

FLUDEX couplings are made up of a few, robust parts.

The inner rotor includes the shaft (106), on which the blade wheel (105) is mounted. The outer housing comprises the cover (102) and the blade shell (101), which are connected via a flanged screw connection.

The outer housing and the inner rotor are double-mounted one inside the other and sealed towards the outside by shaft sealing rings.

In the case of the FN series the hub carrier part (120, carrier hub and large prechamber at the same time) must be attached to the blade shell (101). The prechamber (7) receives part of the oil filling, when the coupling is stationary, and so reduces the effective amount of oil in the working chamber (8) when starting. Thus the starting torque is considerably reduced.

While the coupling is rotating, the oil in the prechamber flows back into the working chamber through small holes time-dependently, so that in continuous operation the entire filling is effective.

For filling the coupling, two filling plugs (153) are located in filling channels offset by 180° (to prevent overfilling) in the blade shell (101). For this, two further holes, offset (approx. 60°) at the periphery, are provided in the cover (102) on the outside diameter of the coupling. In the case of sizes 370 to 755 a fusible safety plug (103) and a screw plug (163) are fitted in these holes. In the case of size 887 screw plugs (163) are located in both holes, the fusible safety plugs (2 x 103) are located next to the flange in the blade shell (101). The screw plugs (163) serve for ventilation during filling and for checking the oil level and draining off the operating fluid.

In the standard design the fusing temperature of the fusible safety screw is 135 °C, and the material used for the seals is Perbunan (for a max. continuous operating temperature of 85 °C). As an extended standard (for continuous operating temperatures of up to 110 °C and/or synthetic oil), the coupling can be fitted with seals made of Viton. In this case a fusible safety plug with a fusing temperature of 160 °C is used in the coupling.



Couplings to be used in potentially explosive areas are designed with the fusible safety plugs (103) permitted for the temperature class.

The fusing temperature of the fusible safety plugs is stamped on their end faces, the **135 °C plugs** also **being marked red**, the **160 °C plugs green**.

For applications where frequent faults or locking of the drive may occur it is recommended that a thermal monitoring system be provided. This will prevent the operating fluid from escaping and being lost and the environment from being polluted and endangered. The transmitter (EOC system) or thermal circuit breaker (mechanical switching equipment) is screwed into the outer diameter of the coupling instead of the screw plug (163). The fusible safety plug (103) remains in the coupling as an emergency safety device.

A thermal circuit breaker with a fusing temperature of 110 °C can be used on a coupling with Perbunan seals and a 135 °C fusible safety plug.

A thermal circuit breaker with a fusing temperature of 140 °C can be used on a coupling with Viton seals and a 160 °C fusible safety plug.

A 160 °C fusible safety plug can be used with an EOC system (transmitter release temperature of 125 °C), including with Perbunan seals.

The type **FNO** includes only the components of the fluid coupling (without attachment) and incorporates a connecting shaft end on the output side. The types **FNA** and **FND** are couplings with an N-EUPEX coupling attached on the output side for connecting two shafts. Types **FNDB** and **FNDS** are FND couplings with brake drum or brake disk, the FNDS-SB being suitable for stopping and holding brakes, the FNDS-HB for holding brakes only.

In the flexible N-EUPEX add-on coupling the H-shaped Perbunan flexible elements are delivered in a standard Shore hardness of 80.



Coupling with worn flexible elements (12) (for wear mark ΔS_v , see section 10) must not be operated in potentially explosive areas.

5.3 Marking the coupling parts for explosion protection

Note: Couplings which are intended for use in potentially explosive areas must bear the following markings:

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FLUDEX coupling - year of construction

Marking of the fusible safety plug:

CE II 2 G c T3 D160 °C II B

- 30 °C ≤ T_a ≤ + 50 °C

CE I M2

T3

The N-EUPEX add-on coupling must have stamped on the hub parts.

5.4 Conditions of use for FLUDEX couplings in potentially explosive areas

The coupling with fusible safety plug with indication  T3 is suited for the service conditions specified in Guideline 94/9/EC:

- Equipment group II (use above ground) temperature class T3 of category 2 and 3 for areas where there are explosible gas, vapour, mist, air mixtures as well as for areas where dust can form explosible atmospheres.
- Equipment group I (underground applications) of the category M2



If it is to be used below ground in potentially explosive areas, the coupling, which is made of aluminium, must be provided with a robust casing to preclude the risk of ignition from e.g. friction, impact or friction sparks. The depositing of heavy metal oxides (rust) on the coupling housing must be precluded by the casing or other suitable precautions.



The FLUDEX coupling can be delivered with attached brake disk. The supplier of the subassembly is responsible for the construction of the brake disk in conformity to the guidelines. To be noted are, amongst other things, hazards from electrostatic charging and hot surfaces.

6. Mounting

The FLUDEX coupling is delivered preferably with finish-bored and grooved hubs.

At the customer's request FLENDER also delivers unbored or prebored coupling parts.

The necessary refinishing must be carried out in strict compliance with the following specifications and with particular care!

Caution!

Responsibility for carrying out the refinishing is borne by the orderer. FLENDER can accept no guarantee claims arising from unsatisfactory refinishing!



Couplings with CE marking for use in potentially explosive areas are delivered exclusively with finish-bored hubs.

6.1 Instructions for applying the finished bore and fitting the axial retaining means, set screws and balancing

6.1.1 Finished bore

- On part 1: Remove flexible elements.
- On part 11; 32: Remove part 10 or parts 8 and 10.
- In the case of hub carrier part (120): demount hub carrier part (120) from the coupling, remove O-rings (117; 138) and retain them for re-use later.
- Remove preservative from coupling parts.



Note manufacturer's instructions for handling solvent.

When machining the finished bore the parts must be carefully aligned. For the permissible radial and axial runout errors and the permissible cylindricity tolerances, see DIN ISO 286. The parts must be mounted on the marked faces ().

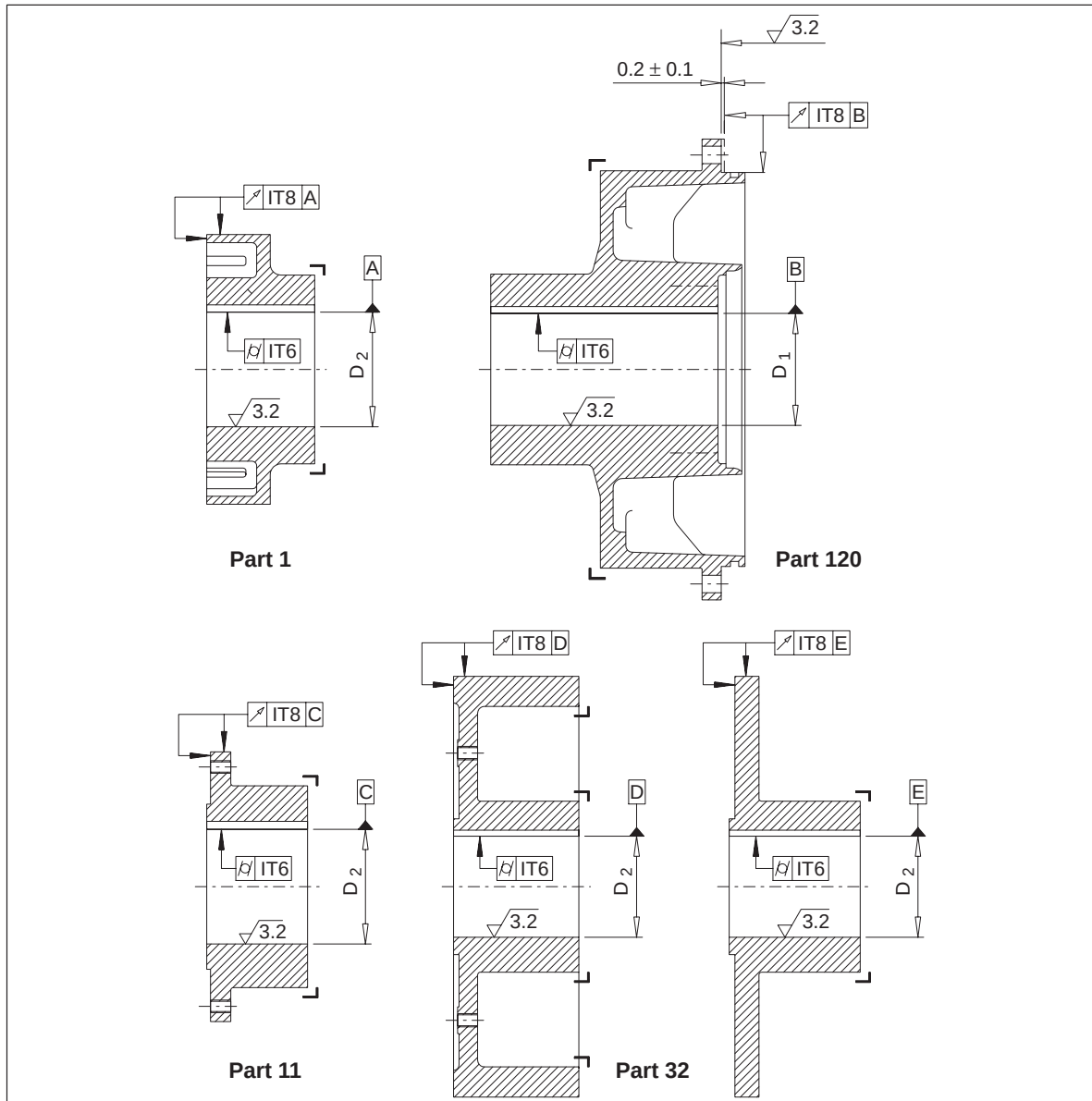
When machining the finished bore in the hub carrier part (120), the axial run-out face of the outer flange must be finish-turned during the same clamping operation (cut rate 0.2 ± 0.1 mm).

Caution!

The maximum permissible bore diameters (see table 6.1.1 a) are designed for drive-type fastenings without taper action to DIN 6885/1 and must not under any circumstances be exceeded. The finish-machined bores must be 100 % checked with suitable measuring equipment.

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If other shaft - hub connections (e.g. taper or stepped bore) are to be used instead of the flanged sleeve connections provided for, FLENDER must be consulted.
Drive-type fastenings with taper action are not permissible.



FLUDEX Size	N-EUPEX Size	max. bore D ₁ Part 120	max. bore D ₂				
			Part 1 FNA	Part 11 FND	Part 11, 32 FNDS SB (32) / HB (11)	Part 32 (Brake drum) FNDB D x B	D ₁
370	180 200	80	75 —	70 —	80	315 x 118 400 x 150	80 90
425	200	100	85	80	80	315 x 118 400 x 150	80 90
490	225 250	110	90 —	90 —	90	400 x 150 500 x 190	90 100
565	250	120	100	100	100	400 x 150 500 x 190	90 100
655	315	135	120	110	100	500 x 190 630 x 236	100 140
755	350	150	140	120	140	630 x 236	140
887	440	170	160	130	140	710 x 265	150

Table 6.1.1 a: max. bores of the hub parts

For drive by means of parallel keys the following fit pairs are prescribed for the bores:

Selection of fit	Hole D ₁		Shaft tolerances	Bore tolerances
	over mm	up to mm		
Shaft tolerances to FLENDER standard		25	k6	H7
	25	100	m6	
	100		n6	
Shaft tolerances to DIN 748/1		50	k6	H7
	50		m6	
System standard shaft		50	h6	K7
	50			M7
	all		h8	N7

Table 6.1.1 b: Fit pairs

Caution!

The assigned fits must be adhered to in order, on the one hand, to keep the play in the shaft-hub connection as low as possible, depending on utilisation of the tolerance zones, or, on the other, to keep the hub tension arising from the oversize within the permissible load limit. Failure to adhere to the fits may impair the shaft-hub connection.

If the tolerance values of the shafts deviate from those in table 6.1.1 b above, FLENDER must be consulted.



Failure to observe these instructions may result in breakage of the coupling. Danger from flying fragments!

The coupling then becomes an explosion hazard.

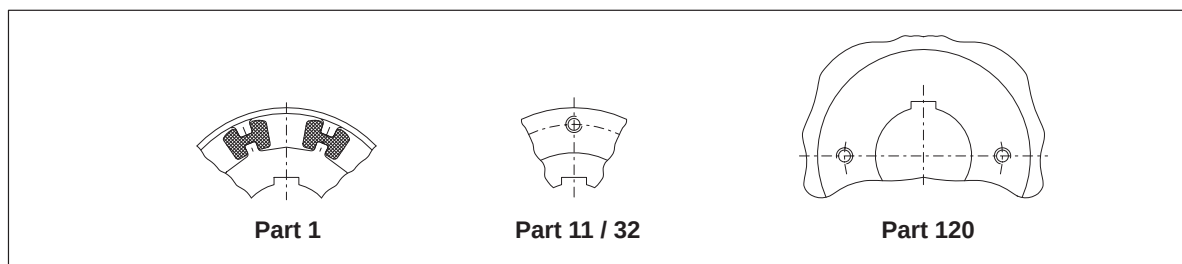
6.1.2 Parallel keyway

The parallel keyways must be designed in accordance with DIN 6885/1. If the keyway geometry deviates, FLENDER must be consulted. Taper keys or nose keys (gib headed keys) are not permissible.

The parallel keyways must be designed to suit the available parallel keys. For parallel keyways the tolerance zone of the hub keyway width **ISO JS 9** must be adhered to.

Caution!

The parallel keyway must be fitted as shown in the illustration.



6.1.3 Axial fastening

The hub carrier part (120) and the brake drum/brake disk (32) are axially secured, preferably with a retaining washer and retaining screw (for tightening torques, see table 6.1.3). Before turning the recess in the brake drum/brake disk, FLENDER must be consulted.

Retaining screw thread	M6	M8	M10	M12	M16	M20	M24	M30
Tightening torque [Nm]	10	25	49	86	160	300	500	800

Table 6.1.3: Tightening torques of the retaining screws

A set screw is provided for axially securing part 1 and 11 and in the case of larger bores in the hub carrier part (120) or in the brake drum/brake disk (32).

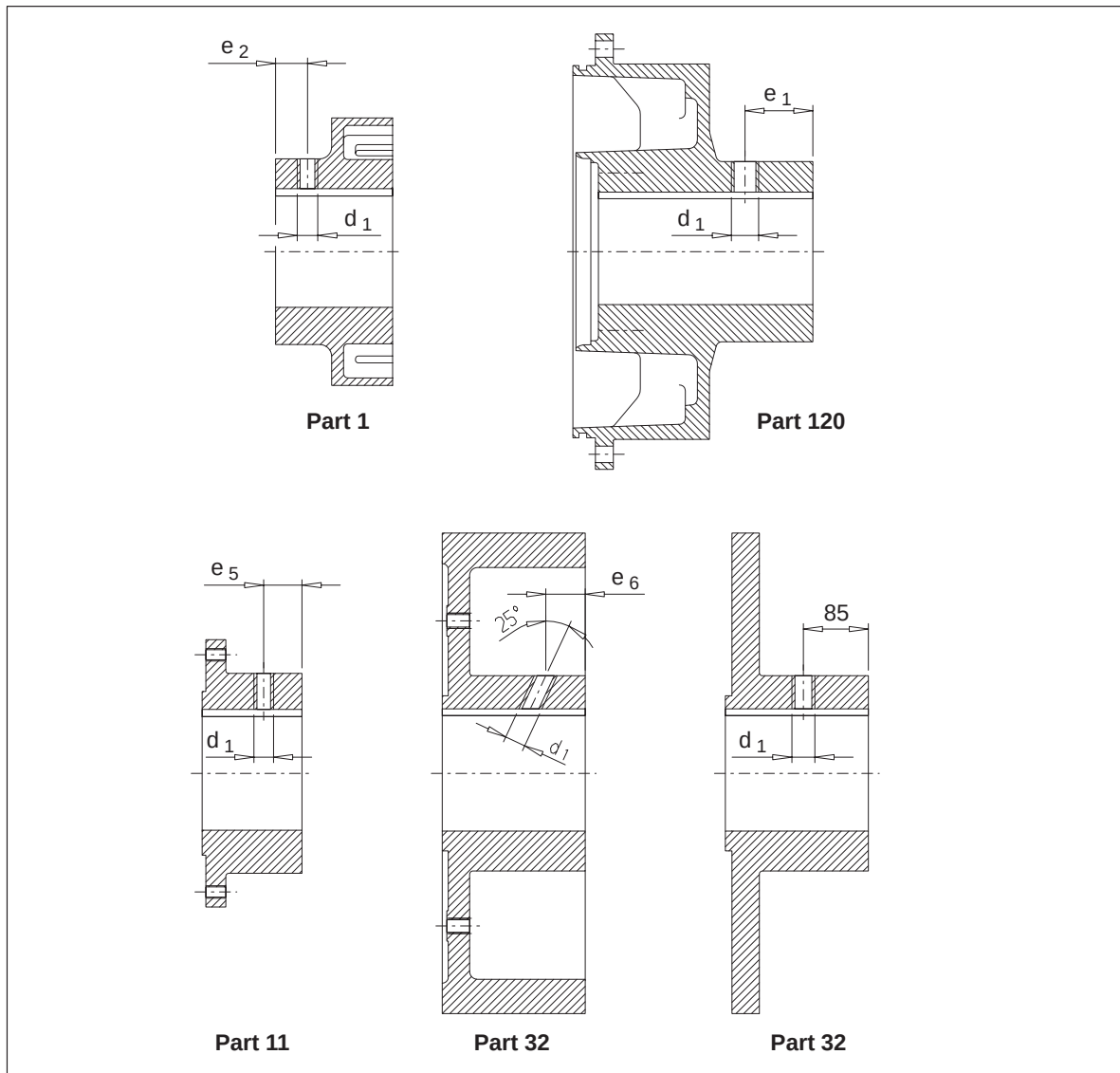
6.1.4 Set screws

Hexagon socket set screws with cup points to DIN 916 must be used for set screws.

The following guidelines must be observed!



The length of the set screw must be selected so that it fills the threaded hole, but does not project from the hub ($L_{\min.} = d_1 \times 1.2$).



FLUDEX Size	N-EUPEX Size	d_1 max.	Part 120 e_1	Part 1 e_2	Part 11 e_5	Part 32 e_6	Tightening torque of the set screws [Nm]
370	180 200	M12	23	16	30	25	25
425	200	M12	25	20	40	25	25
490	225 250	M12	25	22	40	30	25
565	250	M16	23	24	50	50	70
655	315	M16	50	35	60	75	70
755	350	M20	40	40	70	75	130
887	440	M24	50	60	90	75	230

Table 6.1.4 a : Set screw assignment and tightening torques of the set screws

Caution!

The set screws must always be positioned on the keyway.

6.1.5 Balancing

FLUDEX couplings with prebored hub carrier part (120) are delivered balanced.

Prebored add-on coupling parts are delivered unbalanced. It is recommended that these parts are balanced to suit the application after finish-boring (see DIN ISO 1940 and DIN 740/2), but to min. balancing quality G16.

Balancing is normally done by drilling material away.

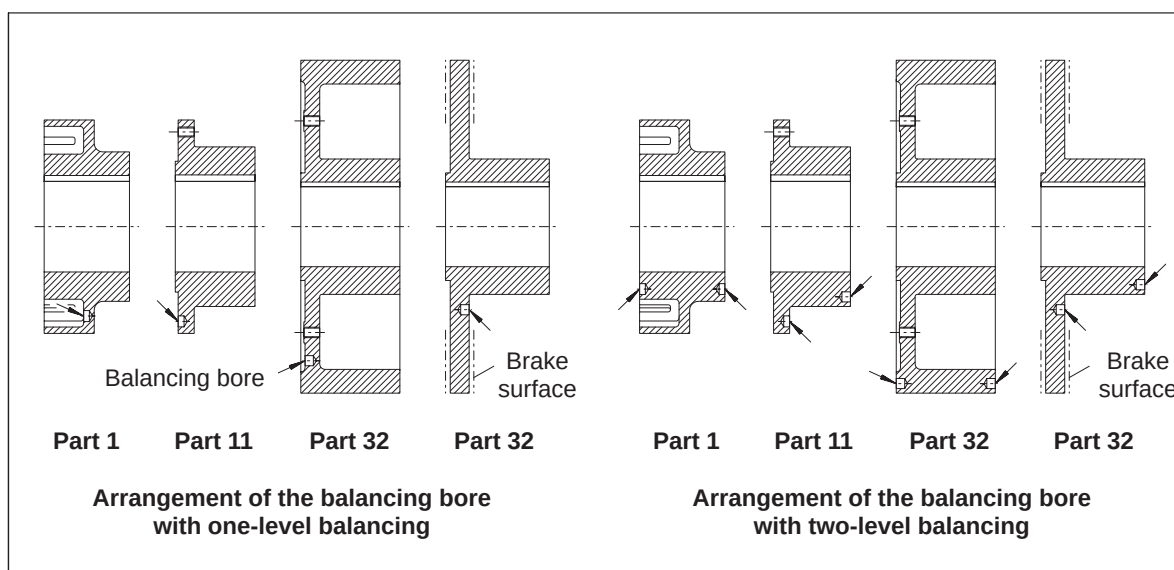
If one-level balancing is required on brake disk / brake drum (32), the material on the front end at the disk's bottom is to be removed. When two-level balancing part 11 or the brake disk / brake drum (32) should be balanced together with part 10 or parts 8 and 10.

Caution!

On part 1 material must be removed from between the webs, the bottom must however not be drilled right through.

On part 32 the brake surface must not under any circumstances be damaged.

Finish-bored couplings or coupling parts are balanced according to the customer's specifications.



6.2 General information on fitting

During fitting, the "Safety Instructions" in Section 3 must be observed.

Fitting work must be done with great care by trained and qualified personnel.

As early as during the planning phase it must be ensured that sufficient space is available for installation and subsequent care and maintenance work.

Adequate lifting equipment must be available before beginning the fitting work.



If lacquered couplings are used in potentially explosive areas, the requirements made of the conductivity of the lacquer and the limitation on the thickness of the lacquer applied must be observed in accordance with EN 13463-1. Where lacquer coatings have a thickness less than 200 µm, no electrostatic charge is to be expected. Where lacquer coatings are thicker than 200 µm, an electrostatic charge must be avoided, e.g. by cleaning the coupling.

6.3 Mounting the coupling parts

FN couplings are delivered with hub carrier part (120) attached. If the delivery included the oil filling, the oil filling must be drained off before assembly is begun (see section 10, item 10.1).

After the screws (139) are undone, the hub carrier part (120) must be demounted from the main coupling and the O-rings (117, 138) removed for re-use later.

The shaft ends, the brake surfaces and the coupling hubs must be carefully cleaned and a lubricant rubbed into the shaft ends. Before cleaning coupling part 1 with solvent the flexible elements must be removed.



Note manufacturer's instructions for handling the solvent.

Caution!

**The coupling hubs must be fitted with the aid of suitable equipment to avoid damaging the shaft bearings through axial joining forces.
Always use suitable lifting equipment.**

If necessary, heating the coupling hubs (max. + 150 °C) will facilitate fitting. With temperatures over + 80 °C the flexible elements must be removed from the coupling part 1 before heating.



Take precautions to avoid burns from hot components!

The coupling hubs must be mounted using the entire length of the bore or until it comes into contact with the shaft shoulder.

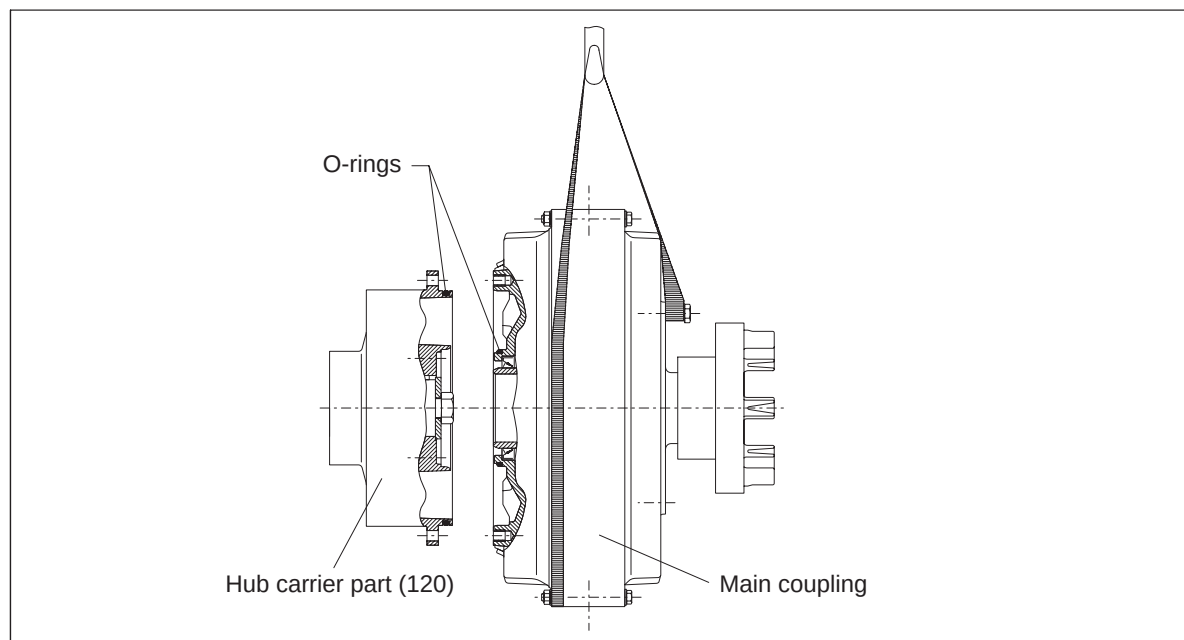
After being pulled on, the coupling hubs must be secured axially by means of a locking plate and a retaining screw or an set screw (see item 6.1.3).

Caution!

Tightening the set screws to a tightening torque in accordance with item 6.1.4.



**Failure to observe these instructions may result in breakage of the coupling.
Danger from flying fragments!
The coupling then becomes an explosion hazard.**



Before the main coupling is flange-fitted to the hub carrier part (120), any dirt must be removed from the interior of the hub carrier part (120) and the corresponding area of the main coupling and the O-rings (117, 138) lightly greased and inserted into their mounting grooves. The sealing and flange surfaces must be clean and undamaged.

To attach the main coupling to the hub carrier part (120), the main coupling must be suspended in suitable lifting gear to ensure as coaxial and as straight a positioning on the centring projection of the hub carrier part (120) as possible.

When flange-fitting, care must be taken that the O-rings (117, 138) and the sealing surfaces are not damaged. The flange may be attached by means of the screws (139). Care must be taken to ensure that the flange surfaces fit closely (for tightening torque of screws, see item 6.6).

On part 1 (1) flexible elements (12) which have been removed must be refitted. It must be ensured that the flexible elements (12) are of identical size and have identical markings. The coupling parts must be only at a maximum temperature of + 80 °C.

Move together the machines to be coupled.



Danger of squeezing!

Dimension S must be adhered to. The tightening torques of the bolts parts 13 and 139 are to be checked (for tightening torques and distance dimension S see item 6.6 and section 1).

Caution!

If components which conduct axial or bending forces into it are placed on the shaft of the FND coupling, FLENDER must be consulted.

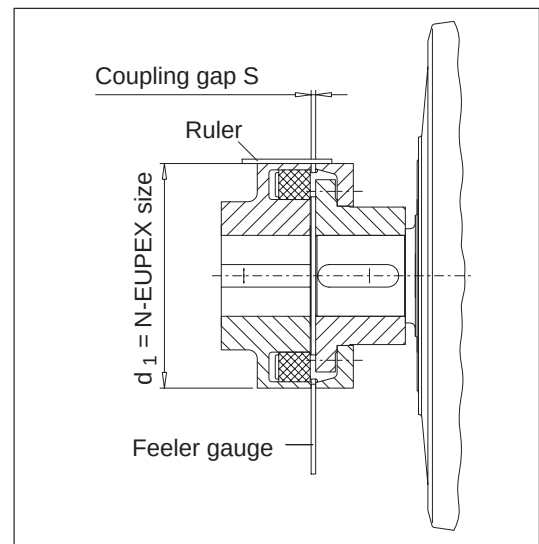
6.4 Alignment

FLUDEX couplings in combination with the flexible N-EUPEX add-on coupling absorb positional deviations of the shaft ends to be connected up to the values shown in item 6.5.

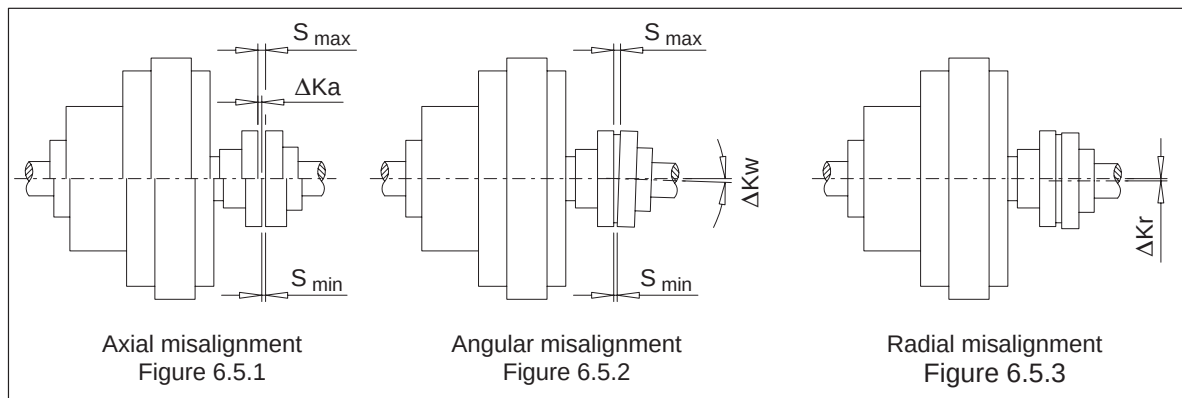
When aligning, the radial and angular misalignment of the shaft ends must be kept as small as possible, because, other conditions being equal, this increases the service life of the flexible elements.

The alignment must be checked once more with the foundation bolts of the motor and gear unit/driven machine tightened.

Any shaft misalignment through heating during operation must be taken into account as far as possible



6.5 Possible misalignments



Misalignments of the coupling parts in relation to each other can be caused by inaccurate alignment during assembly, but also by actual operation of the equipment (expansion due to heat, shaft deflection, insufficiently rigid machine frames, etc.).

Caution!

The following maximum permissible misalignments must by no means be exceeded during operation.

6.5.1 Axial misalignment

Axial misalignment ΔK_a (Fig. 6.5.1) of the coupling parts relative to one another is possible within the "permissible error" for dimension S (see title page).

6.5.2 Angular misalignment

The angular misalignment ΔK_w (Fig. 6.5.2) can usefully be measured as the difference in the gap dimension ($\Delta S = S_{\max.} - S_{\min.}$). For the permissible values for the difference in the gap dimension, refer to item 6.5.4.

If required, the permissible angular misalignment ΔK_w can be calculated as follows:

$\Delta K_{w \text{ perm.}} \text{ in rad} = \frac{\Delta S_{\text{perm.}}}{d_1}$	$\Delta S_{\text{perm.}}$ see item 6.5.4 d_1 corresponds to N-EUPEX size in mm
$\Delta K_{w \text{ perm.}} \text{ in degrees} = \frac{180}{\pi} \times \frac{\Delta S_{\text{perm.}}}{d_1}$	

6.5.3 Radial misalignment

For the permissible radial misalignment $\Delta K_{r \text{ perm.}}$ (Fig. 6.5.3) - depending upon the operating speed -, refer to item 6.5.4.

6.5.4 Permissible shaft misalignment values for radial misalignment $\Delta K_{r \text{ perm.}}$ and difference in gap dimension $\Delta S_{\text{perm.}}$

Values given in mm, rounded off

FLUDEX Size	N-EUPEX Size	Coupling speed in 1/min						
		250	500	750	1000	1500	2000	3000
370	180	0.6	0.5	0.4	0.35	0.3	0.25	0.2
	200	0.8	0.55	0.45	0.4	0.3	0.3	0.2
425	200	0.8	0.55	0.45	0.4	0.3	0.3	0.2
490	225	0.8	0.55	0.5	0.4	0.35	0.3	0.25
	250	0.8	0.6	0.5	0.4	0.35	0.3	
565	250	0.8	0.6	0.5	0.4	0.35	0.3	
655	315	1	0.7	0.6	0.5	0.4	0.35	
755	350	1	0.8	0.6	0.6	0.5		
887	440	1.3	1	0.7	0.7	0.6		

The numerical values of the table can be calculated as follows:

$\Delta K_{r \text{ perm.}} = \Delta S_{\text{perm.}} = \left(0.1 + \frac{d_1}{1000} \right) \times \frac{40}{\sqrt{n}}$	Coupling speed n in 1/min d_1 corresponds to N-EUPEX size in mm Radial misalignment $K_{r \text{ perm.}}$ in mm
---	---

Caution!

Angular and radial misalignment may occur simultaneously.

6.6 Assignment of tightening torques

FLUDEX Size	N-EUPEX Size	Tightening torque T_A and wrench width S_w for screws Part no.															
		13		103		110/142/163		23/24		130/131		139		153		173	
		T_A Nm	S_w mm	T_A Nm	S_w mm	T_A Nm	S_w mm	T_A Nm	S_w mm	T_A Nm	S_w mm	T_A Nm	S_w mm	T_A Nm	S_w mm	T_A Nm	S_w mm
370	180	44	8	60	10	75	19/27/10	49	8/16	21	13	31	8	75	10	25	7
	200	67.5	10	60	10	75	19/27/10	86	10/18	21	13	31	8	75	10	25	7
425	200	67.5	10	60	10	75	19/27/10	86	10/18	21	13	31	8	75	10	25	7
490	225	86	10	60	10	75	19/27/10	86	10/18	40	17	54	10	75	10	25	7
	250	145	14	60	10	75	19/27/10	210	14/24	40	17	54	10	75	10	25	7
565	250	145	14	60	10	75	19/27/10	210	14/24	40	17	54	10	75	10	25	7
655	315	200	14	60	10	75	19/27/10	210	14/24	73	19	135	14	75	10	25	7
755	350	260	17	60	10	75	19/27/10	410	17/30	73	19	135	14	75	10	135	14
887	440	410	17	60	10	75	19/27/10	410	17/30	120	24	135	14	300	17	135	14

Table 6.6: Assignment of tightening torques (for part nos. refer to Section 11)

Note: Tightening torques apply to screws with untreated surfaces which are not or only lightly oiled (coefficient of friction $\mu = 0.14$). The use of lubricant paint or the like, which affects the coefficient of friction μ , is not permitted.

Note: The tightening torques of the set screws are specified in item 6.1.4.

7. Start-up

7.1 Procedure before start-up

Before starting up check the flexible elements for correct seating, i.e. the flexible elements must sit flush with the end face of the hub, and the set screws for tightness, check and, if necessary, adjust the alignment and the gap dimension S and check all screw connections for the specified tightening torques (see section 1 and section 6).

7.2 Fluid filling



Only specifically permitted operating fluids may be used for operation below ground.

The behaviour and the efficiency of the FLUDEX coupling is decisively affected by the quantity of operating fluid put in. As the filling increases, so does the transmitting capacity of the coupling, and the motor load during starting and the max. overload torque on the coupling rises. Operating slip decreases with the identical load.



FLUDEX couplings must not be filled up to more than 80 - 85 % of the total volume (limited by the overfill safety system). As a result of the higher temperature-dependant volume expansion of the operating fluid compared with the aluminium housing higher filling results in a sharp rise in pressure in the coupling, which can result in the coupling being irreparably damaged before the fusing temperature of the fusible safety plug (103) is reached.

The operating fluid must be put in at the filling plug (153). Only these filling holes are provided with a filling channel, which offers protection against accidental overfilling. To better ventilate the inner chamber, the screw plug (163) inserted in the outer flange, or the second fusible safety plug (103), should be unscrewed. It is set at an appropriate angle so that, in the event of imminent overfilling, the excess can escape there.

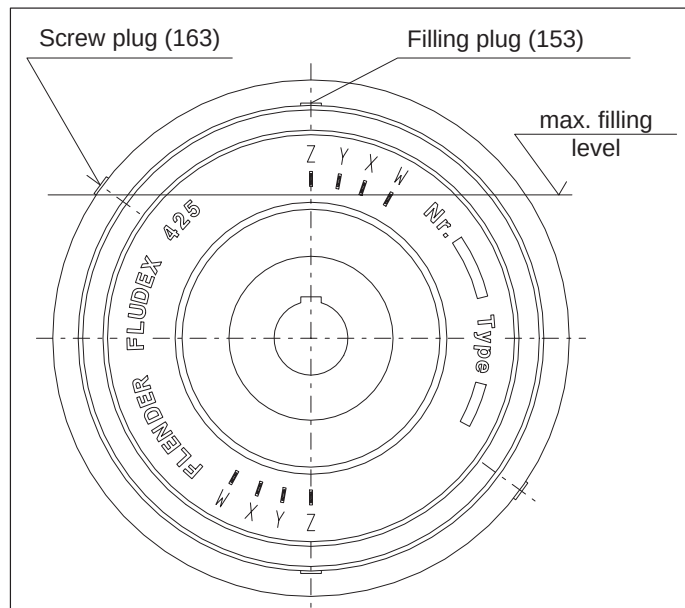
Caution!

The quantity to be put into the coupling is determined in accordance with the order. The filling quantity in litres is stamped on the coupling and specified in section 1.

Marking lines have been cast in raised relief on the coupling housing to assist filling the coupling and checking the oil level. When filling or checking the filling level the mark with the required filling quantity (in the case of intermediate values the corresponding intermediate position) must be turned to the top position (12 o'clock).

In the case of filling by the filling plug the coupling is filled with the required quantity, when the filling level has reached the lip of the ventilation hole (163).

For the filling quantities to be assigned to the marks W to Z, please see table 7.2.



When changing the oil, care must be taken that the used oil is completely drained off. This can be done by way of the screw plug (163). Additionally, the residual oil in the prechamber must be drained off through the hole in the screw plug (173).

Size		370	425	490	565	655	755	887
Filling position	W	5.4	8.4	12.9	20.4	31.7	45.9	76
	X	6.5	10.1	15.5	24.6	38.2	55.4	88
	Y	7.8	12.1	18.6	29.5	45.9	66.5	105
	Z	9.1	14.2	21.7	34.4	53.5	77.5	120

Table 7.2: Filling quantities in litres

After filling tighten filling plug (153) and screw plug (163) and fusible safety plug (103) (for tightening torques, see section 6, item 6.6) and check coupling for leaktightness by carrying out a short test run. This may be done by holding a clean piece of paper parallel to the turning axis and in close to the rotating surface envelope of the coupling. Any operating fluid spurting out will be visible on the paper.

Caution!

FLUDEX couplings must not lose any operating fluid, as this loss may result in a rise in slip and coupling temperature and ultimately to a fusing of the fusible safety plug (103).

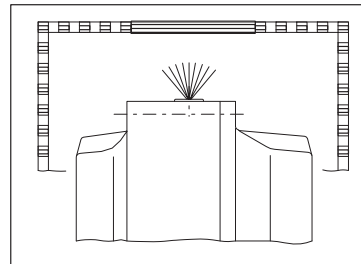
Then fit the coupling guard to prevent unintentional contact.

Faults in the drive (e.g. locking of the output side) may result in the coupling overheating. If the fusing temperature of the fusible safety plug is reached, the insert will melt and the hot operating fluid (oil, water) escape. The coupling protection must therefore be so designed as to also guarantee protection against operating fluid spurting out without appreciably impairing the ventilation of the coupling. The fusible safety plugs (103) and the filling plugs (153) should be accessible.

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Rotating parts must be secured by the purchaser against accidental contact. The coupling protection must also protect against spurting hot operating fluid without appreciably impairing ventilation (see also information on the coupling).



If it is to be used below ground in potentially explosive areas, the coupling, which is made of aluminium, must be provided with a robust casing to preclude the risk of ignition from e.g. friction, impact or friction sparks.
The depositing of heavy metal oxides (rust) on the coupling housing must be precluded by the casing or other suitable precautions.



Couplings to be used in potentially explosive areas are designed with the fusible safety plugs (103) permitted for the temperature class. The coupling is marked with the temperature class of the fusible safety plug (103).
It must be ensured that the drive shuts off not later than 5 minutes after the fusible safety plug (103) is fused.

7.2.1 Operating fluid: oil

In the case of orders without performance data the filling quantity is not entered. In these cases, for operation with oil the filling quantity for the respective coupling size can be obtained from the filling quantity tables in section 10, item 10.9 and item 10.10 independently of the speed and output to be transmitted (and possibly the motor output).

Caution!

The filling quantity tables in section 10 apply solely to oil fillings.

HL or HLP hydraulic oils to DIN 51 524 Part 1 and Part 2 of the VG 22 or VG 32 ISO viscosity classes must be used as operating fluids.

Caution!

The oil quality and purity determine the useful life of FLUDEX bearings and shaft sealing rings.

Company					Mobil		
Description	Aral Degol BG 32	BP Energol HL 22 + HLP 22	Hyspin SP22 + SP32	TORQUE FLUID N 45	Mobil DTE 22	Shell Tegula Öl 32	Rando-Oil 32
	Aral Vitam GF 22	BP Energol HL 32 + HLP 32	Hyspin DSP22 + DSP32	TERESSO 32	Mobil DTE 24	Shell Tellus Öl 22	Torque-Fluid 32
	Aral Vitam GF 32	BP Energol HLP-D 32	Tribol 943 AW22 + AW32	NUTO H 22	Mobil Vactra Oil Light	Shell Tellus Öl C22	
				NUTO H 32	Mobilfluid 125		

Table 7.1: Recommended oil grades



Observe manufacturer's instructions when handling the operating fluid!

7.2.2 Operating fluid: water/water emulsion

Caution!

The coupling must be protected against freezing.



Observe manufacturer's instructions when handling the operating fluid!

Caution!

Not every FLUDEX coupling is suitable for use with water or water emulsion as an operating fluid.

FLUDEX couplings designed for water fillings are marked as follows in the area of the filling plug (153): " ∇ W "



If water or water emulsion is used as operating fluid, only fusible safety plugs with a maximum fusing temperature of 110 °C are permitted. At a higher fusing temperature excessive strain is put on the coupling housing by the operating fluid vapour pressure.

8. Operation

8.1 General operating data

During operation of the coupling watch for:

- changes in running noise
- sudden vibrations

Caution!

If any irregularities are noticed during operation, switch the drive assembly off at once. Determine the cause of the fault, using the table in section 9.

This table contains a list of possible faults, their causes and suggested remedies.

If the cause cannot be identified or the unit repaired with the facilities available, you are advised to contact one of our customer-service offices for specialist assistance (see section 11).

9. Faults, causes and remedy

9.1 General

The following irregularities can serve as a guide for fault tracing.

Where the system is a complex one, all the other component units must be included when tracing faults.

The coupling must run with little noise and without vibration in all operating phases. Irregular behaviour must be treated as a fault requiring immediate remedy.

Caution!

FLENDER will not be bound by the terms of the guarantee or otherwise be responsible in cases of improper use of the coupling, modifications carried out without FLENDER's agreement, or use of spare parts not supplied by FLENDER.



When remedying faults and malfunctions, the gear unit must always be taken out of service.

Secure the drive unit to prevent it from being started up unintentionally. Attach a warning notice to the start switch!

9.2 Possible faults

Malfunctions	Possible causes	Remedy
Sudden changes in the noise level and/or sudden vibrations.	Change in alignment. Flexible elements (12) worn.	Take the system out of service. If necessary, rectify causes of change in alignment (e.g. tighten loose foundation bolts). Check and, if necessary, adjust alignment (see section 6). Wear check, procedure as described in section 10. Take the system out of service. Demount coupling and remove remains of flexible elements (12). Check and replace damaged coupling parts. Flexible elements (12) must be changed in sets; use only identical N-EUPEX flexible elements (12). Assembly of coupling according to section 6 and section 7.
No torque transmission.	The fusible safety screws have fused through overheating or locking and the fluid is escaping from the coupling.	Take the system out of service. Rectify the cause of the overheating or locking. Fit new fusible safety plugs with new sealing rings. Refill the coupling as described in section 7.

Table 9.2: Possible faults



If the torque is being transmitted with worn N-EUPEX flexible elements (12) and metal parts are consequently in contact, proper operation within the meaning of the explosion protection requirements or Guideline 94/9/EC can no longer be guaranteed.

Measurement and assessment of the wear condition of the N-EUPEX flexible elements (12) in accordance with section 10.

9.3 Incorrect use

Experience has shown that the following faults can result in incorrect use of the FLUDEX coupling. In addition to observing the other instructions in these Operating Instructions (BA), care must therefore be taken to avoid these faults. Guideline 94/9/EC requires the manufacturer and user to exercise especial care.



Failure to observe these instructions may result in breakage of the coupling. Danger from flying fragments!



Through incorrect use the coupling may become an explosion hazard.

Caution!

Incorrect use of the FLUDEX coupling can result in damage to the coupling.

Caution!

Coupling damage may result in stoppage of the drive and the entire system.

9.3.1 Possible faults when selecting the coupling or coupling size

- Important information for describing the drive and the environment are not communicated.
- System power too high.
- System speed too high or too low.
- Starting frequency too high.
- The ventilation of the coupling is inadequate.
- Chemically aggressive environment not taken into consideration.
- The ambient temperature is not permissible. See also section 5.
- Machining of a finished bore with incorrect diameter and incorrect fit assignment (see section 6 and title page).
- The transmission capacity of the shaft-hub connection is not appropriate to the operating conditions.

9.3.2 Possible faults when installing the coupling

- Components with transport or other damage are being fitted.
- When mounting coupling parts in a heated condition, already mounted N-EUPEX flexible elements (12) are being excessively heated.
- When attaching the main coupling to the hub carrier part (120), the O-rings (117; 138) are not fitted.
- The shaft diameter is outside the specified tolerance range.
- Coupling parts are being interchanged, i.e. their assignment to the specified direction of drive is incorrect.
- Specified axial fixtures are not fitted.
- Prescribed tightening torques are not being adhered to.
- Alignment or shaft misalignment values do not match the operating instructions.
- The incorrect operating fluid or the incorrect quantity of operating fluid is being put in.
- The coupled machines are not correctly fastened to the foundation, and as a result a shifting of the machines e.g. through loosening of the foundation screw connection is causing excessive displacement of the coupling parts.
- N-EUPEX flexible elements (12) are being omitted or incorrectly positioned.
- The coupling protection does not comply with the guidelines to be applied. It considerably restricts ventilation of the coupling.
- Operating instructions are being changed without authorisation.
- Components are being fitted to the coupling which transmit excessive axial forces or bending moments to the coupling.
- Add-on parts are not designed in compliance with the guideline 94/9/EC and are an explosion hazard.

9.3.3 Possible faults in maintenance

- Maintenance intervals are not being adhered to.
- The incorrect operating fluid or the incorrect quantity of operating fluid is being put in.
- Original FLENDER spare parts are not being used.
- Prescribed tightening torques are not being adhered to.
- Old or damaged N-EUPEX flexible elements (12) are being used.
- Fusible safety plugs (103) with an unsuitable fusing temperature are being used.
- Leakage in the vicinity of the coupling is not being identified and as a result chemically aggressive media are damaging the coupling.

10. Maintenance and repair



All work on the gear unit must be carried out only when it is at a standstill. The drive unit must be secured against being switched on accidentally (e.g. by locking the key switch or removing the fuses from the power supply). A notice should be attached to the ON switch stating clearly that work is in progress.



**Risk of burns after switching off!
Allow the FLUDEX coupling to cool down sufficiently before beginning work!**

10.1 Changing the operating fluid



**Never change the operating fluid immediately after operation!
There is a risk of scalding!
Allow the coupling and the operating fluid to cool down!**

The operating fluid must be changed in accordance with the specifications in table 10.1.

When changing the operating fluid, care must be taken that the old operating fluid is completely drained off. This can be done through the holes in the fusible safety plugs (103) or through the holes in the screw plugs (163) respectively. The residual oil in the prechamber is drained off through the screw plug (173).



Observe the environmental requirements in force!

Operating temperatures / special measures	max. operating hours	max. years of use
max. 80 °C	10 000	5
max. 95 °C or frequent temperature peaks above 100 °C: use of Viton seals necessary	5 000	2
above 95 °C only suitable synthetic oils permitted use of Viton seals necessary	according to the specifications of the operating fluid manufacturer	according to the specifications of the operating fluid manufacturer

Table 10.1: Operating temperatures, operating hours and years of use

Different operating conditions and changing frequencies are permitted with the agreement of the operating fluid manufacturer.

10.2 Changing the shaft sealing rings

Shaft sealing rings are wearing parts subjected to relatively low stresses in the coupling. On couplings which are filled with oil no maintenance interval need be adhered to.

On couplings which are filled with water the shaft sealing rings of the inner seal must be replaced after 12 000 operating hours or after 2 years of operation. We recommend also replacing the other seals and rolling bearings at the same time.

10.3 Maintenance interval of the N-EUPEX add-on coupling

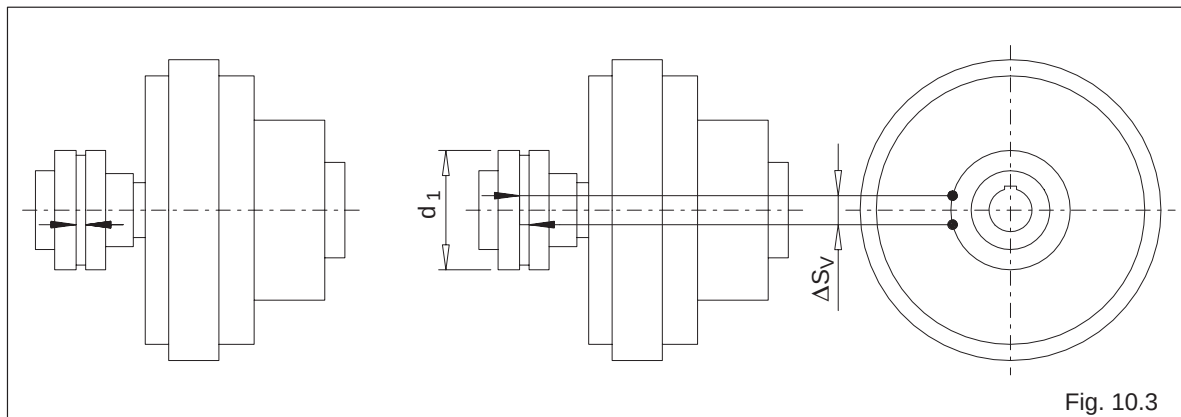


The torsional backlash between the two coupling parts must be checked after three months, then at least once a year.

If an increased coupling backlash does not impair the operation of the coupling, the flexible elements (12) can continue to be used up to a specified wear limit before being replaced. To assess wear, the permitted circumferential backlash, converted to the chord dimension ΔS_V on the outer coupling diameter, is shown in table 10.3. To obtain the dimension ΔS_V , one coupling part is rotated without torque as far as the stop and a mark applied to both side (see Fig. 10.3). If the coupling part is rotated in the opposite direction of rotation as far as the stop, the marks move apart. The distance between the marks is the chord dimension ΔS_V . If the dimension ΔS_V exceeds the value in table 10.3, the flexible elements (12) must be replaced.

Caution!

**The flexible elements must be replaced in sets.
Only identically marked flexible elements must be used.**



FLUDEX Size	370		425	490		565	655	755	887
N-EUPEX Size $[d_1]$	180	200	200	225	250	250	315	350	440
Wear mark ΔS_V [mm]	8.0	8.5	8.5	9.0	10.0	10.0	10.5	11.5	14.0

Table 10.3: Wear mark



**If the above specified maintenance instructions are not adhered to, a correct operation within the meaning of the explosion prevention requirements and Guideline 94/9/EC can no longer be guaranteed.
Use in potentially explosive areas is then not permitted.**

10.4 Replacement of the flexible elements

Only **original N-EUPEX flexible elements** must be used for replacement to guarantee troublefree torque transmission and faultfree operation.

The flexible elements (12) can be replaced without moving the coupled machines. After the screw connection part 2/3 is released, part 3 is shifted axially and turned towards part 2. The flexible elements (12) are now freely accessible.

For re-assembly, the instructions in section 6, "Assembly", and section 7, "Start-up", must be carefully observed.

10.5 Demounting the FLUDEX coupling

For part designations, see section 11.



Risk of burns after switching off!

Allow FLUDEX coupling to cool down sufficiently before beginning work!

To demount the coupling, first, the screw connection (13) must be detached from part 2/3 and part 3 uncoupled axially. The motor with the main coupling can then be demounted radially or axially.

Caution!

Before the main coupling is demounted from the hub carrier part (120) all the operating medium must be drained off (see item 10.1).

In the case of types FND, FNDB and FNDS the coupling can be demounted without having to move the coupled machines. For this, when the operating fluid has been drained off, part 3 must be uncoupled, the attaching position of part 10 marked, the screw connection (23) then released and part 10 demounted radially (see Figure 10.5 a).

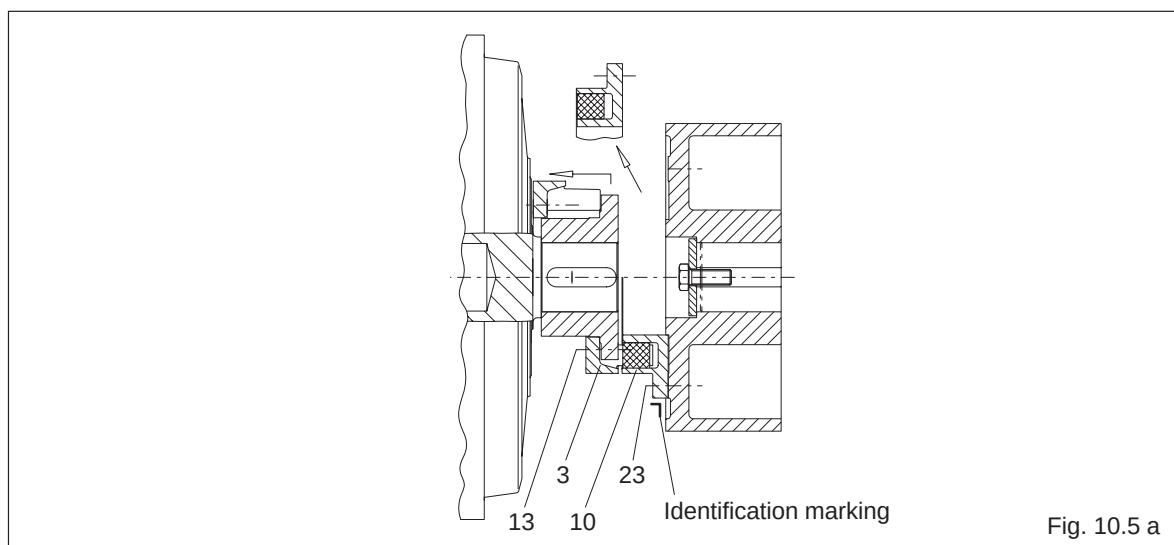


Fig. 10.5 a

The axial space thus obtained is sufficient to detach the main coupling axially from the hub carrier part (120) and demounted radially. At the same time care must be taken that the O-rings (117; 138) are exposed (see Figure 10.5).

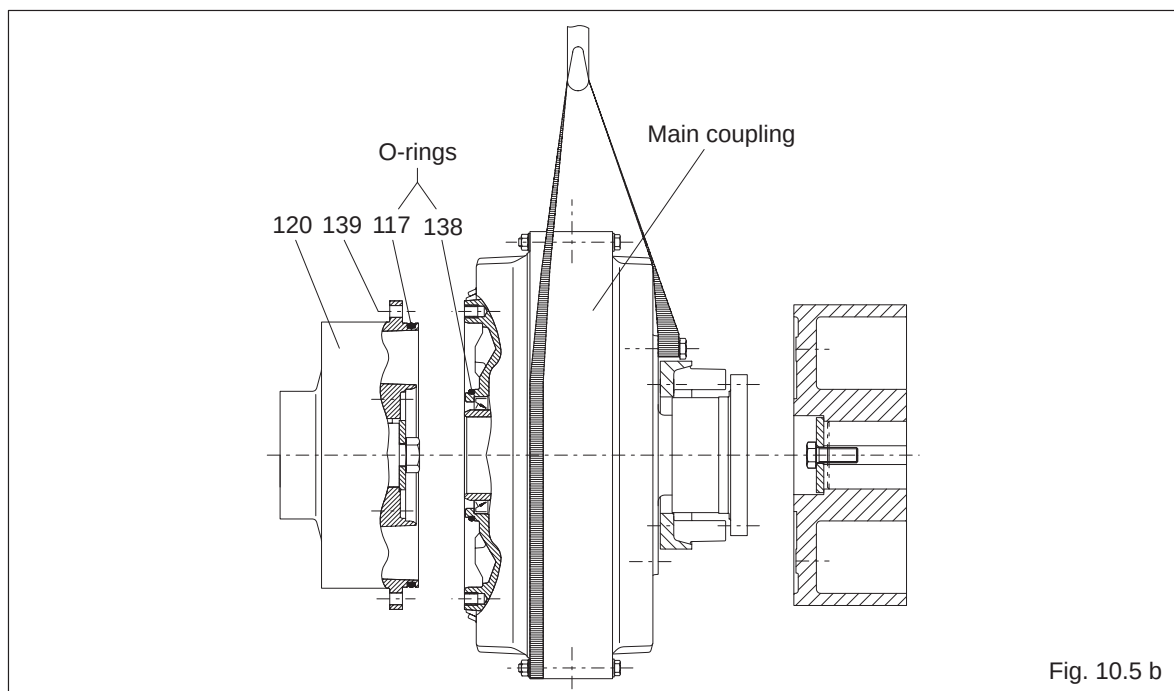
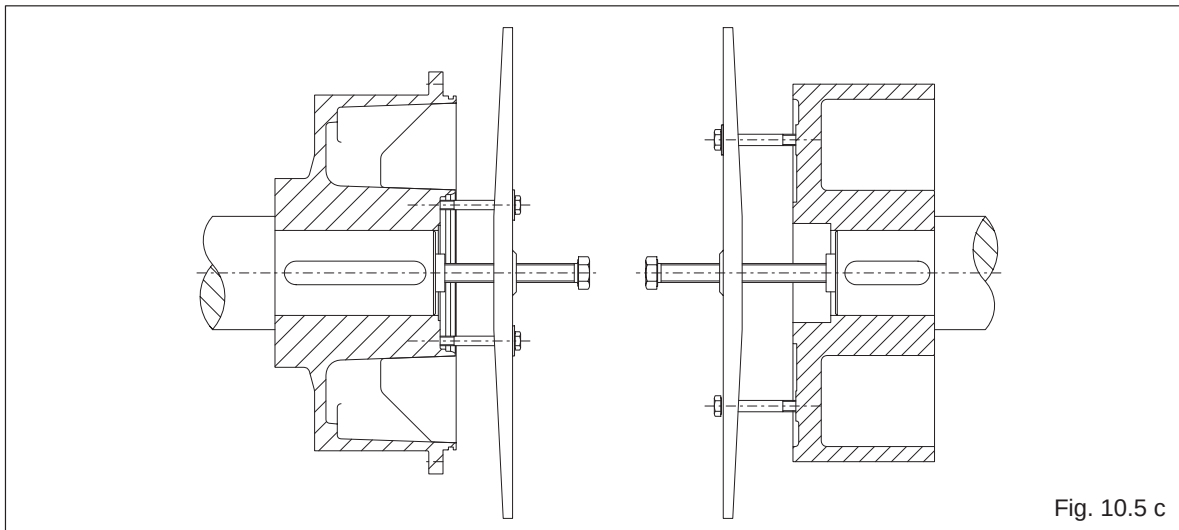


Fig. 10.5 b

The space between the shafts now suffices for the fitted hubs to be drawn off by means of commercially available tools, if necessary after releasing the axial fixture.

Located in the hub carrier part (120) on the end-face side in the hub are two tapped holes set 180° apart, which may be used to insert drawing bolts (see Figure 10.5 c).



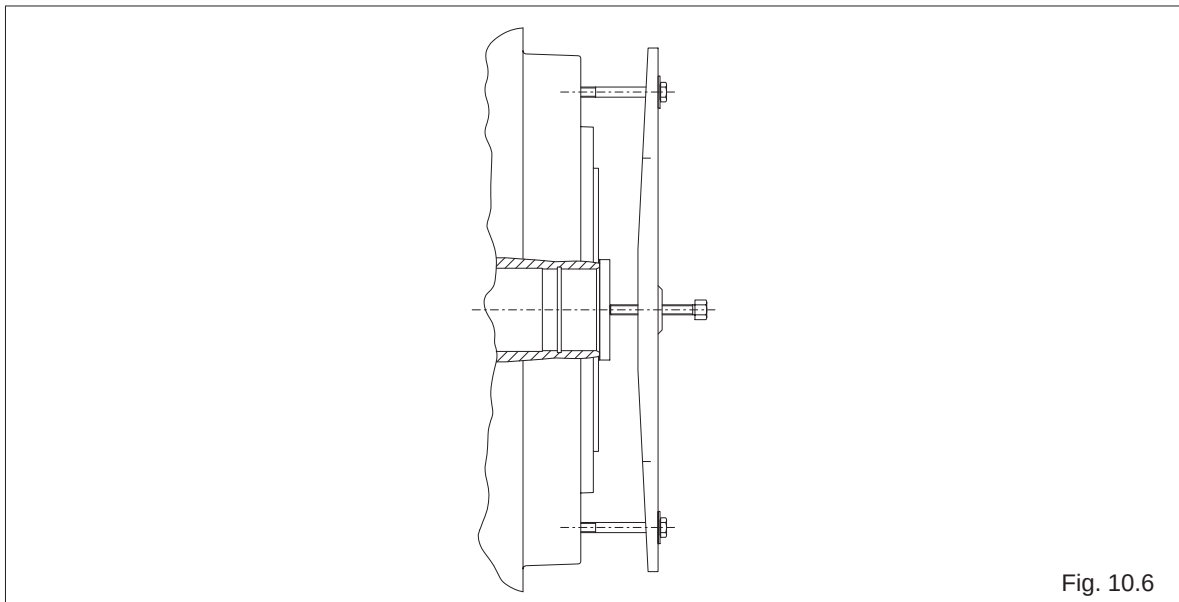
10.6 Disassembling the FLUDEX-coupling

For part designations, see section 11.

FLUDEX couplings should preferably be repaired at the manufacturer's works.

Before disassembly the housing parts must be marked on the flange in their positions relative to one another.

The housing parts of the main coupling, the shell (101) and the cover (102) are joined at the outer flange by screws (130) and nuts (131). For disassembly these screws must first be removed.



To pull off the shell and cover, a pressure spindle with a bridge and pulling screws must be applied as shown in figure 10.6. The pressure acts on the end face of the shaft. The pulling screws are inserted in the flange threads (108) of the shell or cover. After the shell and cover are pulled off, the internal parts (rolling bearings, shaft sealing rings, etc.) will be accessible. The sealing elements should be replaced every time the coupling is disassembled.

10.7 Reassembling the FLUDEX coupling

Reassembly is carried out in the reverse order (note part marking). The O-ring must be lightly greased to assist assembly. The shaft sealing rings (111; 132) are fitted with a grease filling between the dust and sealing lips. Likewise, on the design with an additional Fey laminar ring seal the space between the shaft sealing ring and the plates and the plate mounting groove in the shaft must be filled with grease.

After the old sealing thread has been removed and the sealing surfaces have been cleaned, a fresh sealing thread must be applied to places on the flange which have been sealed with sealing thread. The sealing thread must be placed in a circular form at the place provided on the sealing surface and with the thread ends crossed over. The sealing surface must be undamaged and may be lightly greased for better positioning of the sealing thread.

For tightening torques for screw connections, see section 6, item 6.6.

10.8 Refitting the FLUDEX coupling

For refitting, the instructions in section 6, "Assembly", and section 7, "Start-up", must be carefully observed.

10.9 Filling quantities for FLUDEX FN couplings sizes 370, 490, 655 and 887

Guide values for VG 22 / VG 32 mineral oil

With drive via blade shell (101) (outer wheel), valid for $T_{\max.} = 1.3 \times T_{\text{nom.}}$

Output kW	Speed 1/min										
	600	740	890	980	1180	1470	1770	2300	2950	3550	
	Oil filling quantities in litres										
1.1	5.6										
2.2	7.1	5.7									
3.0	7.9	6.4	5.1								
4.0	8.2	7.0	5.8	5.1							
5.5	14.4	7.8	6.5	5.9							
7.5	16.0	8.2	7.2	6.5	5.3						
11	18.2	14.7	8.2	7.4	6.2						
15	19.0	16.3	13.4	8.2	6.8	5.4					
18	33.5	17.3	14.4	12.9	7.2	5.8					
22	35.4	18.6	15.4	13.9	7.8	6.2	4.9				
30	38.5	19.0	17.0	15.5	12.5	6.9	5.7				
37	41.6	34.3	18.4	16.6	13.7	7.4	6.1	4.4			
45	45.0	36.2	19.0	17.7	14.7	7.9	6.6	4.7			
55	45.0	38.2	32.9	19.0	15.8	12.2	7.0	5.3			
75	76.5	43.0	35.8	33.1	17.4	14.0	7.8	6.0	4.3		
90	80.5	45.0	37.6	34.8	18.7	14.9	11.7	6.4	4.6		
110	85.2	45.0	40.1	36.7	31.8	16.0	13.1	6.8	5.1		
132	89.5	74.7	43.3	38.6	33.2	16.9	14.0	7.2	5.6	4.3	Size 370
160	95.6	80.0	45.0	41.5	35.0	18.1	15.0	10.7	6.0	4.7	
200	105.5	84.5	71.5	45.0	37.1	31.1	16.2	11.8	6.5	5.2	
250	110.0	89.7	76.9	45.0	39.7	33.0	17.4	13.2		5.8	
315		97.5	82.4	76.5	43.8	35.1	30.2	14.5			Size 490
350		102.1	84.6	78.4	45.0	36.1	31.2	15.0			
400		108.9	87.6	81.2	45.0	37.4	32.3				Size 655
500			94.1	86.1	73.3	40.2	34.2				
600			101.4	90.6	78.1	43.5	35.9				
750			110.0	98.5	82.9	66.9	38.2				
900				107.2	86.8	72.7					Size 887
1100					92.1	77.1					
1300					98.2	80.4					
1600						84.9					



With other operating fluids or $T_{\max.}$ not equal to $1.3 \times T_{\text{nom.}}$, observe different filling quantities!

10.10 Filling quantities for FLUDEX FN couplings sizes 425, 565 and 755

Guide values for VG 22 / VG 32 mineral oil

With drive via blade shell (101) (outer wheel), valid for $T_{max.} = 1.3 \times T_{nom.}$

Output kW	Speed 1/min										
	600	740	890	980	1180	1470	1770	2300	2950	3550	
	Oil filling quantities in litres										
3.0	9.7										
4.0	10.7	8.6									
5.5	12.0	9.7									
7.5	12.5	10.7	8.8	7.7							
11	22.6	12.2	10.2	9.2							
15	25.2	12.5	11.2	10.2	8.3						
18	26.6	21.4	12.0	10.8	8.9						
22	28.6	23.1	12.5	11.6	9.6						
30	46.3	25.7	21.1	12.5	10.7	8.5					
37	48.6	27.5	22.9	20.5	11.4	9.2	7.1				
45	51.5	29.0	24.5	22.0	12.3	9.8	7.8				
55	54.0	29.0	26.1	23.7	18.7	10.5	8.6				
75	60.0	49.5	29.0	26.3	21.7	11.6	9.7	6.9			
90	65.0	52.0	29.0	27.9	23.2	12.4	10.3	7.4			
110	69.0	55.0	47.5	29.0	24.9	19.0	11.0	8.3			
132		58.5	50.0	46.4	26.3	20.9	11.7	8.9	6.6		Size 425
160		63.5	52.5	48.5	28.1	22.5	17.4	9.6	6.9		
180		67.5	54.0	50.0	29.0	23.4	18.4	10.0	7.2		
200		69.0	55.5	51.5	44.4	24.3	19.5	10.3	7.6		
250			60.5	54.5	47.0	26.2	21.6	16.0	8.6		
315			67.5	59.0	50.0	28.3	23.5	16.7	9.3		
350				62.0	51.5	43.1	24.4	17.4			Size 565
400				66.0	53.5	44.9	25.5	18.5			
500					57.0	47.5	40.7	20.8			
600					61.0	50.0	42.9	22.3			
750						53.0	45.9				Size 755
900						55.5	48.0				
1100							50.5				
1200							53.0				



With other operating fluids or $T_{max.}$ not equal to $1.3 \times T_{nom.}$, observe different filling quantities!

11. Spare parts, customer-service addresses

By stocking the most important spare and wearing parts on site you can ensure that the coupling is ready for use at any time.

When ordering spare parts, always state the following:

- Part no. and designation (see item 11.2), and, if necessary, fusing temperature of the fusible safety plug
- Type / size / delivery order number (see section 1, item 1.1)
- Quantity

We guarantee only the original spare parts supplied by us.

Caution!

Please note that spare parts and accessories not supplied by us have not been tested or approved by us. The installation or use of such products may therefore impair essential characteristics of the coupling under certain circumstances and so pose an active or passive hazard. FLENDER will assume no liability or guarantee for damage caused by spare parts and accessories not supplied by FLENDER.

Please note that certain components often have special production and supply specifications and that we supply you with spare parts which comply fully with the current state of technical development as well as current legislation.

11.1 Spare parts and customer service addresses

When ordering spare parts or requesting the services of our specialist engineers, please apply first to FLENDER AG.

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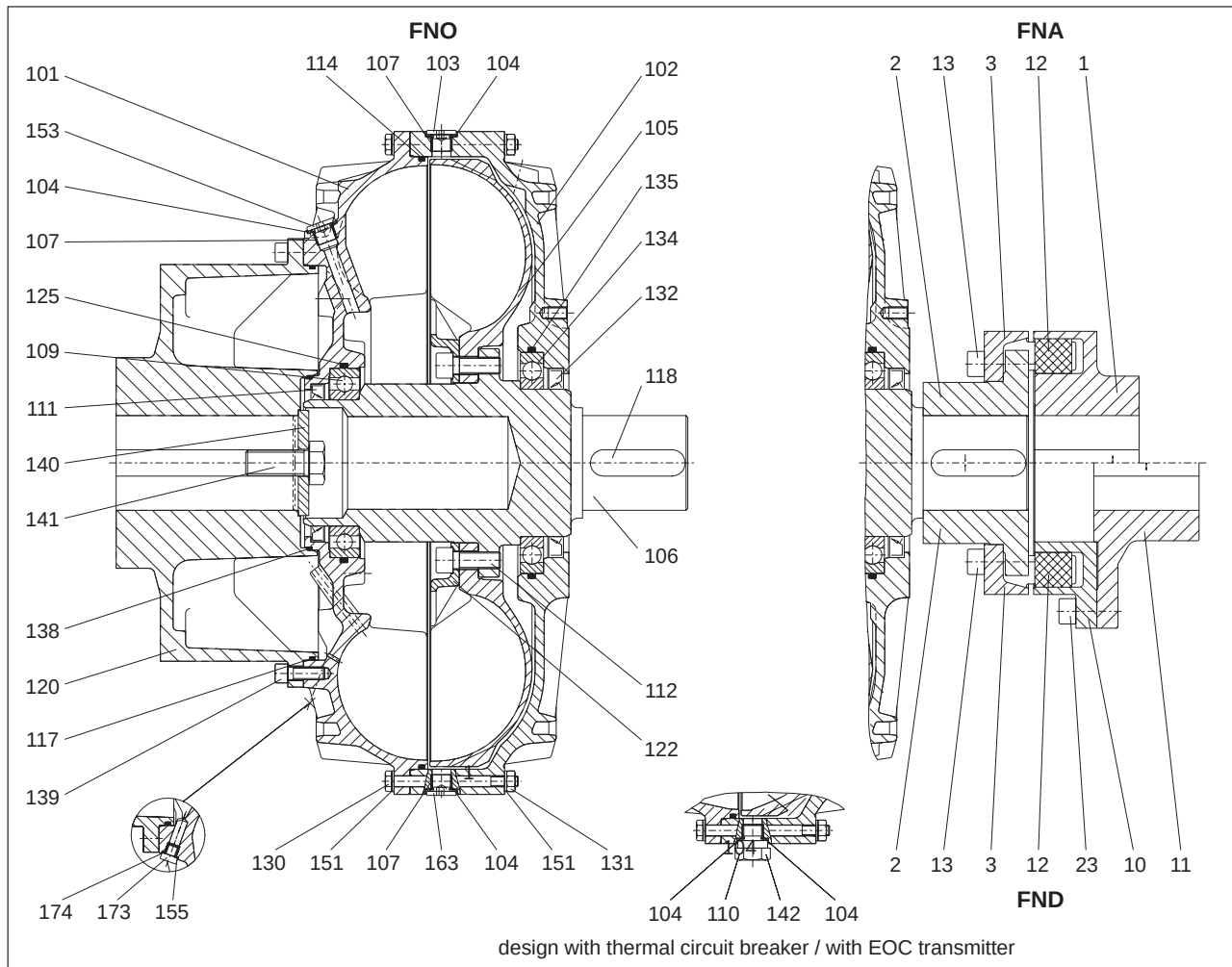
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11.2 Spare parts list Types FNO, FNA, FND

When ordering spare parts, please quote the order number of the original delivery. The order number is stamped on the coupling (blade shell) or stated in section 1.



Part no.	Designation		Part no.	Designation		Part no.	Designation	
1	Part 1 (N-EUPEX)		106	Shaft	G**	132	Shaft seal	D*
2	Part 2 (N-EUPEX)		107	Threaded insert	E	134	Rolling bearing	L
3	Part 3 (N-EUPEX)		109	Rolling bearing	L	135	O-ring	L
10	Part 10 (N-EUPEX)		110	Thermal circuit-breaker	*	138	O-ring	D*
11	Part 11		111	Shaft seal	D*	139	Cheese head screw	
12	Flexible elements (N-EUPEX)		112	Cheese head screw	E	140	Locking plate	
13	Cheese head screw		114	O-ring	D*	141	Retaining screw	
23	Cheese head screw		117	O-ring	D*	142	EOC transmitter	
			118	Parallel key		151	Washer	
101	Blade shell	G	120	Hub carrier part		153	Filling plug	
102	Cover		122	Ring	E	155	Threaded insert	E
103	Fusible safety plug	*	125	O-ring	L	163	Screw plug	
104	Sealing ring		130	Hexagon head screw		173	Drain plug (prechamber)	
105	Bucket wheel	E**	131	Hexagon nut		174	Sealing ring	

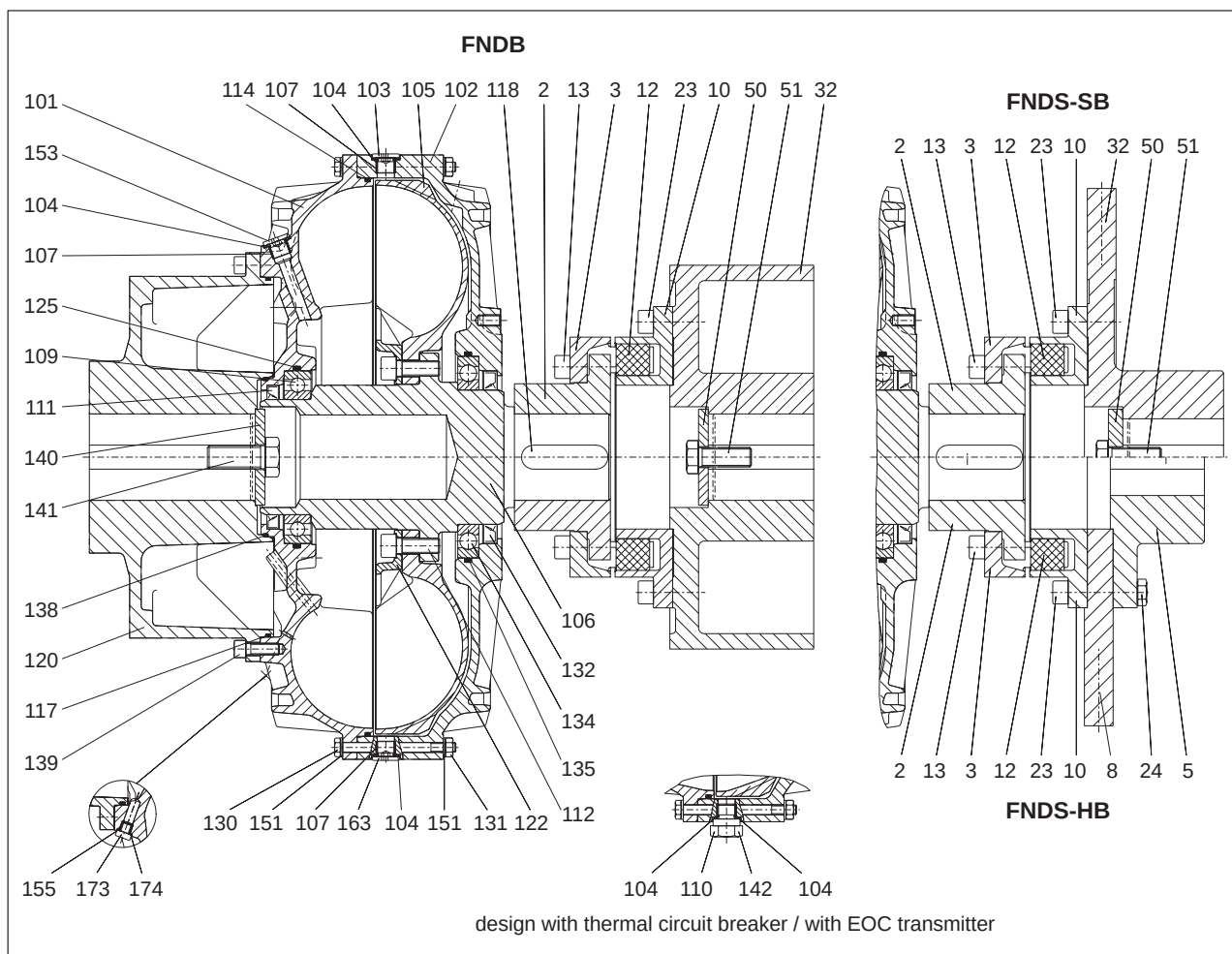
* When ordering, state fusing temperature of the fusible safety plug (stamped on the plug). The temperature specification serves to determine the sealing material (see also item 5.2).

** Shaft and blade wheel available only as assembled and balanced unit

G = subassembly E = part of a subassembly D = part of the seal set L = part of the bearing set

11.3 Spare parts list, Types FNDB, FNDS

When ordering spare parts, please quote the order number of the original delivery. The order number is stamped on the coupling (blade shell) or stated in section 1.



Part no.	Designation		Part no.	Designation		Part no.	Designation	
2	Part 2 (N-EUPEX)		104	Sealing ring		131	Hexagon nut	
3	Part 3 (N-EUPEX)		105	Bucket wheel	E**	132	Shaft seal	D*
5	Part 5		106	Shaft	G**	134	Rolling bearing	L
8	Brake disk		107	Threaded insert	E	135	O-ring	L
10	Part 10 (N-EUPEX)		109	Rolling bearing	L	138	O-ring	D*
12	Flexible elements (N-EUPEX)		110	Thermal circuit-breaker	*	139	Cheese head screw	
13	Cheese head screw		111	Shaft seal	D*	140	Locking plate	
23	Cheese head screw		112	Cheese head screw	E	141	Retaining screw	
24	Hexagon head screw		114	O-ring	D*	142	EOC transmitter	
32	Brake drum / disk		117	O-ring	D*	151	Washer	
50	Locking plate		118	Parallel key		153	Filling plug	
51	Retaining screw		120	Hub carrier part		155	Threaded insert	E
101	Blade shell	G	122	Ring	E	163	Screw plug	
102	Cover		125	O-ring	L	173	Drain plug (prechamber)	
103	Fusible safety plug	*	130	Hexagon head screw		174	Sealing ring	

* When ordering, state fusing temperature of the fusible safety plug (stamped on the plug). The temperature specification serves to determine the sealing material (see also item 5.2).

** Shaft and blade wheel available only as assembled and balanced unit

G = subassembly E = part of a subassembly D = part of the seal set L = part of the bearing set

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Fax: +27 (0) 31 - 7 05 38 72
E-mail: sales@flender.co.za

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E-mail: sales@flender.co.za

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Cnr. Ceramic Curve & Alumina Allee
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Fax: +55 (0) 31 - 33 69 21 66
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conj. 142 - Brooklin Novo
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Fax: +55 (0) 11 - 55 05 30 10
E-mail: flesao@uol.com.br

Flender Brasil Ltda.
Rua Campos Salles, 1095
sala 04 - Centro 14015 - 110,
Ribeirão Preto - SP
Phone: +55 (0) 16 - 6 35 15 90
Fax: +55 (0) 16 - 6 35 11 05
E-mail: flender.ribpreto@uol.com.br

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Markham, Ontario L3R 8V2
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CHILE / ARGENTINA / BOLIVIA ECUADOR / PARAGUAY / URUGUAY

Flender Cono Sur Limitada
Avda. Galvarino Gallardo 1534
Providencia, Santiago
Phone: +56 (0) 2 - 2 35 32 49
Fax: +56 (0) 2 - 2 64 20 25
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www.flender.cl

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Flender Liaison Office Colombia
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Fax: +57 (0) 1 - 5 70 73 35
E-mail: aguerrero@agp.com.co
www.agp.com.co

MEXICO

Flender de Mexico S.A. de C.V.
17, Pte, 713 Centro
72000 Puebla
Phone: +52 (0) 2 22 - 2 37 19 00
Fax: +52 (0) 2 22 - 2 37 11 33
E-mail: szugasti@flendermexico.com
www.flendermexico.com

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Flender de Mexico S.A. de C.V.
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Col. Granada,
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Fax: +52 (0) 55 - 55 31 69 39
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E-mail: info@flendermexico.com

PERU

Potencia Industrial E.I.R.L.
Calle Gonzales Olaechea
110-URB, La Aurora
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Phone: +51 (0) 1 - 2 42 84 68
Fax: +51 (0) 1 - 2 42 08 62
E-mail: cesarzam@potenciaindustrial.com.pe
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Caracas
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ASIA

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Fax: +91 (0) 33 - 2 23 18 57
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PEOPLE'S REPUBLIC OF CHINA

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Fax: +86 (0) 27 - 85 48 68 36
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403 Huanshi Rd. East
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(Tianjin) Co. Ltd.
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Cimaghand Co. Ltd.
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Tehran 15156
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Fax: +98 (0) 21 - 8 73 39 70
E-mail: info@cimaghand.com

ISRAEL

Greenshpon Engineering Works Ltd.
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Fax: +972 (0) 4 - 9 91 34 77
E-mail: sales@greenshpon.com
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JAPAN

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E-mail: adelameen@awalnet.net.sa

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Fax: +961 (0) 1 - 49 49 71
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46393 Bocholt
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Fax: +63 (0) 2 - 8 49 39 17
E-mail: roman@flender.com.ph

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E-mail: meso@flendertr.com

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Phone: +963 (0) 11 - 6 11 67 94
Fax: +963 (0) 11 - 6 11 09 08
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Fax: +61 (0) 2 - 97 56 48 92
E-mail: sales@flender.com.au

12. Declaration by the manufacturer / Declaration of Conformity

Declaration by the manufacturer

in accordance with EC Engineering Guideline 98/37/EC, Appendix II B

We hereby declare that the

**FLUDEX Couplings Types
FNO, FNA, FND, FNDB and FNDS**
including design in accordance with 94/9/EC

described in these Operating Instructions are intended for incorporation in a machine, and that it is prohibited to put them into service before verifying that the machine into which they are incorporated complies with the EC Guidelines (original edition 98/37/EC including any subsequent amendments thereto).

This Manufacturer's Declaration takes into account all the unified standards (inasmuch as they apply to our products) published by the European Commission in the Official Journal of the European Community.



Bocholt, 2004-09-03

Signature (person responsible for products)



Declaration of Conformity

within the meaning of EC Guideline 94/9/EC of 23.03.94
and with the legal requirements laid down for its implementation

The manufacturer A. Friedr. Flender AG, D 46393 Bocholt, declares that the

FLUDEX Couplings Types
FNO, FNA, FND, FNDB and FNDS
including design in accordance with 94/9/EC

equipment described in these operating instructions is in conformity with Article 1 and Article 8, Par graph 1) b) ii) of Guideline 94/9/EC and complies with the requirements of Guideline 94/9/EC and the standards EN 1127-1 : 1997, DIN EN 13463-1 / -5 / -6 / -8 and BGR 132 : 2003.

The technical documentation has been delivered to the named body

EXAM, BBG Prüf- und Zertifizier GmbH, D 44727 Bochum, Code Number: 0158.

Bocholt, 2004-09-03



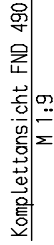
Signature (person responsible for products)



Signature (Director Engineering)



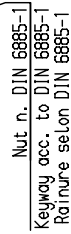
Signature (Couplings Division)



N-EUPEX

FLUDEX

Nabenträger teil
Hub currier
Partie porte-moueu



⑥ Nut n. DIN 6885-1
Keyway acc. to DIN 6885-1
Rainure selon DIN 6885-1

26-JAN-2012 12:03:47

FLENDER BOCHOLT

FLUDEX Ersatzteilzeichnung

Spare parts drawing / Dessin de pièces de rechange /
Plano de repuestos / Disegno pezzi di ricambio /
Wisselstuktekening / Desenho de peças de substituição /
Rysunek części zamiennych / Чертеж запчастей

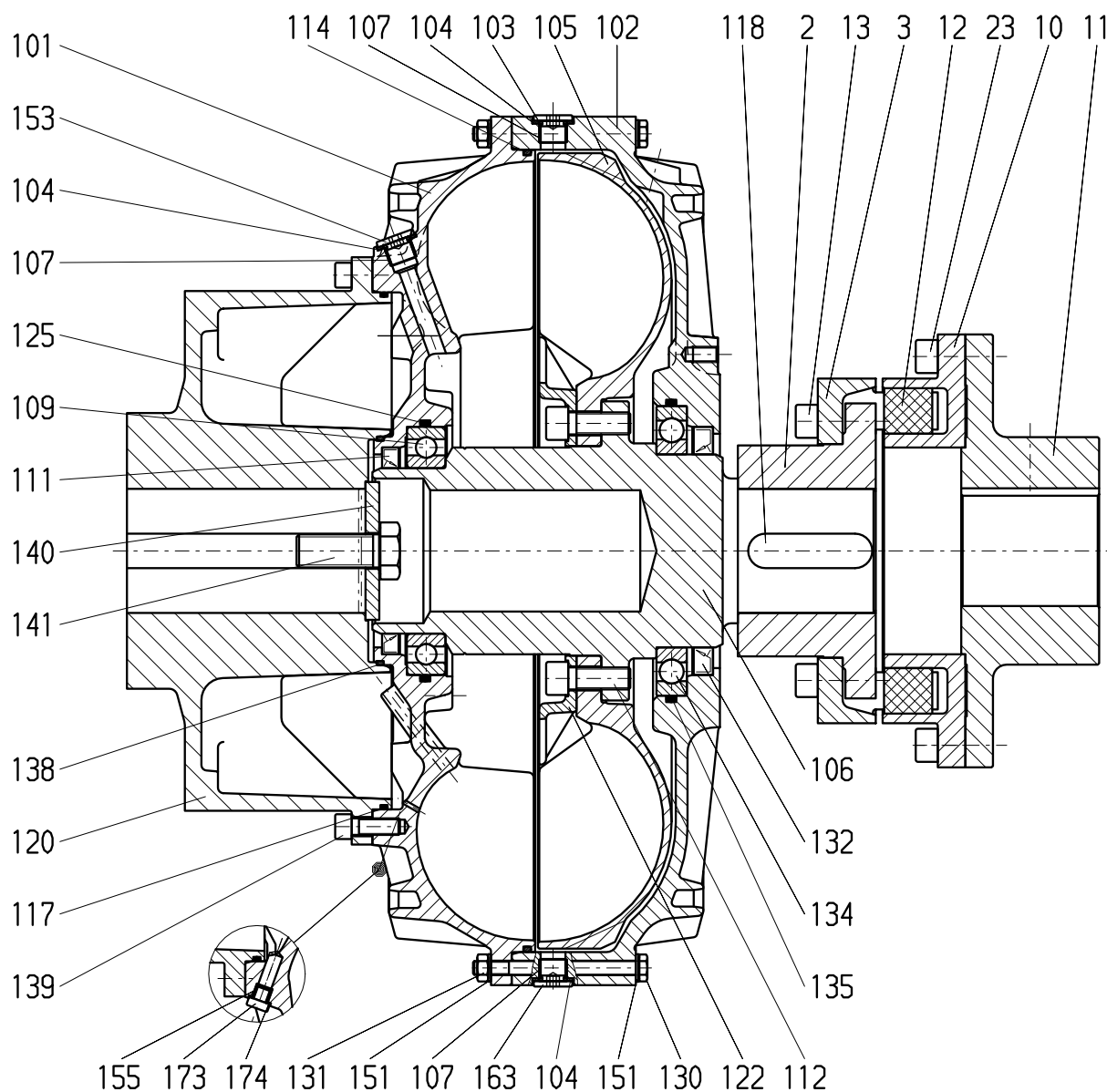
Bauart / Type / Type /
Tipo / Tipo / Constructie /
Tipo / Typ / Серии
Groesse / Size / Taille /
Tamaño / Grandezza / Maat /
Tamanho / Wielkosz / Размер

FND

370-565

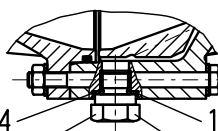
Bei Korrespondenz bitte angeben / Please quote in correspondence / Pour toute
correspondance prière d'indiquer / Indiquese en toda correspondencia / Da indicare
nella corrispondenza / Bij korrespondentie aangeven / Gentileza indicar em corres-
pondencia / Proszę podawać w korespondencji / При переписке указывать

NO.



Ausführung / Design / Modèle
Ejecución / Versione / Uitvoering
Execução / Wykonanie / Исполнение

Therm. Schalter
Thermal switch
Interrupteur thermique
Interruttore termico
Interruttore termico
Thermische schakelaar
Interrupor termico
Wyłącznik termiczny
Термореле



Geber EOC
EOC Transmitter
Transmetteur EOC
Transmissor EOC
Generatore EOC
Sensor EOC
Transmissor EOC
Czujnik EOC
Датчик EOC

104

110

104

142

A. Friedr. Flender AG, D 46395 Bocholt

Tel. +49 (0) 2871/92-0, Fax +49 (0) 2871/922596, <http://www.flender.com>

Datum/ Date

2005-02-09

Name Feldberg

Z-Nr. 6 079 571

KU

V

SIEMENS			Stückliste (SL) Parts list		Bauart FND Type Größe 490,0 Size Übersetzung ratio		Seite Page 1/2	
Bei Korrespondenz bitte angeben					4503614363			
Please quote in correspondence					SL 4637581-020 DE/EN			
Hierzu gehört Zeichnungs-Nr.								
Please refer to DWG No.					5261904,6079571			
Teil-Nr.		Menge	Benennung		Zeichnungs-Nr.		Material-Nr.	Gw(kg)
Part No.		No. off	Description		Drawings No.		Ident no.	Weight
E	0002	1	ST TEIL		5295884/A		000.000.652.528	14,1
			PART					
E*	IW06	1	ST TEIL		9012019/-		000.000.702.126	16,7
			PART					
E**	0002	1	ST TEIL		*****		000.000.501.030	10,4
			PART					
E**	0003	1	ST TEIL		*****		000.000.500.668	6,3
			PART					
E**	0013	8	ST ZYLINDERSCHRAUBE				000.000.322.204	0,0
			HEXAGON SOCKET HEAD CAP SCREW					
E	0011	1	ST FLUDEX TEIL 11 VARI				000.009.900.221	
			FLUDEX PART 11 VARI					
E*	0010	1	ST TEIL		*****		000.000.500.575	7,1
			PART					
E**	IW03	1	ST TEIL		*****		000.000.501.937	7,2
			PART					
E*	0011	1	ST TEIL		6025289/B		000.001.239.214	11,5
			PART					
E*	0023	8	ST ZYLINDERSCHRAUBE				000.000.322.204	0,0
			HEXAGON SOCKET HEAD CAP SCREW					
E	0012	1	ST PAKET (BEI KUPPLUNGEN)				000.001.375.862	0,3
			FLEXIBLES					
E*	0010	8	ST PAKET (BEI KUPPLUNGEN)				000.000.390.227	0,0
			FLEXIBLES					
E	0100	1	ST FLUDEX-KUPPLUNG		5260932/A		000.000.652.878	82,0
			FLUDEX FLUID COUPLING					
G*	0101	1	ST SCHALE		5260689/H		000.001.237.836	20,0
			SHELL					
E**	0107	2	ST GEWINDEEINSATZ (HELI-COIL)				000.000.321.027	0,1
			THREADED INSERT					
E**	0155	2	ST GEWINDEEINSATZ (HELI-COIL)				000.000.321.026	0,1
			THREADED INSERT					
E*	0102	1	ST DECKEL 490		AC43000	5260690/F	000.001.237.837	19,0
			SHELL 490 AC43000					
E**	0107	2	ST GEWINDEEINSATZ (HELI-COIL)				000.000.321.027	0,1
			THREADED INSERT					
E*	0105	1	ST SCHAUFELRAD		5297865/A		000.001.237.838	11,0
			BLADE WHEEL					
G*	0106	1	ST WELLE		5297967/E		000.000.653.165	22,0
			SHAFT					
E*	0109	1	ST WÄLZLAGER				000.000.380.620	1,3
			ROLLING CONTACT BEARING					
E*	0111	1	ST WELLENDICHTRING				000.000.652.144	0,2
			SHAFT SEAL					
E*	0112	16	ST ZYLINDERSCHRAUBE				000.000.782.135	0,0
			HEXAGON SOCKET HEAD CAP SCREW					
Die mit * gekennzeichneten Teile gehören zu einer Baugruppe (G). Die Baugruppe ist nur komplett auszutauschen.								
The parts marked with * belong to a subassembly (G). The subassembly must be replaced complete.								
Siemens Industriegetriebe GmbH, Thierbacher Str.24, 09322 Penig					Datum Date		EBBERT, FRANZ 2746	
Tel. +49(0)37381-61-0, Fax +49(0)37381-80286								

SIEMENS			Stückliste (SL) Parts list		Bauart FND Type Größe 490,0 Size Übersetzung ratio		Seite Page 2/2	
Bei Korrespondenz bitte angeben					4503614363			
Please quote in correspondence					SL 4637581-020 DE/EN			
Hierzu gehört Zeichnungs-Nr.								
Please refer to DWG No.					5261904,6079571			
Teil-Nr.		Menge	Benennung		Zeichnungs-Nr.	Material-Nr.	Gw(kg)	
Part No.		No. off	Description		Drawings No.	Ident no.	Weight	
E*	0114	1	ST	O-RING		000.000.306.509	0,1	
				O-RING				
E*	0114	3,24	M	DICHTFADEN		000.000.343.147	0,1	
				CORD SEAL				
E*	0118	1	ST	PASSFEDER		000.000.300.389	0,1	
				PARALLEL KEY				
E*	0122	1	ST	GESTAFFELTER RING FUER FLUDEX	5297848/A	000.000.653.198	2,9	
				STAGGERED RING				
E*	0125	1	ST	O-RING		000.000.307.156	0,1	
				O-RING				
E*	0130	38	ST	SECHSKANTSCHRAUBE		000.000.321.143	0,1	
				HEXAGON HEAD SCREW				
E*	0131	38	ST	SECHSKANTMUTTER		000.000.340.376	0,0	
				HEXAGON NUT				
E*	0132	1	ST	WELLENDICHTRING		000.000.652.147	0,2	
				SHAFT SEAL				
E*	0134	1	ST	WÄELZLAGER		000.000.380.526	2,4	
				ROLLING CONTACT BEARING				
E*	0135	1	ST	O-RING		000.000.306.513	0,0	
				O-RING				
E*	0151	76	ST	SCHEIBE (NUR UNTERLEGSCHNEIBE ETC.)		000.000.342.012	0,0	
				WASHER				
E*	0173	2	ST	VERSCHLUSSCHRAUBE	5241074/A	000.000.326.951	0,0	
				PLUG SCREW				
E*	0174	2	ST	DICHTRING		000.000.342.251	0,0	
				SEALING RING				
E	0103	1	ST	SCHMELZSICHERUNGSSCHRAUBE	6033005/C	000.001.250.339	0,0	
				FUSIBLE SAFETY BOLT				
E	0117	1	ST	O-RING		000.000.306.518	0,1	
				O-RING				
E	0120	1	ST	NABENTRAEGERTEIL	6136143/C	000.001.345.631	37,0	
				HUB CARRIER				
E	0138	1	ST	O-RING		000.000.307.155	0,1	
				O-RING				
E	0139	12	ST	ZYLINDERSCHRAUBE		000.000.772.521	0,0	
				HEXAGON SOCKET HEAD CAP SCREW				
E	0153	2	ST	VERSCHLUSSCHRAUBE	6127964/A	000.001.337.653	0,0	
				PLUG SCREW				
E	0163	1	ST	VERSCHLUSSCHRAUBE	6127964/A	000.001.337.653	0,0	
				PLUG SCREW				
E	0203	1	ST	SCHMELZSICHERUNGSSCHRAUBE	6033005/C	000.001.250.339	0,0	
				FUSIBLE SAFETY BOLT				
E	0274	2	ST	DICHTRING		000.000.342.251	0,0	
				SEALING RING				
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SIEMENS			Stückliste (SL) Parts list		Bauart FND Type Größe 490,0 Size Übersetzung ratio		Seite Page 1/2	
Bei Korrespondenz bitte angeben					4503614363			
Please quote in correspondence					SL 4637581-130 DE/EN			
Hierzu gehört Zeichnungs-Nr.								
Please refer to DWG No.					5261904,6079571			
Teil-Nr.		Menge	Benennung		Zeichnungs-Nr.	Material-Nr.		Gw(kg)
Part No.		No. off	Description		Drawings No.	Ident no.		Weight
E	0002	1	ST TEIL		5295884/A	000.000.652.528		14,1
			PART					
E*	IW06	1	ST TEIL		9012019/-	000.000.702.126		16,7
			PART					
E**	0002	1	ST TEIL		*****	000.000.501.030		10,4
			PART					
E**	0003	1	ST TEIL		*****	000.000.500.668		6,3
			PART					
E**	0013	8	ST ZYLINDERSCHRAUBE			000.000.322.204		0,0
			HEXAGON SOCKET HEAD CAP SCREW					
E	0011	1	ST FLUDEX TEIL 11 VARI			000.009.900.221		
			FLUDEX PART 11 VARI					
E*	0010	1	ST TEIL		*****	000.000.500.575		7,1
			PART					
E**	IW03	1	ST TEIL		*****	000.000.501.937		7,2
			PART					
E*	0011	1	ST TEIL		6025289/B	000.001.239.214		11,5
			PART					
E*	0023	8	ST ZYLINDERSCHRAUBE			000.000.322.204		0,0
			HEXAGON SOCKET HEAD CAP SCREW					
E	0012	1	ST PAKET (BEI KUPPLUNGEN)			000.001.375.862		0,3
			FLEXIBLES					
E*	0010	8	ST PAKET (BEI KUPPLUNGEN)			000.000.390.227		0,0
			FLEXIBLES					
E	0100	1	ST FLUDEX-KUPPLUNG		5260932/A	000.000.652.878		82,0
			FLUDEX FLUID COUPLING					
G*	0101	1	ST SCHALE		5260689/H	000.001.237.836		20,0
			SHELL					
E**	0107	2	ST GEWINDEEINSATZ (HELI-COIL)			000.000.321.027		0,1
			THREADED INSERT					
E**	0155	2	ST GEWINDEEINSATZ (HELI-COIL)			000.000.321.026		0,1
			THREADED INSERT					
E*	0102	1	ST DECKEL 490		AC43000 5260690/F	000.001.237.837		19,0
			SHELL 490 AC43000					
E**	0107	2	ST GEWINDEEINSATZ (HELI-COIL)			000.000.321.027		0,1
			THREADED INSERT					
E*	0105	1	ST SCHAUFELRAD		5297865/A	000.001.237.838		11,0
			BLADE WHEEL					
G*	0106	1	ST WELLE		5297967/E	000.000.653.165		22,0
			SHAFT					
E*	0109	1	ST WÄLZLAGER			000.000.380.620		1,3
			ROLLING CONTACT BEARING					
E*	0111	1	ST WELLENDICHTRING			000.000.652.144		0,2
			SHAFT SEAL					
E*	0112	16	ST ZYLINDERSCHRAUBE			000.000.782.135		0,0
			HEXAGON SOCKET HEAD CAP SCREW					
Die mit * gekennzeichneten Teile gehören zu einer Baugruppe (G). Die Baugruppe ist nur komplett auszutauschen.								
The parts marked with * belong to a subassembly (G). The subassembly must be replaced complete.								
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Tel. +49(0)37381-61-0, Fax +49(0)37381-80286								

SIEMENS			Stückliste (SL) Parts list		Bauart FND Type Größe 490,0 Size Übersetzung ratio		Seite Page 2/2	
Bei Korrespondenz bitte angeben Please quote in correspondence					4503614363 SL 4637581-130 DE/EN			
Hierzu gehört Zeichnungs-Nr. Please refer to DWG No.					5261904,6079571			
Teil-Nr.		Menge	Benennung		Zeichnungs-Nr.	Material-Nr.	Gw(kg)	
Part No.		No. off	Description		Drawings No.	Ident no.	Weight	
E*	0114	1	ST	O-RING		000.000.306.509	0,1	
				O-RING				
E*	0114	3,24	M	DICHTFADEN		000.000.343.147	0,1	
				CORD SEAL				
E*	0118	1	ST	PASSFEDER		000.000.300.389	0,1	
				PARALLEL KEY				
E*	0122	1	ST	GESTAFFELTER RING FUER FLUDEX	5297848/A	000.000.653.198	2,9	
				STAGGERED RING				
E*	0125	1	ST	O-RING		000.000.307.156	0,1	
				O-RING				
E*	0130	38	ST	SECHSKANTSCHRAUBE		000.000.321.143	0,1	
				HEXAGON HEAD SCREW				
E*	0131	38	ST	SECHSKANTMUTTER		000.000.340.376	0,0	
				HEXAGON NUT				
E*	0132	1	ST	WELLENDICHTRING		000.000.652.147	0,2	
				SHAFT SEAL				
E*	0134	1	ST	WÄELZLAGER		000.000.380.526	2,4	
				ROLLING CONTACT BEARING				
E*	0135	1	ST	O-RING		000.000.306.513	0,0	
				O-RING				
E*	0151	76	ST	SCHEIBE (NUR UNTERLEGSCHIEBE ETC.)		000.000.342.012	0,0	
				WASHER				
E*	0173	2	ST	VERSCHLUSSCHRAUBE	5241074/A	000.000.326.951	0,0	
				PLUG SCREW				
E*	0174	2	ST	DICHTRING		000.000.342.251	0,0	
				SEALING RING				
E	0103	1	ST	SCHMELZSICHERUNGSSCHRAUBE	6033005/C	000.001.250.339	0,0	
				FUSIBLE SAFETY BOLT				
E	0117	1	ST	O-RING		000.000.306.518	0,1	
				O-RING				
E	0120	1	ST	NABENTRAEGERTEIL	6136143/C	000.001.345.631	37,0	
				HUB CARRIER				
E	0138	1	ST	O-RING		000.000.307.155	0,1	
				O-RING				
E	0139	12	ST	ZYLINDERSCHRAUBE		000.000.772.521	0,0	
				HEXAGON SOCKET HEAD CAP SCREW				
E	0153	2	ST	VERSCHLUSSCHRAUBE	6127964/A	000.001.337.653	0,0	
				PLUG SCREW				
E	0163	1	ST	VERSCHLUSSCHRAUBE	6127964/A	000.001.337.653	0,0	
				PLUG SCREW				
E	0203	1	ST	SCHMELZSICHERUNGSSCHRAUBE	6033005/C	000.001.250.339	0,0	
				FUSIBLE SAFETY BOLT				
E	0274	2	ST	DICHTRING		000.000.342.251	0,0	
				SEALING RING				
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SIEMENS	Stückliste (SL) Parts list		Bauart FND Type Größe 490,0 Size Übersetzung ratio		Seite Page 1/2
	Bei Korrespondenz bitte angeben Please quote in correspondence Hierzu gehört Zeichnungs-Nr. Please refer to DWG No.		4503614363 SL 4637581-162 DE/EN 5261904,6079571		
Teil-Nr. Part No.	Menge No. off	Benennung Description	Zeichnungs-Nr. Drawings No.	Material-Nr. Ident no.	Gw(kg) Weight
E 0002	1	ST TEIL PART	5295884/A	000.000.652.528	14,1
E* IW06	1	ST TEIL PART	9012019/-	000.000.702.126	16,7
E** 0002	1	ST TEIL PART	*****	000.000.501.030	10,4
E** 0003	1	ST TEIL PART	*****	000.000.500.668	6,3
E** 0013	8	ST ZYLINDERSCHRAUBE HEXAGON SOCKET HEAD CAP SCREW		000.000.322.204	0,0
E 0011	1	ST FLUDEX TEIL 11 VARI FLUDEX PART 11 VARI		000.009.900.221	
E* 0010	1	ST TEIL PART	*****	000.000.500.575	7,1
E** IW03	1	ST TEIL PART	*****	000.000.501.937	7,2
E* 0011	1	ST TEIL PART	6025289/B	000.001.239.214	11,5
E* 0023	8	ST ZYLINDERSCHRAUBE HEXAGON SOCKET HEAD CAP SCREW		000.000.322.204	0,0
E 0012	1	ST PAKET (BEI KUPPLUNGEN) FLEXIBLES		000.001.375.862	0,3
E* 0010	8	ST PAKET (BEI KUPPLUNGEN) FLEXIBLES		000.000.390.227	0,0
E 0100	1	ST FLUDEX-KUPPLUNG FLUDEX FLUID COUPLING	5260932/A	000.000.652.878	82,0
G* 0101	1	ST SCHALE SHELL	5260689/H	000.001.237.836	20,0
E** 0107	2	ST GEWINDEEINSATZ (HELI-COIL) THREADED INSERT		000.000.321.027	0,1
E** 0155	2	ST GEWINDEEINSATZ (HELI-COIL) THREADED INSERT		000.000.321.026	0,1
E* 0102	1	ST DECKEL 490 SHELL 490 AC43000	AC43000 5260690/F	000.001.237.837	19,0
E** 0107	2	ST GEWINDEEINSATZ (HELI-COIL) THREADED INSERT		000.000.321.027	0,1
E* 0105	1	ST SCHAUFELRAD BLADE WHEEL	5297865/A	000.001.237.838	11,0
G* 0106	1	ST WELLE SHAFT	5297967/E	000.000.653.165	22,0
E* 0109	1	ST WÄLZLAGER ROLLING CONTACT BEARING		000.000.380.620	1,3
E* 0111	1	ST WELLENDICHTRING SHAFT SEAL		000.000.652.144	0,2
E* 0112	16	ST ZYLINDERSCHRAUBE HEXAGON SOCKET HEAD CAP SCREW		000.000.782.135	0,0
Die mit * gekennzeichneten Teile gehören zu einer Baugruppe (G). Die Baugruppe ist nur komplett auszutauschen. The parts marked with * belong to a subassembly (G). The subassembly must be replaced complete.					
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SIEMENS			Stückliste (SL) Parts list		Bauart FND Type Größe 490,0 Size Übersetzung ratio		Seite Page 2/2	
Bei Korrespondenz bitte angeben Please quote in correspondence					4503614363 SL 4637581-162 DE/EN			
Hierzu gehört Zeichnungs-Nr. Please refer to DWG No.					5261904,6079571			
Teil-Nr.		Menge	Benennung		Zeichnungs-Nr.	Material-Nr.	Gw(kg)	
Part No.		No. off	Description		Drawings No.	Ident no.	Weight	
E*	0114	1	ST	O-RING		000.000.306.509	0,1	
				O-RING				
E*	0114	3,24 M	DICHTFADEN			000.000.343.147	0,1	
				CORD SEAL				
E*	0118	1	ST	PASSFEDER		000.000.300.389	0,1	
				PARALLEL KEY				
E*	0122	1	ST	GESTAFFELTER RING FUER FLUDEX	5297848/A	000.000.653.198	2,9	
				STAGGERED RING				
E*	0125	1	ST	O-RING		000.000.307.156	0,1	
				O-RING				
E*	0130	38	ST	SECHSKANTSCHRAUBE		000.000.321.143	0,1	
				HEXAGON HEAD SCREW				
E*	0131	38	ST	SECHSKANTMUTTER		000.000.340.376	0,0	
				HEXAGON NUT				
E*	0132	1	ST	WELLENDICHTRING		000.000.652.147	0,2	
				SHAFT SEAL				
E*	0134	1	ST	WAEHLZLAGER		000.000.380.526	2,4	
				ROLLING CONTACT BEARING				
E*	0135	1	ST	O-RING		000.000.306.513	0,0	
				O-RING				
E*	0151	76	ST	SCHEIBE (NUR UNTERLEGSCHIEBE ETC.)		000.000.342.012	0,0	
				WASHER				
E*	0173	2	ST	VERSCHLUSSCHRAUBE	5241074/A	000.000.326.951	0,0	
				PLUG SCREW				
E*	0174	2	ST	DICHTRING		000.000.342.251	0,0	
				SEALING RING				
E	0103	1	ST	SCHMELZSICHERUNGSSCHRAUBE	6033005/C	000.001.250.339	0,0	
				FUSIBLE SAFETY BOLT				
E	0117	1	ST	O-RING		000.000.306.518	0,1	
				O-RING				
E	0120	1	ST	NABENTRAEGERTEIL	6136143/C	000.001.345.631	37,0	
				HUB CARRIER				
E	0138	1	ST	O-RING		000.000.307.155	0,1	
				O-RING				
E	0139	12	ST	ZYLINDERSCHRAUBE		000.000.772.521	0,0	
				HEXAGON SOCKET HEAD CAP SCREW				
E	0153	2	ST	VERSCHLUSSCHRAUBE	6127964/A	000.001.337.653	0,0	
				PLUG SCREW				
E	0163	1	ST	VERSCHLUSSCHRAUBE	6127964/A	000.001.337.653	0,0	
				PLUG SCREW				
E	0203	1	ST	SCHMELZSICHERUNGSSCHRAUBE	6033005/C	000.001.250.339	0,0	
				FUSIBLE SAFETY BOLT				
E	0274	2	ST	DICHTRING		000.000.342.251	0,0	
				SEALING RING				
Siemens Industriegetriebe GmbH, Thierbacher Str.24, 09322 Penig Tel. +49(0)37381-61-0, Fax +49(0)37381-80286					Datum Date	EBBERT, FRANZ 2746		

FLENDER N-EUPEX® and FLENDER N-EUPEX-DS® couplings

Types A, B
and ADS, BDS

Operating instructions
BA 3100 EN 08/2011



FLENDER couplings

SIEMENS

FLENDER N-EUPEX® and FLENDER N-EUPEX-DS® couplings

Types A, B
and ADS, BDS

Operating instructions

Translation of the original operating instructions

Technical data

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Notes

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Fitting

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Start-up
and operation

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Faults, causes
and remedy

5

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and repair

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Notes and symbols in these operating instructions

Note: The term "Operating instructions" will in the following also be shortened to "instructions" or "manual".

Legal notes

Warning-note concept

This manual comprises notes which must be observed for your personal safety and for preventing material damage. Notes for your personal safety are marked with a warning triangle or an "Ex" symbol (when applying Directive 94/9/EC), those only for preventing material damage with a "STOP" sign.



WARNING! Imminent explosion!

The notes indicated by this symbol are given to prevent **explosion damage**.
Disregarding these notes may result in serious injury or death.



WARNING! Imminent personal injury!

The notes indicated by this symbol are given to prevent **personal injury**.
Disregarding these notes may result in serious injury or death.



WARNING! Imminent damage to the product!

The notes indicated by this symbol are given to prevent **damage to the product**.
Disregarding these notes may result in material damage.



NOTE!

The notes indicated by this symbol must be treated as general **operating information**.
Disregarding these notes may result in undesirable results or conditions.



WARNING! Hot surfaces!

The notes indicated by this symbol are made to prevent **risk of burns due to hot surfaces** and must always be observed.
Disregarding these notes may result in light or serious injury.

Where there is more than one hazard, the warning note for whichever hazard is the most serious is always used. If in a warning note a warning triangle is used to warn of possible personal injury, a warning of material damage may be added to the same warning note.

Qualified personnel

The product or system to which these instructions relate may be handled only by persons qualified for the work concerned and in accordance with the instructions relating to the work concerned, particularly the safety and warning notes contained in those instructions. Qualified personnel must be specially trained and have the experience necessary to recognise risks associated with these products or systems and to avoid possible hazards.

Intended use of Siemens products

Observe also the following:



Siemens products must be used only for the applications provided for in the catalogue and the relevant technical documentation. If products and components of other makes are used, they must be recommended or approved by Siemens. The faultfree, safe operation of the products calls for proper transport, proper storage, erection, assembly, installation, start-up, operation and maintenance. The permissible ambient conditions must be adhered to. Notes in the relevant documentations must be observed.

Trademarks

All designations indicated with the registered industrial property mark ® are registered trademarks of Siemens AG. Other designations used in these instructions may be trademarks the use of which by third parties for their own purposes may infringe holders' rights.

Exclusion of liability

We have checked the content of the instructions for compliance with the hard- and software described. Nevertheless, variances may occur, and so we can offer no warranty for complete agreement. The information given in these instructions is regularly checked, and any necessary corrections are included in subsequent editions.

Note on the EC Machinery Directive 2006/42/EC

Siemens couplings in the "FLENDER couplings" product range must be treated as "components" in the sense of the EC Machinery Directive 2006/42/EC.

Therefore, Siemens needs not issue a declaration of incorporation.

Information on safe fitting, safe startup and safe operation can be found in this instruction manual; in addition the "warning-note concept" therein must be observed.

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1. Technical data

The instructions describe the coupling in horizontal mounting position with shaft-to-hub connection by cylindrical or conical bore with parallel key. If a vertical or inclined arrangement or other shaft-to-hub connections, such as shrink fit or splines to DIN 5480, are to be used, Siemens must be consulted.

The coupling described below may be used in potentially explosive areas. The couplings must have a CE marking (for marking, see item 2.2).



Couplings which do not have a CE marking, must not be used in potentially explosive areas.

If a dimensioned drawing has been made out for the coupling, the data in this drawing must be given priority. The dimensioned drawing including any other documents should be made available to the user of the system.

For part numbers and part designations, see the corresponding spare-parts drawing in section 7 or the dimensioned drawing.

1.1 Speeds, geometric data and weights

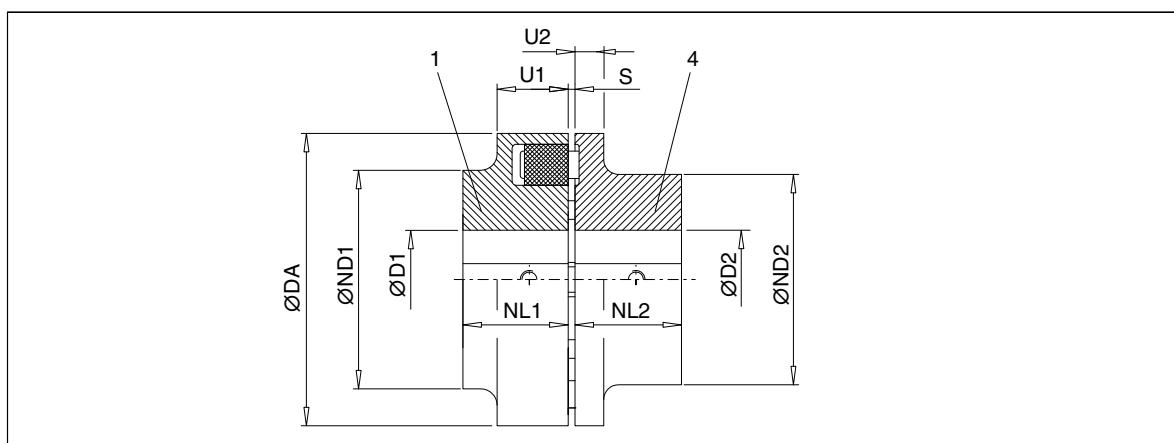


Fig. 1: Type B

Table 1: Speeds, geometric data and weights of type B

Size	Speed $n_{\max.}$ 1/min	Maximum bore 1)		DA mm	ND1 mm	ND2 mm	NL1 / NL2 mm	S mm	U1 mm	U2 mm	Weight 2) m kg
		D1 mm	D2 mm								
58	7500	19	24	58	58	40	20	2... 4	20	8	0.4
68	7000	24	28	68	68	50	20	2... 4	20	8	0.54
80	6000	30	38	80	80	68	30	2... 4	30	10	1.3
95	5500	42	42	95	76	76	35	2... 4	30	12	2.2
110	5300	48	48	110	86	86	40	2... 4	34	14	3.3
125	5100	55	55	125	100	100	50	2... 4	36	18	5.2
140	4900	60	60	140	100	100	55	2... 4	34	20	5.6
160	4250	65	65	160	108	108	60	2... 6	39	20	7.8
180	3800	75	75	180	125	125	70	2... 6	42	20	11.5
200	3400	85	85	200	140	140	80	2... 6	47	24	16
225	3000	90	90	225	150	150	90	2... 6	52	18	20
250	2750	100	100	250	165	165	100	3... 8	60	18	29
280	2450	110	110	280	180	180	110	3... 8	65	20	38

1) Maximum bore with keyway to DIN 6885/1.

2) Weights apply to maximum bores.

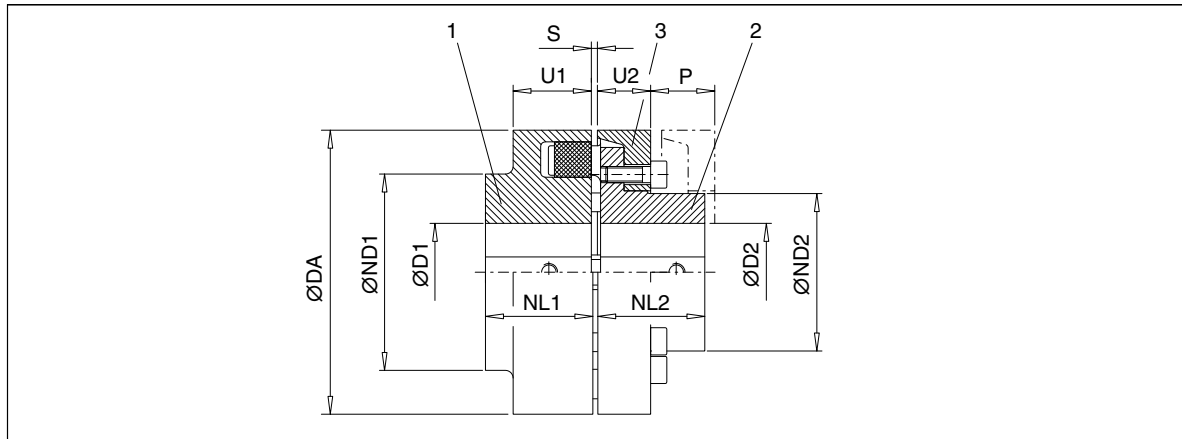


Fig. 2: Type A

Table 2: Speeds, geometric data and weights of type A

Size	Speed $n_{\max}$ 1/min	Maximum bore 1)		DA mm	ND1 mm	ND2 mm	NL1 / NL2 mm	S mm	U1 mm	U2 mm	P mm	Weight 2) m kg
		D1 mm	D2 mm									
110	5300	48	38	110	86	62	40	2... 4	34	20	33	3
125	5100	55	45	125	100	75	50	2... 4	36	23	38	4.8
140	4900	60	50	140	100	82	55	2... 4	34	28	43	6
160	4250	65	58	160	108	95	60	2... 6	39	28	47	8.4
180	3800	75	65	180	125	108	70	2... 6	42	30	50	12
200	3400	85	75	200	140	122	80	2... 6	47	32	53	17
225	3000	90	85	225	150	138	90	2... 6	52	38	61	23
250	2750	100	95	250	165	155	100	3... 8	60	42	69	31
280	2450	110	105	280	180	172	110	3... 8	65	42	73	41
315	2150	100 120	100 120	315	165 200	165 200	125	3... 8	70	47	78	57 61
350	2000	110 140	110 140	350	180 230	180 230	140	3... 8	74	51	83	78 82
400	1700	120 150	120 150	400	200 250	200 250	160	3... 8	78	56	88	112 117
440	1550	130 160	130 160	440	215 265	215 265	180	5...10	86	64	99	147 155
480	1400	145 180	145 180	480	240 300	240 300	190	5...10	90	65	104	184 200
520	1300	150 190	150 190	520	250 315	250 315	210	5...10	102	68	115	234 254
560	1200	200	200	560	320	320	220	6...12	115	80	125	329
610	1100	220	220	610	352	352	240	6...12	121	88	135	416
660	1000	240	240	660	384	384	260	6...12	132	96	145	546
710	1000	260	260	710	416	416	290	6...12	138	102	155	680

1) Maximum bore with keyway to DIN 6885/1.

2) Weights apply to maximum bores.

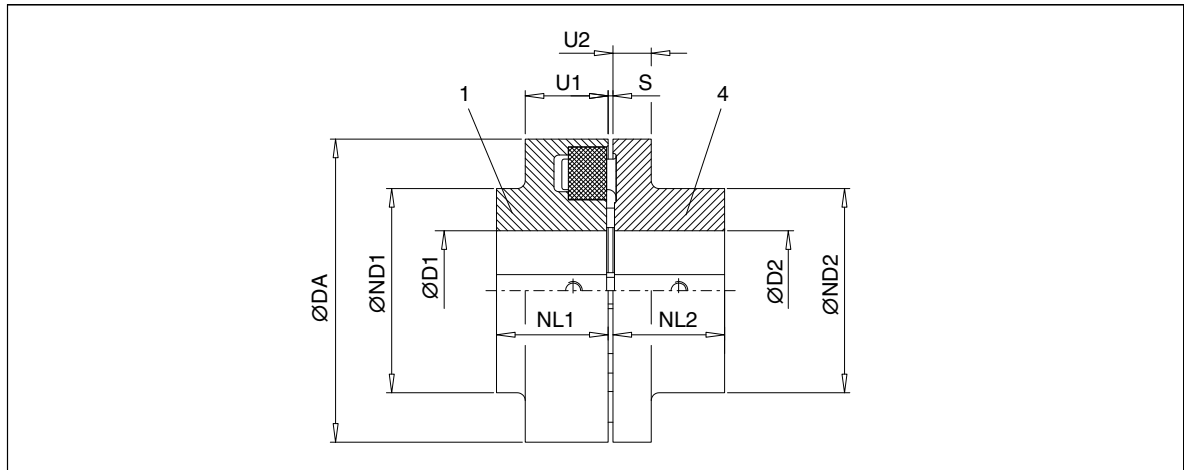


Fig. 3: Type BDS

Table 3: Speeds, geometric data and weights of type BDS

Size	Speed $n_{max.}$ 1/min	Maximum bore 1)		DA mm	ND1 mm	ND2 mm	NL1 / NL2 mm	S mm	U1 mm	U2 mm	Weight 2) m kg
		D1 mm	D2 mm								
66	7500	19	24	66	66	40	20	2.... 4	20	8	0.5
76	7000	24	28	76	76	50	20	2.... 4	20	8	0.65
88	6000	30	38	88	88	68	30	2.... 4	30	10	1.8
103	5500	42	42	103	76	76	35	2.... 4	30	12	3
118	5300	48	48	118	86	86	40	2.... 4	34	14	3.7
135	5100	55	55	135	100	100	50	2.... 4	36	18	6.1
152	4900	60	60	152	108	100	55	2.... 4	36	20	7
172	4250	65	65	172	118	108	60	2.... 6	41	20	11
194	3800	75	75	194	135	125	70	2.... 6	44	20	17
218	3400	85	85	218	150	140	80	2.... 6	47	24	23
245	3000	90	90	245	150	150	90	2.... 6	52	18	27
272	2750	100	100	272	165	165	100	3.... 8	60	18	36
305	2450	110	110	305	180	180	110	3.... 8	65	20	47

1) Maximum bore with keyway to DIN 6885/1.

2) Weights apply to maximum bores.

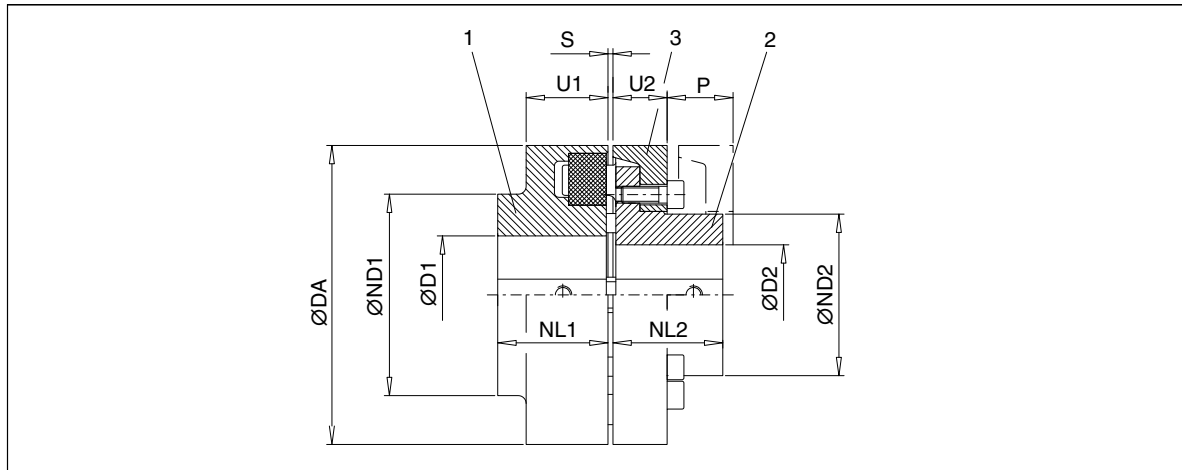


Fig. 4: Type ADS

Table 4: Speeds, geometric data and weights of type ADS

Size	Speed $n_{max.}$ 1/min	Maximum bore 1)		DA mm	ND1 mm	ND2 mm	NL1 / NL2 mm	S mm	U1 mm	U2 mm	P mm	Weight 2) m kg
		D1 mm	D2 mm									
118	5300	48	38	118	86	62	40	2... 4	34	20	33	3.5
135	5100	55	45	135	100	75	50	2... 4	36	23	38	5.5
152	4900	60	50	152	108	82	55	2... 4	36	28	43	7.7
172	4250	65	58	172	118	95	60	2... 6	41	28	47	10.5
194	3800	75	65	194	135	108	70	2... 6	44	30	50	15
218	3400	85	75	218	150	122	80	2... 6	47	32	53	21
245	3000	90	85	245	150	138	90	2... 6	52	38	61	28
272	2750	100	95	272	165	155	100	3... 8	60	42	69	40
305	2450	110	105	305	180	172	110	3... 8	65	42	73	50
340	2150	120	100 120	340	200	165 200	125	3... 8	70	47	78	72 73
380	2000	140	110 140	380	230	180 230	140	3... 8	74	51	83	100 104
430	1700	150	120 150	430	250	200 250	160	3... 8	78	56	88	135 140
472	1550	160	130 160	472	265	215 265	180	5...10	86	64	99	174 180
514	1400	180	145 180	514	300	240 300	190	5...10	90	65	104	220 237
556	1300	190	150 190	556	315	250 315	210	5...10	102	68	115	281 290

1) Maximum bore with keyway to DIN 6885/1.

2) Weights apply to maximum bores.

1.2 Flexible elements (12)

- Flexible elements may be stored for up to 5 years.
- Flexible elements must be protected against direct sunlight, artificial light with a high ultraviolet content and extreme temperatures.
- Flexible elements must not come into contact with aggressive media.
- Flexible elements must not be heated up to impermissible temperatures during fitting work (see table 5).
- Flexible elements must be replaced in sets; only identical flexible elements may be used in one coupling.

Table 5: N-EUPEX flexible elements

Material	Hardness	Remark	Mark	Temperature range
NBR	80 Shore A	Standard	black flexible elements with blue stripes	- 30 °C to + 80 °C
NBR	65 Shore A	Special, soft, shift of rotary resonance speed, nominal torque reduced	black flexible elements with green stripe	- 30 °C to + 80 °C
NBR	90 Shore A	Special, hard, shift of rotary resonance speed	black flexible elements with magenta stripe	- 30 °C to + 80 °C
NBR	80 Shore A	Special, increased (low-backlash)	black flexible elements with yellow stripe	- 30 °C to + 80 °C
NBR	65 Shore A	Special, increased (low-backlash), shift of rotary resonance speed, nominal torque reduced	black flexible elements with white stripe	- 30 °C to + 80 °C
NR	80 Shore A	Special, use at low temperature	black flexible elements with orange stripe	- 50 °C to + 50 °C
HNBR	80 Shore A	Special, use at high temperature	black flexible elements with red stripe	- 10 °C to + 100 °C
NBR	80 Shore A	Special, electrically insulating	green flexible elements	- 30 °C to + 80 °C



Electrically insulation flexible elements (green) are approved for explosion groups IIA and IIB.

High-temperature flexible elements (red marking) are not approved for use in potentially explosive areas.

Table 6: N-EUPEX-DS flexible elements

Material	Hardness	Remark	Mark	Temperature range
NBR	80/90 Shore A	Standard, 2-components, sizes 66 to 272	black flexible elements	- 30 °C to + 80 °C
NBR	90 Shore A	Standard, Sizes 305 to 556	black flexible elements	- 30 °C to + 80 °C
PU	95 Shore A	Special, Electrically insulating	blue / green flexible elements	- 30 °C to + 50 °C



Electrically insulation flexible elements (blue / green) are approved for explosion groups IIA and IIB.

2. Notes

2.1 Safety instructions and general notes



All persons involved in the installation, operation, maintenance and repair of the coupling or clutch must have read and understood these instructions and must comply with them at all times. Disregarding these instructions may cause damage to the product and material and/or injury to persons. Damage caused by disregard of these instructions will result in exclusion of liability.

During transport, installation, dismantling, operation and maintenance of the unit, the relevant safety and environmental regulations must be complied with at all times.



Lifting gears and load equipment for handling the components must be suitable for the weight of the coupling.

Depending on national regulations, coupling and clutch components may have to be disposed of separately or separated for recycling.

The coupling must be stored in a dry environment. Adequate preservation must be carried out.

Operators and users must not make any changes to the coupling themselves over and above the treatment specified in these instructions.



If there is any visible damage the coupling or clutch must not be fitted or put into operation!

The coupling must not be operated unless housed in a suitable enclosure in accordance with the standards applying. This also applies to test runs and when checking the direction of rotation.

All work on the coupling must be carried out only when it is at a standstill. Secure the drive unit to prevent unintentional switch-on. A notice should be attached to the ON switch stating clearly that work is in progress.

In addition to any generally prescribed personal safety equipment (such as safety shoes, safety clothing, helmet) **suitable safety gloves** and **suitable safety glasses** must be worn when handling the coupling or clutch!

Only spare parts made by the manufacturer Siemens must be used.

Any enquiries should be addressed to:

Siemens AG
Schlavenhorst 100
46395 Bocholt

Tel.: +49 (0)2871 / 92-0
Fax: +49 (0)2871 / 92-2596

2.2 Marking of the coupling parts for use in potentially explosive zones

Couplings which are ordered in Atex configuration, have the following marking on the outer circumference of coupling part 1:

Siemens AG	 	II 2G T4 / T5 / T6 D120 °C
46393 Bocholt - Germany		(- 50 °C) - 30 °C ≤ T _a ≤ + 80 °C / + 50 °C / + 40 °C
FLENDER couplings N-EUPEX <year built>	 	I M2

Coupling part 2 or coupling part 4 bear the marking  .

The marking is in one or two lines.

If, in addition to the CE mark, the letter "U" together with the Siemens order number has been stamped on, the coupling part has been delivered by Siemens un- or prebored.



Siemens supplies unbored and prebored couplings with CE marking only under the condition that the customer assumes the responsibility and liability for correct refinishing in a declaration of exemption.

When using electrically insulating flexible elements, the marking of the explosion group IIA, IIB is added.

2.3 Service conditions

The coupling is suited for service conditions in accordance with Directive 94/9/EC:

- Equipment group II (use above ground) of categories 2 and 3 for areas where there are explosible gas, vapour, mist, air mixtures as well as for areas where dust can form explosible atmospheres.
- The permissible temperature classes and/or maximum surface temperatures are assigned as a function of the maximum ambient temperature occurring in the immediate vicinity of the coupling (see Table 7).

Table 7: Temperature classes

Ambient temperature	Temperature class	max. surface temperature
max. 80 °C	T4	< 108 °C
max. 50 °C	T5	< 80 °C
max. 40 °C	T6	< 68 °C

- Equipment group I (underground applications) of the category M2.
- Explosion group IIA or IIB in case of electrically insulating flexible elements.



If they are to be used below ground in potentially explosive areas the couplings must only be used with drive motors, which can be switched off on occurring of an explosible atmosphere.

The machines connected by the coupling must be earthed by an earth leakage resistance < 10⁶ Ω.

If lacquered couplings are used in potentially explosive areas, the requirements made of the conductivity of the lacquer and the limitation on the thickness of the lacquer applied must be observed in accordance with EN 13463-1. Where lacquer coatings have a thickness less than 200 µm, no electrostatic charge is to be expected.

3. Fitting

3.1 Machining the finished bore

Remove the flexible elements (12).

Depreserve and clean coupling parts (1: 2; 2/3; 4).

Clamp on surfaces marked with , and align.



In case of coupling part 2/3 and coupling part 4 great caution is necessary owing to the rotating cams.

Machine the finished bore, observe maximum bore described in section 1.

Check finished bore as described in section 5.

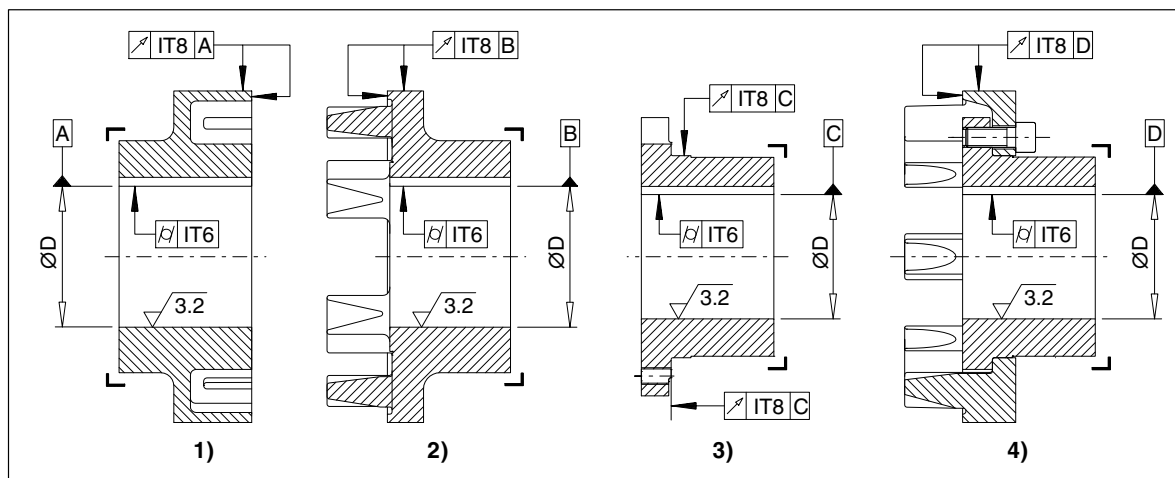


Fig. 5: Machining the finished bore

- 1) Coupling part 1
- 2) Coupling part 4
- 3) Coupling part 2
- 4) Coupling part 2/3

Table 8: Fit recommendation for bores with parallel-key connection

Description	Push fit not suitable for reversing operation		Press fit		Interference fit suitable for reversing operation		
	j6	h6	h6	k6	m6	n6	h6
Shaft tolerance	j6	h6	h6	k6	m6	n6	h6
Bore tolerance	H7	J7	K7	H7	H7	H7	M7

The fit assignment m6 / H7 is particularly suitable for many applications.



**Failure to observe these instructions may result in breakage of the coupling.
Danger from flying fragments!
The coupling may then become an explosion hazard.**

3.2 Machining the parallel keyway

- Parallel keyway to DIN 6885/1 ISO JS9 with usual operating conditions.
- Width of parallel keyway ISO P9 with reversing operation.

Arrangement of parallel keyway:

- with coupling part 1: centrally between the lands of the flexible elements
- with coupling part 1 DS: centrally between the pockets of the flexible elements
- with coupling part 2: centrally between the threaded holes
- with coupling part 2 DS: centrally between the treaded holes and offset to the recesses for the exchange of flexible elements
- with coupling part 4 / 4 DS: underneath a cam

3.3 Axial fastening

Arrange set screw on the parallel keyway. An exception are the following coupling parts:

Part 1: Size 58 / 66 : Bore $D_1 \geq 15$ mm set screw displaced by 180° relative to the keyway.
Size 68 / 76 : Bore $D_1 \geq 20$ mm set screw displaced by 144° relative to the keyway.
Size 80 / 88 : Bore $D_1 \geq 25$ mm set screw displaced by 180° relative to the keyway.
Size 95 / 103 : Bore $D_1 \geq 38$ mm set screw displaced by 180° relative to the keyway.

Part 2: Size 110 / 118 : Bore $D_1 \geq 30$ mm set screw displaced by 180° relative to the keyway.

Part 4: Size 58 / 66 : Bore $D_1 \geq 18$ mm set screw displaced by 180° relative to the keyway.
Size 68 / 76 : Bore $D_1 \geq 20$ mm set screw displaced by 180° relative to the keyway.

Position of the set screw to table 9.

Use threaded studs to DIN 916 with cup points as set screws (set-screw size to table 9).

The set screw is intended to fill out the screw thread as much as possible and must not project beyond the hub.

Alternatively use end plate; as regards recess contact Siemens.

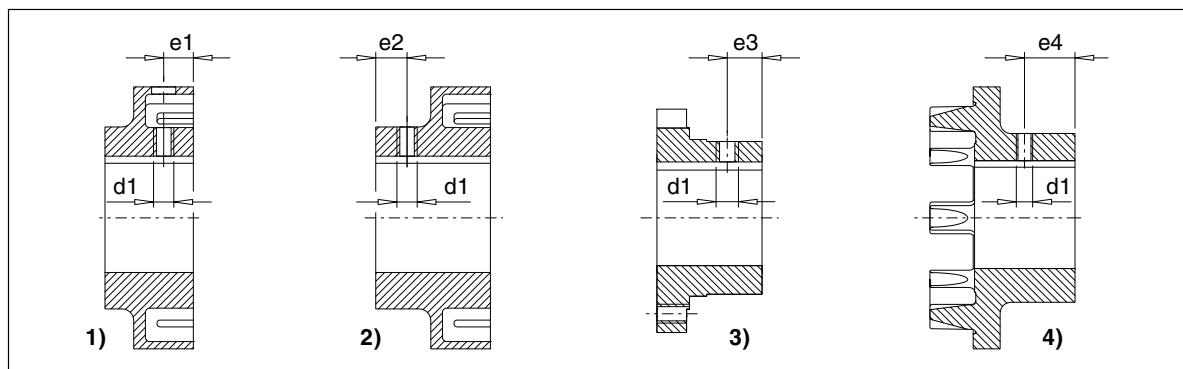


Fig. 6: Position of the set screw

- 1) Coupling part 1; position of the set screw up to size 125 / 135
- 2) Coupling part 1; position of the set screw from size 140 / 152 upwards
- 3) Coupling part 2
- 4) Coupling part 4

Table 9: Set-screw assignment, set-screw position and tightening torques

Size	58	68	80	95	110	125	140	160	180	200	225	250	280	315	350	400	440	480	520	560	610	660	710
	66	76	88	103	118	135	152	172	194	218	245	272	305	340	380	430	472	514	556	-	-	-	-
d1	M5	M6	M6	M6	M6	M8	M8	M10	M12	M12	M12	M16	M16	M16	M20	M20	M24	M24	M24	M24	M24	M24	M24
e1	10	10	11	15	18	20	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
e2	-	-	-	-	-	-	13	13	16	20	22	24	28	35	40	50	60	70	80	75	85	100	115
e3	-	-	-	-	9	12	15	20	30	30	35	40	45	50	60	70	80	90	100	100	110	130	140
e4	8	8	12	15	18	20	22	25	32	40	40	45	45	-	-	-	-	-	-	-	-	-	-
1)	3	4	4	4	4	8	8	15	25	25	25	70	70	70	130	130	230	230	230	230	230	230	230

1) Tightening torques of the set screws in Nm

Tightening torques apply to bolts with untreated surfaces which are not or only lightly oiled (coefficient of friction $\mu = 0.14$). The use of lubricant paint or lubricant, which affects the coefficient of friction " μ ", is not permitted.

The specified tightening torques T_A must be complied with, applying DIN 25202 Screw-Connection Class "C", with an output-torque scatter of $\pm 5\%$.

3.4 Balancing after machining the finished bore

The balancing quality is to be specified in accordance with the specific application (however min. G16 to DIN ISO 1940).

Balancing prescription to DIN ISO 8821 of the shaft must be observed.



Balancing bores must not affect the load-bearing capacity of the coupling parts .

The balancing bores must be applied on a large radius with sufficient distance to the lands / pockets of the flexible elements, cams and the outer circumference.



In case of coupling part 1 the bottom of the pockets of the flexible elements must not be completely drilled through.

3.5 Installation of the coupling parts

Unscrew the set screw.

Clean holes and shaft ends.

Coat the bores of the coupling parts (1; 2; 4) and the shafts with MoS₂ mounting paste (e.g. Microgleit LP 405).

If demounted, position the coupling part 3 on the shaft before pulling coupling part 2 on.



Coupling parts (1; 2; 4) with tapered bore and parallel-key connection must be fitted in cold condition and secured with suitable end plates, without drawing the coupling parts (1; 2; 4) further onto the taper (fitting dimension = 0).

Place coupling parts (1; 2; 4), with cylindrical bore heat up to max. + 150 °C, if necessary. When heating up observe the temperature range of the flexible elements (12) (see table 5 or 6), if necessary demount the flexible elements (12).



Heated coupling parts form an explosion hazard, therefore a non-explosive environment must be ensured.

Axial securing is effected by means of the set screw or end plate. When securing by set screw the shaft must not project or be set back from the inner sides of the hub.

Fit the set screw or end plate (tightening torques of the set screw to table 9).



**Failure to observe these instructions may result in breakage of the coupling.
Danger from flying fragments!
The coupling may then become an explosion hazard.**

If necessary, re-fit flexible elements (12). Observe the temperature range (see table 5 or 6).

Align the coupling as described in item 3.7.

3.6 Possible misalignments

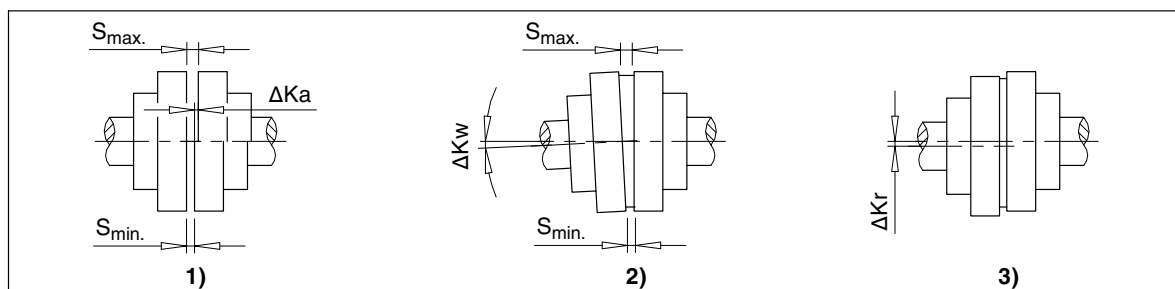


Fig. 7: Possible misalignments

- 1) Axial misalignment (ΔK_a)
- 2) Angular misalignment (ΔK_w)
- 3) Radial misalignment (ΔK_r)

3.6.1 Axial misalignment

The gap dimension ΔK_a should be set within the deviation permitted for the dimension "S" (see section 1).

3.6.2 Angular misalignment

The angular misalignment ΔK_w can be measured as difference of the gap dimension ($\Delta S = S_{\max.} - S_{\min.}$). $\Delta S_{\text{perm.}}$, see table 10.

If required, the permissible angular misalignment ΔK_w can be calculated as follows:

$$\Delta K_{w_{\text{perm.}}} \text{ in RAD} = \Delta S_{\text{perm.}} / DA \quad \Delta S_{\text{perm.}}, \text{ see table 10.}$$

$$\Delta K_{w_{\text{perm.}}} \text{ in GRAD} = (\Delta S_{\text{perm.}} / DA) \times (180 / \pi) \quad "DA" \text{ in mm see section 1.}$$

3.6.3 Radial misalignment

The permissible radial misalignment $\Delta K_{r_{\text{perm.}}}$ can be found in table 10 (depending on the operating speed).

3.7 Alignment



When aligning the angular and radial misalignment should be kept as low as possible.

Misalignment values specified in table 10 are maximum permissible overall values in operation, resulting from mispositioning through imprecision during alignment and misalignment through operation (e.g. deformation through load, heat expansion).

Reduced misalignment in the coupling minimises expected wear on the flexible elements. Misalignment in the coupling gives rise to restorative forces which may impose inadmissible stress on adjacent machine parts (e.g. bearings).

3.8 Shaft-displacement values during operation



The following maximum permissible misalignments must by no means be exceeded during operation.

When aligning the angular and radial misalignment should be kept appreciably smaller (tending towards zero).

Table 10: Shaft-displacement values $\Delta S_{perm.}$ and $\Delta K_{perm.}$ maximum permissible during operation, stated in mm (rounded)

Type, size		Coupling speed in 1/min								
A, B	ADS, BDS	250	500	750	1000	1500	2000	3000	4000	5000
58	66	0.4	0.3	0.25	0.2	0.2	0.15	0.15	0.1	0.1
68	76	0.4	0.3	0.25	0.2	0.2	0.15	0.15	0.1	0.1
80	88	0.4	0.3	0.25	0.2	0.2	0.15	0.15	0.1	0.1
95	103	0.5	0.35	0.25	0.25	0.2	0.2	0.15	0.1	0.1
110	118	0.5	0.35	0.3	0.25	0.2	0.2	0.15	0.1	0.1
125	135	0.5	0.4	0.3	0.25	0.25	0.2	0.15	0.15	0.1
140	152	0.6	0.4	0.35	0.3	0.25	0.2	0.2	0.15	
160	172	0.6	0.5	0.4	0.35	0.3	0.25	0.2	0.15	
180	194	0.6	0.5	0.4	0.35	0.3	0.25	0.2		
200	218	0.8	0.55	0.45	0.4	0.3	0.3	0.2		
225	245	0.8	0.55	0.5	0.4	0.35	0.3	0.25		
250	272	0.8	0.6	0.5	0.4	0.35	0.3			
280	305	1	0.7	0.6	0.5	0.4	0.35			
315	340	1	0.7	0.6	0.5	0.4	0.35			
350	380	1	0.8	0.6	0.6	0.5				
400	430	1.2	0.9	0.7	0.6	0.5				
440	472	1.3	1	0.7	0.7	0.6				
480	514	1.4	1	0.8	0.7					
520	556	1.5	1.1	0.9	0.8					
560		1.6	1.2	1	0.8					
610		1.8	1.3	1	0.9					
660		1.9	1.4	1.1	1					
710		2	1.5	1.2						

The numerical values of the table, as well as the intermediate values, can be calculated as follows:

$$\Delta K_{rperm.} = \Delta S_{perm.} = (0.1 + DA / 1000) \times 40 / \sqrt{n}$$

Coupling speed "n" in 1/min
"DA" in mm, see section 1.
Radial misalignment $\Delta K_{rperm.}$ in mm

For speeds < 250 1/min the values in the colon "250 1/min" in table 10 apply.

3.9 Assignment of the tightening torques and wrench widths



The use of an impact screwdriver is not permissible!

Tightening torques apply to bolts with untreated surfaces which are not or only lightly oiled (coefficient of friction $\mu = 0.14$). The use of lubricant paint or lubricant, which affects the coefficient of friction " μ ", is not permitted.

The specified tightening torques T_A must be complied with, applying DIN 25202 Screw-Connection Class "C", with an output-torque scatter of $\pm 5\%$.

The tightening torques and wrench widths of the set screws are specified in table 9.

Table 11: Tightening torques for part 13 of Types A and ADS

N-EUPEX coupling	N-EUPEX-DS coupling	Tightening torque T_A and wrench width SW for hexagon socket screws to DIN EN ISO 4762	
		T_A Nm	SW mm
110	118	14	6
125	135	17.5	6
140	152	29	8
160	172	35	8
180	194	44	8
200	218	67.5	10
225	245	86	10
250	272	145	14
280	305	185	14
315	340	200	14
350	380	260	17
400	430	340	17
440	472	410	17
480	514	550	19
520	556	670	19
560		710	19
610		1450	22
660		1450	22
710		1450	22

4. Start-up and operation



Bolt-tightening torques for the coupling and tightening torques for the foundation bolts of the coupled machine must be checked before start-up. Enclosures (coupling protection, contact guard) must be fitted!

Overload conditions during start-up cannot be excluded. If the coupling breaks through overload, metal parts may fly off and cause personal injury and/or material damage.



If it is to be used below ground in potentially explosive areas, the coupling, which is made of cast iron or steel, must be provided with a robust casing to preclude the risk of ignition from e.g. friction, impact or friction sparks.

The depositing of heavy metal oxides (rust) on the coupling must be precluded by the casing or other suitable precautions.

The coupling must run with little noise and without vibration. Irregular behaviour must be treated as a fault requiring immediate remedy. In case of fault the drive must be stopped at once. The necessary measures for repair must be taken in accordance with the safety regulations applying.

5. Faults, causes and remedy

5.1 Possible cause of fault

Change in alignment:

- Rectify the cause of the change in alignment (e.g. loose foundation bolts).
- Align the coupling.
- Check the axial fastening and, if necessary, adjust.
- Wear check of the flexible elements (12) as described in section 6.

Flexible elements (12) worn:

- Wear check of the flexible elements (12) as described in section 6, if necessary replace flexible elements (12).

5.2 Incorrect use



**Failure to observe these instructions may result in breakage of the coupling.
Danger from flying fragments!
Through incorrect use the coupling may become an explosion hazard.**

5.2.1 Frequent faults when selecting the coupling and/or coupling size

- Important information for describing the drive and the environment are not communicated.
- System torque too high.
- System speed too high.
- Application factor not correctly selected.
- Chemically aggressive environment not taken into consideration.
- The ambient temperature is not permissible.
- Finished bore with inadmissible diameter and/or inadmissible assigned fits.
- Machining of parallel keyways of which the width across corners is greater than the width across corners of the parallel keyways to DIN 6885/1 with a maximum permissible bore.
- The transmission capacity of the shaft-to-hub connection is not appropriate to the operating conditions.
- Maximum load or overload conditions are not being taken into consideration.
- Dynamic load conditions are not being taken into consideration.
- Shaft-to-hub connection resulting in impermissible material stress on the coupling.
- Operating conditions are being changed without authorisation.
- Coupling and machine / drive train form a critical torsional, axial and bending vibration system.
- Fatigue torque load too high.

5.2.2 Frequent faults when fitting the coupling

- Components with transport or other damage are being fitted.
- When fitting coupling parts in a heated condition, already fitted N-EUPEX flexible elements (12) are being excessively heated.
- The shaft diameter is beyond the specified tolerance range.
- Coupling parts are being interchanged, i.e. their assignment to the specified shaft is incorrect.

- Specified axial fastenings are not fitted.
- Specified tightening torques are not being adhered to.
- Bolts are inserted dry or greased.
- Flange surfaces of screwed connections have not been cleaned.
- Alignment and/or shaft-misalignment values do not match the specifications in the instructions manual.
- The coupled machines are not correctly fastened to the foundation, and as a result shifting of the machines, e.g. through loosening of the foundation-screw connection, is causing excessive displacement of the coupling parts.
- The coupled machines are not sufficiently earthed.
- N-EUPEX flexible elements (12) are not fitted.
- The coupling guard used is not suitable.

5.2.3 Frequent faults in maintenance

- Maintenance intervals are not being adhered to.
- Original N-EUPEX spare parts are not being used.
- Old or damaged N-EUPEX spare parts are being used.
- Different N-EUPEX flexible elements (12) are being used.
- Leakage in the vicinity of the coupling is not being identified and as a result chemically aggressive media are damaging the coupling.
- Fault indications (noise, vibration, etc.) are not being observed.
- Specified tightening torques are not being adhered to.
- Alignment and/or shaft-misalignment values do not match the specifications in the instructions manual.

6. Maintenance and repair

6.1 Maintenance interval



On Types A and B the torsional backlash between the two coupling parts must be checked after three months, then at least once a year.

On Types ADS and BDS the torsional backlash of the flexible elements (12) need be checked against Directive 94/9/EC only if a failure of the flexible elements (12) and as a result a stoppage of the drive result in an explosion hazard. It is recommended that preventive maintenance, also on coupling types ADS and BDS, include the regular check of the torsional backlash.

The flexible elements (12) must be replaced, when the torsional backlash exceeds the value stated in table 12.

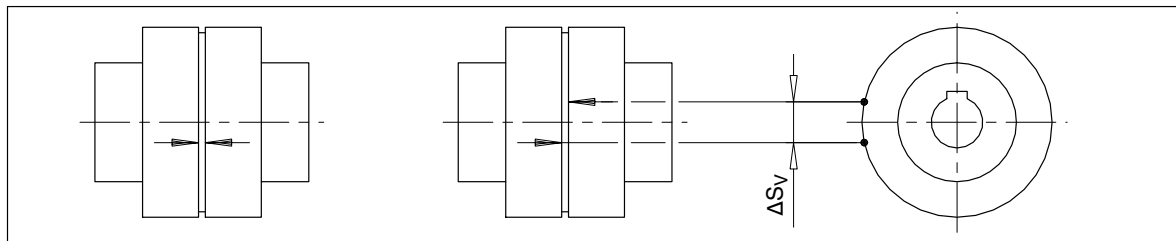


Fig. 8: Wear mark

Table 12: Wear mark for the torsional backlash, Types A and B

Size	58	68	80	95	110	125	140	160	180	200	225	250	280	315	350	400	440	480	520	560	610	660	710
Wear mark ΔS_V (mm)	5.5	5.5	5.0	6.0	7.0	8.0	8.0	8.0	8.0	8.5	9.0	10.0	11.5	10.5	11.5	13.0	14.0	15.5	17.5	17.5	19.5	21.0	22.5

Table 13: Wear mark for the torsional backlash, Types ADS and BDS

Size	66	76	88	103	118	135	152	172	194	218	245	272	305	340	380	430	472	514	556
Wear mark ΔS_V (mm)	6.0	7.0	5.0	7.0	9.0	10.5	11.5	9.0	8.0	7.0	6.5	7.0	8.0	6.5	7.0	10.0	12.0	14.0	16.0



If the above specified maintenance instructions are not adhered to, a correct operation within the meaning of the explosion prevention requirements and/or Directive 94/9/EC can no longer be guaranteed on Types A and B (types with form fitting of the metal parts).

Use in potentially explosive areas is then not permitted.



Failure to observe these instructions may result in breakage of the coupling.
Danger from flying fragments!

6.2 Replacement of wearing parts

Replacement of the flexible elements (12) is possible without shifting the coupled machines only on Types A and ADS.

After releasing the connection of coupling parts 2/3, coupling part 3 is shifted axially. The flexible elements (12) will then be made freely accessible by rotating coupling part 2. To facilitate the release of coupling part 3, on sizes 225 to 430 forcing-off threads are provided in coupling part 1. From size 440 onwards the forcing-off thread is located within coupling part 3.

**Fig. 9:** Releasing coupling part 3

1) Forcing-off thread in coupling part 1

2) Forcing-off thread in coupling part 3

The flexible elements (12) must be replaced in sets. Only identical flexible elements (12) may be used.

For re-assembly, the instructions in sections 3 and 4 must be observed.

6.3 Demounting the coupling parts in case of shaft-to-hub connection with parallel key

Move the coupled machines apart.

Remove the axial fastening (set screw, end plate). Mount suitable detaching device. Using a burner, heat coupling part (1; 2; 4) along its length and above the parallel keyway (max. + 80 °C). When heating up observe the temperature range of the flexible elements (12) (see tables 5 and 6), if necessary demount the flexible elements (12).



Burner and heated coupling parts form an explosion hazard; therefore a non-explosive environment must be ensured.

Pull the coupling part off. Examine the hub bore and the shaft for damage, and protect against rust. Damaged parts must be replaced.

For re-assembly, the instructions in sections 3 and 4 must be observed.

7. Stocking spare parts

7.1 Spare parts

For ordering spare parts state the following data, as far as possible:

- Siemens order number and position
- Drawing number
- Coupling type and coupling size
- Part number (see spare-parts list)
- Bore, bore tolerance, keyway and balancing as well as particular characteristics such as flange-connection dimensions, intermediate-shaft length, brake-drum dimensions
- Any special details such as temperature, electrically insulating

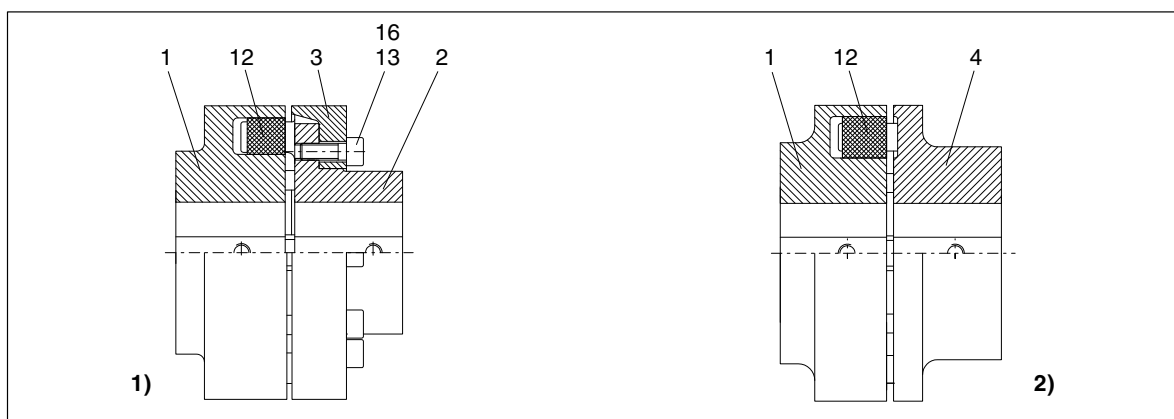


Fig. 10: Spare-parts drawing

1) Types A and ADS

2) Types B and BDS

Table 14: Spare-parts list

Type A, ADS		Type B, BDS	
Part number	Designation	Part number	Designation
1	Coupling part 1	1	Coupling part 1
2	Coupling part 2	4	Coupling part 4
3	Coupling part 3	12	Flexible element
12	Flexible element		
13	Cheese-head bolt		
16	Parallel pin only on Type A Size 560 to 710		

8. Declarations

8.1 EC-Declaration of conformity



EC-Declaration of conformity

within the meaning of EC Directive 94/9/EC of 23.03.1994 and the legal requirements laid down for its implementation

The manufacturer, Siemens AG, 46395 Bocholt, Germany, declares that the equipment described in these operating instructions:

FLENDER N-EUPEX® and FLENDER N-EUPEX-DS® couplings

Types A, B and ADS, BDS

is in conformity with Article 1 and Article 8, Paragraph 1 b) ii) of Directive 94/9/EC and complies with the requirements of Directive 94/9/EC and the following standards:

DIN EN 1127-1 : 02-2008
DIN EN 13463-1 : 07-2009

The technical documentation has been delivered to the body named below:

DEKRA EXAM GmbH, D - 44727 Bochum, code number: 0158.

Bocholt, 2011-08-22

Andre Jansen
(Director Engineering KUE)

Bocholt, 2011-08-22

Nicola Warning
(Director Business Subsegment KU)

Further Information:

"FLENDER gear units" on the Internet

www.siemens.com/gearunits

"FLENDER couplings" on the Internet

www.siemens.com/couplings

Service & Support:

<http://support.automation.siemens.com/WW/view/en/10803928/133300>

Lubricants:

<http://support.automation.siemens.com/WW/view/en/42961591/133000>

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Subject to modifications

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